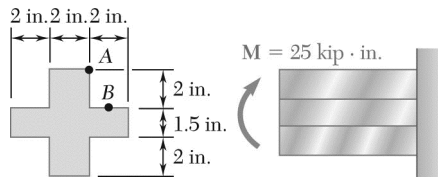


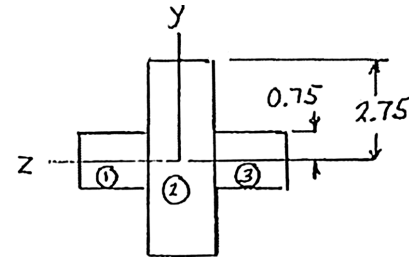
CHAPTER 4



PROBLEM 4.1

Knowing that the couple shown acts in a vertical plane, determine the stress at (a) point A, (b) point B.

SOLUTION



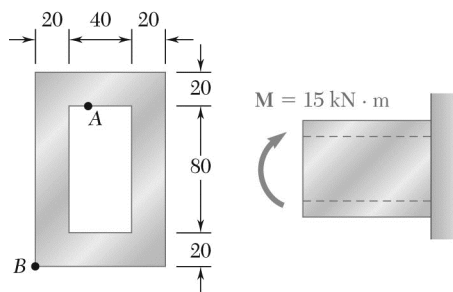
For rectangle: $I = \frac{1}{12}bh^3$

For cross sectional area:

$$I = I_1 + I_2 + I_3 = \frac{1}{12}(2)(1.5)^3 + \frac{1}{12}(2)(5.5)^3 + \frac{1}{12}(2)(1.5)^3 = 28.854 \text{ in}^4$$

(a) $y_A = 2.75 \text{ in.}$ $\sigma_A = -\frac{My_A}{I} = -\frac{(25)(2.75)}{28.854}$ $\sigma_A = -2.38 \text{ ksi} \blacktriangleleft$

(b) $y_B = 0.75 \text{ in.}$ $\sigma_B = -\frac{My_B}{I} = -\frac{(25)(0.75)}{28.854}$ $\sigma_B = -0.650 \text{ ksi} \blacktriangleleft$



Dimensions in mm

PROBLEM 4.2

Knowing that the couple shown acts in a vertical plane, determine the stress at (a) point *A*, (b) point *B*.

SOLUTION

For rectangle: $I = \frac{1}{12}bh^3$

Outside rectangle: $I_1 = \frac{1}{12}(80)(120)^3$

$$I_1 = 11.52 \times 10^6 \text{ mm}^4 = 11.52 \times 10^{-6} \text{ m}^4$$

Cutout: $I_2 = \frac{1}{12}(40)(80)^3$

$$I_2 = 1.70667 \times 10^6 \text{ mm}^4 = 1.70667 \times 10^{-6} \text{ m}^4$$

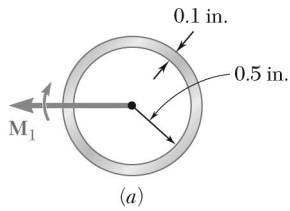
Section: $I = I_1 - I_2 = 9.81333 \times 10^{-6} \text{ m}^4$

(a) $y_A = 40 \text{ mm} = 0.040 \text{ m}$ $\sigma_A = -\frac{My_A}{I} = -\frac{(15 \times 10^3)(0.040)}{9.81333 \times 10^{-6}} = -61.6 \times 10^6 \text{ Pa}$

$$\sigma_A = -61.6 \text{ MPa} \quad \blacktriangleleft$$

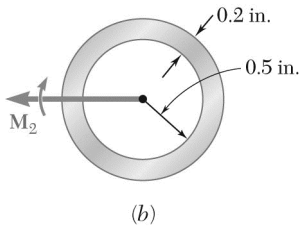
(b) $y_B = -60 \text{ mm} = -0.060 \text{ m}$ $\sigma_B = -\frac{My_B}{I} = -\frac{(15 \times 10^3)(-0.060)}{9.81333 \times 10^{-6}} = 91.7 \times 10^6 \text{ Pa}$

$$\sigma_B = 91.7 \text{ MPa} \quad \blacktriangleleft$$



PROBLEM 4.3

Using an allowable stress of 16 ksi, determine the largest couple that can be applied to each pipe.



SOLUTION

$$(a) \quad I = \frac{\pi}{4} (r_o^4 - r_i^4) = \frac{\pi}{4} (0.6^4 - 0.5^4) = 52.7 \times 10^{-3} \text{ in}^4$$

$$c = 0.6 \text{ in.}$$

$$\sigma = \frac{Mc}{I} : \quad M = \frac{\sigma I}{c} = \frac{(16)(52.7 \times 10^{-3})}{0.6}$$

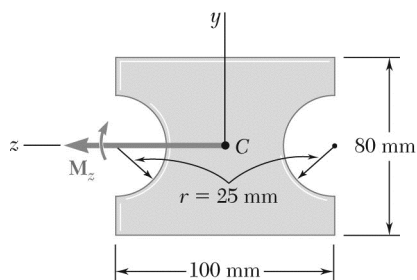
$$M = 1.405 \text{ kip} \cdot \text{in} \quad \blacktriangleleft$$

$$(b) \quad I = \frac{\pi}{4} (0.7^4 - 0.5^4) = 139.49 \times 10^{-3} \text{ in}^4$$

$$c = 0.7 \text{ in.}$$

$$\sigma = \frac{Mc}{I} : \quad M = \frac{\sigma I}{c} = \frac{(16)(139.49 \times 10^{-3})}{0.7}$$

$$M = 3.19 \text{ kip} \cdot \text{in} \quad \blacktriangleleft$$



PROBLEM 4.4

A nylon spacing bar has the cross section shown. Knowing that the allowable stress for the grade of nylon used is 24 MPa, determine the largest couple $\mathbf{M}_z$ that can be applied to the bar.

SOLUTION

$$I = I_{\text{rect}} - I_{\text{circle}} = \frac{1}{12}bh^3 - \frac{\pi}{4}r^4$$

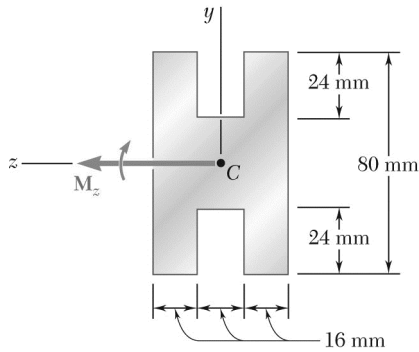
$$= \frac{1}{12}(100)(80)^3 - \frac{\pi}{4}(25)^4 = 3.9599 \times 10^6 \text{ mm}^4$$

$$= 3.9599 \times 10^{-6} \text{ m}^4$$

$$c = \frac{80}{2} = 40 \text{ mm} = 0.040 \text{ m}$$

$$\sigma = \frac{Mc}{I} : \quad M_z = \frac{\sigma I}{c} = \frac{(24 \times 10^6)(3.9599 \times 10^{-6})}{0.040} = 2.38 \times 10^3 \text{ N} \cdot \text{m}$$

$$M_z = 2.38 \text{ kN} \cdot \text{m} \quad \blacktriangleleft$$



PROBLEM 4.5

A beam of the cross section shown is extruded from an aluminum alloy for which $\sigma_Y = 250$ MPa and $\sigma_U = 450$ MPa. Using a factor of safety of 3.00, determine the largest couple that can be applied to the beam when it is bent about the z -axis.

SOLUTION

Allowable stress.
$$= \frac{\sigma_U}{F.S.} = \frac{450}{3} = 150 \text{ MPa}$$
$$= 150 \times 10^6 \text{ Pa}$$

Moment of inertia about z -axis.

$$I_1 = \frac{1}{12}(16)(80)^3 = 682.67 \times 10^3 \text{ mm}^4$$

$$I_2 = \frac{1}{12}(16)(32)^3 = 43.69 \times 10^3 \text{ mm}^4$$

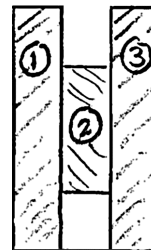
$$I_3 = I_1 = 682.67 \times 10^3 \text{ mm}^4$$

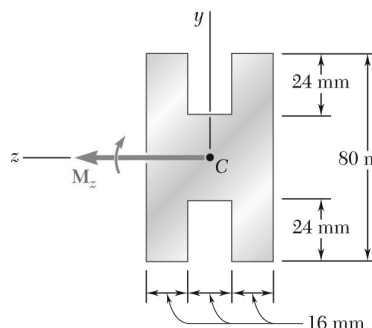
$$I = I_1 + I_2 + I_3 = 1.40902 \times 10^6 \text{ mm}^4 = 1.40902 \times 10^{-6} \text{ m}^4$$

$$\sigma = \frac{Mc}{I} \quad \text{with} \quad c = \frac{1}{2}(80) = 40 \text{ mm} = 0.040 \text{ m}$$

$$M = \frac{I\sigma}{c} = \frac{(1.40902 \times 10^{-6})(150 \times 10^6)}{0.040} = 5.28 \times 10^3 \text{ N} \cdot \text{m}$$

$$M = 5.28 \text{ kN} \cdot \text{m} \quad \blacktriangleleft$$





PROBLEM 4.6

Solve Prob. 4.5, assuming that the beam is bent about the y -axis.

PROBLEM 4.5 A beam of the cross section shown is extruded from an aluminum alloy for which $\sigma_Y = 250$ MPa and $\sigma_U = 450$ MPa. Using a factor of safety of 3.00, determine the largest couple that can be applied to the beam when it is bent about the z -axis.

SOLUTION

$$\begin{aligned} \text{Allowable stress:} \quad &= \frac{\sigma_U}{F.S.} = \frac{450}{3.00} = 150 \text{ MPa} \\ &= 150 \times 10^6 \text{ Pa} \end{aligned}$$

Moment of inertia about y -axis.

$$I_1 = \frac{1}{12}(80)(16)^3 + (80)(16)(16)^2 = 354.987 \times 10^3 \text{ mm}^4$$

$$I_2 = \frac{1}{12}(32)(16)^3 = 10.923 \times 10^3 \text{ mm}^4$$

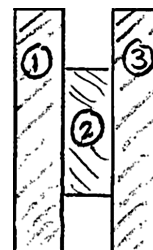
$$I_3 = I_1 = 354.987 \times 10^3 \text{ mm}^4$$

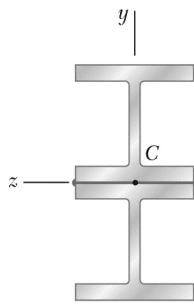
$$I = I_1 + I_2 + I_3 = 720.9 \times 10^3 \text{ mm}^4 = 720.9 \times 10^{-9} \text{ m}^4$$

$$\sigma = \frac{Mc}{I} \quad \text{with} \quad c = \frac{1}{2}(48) = 24 \text{ mm} = 0.024 \text{ m}$$

$$M = \frac{I\sigma}{c} = \frac{(720.9 \times 10^{-9})(150 \times 10^6)}{0.024} = 4.51 \times 10^3 \text{ N} \cdot \text{m}$$

$$M = 4.51 \text{ kN} \cdot \text{m} \quad \blacktriangleleft$$





PROBLEM 4.7

Two W4×13 rolled sections are welded together as shown. Knowing that for the steel alloy used $\sigma_Y = 36$ ksi and $\sigma_U = 58$ ksi and using a factor of safety of 3.0, determine the largest couple that can be applied when the assembly is bent about the z axis.

SOLUTION

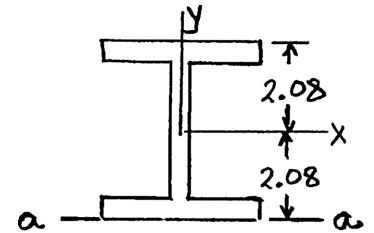
Properties of W4 × 13 rolled section.

(See Appendix C.)

$$\text{Area} = 3.83 \text{ in}^2$$

$$\text{Depth} = 4.16 \text{ in.}$$

$$I_x = 11.3 \text{ in}^4$$



For one rolled section, moment of inertia about axis $a-a$ is

$$I_a = I_x + Ad^2 = 11.3 + (3.83)(2.08)^2 = 27.87 \text{ in}^4$$

For both sections,

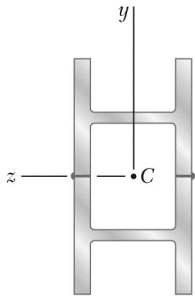
$$I_z = 2I_a = 55.74 \text{ in}^4$$

$$c = \text{depth} = 4.16 \text{ in.}$$

$$\sigma_{\text{all}} = \frac{\sigma_U}{F.S.} = \frac{58}{3.0} = 19.333 \text{ ksi} \quad \sigma = \frac{Mc}{I}$$

$$M_{\text{all}} = \frac{\sigma_{\text{all}} I}{c} = \frac{(19.333)(55.74)}{4.16}$$

$$M_{\text{all}} = 259 \text{ kip} \cdot \text{in} \quad \blacktriangleleft$$



PROBLEM 4.8

Two W4×13 rolled sections are welded together as shown. Knowing that for the steel alloy used $\sigma_Y = 36$ ksi and $\sigma_U = 58$ ksi and using a factor of safety of 3.0, determine the largest couple that can be applied when the assembly is bent about the z axis.

SOLUTION

Properties of W4 × 13 rolled section.

(See Appendix C.)

$$\text{Area} = 3.83 \text{ in}^2$$

$$\text{Width} = 4.060 \text{ in.}$$

$$I_y = 3.86 \text{ in}^4$$

For one rolled section, moment of inertia about axis b - b is

$$I_b = I_y + Ad^2 = 3.86 + (3.83)(2.030)^2 = 19.643 \text{ in}^4$$

For both sections,

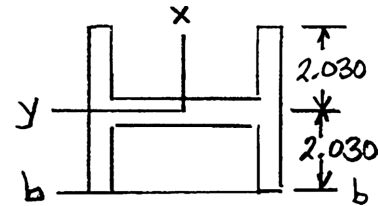
$$I_z = 2I_b = 39.286 \text{ in}^4$$

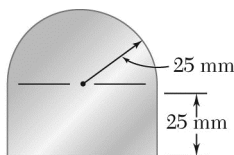
$$c = \text{width} = 4.060 \text{ in.}$$

$$\sigma_{\text{all}} = \frac{\sigma_U}{F.S.} = \frac{58}{3.0} = 19.333 \text{ ksi} \quad \sigma = \frac{Mc}{I}$$

$$M_{\text{all}} = \frac{\sigma_{\text{all}} I}{c} = \frac{(19.333)(39.286)}{4.060}$$

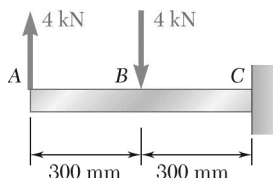
$$M_{\text{all}} = 187.1 \text{ kip} \cdot \text{in} \quad \blacktriangleleft$$



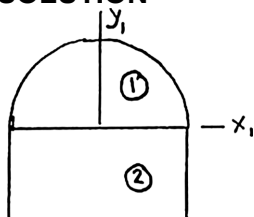


PROBLEM 4.9

Two vertical forces are applied to a beam of the cross section shown. Determine the maximum tensile and compressive stresses in portion BC of the beam.



SOLUTION



$$A_1 = \frac{\pi}{2} r^2 = \frac{\pi}{2} (25)^2 = 981.7 \text{ mm}^2 \quad \bar{y}_1 = \frac{4r}{3\pi} = \frac{(4)(25)}{3\pi} = 10.610 \text{ mm}$$

$$A_2 = bh = (50)(25) = 1250 \text{ mm}^2 \quad \bar{y}_2 = -\frac{h}{2} = -\frac{25}{2} = -12.5 \text{ mm}$$

$$\bar{y} = \frac{A_1 \bar{y}_1 + A_2 \bar{y}_2}{A_1 + A_2} = \frac{(981.7)(10.610) + (1250)(-12.5)}{981.7 + 1250} = -2.334 \text{ mm}$$

$$\bar{I}_1 = I_{x_1} - A_1 \bar{y}_1^2 = \frac{\pi}{8} r^4 - A_1 \bar{y}_1^2 = \frac{\pi}{8} (25)^4 - (981.7)(10.610)^2 = 42.886 \times 10^6 \text{ mm}^4$$

$$d_1 = \bar{y}_1 - \bar{y} = 10.610 - (-2.334) = 12.944 \text{ mm}$$

$$I_1 = \bar{I}_1 + A_1 d_1^2 = 42.886 \times 10^6 + (981.7)(12.944)^2 = 207.35 \times 10^6 \text{ mm}^4$$

$$\bar{I}_2 = \frac{1}{12} b h^3 = \frac{1}{12} (50)(25)^3 = 65.104 \times 10^6 \text{ mm}^4$$

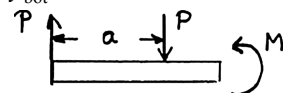
$$d_2 = |\bar{y}_2 - \bar{y}| = |-12.5 - (-2.334)| = 10.166 \text{ mm}$$

$$I_2 = \bar{I}_2 + A_2 d_2^2 = 65.104 \times 10^6 + (1250)(10.166)^2 = 194.288 \times 10^6 \text{ mm}^4$$

$$I = I_1 + I_2 = 401.64 \times 10^6 \text{ mm}^4 = 401.64 \times 10^{-9} \text{ m}^4$$

$$y_{\text{top}} = 25 + 2.334 = 27.334 \text{ mm} = 0.027334 \text{ m}$$

$$y_{\text{bot}} = -25 + 2.334 = -22.666 \text{ mm} = -0.022666 \text{ m}$$



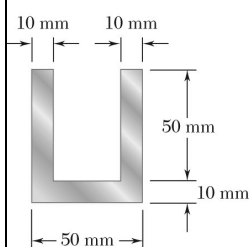
$$M - Pa = 0 : \quad M = Pa = (4 \times 10^3)(300 \times 10^{-3}) = 1200 \text{ N} \cdot \text{m}$$

$$\sigma_{\text{top}} = \frac{-M y_{\text{top}}}{I} = -\frac{(1200)(0.027334)}{401.64 \times 10^{-9}} = -81.76 \times 10^6 \text{ Pa}$$

$$\sigma_{\text{top}} = -81.8 \text{ MPa} \quad \blacktriangleleft$$

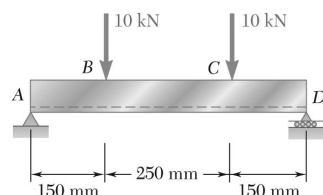
$$\sigma_{\text{bot}} = \frac{-M y_{\text{bot}}}{I} = -\frac{(1200)(-0.022666)}{401.64 \times 10^{-9}} = 67.80 \times 10^6 \text{ Pa}$$

$$\sigma_{\text{bot}} = 67.8 \text{ MPa} \quad \blacktriangleleft$$

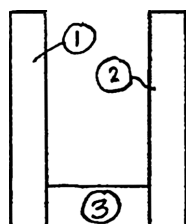


PROBLEM 4.10

Two vertical forces are applied to a beam of the cross section shown. Determine the maximum tensile and compressive stresses in portion BC of the beam.



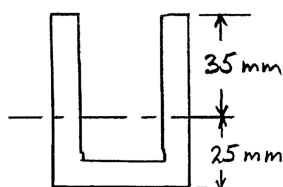
SOLUTION



	$A, \text{ mm}^2$	$\bar{y}_0, \text{ mm}$	$A\bar{y}_0, \text{ mm}^3$
①	600	30	18×10^3
②	600	30	18×10^3
③	300	5	1.5×10^3
	1500		37.5×10^3

$$\bar{Y}_0 = \frac{37.5 \times 10^3}{1500} = 25 \text{ mm}$$

Neutral axis lies 25 mm above the base.



$$I_1 = \frac{1}{12}(10)(60)^3 + (600)(5)^2 = 195 \times 10^3 \text{ mm}^4 \quad I_2 = I_1 = 195 \text{ mm}^4$$

$$I_3 = \frac{1}{12}(30)(10)^3 + (300)(20)^2 = 122.5 \times 10^3 \text{ mm}^4$$

$$I = I_1 + I_2 + I_3 = 512.5 \times 10^3 \text{ mm}^4 = 512.5 \times 10^{-9} \text{ m}^4$$

$$y_{\text{top}} = 35 \text{ mm} = 0.035 \text{ m} \quad y_{\text{bot}} = -25 \text{ mm} = -0.025 \text{ m}$$

$$a = 150 \text{ mm} = 0.150 \text{ m} \quad P = 10 \times 10^3 \text{ N}$$

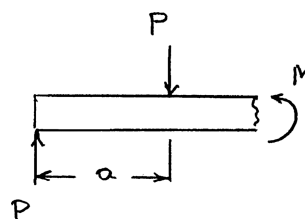
$$M = Pa = (10 \times 10^3)(0.150) = 1.5 \times 10^3 \text{ N} \cdot \text{m}$$

$$\sigma_{\text{top}} = -\frac{My_{\text{top}}}{I} = -\frac{(1.5 \times 10^3)(0.035)}{512.5 \times 10^{-9}} = -102.4 \times 10^6 \text{ Pa}$$

$$\sigma_{\text{top}} = -102.4 \text{ MPa (compression)} \quad \blacktriangleleft$$

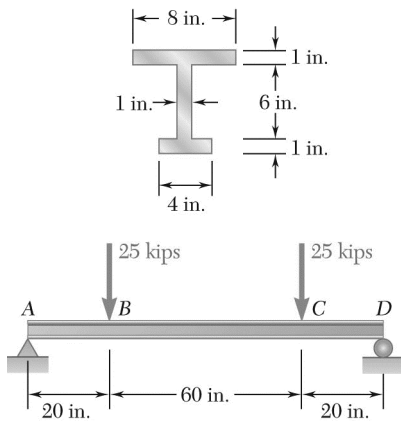
$$\sigma_{\text{bot}} = -\frac{My_{\text{bot}}}{I} = -\frac{(1.5 \times 10^3)(-0.025)}{512.5 \times 10^{-9}} = 73.2 \times 10^6 \text{ Pa}$$

$$\sigma_{\text{bot}} = 73.2 \text{ MPa (tension)} \quad \blacktriangleleft$$

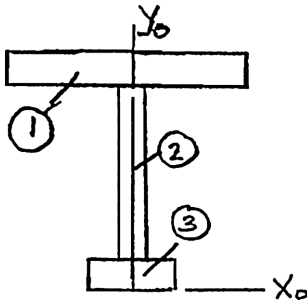


PROBLEM 4.11

Two vertical forces are applied to a beam of the cross section shown. Determine the maximum tensile and compressive stresses in portion BC of the beam.



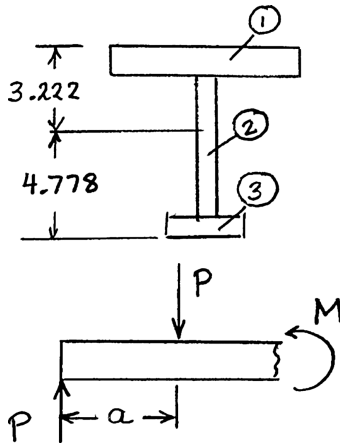
SOLUTION



	A	$\bar{y}_0$	$A\bar{y}_0$
①	8	7.5	60
②	6	4	24
③	4	0.5	2
Σ	18		86

$$\bar{Y}_0 = \frac{86}{18} = 4.778 \text{ in.}$$

Neutral axis lies 4.778 in. above the base.



$$I_1 = \frac{1}{12} b_1 h_1^3 + A_1 d_1^2 = \frac{1}{12} (8)(1)^3 + (8)(2.772)^2 = 59.94 \text{ in}^4$$

$$I_2 = \frac{1}{12} b_2 h_2^3 + A_2 d_2^2 = \frac{1}{12} (1)(6)^3 + (6)(0.778)^2 = 21.63 \text{ in}^4$$

$$I_3 = \frac{1}{12} b_3 h_3^3 + A_3 d_3^2 = \frac{1}{12} (4)(1)^3 + (4)(4.278)^2 = 73.54 \text{ in}^4$$

$$I = I_1 + I_2 + I_3 = 59.94 + 21.63 + 73.57 = 155.16 \text{ in}^4$$

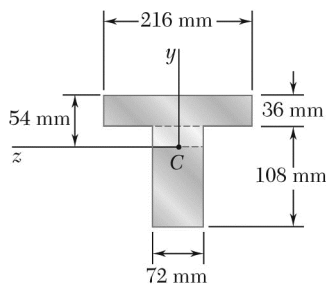
$$y_{\text{top}} = 3.222 \text{ in.} \quad y_{\text{bot}} = -4.778 \text{ in.}$$

$$M - Pa = 0$$

$$M = Pa = (25)(20) = 500 \text{ kip} \cdot \text{in}$$

$$\sigma_{\text{top}} = -\frac{My_{\text{top}}}{I} = -\frac{(500)(3.222)}{155.16} \quad \sigma_{\text{top}} = -10.38 \text{ ksi (compression)} \quad \blacktriangleleft$$

$$\sigma_{\text{bot}} = -\frac{My_{\text{bot}}}{I} = -\frac{(500)(-4.778)}{155.16} \quad \sigma_{\text{bot}} = 15.40 \text{ ksi (tension)} \quad \blacktriangleleft$$



PROBLEM 4.12

Knowing that a beam of the cross section shown is bent about a horizontal axis and that the bending moment is $6 \text{ kN} \cdot \text{m}$, determine the total force acting on the top flange.

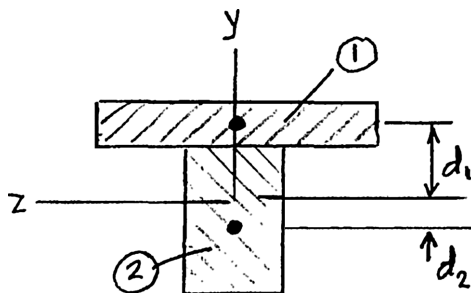
SOLUTION

The stress distribution over the entire cross section is given by the bending stress formula:

$$\sigma_x = -\frac{My}{I}$$

where y is a coordinate with its origin on the neutral axis and I is the moment of inertia of the entire cross sectional area. The force on the shaded portion is calculated from this stress distribution. Over an area element dA , the force is

$$dF = \sigma_x dA = -\frac{My}{I} dA$$



The total force on the shaded area is then

$$F = \int dF = -\int \frac{My}{I} dA = -\frac{M}{I} \int y dA = -\frac{M}{I} \bar{y}^* A^*$$

where $\bar{y}^*$ is the centroidal coordinate of the shaded portion and A^* is its area.

$$d_1 = 54 - 18 = 36 \text{ mm}$$

$$d_2 = 54 + 36 - 54 = 36 \text{ mm}$$

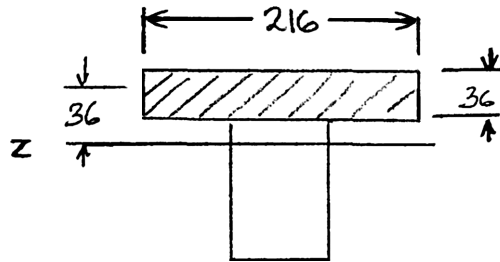
PROBLEM 4.12 (Continued)

Moment of inertia of entire cross section:

$$I_1 = \frac{1}{12} b_1 h_1^3 + A_1 d_1^2 = \frac{1}{12} (216)(36)^3 + (216)(36)(36)^2 = 10.9175 \times 10^6 \text{ mm}^4$$

$$I_2 = \frac{1}{12} b_2 h_2^3 + A_2 d_2^2 = \frac{1}{12} (72)(108)^3 + (72)(108)(36)^2 = 17.6360 \times 10^6 \text{ mm}^4$$

$$I = I_1 + I_2 = 28.5535 \times 10^6 \text{ mm}^4 = 28.5535 \times 10^{-6} \text{ m}^4$$



For the shaded area,

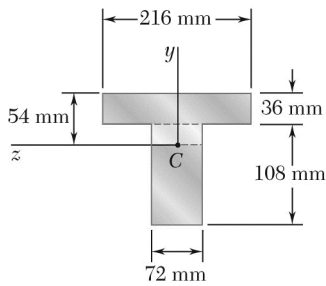
$$A^* = (216)(36) = 7776 \text{ mm}^2$$

$$\bar{y}^* = 36 \text{ mm}$$

$$A^* \bar{y}^* = 279.936 \times 10^3 \text{ mm}^3 = 279.936 \times 10^{-6} \text{ m}^3$$

$$F = - \left| \frac{MA^* \bar{y}^*}{I} \right| = \frac{(6 \times 10^3)(279.936 \times 10^{-6})}{28.5535 \times 10^{-6}}$$
$$= 58.8 \times 10^3 \text{ N}$$

$$F = 58.8 \text{ kN} \quad \blacktriangleleft$$



PROBLEM 4.13

Knowing that a beam of the cross section shown is bent about a horizontal axis and that the bending moment is $6 \text{ kN} \cdot \text{m}$, determine the total force acting on the shaded portion of the web.

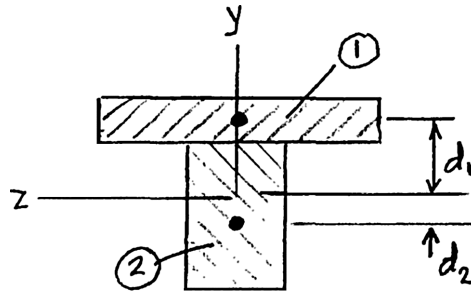
SOLUTION

The stress distribution over the entire cross section is given by the bending stress formula:

$$\sigma_x = -\frac{My}{I}$$

where y is a coordinate with its origin on the neutral axis and I is the moment of inertia of the entire cross sectional area. The force on the shaded portion is calculated from this stress distribution. Over an area element dA , the force is

$$dF = \sigma_x dA = -\frac{My}{I} dA$$



The total force on the shaded area is then

$$F = \int dF = -\int \frac{My}{I} dA = -\frac{M}{I} \int y dA = -\frac{M}{I} \bar{y}^* A^*$$

where $\bar{y}^*$ is the centroidal coordinate of the shaded portion and A^* is its area.

$$d_1 = 54 - 18 = 36 \text{ mm}$$

$$d_2 = 54 + 36 - 54 = 36 \text{ mm}$$

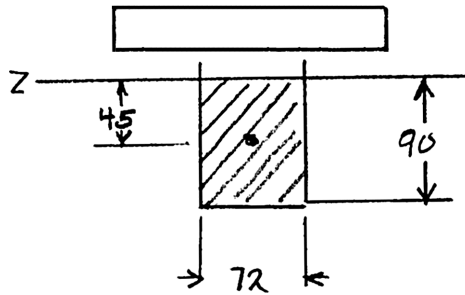
PROBLEM 4.13 (Continued)

Moment of inertia of entire cross section:

$$I_1 = \frac{1}{12} b_1 h_1^3 + A_1 d_1^2 = \frac{1}{12} (216)(36)^3 + (216)(36)(36)^2 = 10.9175 \times 10^6 \text{ mm}^4$$

$$I_2 = \frac{1}{12} b_2 h_2^3 + A_2 d_2^2 = \frac{1}{12} (72)(108)^3 + (72)(108)(36)^2 = 17.6360 \times 10^6 \text{ mm}^4$$

$$I = I_1 + I_2 = 28.5535 \times 10^6 \text{ mm}^4 = 28.5535 \times 10^{-6} \text{ m}^4$$



For the shaded area,

$$A^* = (72)(90) = 6480 \text{ mm}^2$$

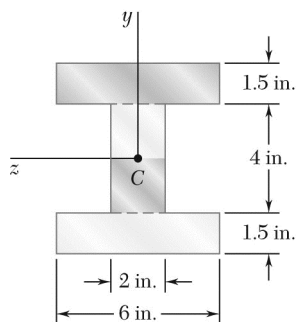
$$\bar{y}^* = 45 \text{ mm}$$

$$A^* \bar{y}^* = 291.6 \times 10^3 \text{ mm}^3 = 291.6 \times 10^{-6} \text{ m}$$

$$F = \left| \frac{MA^* \bar{y}^*}{I} \right| = \frac{(6 \times 10^3)(291.6 \times 10^{-6})}{28.5535 \times 10^{-6}}$$

$$= 61.3 \times 10^3 \text{ N}$$

$$F = 61.3 \text{ kN} \quad \blacktriangleleft$$

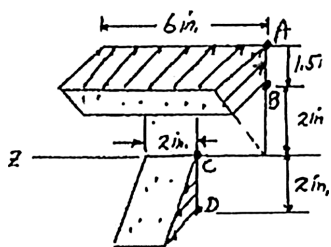


PROBLEM 4.14

Knowing that a beam of the cross section shown is bent about a horizontal axis and that the bending moment is 50 kip · in., determine the total force acting (a) on the top flange, (b) on the shaded portion of the web.

SOLUTION

The stress distribution over the entire cross-section is given by the bending stress formula: $\sigma_x = -\frac{My}{I}$



where y is a coordinate with its origin on the neutral axis and I is the moment of inertia of the entire cross sectional area. The force on the shaded portion is calculated from this stress distribution. Over an area element dA , the force is

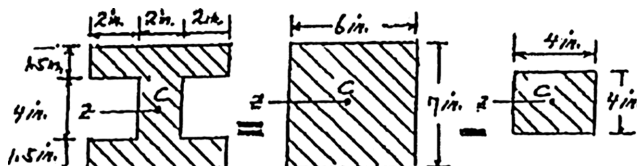
$$dF = \sigma_x dA = -\frac{My}{I} dA$$

The total force on the shaded area is then

$$F = \int dF = -\int \frac{My}{I} dA = -\frac{M}{I} \int y dA = -\frac{M}{I} \bar{y}^* A^*$$

where $\bar{y}^*$ is the centroidal coordinate of the shaded portion and A^* is its area.

Calculate the moment of inertia.



$$I = \frac{1}{12} (6 \text{ in.})(7 \text{ in.})^3 - \frac{1}{12} (4 \text{ in.})(4 \text{ in.})^3 = 150.17 \text{ in}^4$$

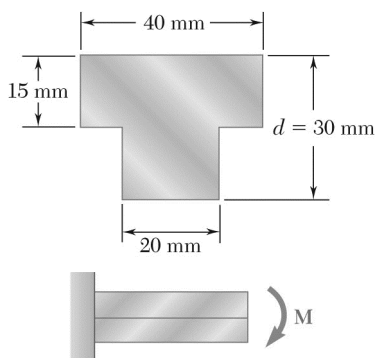
$$M = 50 \text{ kip} \cdot \text{in}$$

(a) Top flange: $A^* = (6 \text{ in.})(1.5 \text{ in.}) = 9 \text{ in}^2$ $\bar{y}^* = 2 \text{ in.} + 0.75 \text{ in.} = 2.75 \text{ in.}$

$$F = \frac{50 \text{ kip} \cdot \text{in}}{150.17 \text{ in}^4} (9 \text{ in}^2)(2.75 \text{ in.}) = 8.24 \text{ kips} \quad F = 8.24 \text{ kips} \quad \blacktriangleleft$$

(b) Half web: $A^* = (2 \text{ in.})(2 \text{ in.}) = 4 \text{ in}^2$ $\bar{y}^* = 1 \text{ in.}$

$$F = \frac{50 \text{ kip} \cdot \text{in}}{150.17 \text{ in}^4} (4 \text{ in}^2)(1 \text{ in.}) = 1.332 \text{ kips} \quad F = 1.332 \text{ kips} \quad \blacktriangleleft$$

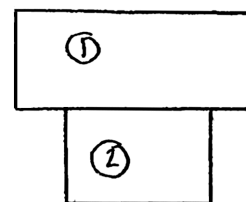


PROBLEM 4.15

The beam shown is made of a nylon for which the allowable stress is 24 MPa in tension and 30 MPa in compression. Determine the largest couple **M** that can be applied to the beam.

SOLUTION

	$A, \text{ mm}^2$	$\bar{y}_0, \text{ mm}$	$A\bar{y}_0, \text{ mm}^3$
①	600	22.5	13.5×10^3
②	300	7.5	2.25×10^3
Σ	900		15.75×10^3



$$\bar{Y}_0 = \frac{15.75 \times 10^3}{900} = 17.5 \text{ mm} \quad \text{The neutral axis lies 17.5 mm above the bottom.}$$

$$y_{\text{top}} = 30 - 17.5 = 12.5 \text{ mm} = 0.0125 \text{ m}$$

$$y_{\text{bot}} = -17.5 \text{ mm} = -0.0175 \text{ m}$$

$$I_1 = \frac{1}{12} b_1 h_1^3 + A_1 d_1^2 = \frac{1}{12} (40)(15)^3 + (600)(5)^2 = 26.25 \times 10^3 \text{ mm}^4$$

$$I_2 = \frac{1}{12} b_2 h_2^3 + A_2 d_2^2 = \frac{1}{12} (20)(15)^3 + (300)(10)^2 = 35.625 \times 10^3 \text{ mm}^4$$

$$I = I_1 + I_2 = 61.875 \times 10^3 \text{ mm}^4 = 61.875 \times 10^{-9} \text{ m}^4$$

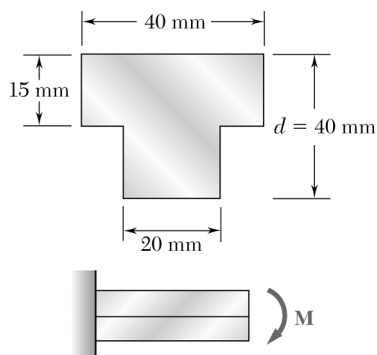
$$|\sigma| = \left| \frac{My}{I} \right| \quad M = \left| \frac{\sigma I}{y} \right|$$

Top: (tension side) $M = \frac{(24 \times 10^6)(61.875 \times 10^{-9})}{0.0125} = 118.8 \text{ N} \cdot \text{m}$

Bottom: (compression) $M = \frac{(30 \times 10^6)(61.875 \times 10^{-9})}{0.0175} = 106.1 \text{ N} \cdot \text{m}$

Choose smaller value.

$$M = 106.1 \text{ N} \cdot \text{m} \quad \blacktriangleleft$$



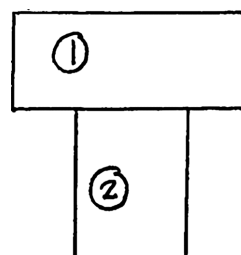
PROBLEM 4.16

Solve Prob. 4.15, assuming that $d = 40$ mm.

PROBLEM 4.15 The beam shown is made of a nylon for which the allowable stress is 24 MPa in tension and 30 MPa in compression. Determine the largest couple M that can be applied to the beam.

SOLUTION

	$A, \text{ mm}^2$	$\bar{y}_0, \text{ mm}$	$A\bar{y}_0, \text{ mm}^3$
①	600	32.5	19.5×10^3
②	500	12.5	6.25×10^3
Σ	1100		25.75×10^3



$$\bar{Y}_0 = \frac{25.75 \times 10^3}{1100} = 23.41 \text{ mm} \quad \text{The neutral axis lies 23.41 mm above the bottom.}$$

$$y_{\text{top}} = 40 - 23.41 = 16.59 \text{ mm} = 0.01659 \text{ m}$$

$$y_{\text{bot}} = -23.41 \text{ mm} = -0.02341 \text{ m}$$

$$I_1 = \frac{1}{12} b_1 h_1^3 + A_1 d_1^2 = \frac{1}{12} (40)(15)^3 + (600)(9.09)^2 = 60.827 \times 10^3 \text{ mm}^4$$

$$I_2 = \frac{1}{12} b_2 h_2^3 + A_2 d_2^2 = \frac{1}{12} (20)(25)^3 + (500)(10.91)^2 = 85.556 \times 10^3 \text{ mm}^4$$

$$I = I_1 + I_2 = 146.383 \times 10^3 \text{ mm}^4 = 146.383 \times 10^{-9} \text{ m}^4$$

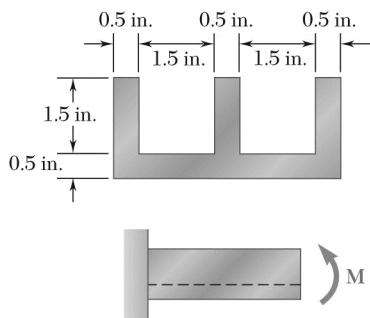
$$|\sigma| = \left| \frac{My}{I} \right| \quad M = \left| \frac{\sigma I}{y} \right|$$

$$\text{Top: (tension side)} \quad M = \frac{(24 \times 10^6)(146.383 \times 10^{-9})}{0.01659} = 212 \text{ N} \cdot \text{m}$$

$$\text{Bottom: (compression)} \quad M = \frac{(30 \times 10^6)(146.383 \times 10^{-9})}{0.02341} = 187.6 \text{ N} \cdot \text{m}$$

Choose smaller value.

$$M = 187.6 \text{ N} \cdot \text{m} \quad \blacktriangleleft$$



PROBLEM 4.17

Knowing that for the extruded beam shown the allowable stress is 12 ksi in tension and 16 ksi in compression, determine the largest couple **M** that can be applied.

SOLUTION

	A	$\bar{y}_0$	$A\bar{y}_0$
①	2.25	1.25	2.8125
②	2.25	0.25	0.5625
	4.50		3.375

$$\bar{Y} = \frac{3.375}{4.50} = 0.75 \text{ in.}$$

The neutral axis lies 0.75 in. above bottom.

$$y_{\text{top}} = 2.0 - 0.75 = 1.25 \text{ in.}, \quad y_{\text{bot}} = -0.75 \text{ in.}$$

$$I_1 = \frac{1}{12} b_1 h_1^3 + A_1 d_1^2 = \frac{1}{12} (1.5)(1.5)^3 + (2.25)(0.5)^2 = 0.984375 \text{ in}^4$$

$$I_2 = \frac{1}{12} b_2 h_2^3 + A_2 d_2^2 = \frac{1}{12} (4.5)(0.5)^3 + (2.25)(0.5)^2 = 0.609375 \text{ in}^4$$

$$I = I_1 + I_2 = 1.59375 \text{ in}^4$$

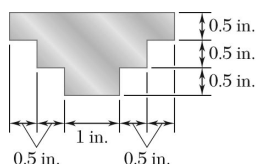
$$|\sigma| = \left| \frac{My}{I} \right| \quad |M| = \left| \frac{\sigma I}{y} \right|$$

Top: (compression) $M = \frac{(16)(1.59375)}{1.25} = 20.4 \text{ kip} \cdot \text{in}$

Bottom: (tension) $M = \frac{(12)(1.59375)}{0.75} = 25.5 \text{ kip} \cdot \text{in}$

Choose the smaller as M_{all} .

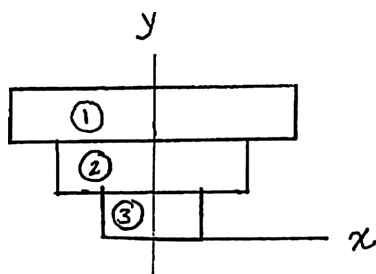
$$M_{\text{all}} = 20.4 \text{ kip} \cdot \text{in} \quad \blacktriangleleft$$



PROBLEM 4.18

Knowing that for the casting shown the allowable stress is 5 ksi in tension and 18 ksi in compression, determine the largest couple **M** that can be applied.

SOLUTION



Locate the neutral axis and compute the moment of inertia.

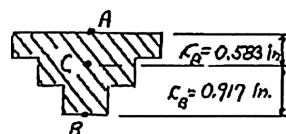
$$\bar{Y} = \frac{\sum A_i \bar{y}_i}{\sum A_i} \quad \bar{I} = \frac{1}{12} b_i h_i^3 \quad \text{for rectangle}$$

$$d_i = |\bar{y}_i - \bar{Y}| \quad I = \sum (A_i d_i^2 + \bar{I})$$

Part	A, in^2	$\bar{y}_i, \text{in.}$	$A_i \bar{y}_i, \text{in}^3$	$d_i, \text{in.}$	$A_i d_i^2, \text{in}^4$	$\bar{I}, \text{in}^4$
1	1.5	1.25	1.875	0.3333	0.1667	0.03125
2	1.0	0.75	0.75	0.1667	0.0277	0.02083
3	0.5	0.25	0.125	0.6667	0.2222	0.01042
Σ	3.0		2.75		0.4166	0.0625

$$\bar{Y} = \frac{\sum A \bar{y}}{\sum A} = \frac{2.75}{3.0} = 0.9167 \text{ in.} \quad I = \sum (\bar{I} + A d^2) = 0.4166 + 0.0625 = 0.479 \text{ in}^4$$

Allowable bending moment.



$$\sigma = \frac{Mc}{I} \quad \text{or} \quad M = \frac{\sigma I}{c}$$

Tension at A:

$$\sigma_A \leq 5 \text{ ksi}$$

$$c_A = 0.583 \text{ in.}$$

$$M \leq \frac{(5)(0.479)}{0.583} = 4.11 \text{ kip} \cdot \text{in}$$

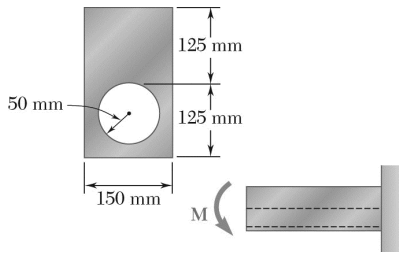
Compression at B:

$$\sigma_B \leq 18 \text{ ksi} \quad c_B = 0.917 \text{ in.}$$

$$M \leq \frac{(18)(0.479)}{0.917} = 9.40 \text{ kip} \cdot \text{in}$$

The smaller value is the allowable value of **M**.

$$M = 4.11 \text{ kip} \cdot \text{in} \quad \blacktriangleleft$$



PROBLEM 4.19

Knowing that for the extruded beam shown the allowable stress is 120 MPa in tension and 150 MPa in compression, determine the largest couple **M** that can be applied.

SOLUTION

- ① rectangle ② circular cutout

$$A_1 = (150)(250) = 37.5 \times 10^3 \text{ mm}^2$$

$$A_2 = -\pi(50)^2 = -7.85398 \times 10^3 \text{ mm}^2$$

$$A = A_1 + A_2 = 29.64602 \times 10^3 \text{ mm}^2$$

$$\bar{y}_1 = 0 \text{ mm}$$

$$\bar{y}_2 = -50 \text{ mm}$$

$$\bar{Y} = \frac{\Sigma A \bar{y}}{\Sigma A}$$

$$\begin{aligned} \bar{Y} &= \frac{(37.5 \times 10^3)(0) + (-7.85398 \times 10^3)(-50)}{29.64602 \times 10^3} \\ &= 13.2463 \text{ mm} \end{aligned}$$

$$\begin{aligned} I_{x'} &= \Sigma(I + Ad^2) = I_1 - I_2 \\ &= \left[\frac{1}{12}(150)(250)^3 + (37.5 \times 10^3)(13.2463)^2 \right] \\ &\quad - \left[\frac{\pi}{4}(50)^4 + (7.85398 \times 10^3)(50 + 13.2463)^2 \right] \\ &= 201.892 \times 10^6 - 36.3254 \times 10^6 = 165.567 \times 10^6 \text{ mm}^4 \\ &= 165.567 \times 10^{-6} \text{ m}^4 \end{aligned}$$

Top: (tension side)

$$c = 125 - 13.2463 = 111.7537 \text{ mm} = 0.11175 \text{ m}$$

$$\begin{aligned} \sigma &= \frac{Mc}{I} & M &= \frac{I\sigma}{c} = \frac{(165.567 \times 10^{-6})(120 \times 10^6)}{0.11175} \\ & & &= 177.79 \times 10^3 \text{ N} \cdot \text{m} \end{aligned}$$

Bottom: (compression side)

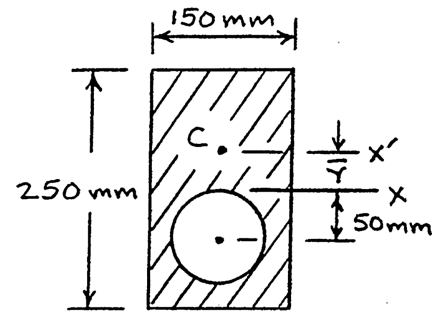
$$\begin{aligned} c &= 125 + 13.2463 = 138.2463 \text{ mm} \\ &= 0.13825 \text{ m} \end{aligned}$$

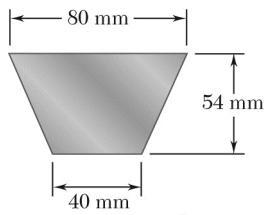
$$\begin{aligned} \sigma &= \frac{Mc}{I} & M &= \frac{I\sigma}{c} = \frac{(165.567 \times 10^{-6})(150 \times 10^6)}{0.13825} \\ & & &= 179.64 \times 10^3 \text{ N} \cdot \text{m} \end{aligned}$$

Choose the smaller.

$$M = 177.8 \times 10^3 \text{ N} \cdot \text{m}$$

$$M = 177.8 \text{ kN} \cdot \text{m} \quad \blacktriangleleft$$



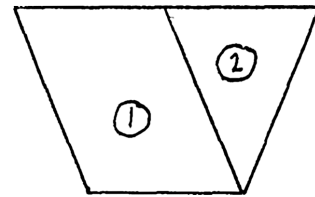


PROBLEM 4.20

Knowing that for the extruded beam shown the allowable stress is 120 MPa in tension and 150 MPa in compression, determine the largest couple **M** that can be applied.

SOLUTION

	A, mm^2	$\bar{y}_0, \text{mm}$	$A\bar{y}_0, \text{mm}^3$	d, mm
①	2160	27	58320	3
②	1080	36	38880	3
Σ	3240		97200	



$$\bar{Y} = \frac{97200}{3240} = 30 \text{ mm} \quad \text{The neutral axis lies 30 mm above the bottom.}$$

$$y_{\text{top}} = 54 - 30 = 24 \text{ mm} = 0.024 \text{ m} \quad y_{\text{bot}} = -30 \text{ mm} = -0.030 \text{ m}$$

$$I_1 = \frac{1}{12} b_1 h_1^3 + A_1 d_1^2 = \frac{1}{12} (40)(54)^3 + (40)(54)(3)^2 = 544.32 \times 10^3 \text{ mm}^4$$

$$I_2 = \frac{1}{36} b_2 h_2^2 + A_2 d_2^2 = \frac{1}{36} (40)(54)^3 + \frac{1}{2} (40)(54)(6)^2 = 213.84 \times 10^3 \text{ mm}^4$$

$$I = I_1 + I_2 = 758.16 \times 10^3 \text{ mm}^4 = 758.16 \times 10^{-9} \text{ m}^4$$

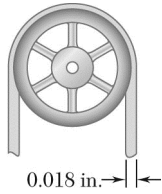
$$|\sigma| = \left| \frac{My}{I} \right| \quad |M| = \left| \frac{\sigma I}{y} \right|$$

$$\text{Top: (tension side)} \quad M = \frac{(120 \times 10^6)(758.16 \times 10^{-9})}{0.024} = 3.7908 \times 10^3 \text{ N} \cdot \text{m}$$

$$\text{Bottom: (compression)} \quad M = \frac{(150 \times 10^6)(758.16 \times 10^{-9})}{0.030} = 3.7908 \times 10^3 \text{ N} \cdot \text{m}$$

$$\text{Choose the smaller as } M_{\text{all}}. \quad M_{\text{all}} = 3.7908 \times 10^3 \text{ N} \cdot \text{m}$$

$$M_{\text{all}} = 3.79 \text{ kN} \cdot \text{m} \quad \blacktriangleleft$$



PROBLEM 4.21

A steel band saw blade, that was originally straight, passes over 8-in.-diameter pulleys when mounted on a band saw. Determine the maximum stress in the blade, knowing that it is 0.018 in. thick and 0.625 in. wide. Use $E = 29 \times 10^6$ psi.

SOLUTION

Band blade thickness: $t = 0.018$ in.

Radius of pulley: $r = \frac{1}{2}d = 4.000$ in.

Radius of curvature of centerline of blade:

$$\rho = r + \frac{1}{2}t = 4.009 \text{ in.}$$

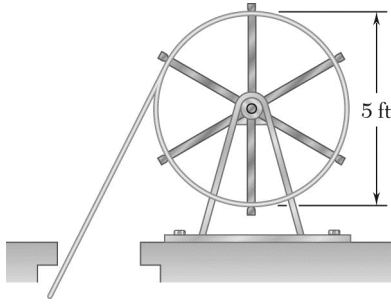
$$c = \frac{1}{2}t = 0.009 \text{ in.}$$

Maximum strain: $\epsilon_m = \frac{c}{\rho} = \frac{0.009}{4.009} = 0.002245$

Maximum stress: $\sigma_m = E\epsilon_m = (29 \times 10^6)(0.002245)$

$$\sigma_m = 65.1 \times 10^3 \text{ psi}$$

$$\sigma_m = 65.1 \text{ ksi} \quad \blacktriangleleft$$



PROBLEM 4.22

Straight rods of 0.30-in. diameter and 200-ft length are sometimes used to clear underground conduits of obstructions or to thread wires through a new conduit. The rods are made of high-strength steel and, for storage and transportation, are wrapped on spools of 5-ft diameter. Assuming that the yield strength is not exceeded, determine (a) the maximum stress in a rod, when the rod, which is initially straight, is wrapped on a spool, (b) the corresponding bending moment in the rod. Use $E = 29 \times 10^6$ psi.

SOLUTION

Radius of cross section: $r = \frac{1}{2}d = \frac{1}{2}(0.30) = 0.15$ in.

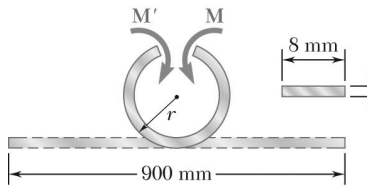
Moment of inertia: $I = \frac{\pi}{4}r^4 = \frac{\pi}{4}(0.15)^4 = 397.61 \times 10^{-6}$ in⁴

$D = 5$ ft = 60 in. $\rho = \frac{1}{2}D = 30$ in.

$c = r = 0.15$ in.

(a) $\sigma_{\max} = \frac{Ec}{\rho} = \frac{(29 \times 10^6)(0.15)}{30} = 145 \times 10^3$ psi $\sigma_{\max} = 145$ ksi ◀

(b) $M = \frac{EI}{\rho} = \frac{(29 \times 10^6)(397.61 \times 10^{-6})}{30}$ $M = 384$ lb · in ◀



PROBLEM 4.23

A 900-mm strip of steel is bent into a full circle by two couples applied as shown. Determine (a) the maximum thickness t of the strip if the allowable stress of the steel is 420 MPa, (b) the corresponding moment M of the couples. Use $E = 200$ GPa.

SOLUTION

When the rod is bent into a full circle, the circumference is 900 mm. Since the circumference is equal to 2π times ρ , the radius of curvature, we get

$$\rho = \frac{900 \text{ mm}}{2\pi} = 143.24 \text{ mm} = 0.14324 \text{ m}$$

$$\text{Stress: } \sigma = E\varepsilon = \frac{Ec}{\rho} \quad \text{or} \quad c = \frac{\rho\sigma}{E}$$

For $\sigma = 420$ MPa and $E = 200$ GPa,

$$c = \frac{(0.14324)(420 \times 10^6)}{200 \times 10^9} = 0.3008 \times 10^{-3} \text{ m}$$

(a) Maximum thickness: $t = 2c = 0.6016 \times 10^{-3} \text{ m}$

$$t = 0.602 \text{ mm} \quad \blacktriangleleft$$

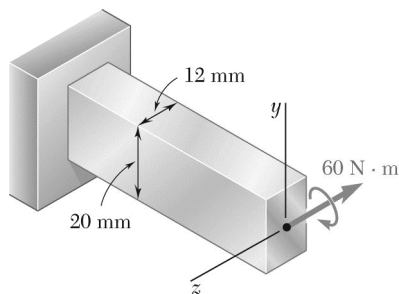
Moment of inertia for a rectangular section.

$$I = \frac{bt^3}{12} = \frac{(8 \times 10^{-3})(0.6016 \times 10^{-3})^3}{12} = 145.16 \times 10^{-15} \text{ m}^4$$

(b) Bending moment: $M = \frac{EI}{\rho}$

$$M = \frac{(200 \times 10^9)(145.16 \times 10^{-15})}{0.14324} = 0.203 \text{ N} \cdot \text{m}$$

$$M = 0.203 \text{ N} \cdot \text{m} \quad \blacktriangleleft$$



PROBLEM 4.24

A $60 \text{ N} \cdot \text{m}$ couple is applied to the steel bar shown. (a) Assuming that the couple is applied about the z axis as shown, determine the maximum stress and the radius of curvature of the bar. (b) Solve part *a*, assuming that the couple is applied about the y axis. Use $E = 200 \text{ GPa}$.

SOLUTION

(a) Bending about z -axis.

$$I = \frac{1}{12}bh^3 = \frac{1}{12}(12)(20)^3 = 8 \times 10^3 \text{ mm}^4 = 8 \times 10^{-9} \text{ m}^4$$

$$c = \frac{20}{2} = 10 \text{ mm} = 0.010 \text{ m}$$

$$\sigma = \frac{Mc}{I} = \frac{(60)(0.010)}{8 \times 10^{-9}} = 75.0 \times 10^6 \text{ Pa} \quad \sigma = 75.0 \text{ MPa} \quad \blacktriangleleft$$

$$\frac{1}{\rho} = \frac{M}{EI} = \frac{60}{(200 \times 10^9)(8 \times 10^{-9})} = 37.5 \times 10^{-3} \text{ m}^{-1} \quad \rho = 26.7 \text{ m} \quad \blacktriangleleft$$

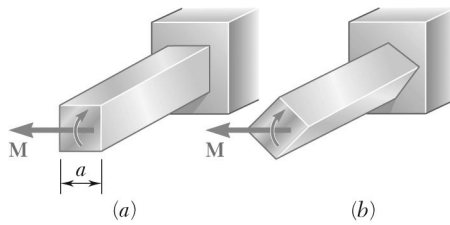
(b) Bending about y -axis.

$$I = \frac{1}{12}bh^3 = \frac{1}{12}(20)(12)^3 = 2.88 \times 10^3 \text{ mm}^4 = 2.88 \times 10^{-9} \text{ m}^4$$

$$c = \frac{12}{2} = 6 \text{ mm} = 0.006 \text{ m}$$

$$\sigma = \frac{Mc}{I} = \frac{(60)(0.006)}{2.88 \times 10^{-9}} = 125.0 \times 10^6 \text{ Pa} \quad \sigma = 125.0 \text{ MPa} \quad \blacktriangleleft$$

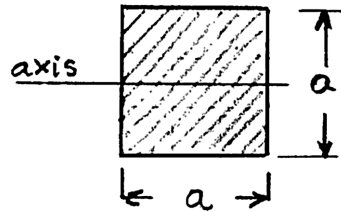
$$\frac{1}{\rho} = \frac{M}{EI} = \frac{60}{(200 \times 10^9)(2.88 \times 10^{-9})} = 104.17 \times 10^{-3} \text{ m}^{-1} \quad \rho = 9.60 \text{ m} \quad \blacktriangleleft$$



PROBLEM 4.25

A couple of magnitude **M** is applied to a square bar of side **a**. For each of the orientations shown, determine the maximum stress and the curvature of the bar.

SOLUTION



$$I = \frac{1}{12}bh^3 = \frac{1}{12}aa^3 = \frac{a^4}{12}$$

$$c = \frac{a}{2}$$

$$\sigma_{\max} = \frac{Mc}{I} = \frac{M \frac{a}{2}}{\frac{a^4}{12}}$$

$$\sigma_{\max} = \frac{6M}{a^3} \quad \blacktriangleleft$$

$$\frac{1}{\rho} = \frac{M}{EI} = \frac{M}{E \frac{a^4}{12}}$$

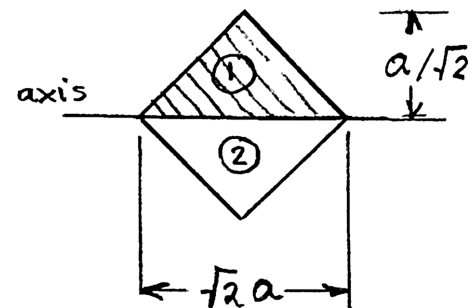
$$\frac{1}{\rho} = \frac{12M}{Ea^4} \quad \blacktriangleleft$$

For one triangle, the moment of inertia about its base is

$$I_1 = \frac{1}{12}bh^3 = \frac{1}{12}(\sqrt{2}a)\left(\frac{a}{\sqrt{2}}\right)^3 = \frac{a^4}{24}$$

$$I_2 = I_1 = \frac{a^4}{24}$$

$$I = I_1 + I_2 = \frac{a^4}{12}$$

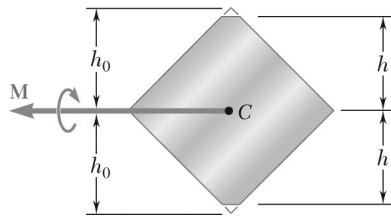


$$c = \frac{a}{\sqrt{2}} \quad \sigma_{\max} = \frac{Mc}{I} = \frac{Ma/\sqrt{2}}{a^4/12} = \frac{6\sqrt{2}M}{a^3}$$

$$\sigma_{\max} = \frac{8.49M}{a^3} \quad \blacktriangleleft$$

$$\frac{1}{\rho} = \frac{M}{EI} = \frac{M}{E \frac{a^4}{12}}$$

$$\frac{1}{\rho} = \frac{12M}{Ea^4} \quad \blacktriangleleft$$

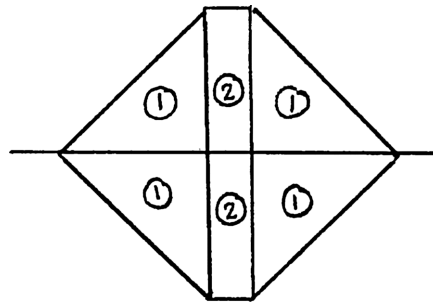


PROBLEM 4.26

A portion of a square bar is removed by milling so that its cross section is as shown. The bar is then bent about its horizontal axis by a couple $\mathbf{M}$. Considering the case where $h = 0.9h_0$, express the maximum stress in the bar in the form $\sigma_m = k\sigma_0$, where σ_0 is the maximum stress that would have occurred if the original square bar had been bent by the same couple $\mathbf{M}$, and determine the value of k .

SOLUTION

$$\begin{aligned}
 I &= 4I_1 + 2I_2 \\
 &= (4)\left(\frac{1}{12}\right)h h^3 + (2)\left(\frac{1}{3}\right)(2h_0 - 2h)(h^3) \\
 &= \frac{1}{3}h^4 + \frac{4}{3}h_0 h^3 - \frac{4}{3}h h^3 = \frac{4}{3}h_0 h^3 - h^4 \\
 c &= h \\
 \sigma &= \frac{Mc}{I} = \frac{Mh}{\frac{4}{3}h_0 h^3 - h^4} = \frac{3M}{(4h_0 - 3h)h^2}
 \end{aligned}$$

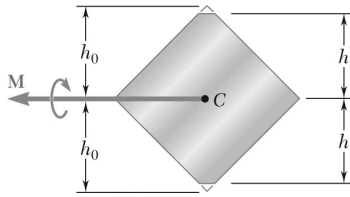


For the original square,

$$\begin{aligned}
 h &= h_0, \quad c = h_0. \\
 \sigma_0 &= \frac{3M}{(4h_0 - 3h_0)h_0^2} = \frac{3M}{h_0^3} \\
 \frac{\sigma}{\sigma_0} &= \frac{h_0^3}{(4h_0 - 3h)h^2} = \frac{h_0^3}{(4h_0 - (3)(0.9)h_0)(0.9h_0^2)} = 0.950 \\
 \sigma &= 0.950\sigma_0
 \end{aligned}$$

$$k = 0.950 \quad \blacktriangleleft$$

PROBLEM 4.27

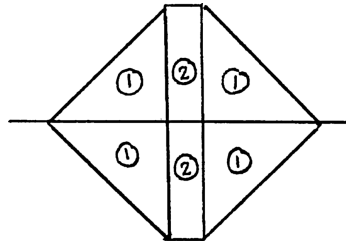


In Prob. 4.26, determine (a) the value of h for which the maximum stress σ_m is as small as possible, (b) the corresponding value of k .

PROBLEM 4.26 A portion of a square bar is removed by milling so that its cross section is as shown. The bar is then bent about its horizontal axis by a couple $\mathbf{M}$. Considering the case where $h = 0.9h_0$, express the maximum stress in the bar in the form $\sigma_m = k\sigma_0$, where σ_0 is the maximum stress that would have occurred if the original square bar had been bent by the same couple $\mathbf{M}$, and determine the value of k .

SOLUTION

$$\begin{aligned}
 I &= 4I_1 + 2I_2 \\
 &= (4)\left(\frac{1}{12}\right)hh^3 + (2)\left(\frac{1}{3}\right)(2h_0 - 2h)h^3 \\
 &= \frac{1}{3}h^4 - \frac{4}{3}h_0h^3 + \frac{4}{3}h^3 = \frac{4}{3}h_0h^3 - h^4 \\
 c &= h \quad \frac{I}{c} = \frac{4}{3}h_0h^2 - h^3
 \end{aligned}$$



$$\frac{I}{c} \text{ is maximum at } \frac{d}{dh} \left[\frac{4}{3}h_0h^2 - h^3 \right] = 0.$$

$$\frac{8}{3}h_0h - 3h^2 = 0$$

$$h = \frac{8}{9}h_0 \quad \blacktriangleleft$$

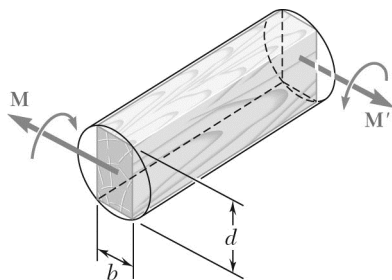
$$\frac{I}{c} = \frac{4}{3}h_0 \left(\frac{8}{9}h_0 \right)^2 - \left(\frac{8}{9}h_0 \right)^3 = \frac{256}{729}h_0^3 \quad \sigma = \frac{Mc}{I} = \frac{729M}{256h_0^3}$$

$$\text{For the original square,} \quad h = h_0 \quad c = h_0 \quad \frac{I_0}{c_0} = \frac{1}{3}h_0^3$$

$$\sigma_0 = \frac{Mc_0}{I_0} = \frac{3M}{h_0^2}$$

$$\frac{\sigma}{\sigma_0} = \frac{729}{256} \cdot \frac{1}{3} = \frac{729}{768} = 0.949$$

$$k = 0.949 \quad \blacktriangleleft$$



PROBLEM 4.28

A couple M will be applied to a beam of rectangular cross section that is to be sawed from a log of circular cross section. Determine the ratio d/b , for which (a) the maximum stress σ_m will be as small as possible, (b) the radius of curvature of the beam will be maximum.

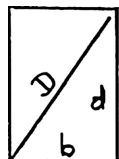
SOLUTION

Let D be the diameter of the log.

$$D^2 = b^2 + d^2 \quad d^2 = D^2 - b^2$$

$$I = \frac{1}{12}bd^3 \quad c = \frac{1}{2}d \quad \frac{I}{c} = \frac{1}{6}bd^2$$

(a) σ_m is the minimum when $\frac{I}{c}$ is maximum.



$$\frac{I}{c} = \frac{1}{6}b(D^2 - b^2) = \frac{1}{6}D^2b - \frac{1}{6}b^3$$

$$\frac{d}{db}\left(\frac{I}{c}\right) = \frac{1}{6}D^2 - \frac{3}{6}b^2 = 0 \quad b = \frac{1}{\sqrt{3}}D$$

$$d = \sqrt{D^2 - \frac{1}{3}D^2} = \sqrt{\frac{2}{3}}D \quad \frac{d}{b} = \sqrt{2} \quad \blacktriangleleft$$

(b) $\rho = \frac{EI}{M}$

ρ is maximum when I is maximum, $\frac{1}{12}bd^3$ is maximum, or b^2d^6 is maximum.

$(D^2 - d^2)d^6$ is maximum.

$$6D^2d^5 - 8d^7 = 0 \quad d = \frac{\sqrt{3}}{2}D$$

$$b = \sqrt{D^2 - \frac{3}{4}D^2} = \frac{1}{2}D \quad \frac{d}{b} = \sqrt{3} \quad \blacktriangleleft$$

PROBLEM 4.29

For the aluminum bar and loading of Sample Prob. 4.1, determine (a) the radius of curvature ρ' of a transverse cross section, (b) the angle between the sides of the bar that were originally vertical. Use $E = 10.6 \times 10^6$ psi and $\nu = 0.33$.

SOLUTION

From Sample Prob. 4.1, $I = 12.97 \text{ in}^4$ $M = 103.8 \text{ kip} \cdot \text{in}$

$$\frac{1}{\rho} = \frac{M}{EI} = \frac{103.8 \times 10^3}{(10.6 \times 10^6)(12.97)} = 755 \times 10^{-6} \text{ in}^{-1}$$

$$(a) \quad \frac{1}{\rho'} = \nu \frac{1}{\rho} = (0.33)(755 \times 10^{-6}) = 249 \times 10^{-6} \text{ in}^{-1}$$

$$\rho' = 4010 \text{ in.}$$

$$\rho' = 334 \text{ ft} \quad \blacktriangleleft$$

$$(b) \quad \theta = \frac{\text{length of arc}}{\text{radius}} = \frac{b}{\rho'} = \frac{3.25}{4010} = 810 \times 10^{-6} \text{ rad}$$

$$\theta = 0.0464^\circ \quad \blacktriangleleft$$

PROBLEM 4.30

For the bar and loading of Example 4.01, determine (a) the radius of curvature ρ , (b) the radius of curvature ρ' of a transverse cross section, (c) the angle between the sides of the bar that were originally vertical. Use $E = 29 \times 10^6$ psi and $\nu = 0.29$.

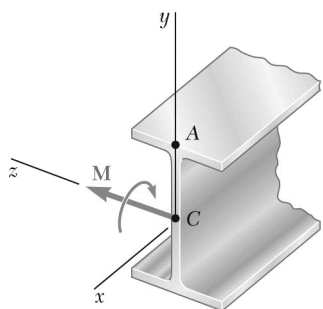
SOLUTION

From Example 4.01, $M = 30 \text{ kip} \cdot \text{in.}$, $I = 1.042 \text{ in}^4$

$$(a) \quad \frac{1}{\rho} = \frac{M}{EI} = \frac{(30 \times 10^3)}{(29 \times 10^6)(1.042)} = 993 \times 10^{-6} \text{ in}^{-1} \quad \rho = 1007 \text{ in.} \quad \blacktriangleleft$$

$$(b) \quad \varepsilon^1 = \nu \varepsilon = \frac{\nu c}{\rho} = \nu \frac{c}{\rho'} \\ \frac{1}{\rho'} = \nu \frac{1}{\rho} = (0.29)(993 \times 10^{-6}) \text{ in.}^{-1} = 288 \times 10^{-6} \text{ in.}^{-1} \quad \rho' = 3470 \text{ in.} \quad \blacktriangleleft$$

$$(c) \quad \theta = \frac{\text{length of arc}}{\text{radius}} = \frac{b}{\rho'} = \frac{0.8}{3470} = 230 \times 10^{-6} \text{ rad} \quad \theta = 0.01320^\circ \quad \blacktriangleleft$$



PROBLEM 4.31

A W200×31.3 rolled-steel beam is subjected to a couple $\mathbf{M}$ of moment $45 \text{ kN} \cdot \text{m}$. Knowing that $E = 200 \text{ GPa}$ and $\nu = 0.29$, determine (a) the radius of curvature ρ , (b) the radius of curvature ρ' of a transverse cross section.

SOLUTION

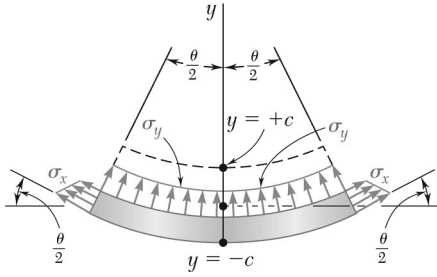
For W 200×31.3 rolled steel section,

$$\begin{aligned} I &= 31.4 \times 10^6 \text{ mm}^4 \\ &= 31.4 \times 10^{-6} \text{ m}^4 \end{aligned}$$

$$(a) \quad \frac{1}{\rho} = \frac{M}{EI} = \frac{45 \times 10^3}{(200 \times 10^9)(31.4 \times 10^{-6})} = 7.17 \times 10^{-3} \text{ m}^{-1} \quad \rho = 139.6 \text{ m} \quad \blacktriangleleft$$

$$(b) \quad \frac{1}{\rho'} = \nu \frac{1}{\rho} = (0.29)(7.17 \times 10^{-3}) = 2.07 \times 10^{-3} \text{ m}^{-1} \quad \rho' = 481 \text{ m} \quad \blacktriangleleft$$

PROBLEM 4.32



It was assumed in Sec. 4.3 that the normal stresses σ_y in a member in pure bending are negligible. For an initially straight elastic member of rectangular cross section, (a) derive an approximate expression for σ_y as a function of y , (b) show that $(\sigma_y)_{\max} = -(c/2\rho)(\sigma_x)_{\max}$ and, thus, that σ_y can be neglected in all practical situations. (*Hint:* Consider the free-body diagram of the portion of beam located below the surface of ordinate y and assume that the distribution of the stress σ_x is still linear.)

SOLUTION

Denote the width of the beam by b and the length by L .

$$\theta = \frac{L}{\rho}$$

Using the free body diagram above, with $\cos \frac{\theta}{2} \approx 1$

$$\Sigma F_y = 0: \quad \sigma_y bL + 2 \int_{-c}^y \sigma_x b dy \sin \frac{\theta}{2} = 0$$

$$\sigma_y = -\frac{2}{L} \sin \frac{\theta}{2} \int_{-c}^y \sigma_x dy \approx -\frac{\theta}{L} \int_{-c}^y \sigma_x dy = -\frac{1}{\rho} \int_{-c}^y \sigma_x dy$$

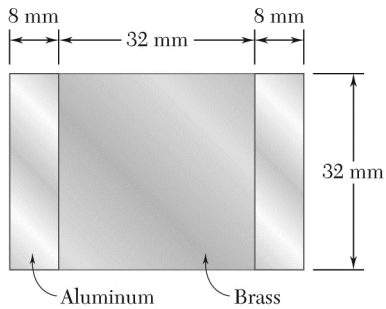
But,
$$\sigma_x = -(\sigma_x)_{\max} \frac{y}{c}$$

$$(a) \quad \sigma_y = \frac{(\sigma_x)_{\max}}{\rho c} \int_{-c}^y y dy = \frac{(\sigma_x)_{\max}}{\rho c} \frac{y^2}{2} \Big|_{-c}^y \quad \sigma_y = \frac{(\sigma_x)_{\max}}{2\rho c} (y^2 - c^2) \quad \blacktriangleleft$$

The maximum value σ_y occurs at $y = 0$.

$$(b) \quad (\sigma_y)_{\max} = -\frac{(\sigma_x)_{\max} c^2}{2\rho c} = -\frac{(\sigma_x)_{\max} c}{2\rho} \quad \blacktriangleleft$$

PROBLEM 4.33



A bar having the cross section shown has been formed by securely bonding brass and aluminum stock. Using the data given below, determine the largest permissible bending moment when the composite bar is bent about a horizontal axis.

	Aluminum	Brass
Modulus of elasticity	70 GPa	105 GPa
Allowable stress	100 MPa	160 MPa

SOLUTION

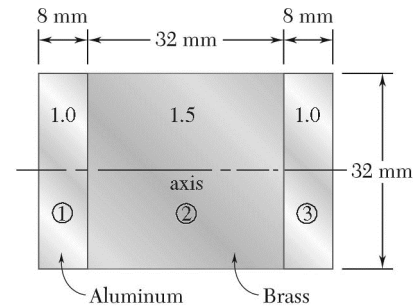
Use aluminum as the reference material.

For aluminum, $n = 1.0$

For brass, $n = E_b/E_a = 105/70 = 1.5$

Values of n are shown on the figure.

For the transformed section,



$$I_1 = \frac{n_1}{12} b_1 h_1^3 = \frac{1.0}{12} (8)(32)^3 = 21.8453 \times 10^3 \text{ mm}^4$$

$$I_2 = \frac{n_2}{12} b_2 h_2^3 = \frac{1.5}{12} (32)(32)^3 = 131.072 \times 10^3 \text{ mm}^4$$

$$I_3 = I_1 = 21.8453 \times 10^3 \text{ mm}^4$$

$$I = I_1 + I_2 + I_3 = 174.7626 \times 10^3 \text{ mm}^4 = 174.7626 \times 10^{-9} \text{ m}^4$$

$$|\sigma| = \left| \frac{n M y}{I} \right| \quad M = \left| \frac{\sigma I}{n y} \right|$$

Aluminum: $n = 1.0$, $|y| = 16 \text{ mm} = 0.016 \text{ m}$, $\sigma = 100 \times 10^6 \text{ Pa}$

$$M = \frac{(100 \times 10^6)(174.7626 \times 10^{-9})}{(1.0)(0.016)} = 1.0923 \times 10^3 \text{ N} \cdot \text{m}$$

Brass: $n = 1.5$, $|y| = 16 \text{ mm} = 0.016 \text{ m}$, $\sigma = 160 \times 10^6 \text{ Pa}$

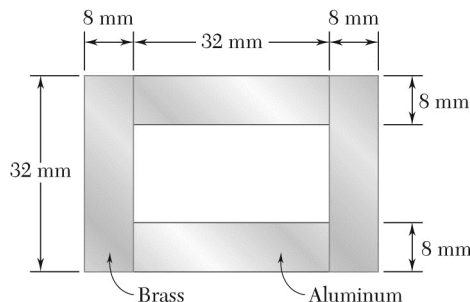
$$M = \frac{(160 \times 10^6)(174.7626 \times 10^{-9})}{(1.5)(0.016)} = 1.1651 \times 10^3 \text{ N} \cdot \text{m}$$

Choose the smaller value.

$$M = 1.092 \times 10^3 \text{ N} \cdot \text{m}$$

$$M = 1.092 \text{ kN} \cdot \text{m} \quad \blacktriangleleft$$

PROBLEM 4.34



A bar having the cross section shown has been formed by securely bonding brass and aluminum stock. Using the data given below, determine the largest permissible bending moment when the composite bar is bent about a horizontal axis.

	Aluminum	Brass
Modulus of elasticity	70 GPa	105 GPa
Allowable stress	100 MPa	160 MPa

SOLUTION

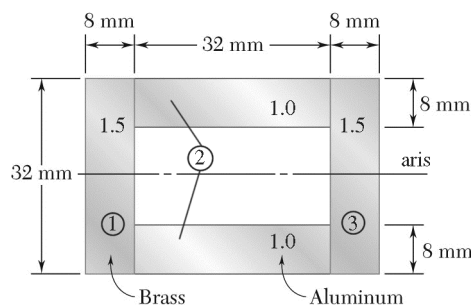
Use aluminum as the reference material.

For aluminum, $n = 1.0$

For brass, $n = E_b/E_a = 105/70 = 1.5$

Values of n are shown on the sketch.

For the transformed section,



$$I_1 = \frac{n_1}{12} b_1 h_1^3 = \frac{1.5}{12} (8)(32)^3 = 32.768 \times 10^3 \text{ mm}^4$$

$$I_2 = \frac{n_2}{12} b_2 (H_2^3 - h_2^3) = \frac{1.0}{12} (32)(32^3 - 16^3) = 76.459 \times 10^3 \text{ mm}^4$$

$$I_3 = I_1 = 32.768 \times 10^3 \text{ mm}^4$$

$$I = I_1 + I_2 + I_3 = 141.995 \times 10^3 \text{ mm}^4 = 141.995 \times 10^{-9} \text{ m}^4$$

$$|\sigma| = \left| \frac{nMy}{I} \right| \quad M = \left| \frac{\sigma I}{ny} \right|$$

Aluminum: $n = 1.0$, $|y| = 16 \text{ mm} = 0.016 \text{ m}$, $\sigma = 100 \times 10^6 \text{ Pa}$

$$M = \frac{(100 \times 10^6)(141.995 \times 10^{-9})}{(1.0)(0.016)} = 887.47 \text{ N} \cdot \text{m}$$

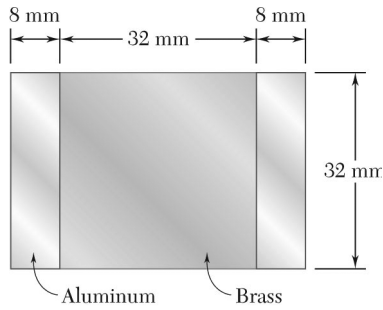
Brass: $n = 1.5$, $|y| = 16 \text{ mm} = 0.016 \text{ m}$, $\sigma = 160 \times 10^6 \text{ Pa}$

$$M = \frac{(160 \times 10^6)(141.995 \times 10^{-9})}{(1.5)(0.016)} = 946.63 \text{ N} \cdot \text{m}$$

Choose the smaller value.

$$M = 887 \text{ N} \cdot \text{m} \quad \blacktriangleleft$$

PROBLEM 4.35



For the composite bar indicated, determine the largest permissible bending moment when the bar is bent about a vertical axis.

PROBLEM 4.35 Bar of Prob. 4.33.

	Aluminum	Brass
Modulus of elasticity	70 GPa	105 GPa
Allowable stress	100 MPa	160 MPa

SOLUTION

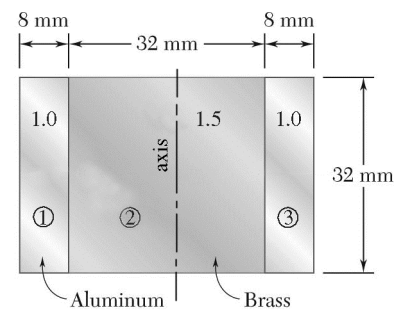
Use aluminum as the reference material.

For aluminum, $n = 1.0$

For brass, $n = E_b/E_a = 105/70 = 1.5$

Values of n are shown on the figure.

For the transformed section,



$$I_1 = \frac{n_1}{12} h_1 b_1^3 + n_1 A_1 d_1^2 = \frac{1.0}{12} (32)(8)^3 + (1.0)[(32)(8)](20)^2 = 103.7653 \times 10^3 \text{ mm}^4$$

$$I_2 = \frac{n_2}{12} h_2 b_2^3 = \frac{1.5}{12} (32)(32)^3 = 131.072 \times 10^3 \text{ mm}^4$$

$$I_3 = I_1 = 103.7653 \times 10^3 \text{ mm}^4$$

$$I = I_1 + I_2 + I_3 = 338.58 \times 10^3 \text{ mm}^4 = 338.58 \times 10^{-9} \text{ m}^4$$

$$|\sigma| = \left| \frac{nMy}{I} \right| \quad M = \left| \frac{\sigma I}{ny} \right|$$

Aluminum: $n = 1.0$, $|y| = 24 \text{ mm} = 0.024 \text{ m}$, $\sigma = 100 \times 10^6 \text{ Pa}$

$$M = \frac{(100 \times 10^6)(338.58 \times 10^{-9})}{(1.0)(0.024)} = 1.411 \times 10^3 \text{ N} \cdot \text{m}$$

Brass: $n = 1.5$, $|y| = 16 \text{ mm} = 0.016 \text{ m}$, $\sigma = 160 \times 10^6 \text{ Pa}$

$$M = \frac{(160 \times 10^6)(338.58 \times 10^{-9})}{(1.5)(0.016)} = 2.257 \times 10^3 \text{ N} \cdot \text{m}$$

Choose the smaller value.

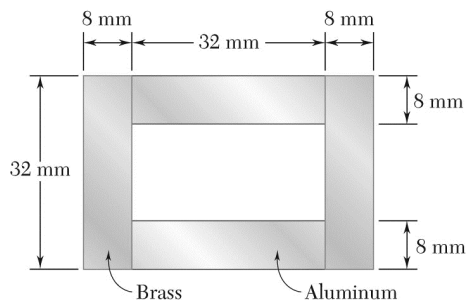
$$M = 1.411 \times 10^3 \text{ N} \cdot \text{m}$$

$$M = 1.411 \text{ kN} \cdot \text{m} \quad \blacktriangleleft$$

PROBLEM 4.36

For the composite bar indicated, determine the largest permissible bending moment when the bar is bent about a vertical axis.

PROBLEM 4.36 Bar of Prob. 4.34.



	Aluminum	Brass
Modulus of elasticity	70 GPa	105 GPa
Allowable stress	100 MPa	160 MPa

SOLUTION

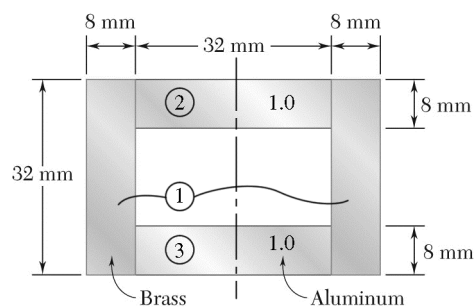
Use aluminum as the reference material.

For aluminum, $n = 1.0$

For brass, $n = E_b/E_a = 105/70 = 1.5$

Values of n are shown on the sketch.

For the transformed section,



$$I_1 = \frac{n_1}{12} h_1 (B_1^3 - b_1^3) = \frac{1.5}{12} (32)(48^3 - 32^3) = 311.296 \times 10^3 \text{ mm}^4$$

$$I_2 = \frac{n_2}{12} h_2 b_2^3 = \frac{1.0}{12} (8)(32)^3 = 21.8453 \times 10^3 \text{ mm}^4$$

$$I_3 = I_2 = 21.8453 \times 10^3 \text{ mm}^4$$

$$I = I_1 + I_2 + I_3 = 354.99 \times 10^3 \text{ mm}^4 = 354.99 \times 10^{-9} \text{ m}^4$$

$$|\sigma| = \left| \frac{nMy}{I} \right| \quad M = \left| \frac{\sigma I}{ny} \right|$$

Aluminum:

$$n = 1.0, \quad |y| = 16 \text{ mm} = 0.016 \text{ m}, \quad \sigma = 100 \times 10^6 \text{ Pa}$$

$$M = \frac{(100 \times 10^6)(354.99 \times 10^{-9})}{(1.0)(0.016)} = 2.2187 \times 10^3 \text{ N} \cdot \text{m}$$

Brass:

$$n = 1.5, \quad |y| = 24 \text{ mm} = 0.024 \text{ m}, \quad \sigma = 160 \times 10^6 \text{ Pa}$$

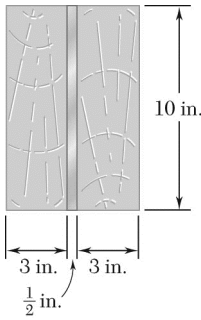
$$M = \frac{(160 \times 10^6)(354.99 \times 10^{-9})}{(1.5)(0.024)} = 1.57773 \times 10^3 \text{ N} \cdot \text{m}$$

Choose the smaller value.

$$M = 1.57773 \times 10^3 \text{ N} \cdot \text{m}$$

$$M = 1.578 \text{ kN} \cdot \text{m} \quad \blacktriangleleft$$

PROBLEM 4.37



Wooden beams and steel plates are securely bolted together to form the composite member shown. Using the data given below, determine the largest permissible bending moment when the member is bent about a horizontal axis.

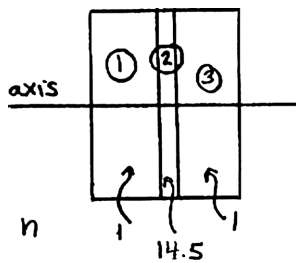
	Wood	Steel
Modulus of elasticity:	2×10^6 psi	29×10^6 psi
Allowable stress:	2000 psi	22 ksi

SOLUTION

Use wood as the reference material.

$$n = 1.0 \text{ in wood}$$

$$n = E_s/E_w = 29/2 = 14.5 \text{ in steel}$$



For the transformed section,

$$I_1 = \frac{n_1}{12} b_1 h_1^3 = \frac{1.0}{12} (3)(10)^3 = 250 \text{ in}^4$$

$$I_2 = \frac{n_2}{12} b_2 h_2^3 = \frac{14.5}{12} \left(\frac{1}{2}\right) (10)^3 = 604.17 \text{ in}^4$$

$$I_3 = I_1 = 250 \text{ in}^4$$

$$I = I_1 + I_2 + I_3 = 1104.2 \text{ in}^4$$

$$|\sigma| = \left| \frac{nMy}{I} \right| \quad \therefore M = \frac{\sigma I}{ny}$$

Wood: $n = 1.0, \quad y = 5 \text{ in.}, \quad \sigma = 2000 \text{ psi}$

$$M = \frac{(2000)(1104.2)}{(1.0)(5)} = 441.7 \times 10^3 \text{ lb} \cdot \text{in}$$

Steel: $n = 14.5, \quad y = 5 \text{ in.}, \quad \sigma = 22 \text{ ksi} = 22 \times 10^3 \text{ psi}$

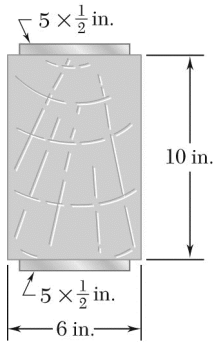
$$M = \frac{(22 \times 10^3)(1104.2)}{(14.5)(5)} = 335.1 \times 10^3 \text{ lb} \cdot \text{in}$$

Choose the smaller value.

$$M = 335 \times 10^3 \text{ lb} \cdot \text{in}$$

$$M = 335 \text{ kip} \cdot \text{in} \quad \blacktriangleleft$$

PROBLEM 4.38



Wooden beams and steel plates are securely bolted together to form the composite member shown. Using the data given below, determine the largest permissible bending moment when the member is bent about a horizontal axis.

	Wood	Steel
Modulus of elasticity:	2×10^6 psi	29×10^6 psi
Allowable stress:	2000 psi	22 ksi

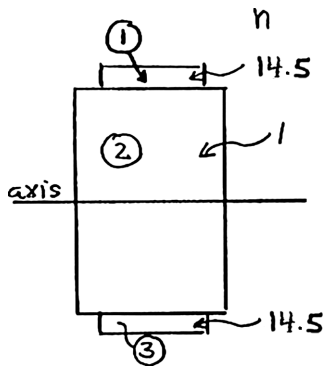
SOLUTION

Use wood as the reference material.

$$n = 1.0 \text{ in wood}$$

$$n = E_s/E_w = 29/2 = 14.5 \text{ in steel}$$

For the transformed section,



$$I_1 = \frac{n_1}{12} b_1 h_1^3 + n_1 A_1 d_1^2$$

$$= \frac{14.5}{12} (5) \left(\frac{1}{2} \right)^3 + (14.5)(5) \left(\frac{1}{2} \right) (5.25)^2 = 999.36 \text{ in}^4$$

$$I_2 = \frac{n_2}{12} b_2 h_2^3 = \frac{1.0}{12} (6)(10)^3 = 500 \text{ in}^4$$

$$I_3 = I_1 = 999.36 \text{ in}^4$$

$$I = I_1 + I_2 + I_3 = 2498.7 \text{ in}^4$$

$$|\sigma| = \left| \frac{nMy}{I} \right| \quad \therefore M = \frac{\sigma I}{ny}$$

Wood: $n = 1.0$, $y = 5 \text{ in.}$, $\sigma = 2000 \text{ psi}$

$$M = \frac{(2000)(2499)}{(1.0)(5)} = 999.5 \times 10^3 \text{ lb} \cdot \text{in}$$

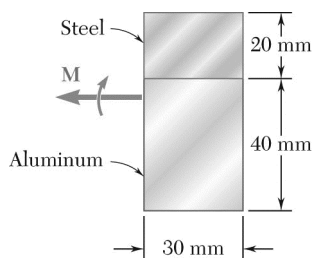
Steel: $n = 14.5$, $y = 5.5 \text{ in.}$, $\sigma = 22 \text{ ksi} = 22 \times 10^3 \text{ psi}$

$$M = \frac{(22 \times 10^3)(2499)}{(14.5)(5.5)} = 689.3 \times 10^3 \text{ lb} \cdot \text{in}$$

Choose the smaller value.

$$M = 689 \times 10^3 \text{ lb} \cdot \text{in}$$

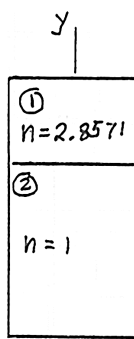
$$M = 689 \text{ kip} \cdot \text{in} \quad \blacktriangleleft$$



PROBLEM 4.39

A steel bar and an aluminum bar are bonded together to form the composite beam shown. The modulus of elasticity for aluminum is 70 GPa and for steel is 200 GPa. Knowing that the beam is bent about a horizontal axis by a couple of moment $M = 1500 \text{ N} \cdot \text{m}$, determine the maximum stress in (a) the aluminum, (b) the steel.

SOLUTION



Use aluminum as the reference material.

For aluminum, $n = 1$

For steel, $n = E_s/E_a = 200/70 = 2.8571$

Transformed section:

Part	A, mm^2	nA, mm^2	$\bar{y}_o, \text{mm}$	$nA\bar{y}_o, \text{mm}^3$	d, mm
1	600	1714.3	50	85714	12.35
2	1200	1200	20	24000	17.65
Σ		2914.3		109714	

$$\bar{Y}_0 = \frac{109714}{2914.3} = 37.65 \text{ mm} \quad d = |\bar{y}_0 - \bar{Y}_0|$$

$$I_1 = \frac{n_1}{12} b_1 h_1^3 + n_1 A_1 d_1^2 = \frac{2.8571}{12} (30)(20)^3 + (1714.3)(12.35)^2 = 318.61 \times 10^3 \text{ mm}^4$$

$$I_2 = \frac{n_2}{12} b_2 h_2^3 + n_2 A_2 d_2^2 = \frac{1}{12} (30)(40)^3 + (1200)(17.65)^2 = 533.83 \times 10^3 \text{ mm}^4$$

$$I = I_1 + I_2 = 852.44 \times 10^3 \text{ mm}^4 = 852.44 \times 10^{-9} \text{ m}^4$$

$$M = 1500 \text{ N} \cdot \text{m}$$

Stress: $\sigma = -\frac{nMy}{I}$

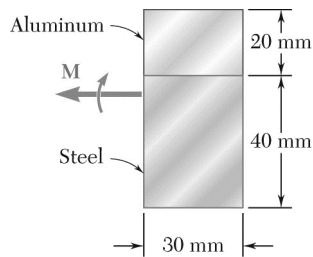
(a) Aluminum: $n = 1, \quad y = -37.65 \text{ mm} = -0.03765 \text{ m}$

$$\sigma_a = -\frac{(1)(1500)(-0.03765)}{852.44 \times 10^{-9}} = 66.2 \times 10^6 \text{ Pa} \quad \sigma_a = 66.2 \text{ MPa} \quad \blacktriangleleft$$

(b) Steel: $n = 2.8571, \quad y = 60 - 37.65 = 22.35 \text{ mm} = 0.02235 \text{ m}$

$$\sigma_s = -\frac{nMy}{I} = -\frac{(2.8571)(1500)(0.02235)}{852.44 \times 10^{-9}} = -112.4 \times 10^6 \text{ Pa}$$

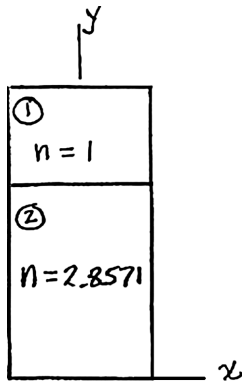
$$\sigma_s = -112.4 \text{ MPa} \quad \blacktriangleleft$$



PROBLEM 4.40

A steel bar and an aluminum bar are bonded together to form the composite beam shown. The modulus of elasticity for aluminum is 70 GPa and for steel is 200 GPa. Knowing that the beam is bent about a horizontal axis by a couple of moment $M = 1500 \text{ N} \cdot \text{m}$, determine the maximum stress in (a) the aluminum, (b) the steel.

SOLUTION



Use aluminum as the reference material.

For aluminum, $n = 1$

For steel, $n = E_s/E_a = 200/70 = 2.8571$

Transformed section:

Part	A, mm^2	nA, mm^2	$\bar{y}_o, \text{mm}$	$nA\bar{y}_o, \text{mm}^3$	d, mm
1	600	600	50	30000	25.53
2	1200	3428.5	20	68570	4.47
Σ		4028.5		98570	

$$\bar{Y}_0 = \frac{98570}{4028.5} = 24.47 \text{ mm} \quad d = |\bar{y}_o - \bar{Y}_0|$$

$$I_1 = \frac{n_1}{12} b_1 h_1^3 + n_1 A_1 d_1^2 = \frac{1}{12} (30)(20)^3 + (600)(25.53)^2 = 411.07 \times 10^3 \text{ mm}^4$$

$$I_2 = \frac{n_2}{12} b_2 h_2^3 + n_2 A_2 d_2^2 = \frac{2.8571}{12} (30)(40)^3 + (3428.5)(4.47)^2 = 525.64 \times 10^3 \text{ mm}^4$$

$$I = I_1 + I_2 = 936.71 \times 10^3 \text{ mm}^4 = 936.71 \times 10^{-9} \text{ m}^4$$

$$M = 1500 \text{ N} \cdot \text{m}$$

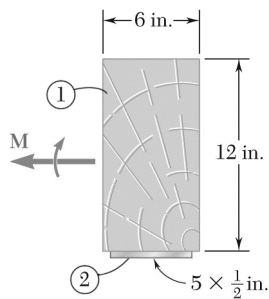
$$\text{Stress:} \quad \sigma = -\frac{nMy}{I}$$

(a) Aluminum: $n = 1, \quad y = 60 - 24.47 = 35.53 \text{ mm} = 0.03553 \text{ m}$

$$\sigma_a = -\frac{(1)(1500)(0.03553)}{936.71 \times 10^{-9}} = -56.9 \times 10^6 \text{ Pa} \quad \sigma_a = -56.9 \text{ MPa} \quad \blacktriangleleft$$

(b) Steel: $n = 2.8571, \quad y = -24.47 \text{ mm} = -0.02447 \text{ m}$

$$\sigma_s = -\frac{(2.8571)(1500)(-0.02447)}{936.71 \times 10^{-9}} = 111.9 \times 10^6 \text{ Pa} \quad \sigma_s = 111.9 \text{ MPa} \quad \blacktriangleleft$$



PROBLEM 4.41

The 6×12-in. timber beam has been strengthened by bolting to it the steel reinforcement shown. The modulus of elasticity for wood is 1.8×10^6 psi and for steel, 29×10^6 psi. Knowing that the beam is bent about a horizontal axis by a couple of moment $M = 450$ kip·in., determine the maximum stress in (a) the wood, (b) the steel.

SOLUTION

Use wood as the reference material.

For wood, $n = 1$

For steel, $n = E_s / E_w = 29 / 1.8 = 16.1111$

Transformed section: ① = wood ② = steel

	$A, \text{ in}^2$	$nA, \text{ in}^2$	$\bar{y}_o$	$nA\bar{y}_o, \text{ in}^3$
①	72	72	6	432
②	2.5	40.278	-0.25	-10.069
		112.278		421.931

The neutral axis lies 3.758 in. above the wood-steel interface.

$$I_1 = \frac{n_1}{12} b_1 h_1^3 + n_1 A_1 d_1^2 = \frac{1}{12} (6)(12)^3 + (72)(6 - 3.758)^2 = 1225.91 \text{ in}^4$$

$$I_2 = \frac{n_2}{12} b_2 h_2^3 + n_2 A_2 d_2^2 = \frac{16.1111}{12} (5)(0.5)^3 + (40.278)(3.758 + 0.25)^2 = 647.87 \text{ in}^4$$

$$I = I_1 + I_2 = 1873.77 \text{ in}^4$$

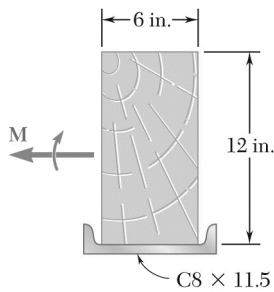
$$M = 450 \text{ kip} \cdot \text{in} \quad \sigma = -\frac{nMy}{I}$$

(a) Wood: $n = 1, \quad y = 12 - 3.758 = 8.242 \text{ in}$

$$\sigma_w = -\frac{(1)(450)(8.242)}{1873.77} = -1.979 \text{ ksi} \quad \sigma_w = -1.979 \text{ ksi} \quad \blacktriangleleft$$

(b) Steel: $n = 16.1111, \quad y = -3.758 - 0.5 = -4.258 \text{ in}$

$$\sigma_s = -\frac{(16.1111)(450)(-4.258)}{1873.77} = 16.48 \text{ ksi} \quad \sigma_s = 16.48 \text{ ksi} \quad \blacktriangleleft$$



PROBLEM 4.42

The 6×12-in. timber beam has been strengthened by bolting to it the steel reinforcement shown. The modulus of elasticity for wood is 1.8×10^6 psi and for steel, 29×10^6 psi. Knowing that the beam is bent about a horizontal axis by a couple of moment $M = 450 \text{ kip} \cdot \text{in.}$, determine the maximum stress in (a) the wood, (b) the steel.

SOLUTION

Use wood as the reference material.

For wood, $n = 1$

For steel, $n = \frac{E_s}{E_w} = \frac{29 \times 10^6}{1.8 \times 10^6} = 16.1111$

For C8 × 11.5 channel section,

$$A = 3.38 \text{ in}^2, \quad t_w = 0.220 \text{ in.}, \quad \bar{x} = 0.571 \text{ in.}, \quad I_y = 1.32 \text{ in}^4$$

For the composite section, the centroid of the channel (part 1) lies 0.571 in. above the bottom of the section. The centroid of the wood (part 2) lies $0.220 + 6.00 = 6.22$ in. above the bottom.

Transformed section:

Part	A, in^2	nA, in^2	$\bar{y}, \text{in.}$	$nA\bar{y}, \text{in}^3$	$d, \text{in.}$
1	3.38	54.456	0.571	31.091	3.216
2	72	72	6.22	447.84	2.433
Σ		126.456		478.93	

$$\bar{Y}_0 = \frac{478.93 \text{ in}^3}{126.456 \text{ in}^2} = 3.787 \text{ in.}$$

$$d = |\bar{y}_0 - \bar{Y}_0|$$

The neutral axis lies 3.787 in. above the bottom of the section.

$$I_1 = n_1 \bar{I}_1 + n_1 A_1 d_1^2 = (16.1111)(1.32) + (54.456)(3.216)^2 = 584.49 \text{ in}^4$$

$$I_2 = \frac{n_2}{12} b_2 h_2^3 + n_2 A_2 d_2^2 = \frac{1}{12} (6)(12)^3 + (72)(2.433)^2 = 1290.20 \text{ in}^4$$

$$I = I_1 + I_2 = 1874.69 \text{ in}^4$$

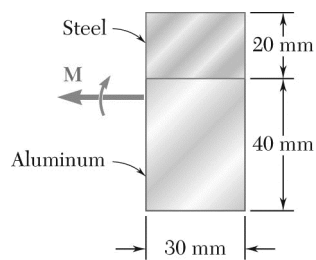
$$M = 450 \text{ kip} \cdot \text{in.} \quad \sigma = -\frac{nMy}{I}$$

(a) Wood: $n = 1, \quad y = 12 + 0.220 - 3.787 = 8.433 \text{ in.}$

$$\sigma_w = -\frac{(1)(450)(8.433)}{1874.69} = -2.02 \text{ ksi} \quad \sigma_w = -2.02 \text{ ksi} \quad \blacktriangleleft$$

(b) Steel: $n = 16.1111, \quad y = -3.787 \text{ in.}$

$$\sigma_s = -\frac{(16.1111)(450)(-3.787)}{1874.67} = 14.65 \text{ ksi} \quad \sigma_s = 14.65 \text{ ksi} \quad \blacktriangleleft$$



PROBLEM 4.43

For the composite beam indicated, determine the radius of curvature caused by the couple of moment 1500 N · m.

Beam of Prob. 4.39.

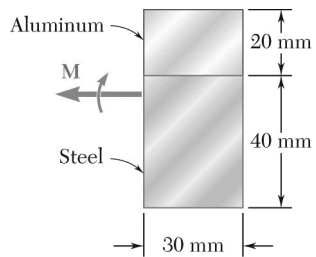
SOLUTION

See solution to Prob. 4.39 for the calculation of I .

$$I = 852.44 \times 10^{-9} \text{ m}^4 \quad E_a = 70 \times 10^9 \text{ Pa}$$

$$\frac{1}{\rho} = \frac{M}{EI} = \frac{1500}{(70 \times 10^9)(852.44 \times 10^{-9})} = 0.02513 \text{ m}^{-1}$$

$$\rho = 39.8 \text{ m} \quad \blacktriangleleft$$



PROBLEM 4.44

For the composite bar indicated, determine the radius of curvature caused by the couple of moment $1500 \text{ N} \cdot \text{m}$.

Beam of Prob. 4.40.

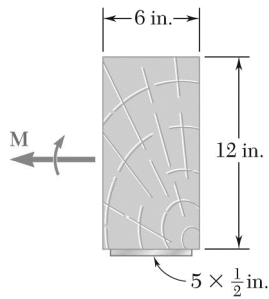
SOLUTION

See solution to Prob. 4.40 for calculation of I .

$$I = 936.71 \times 10^{-9} \text{ m}^4 \quad E_a = 70 \times 10^9 \text{ Pa}$$

$$\frac{1}{\rho} = \frac{M}{EI} = \frac{1500}{(70 \times 10^9)(936.71 \times 10^{-9})} = 0.02288 \text{ m}^{-1}$$

$$\rho = 43.7 \text{ m} \quad \blacktriangleleft$$



PROBLEM 4.45

For the composite beam indicated, determine the radius of curvature caused by the couple of moment 450 kip · in.

Beam of Prob. 4.41.

SOLUTION

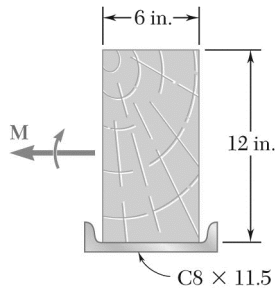
See solution to Prob. 4.41 for calculation of I .

$$I = 1873.77 \text{ in}^4 \quad E_w = 1.8 \times 10^6 \text{ psi}$$

$$M = 450 \text{ kip} \cdot \text{in} = 450 \times 10^3 \text{ lb} \cdot \text{in}$$

$$\frac{1}{\rho} = \frac{M}{EI} = \frac{450 \times 10^3}{(1.8 \times 10^6)(1873.77)} = 133.42 \times 10^{-6} \text{ in}^{-1}$$

$$\rho = 7495 \text{ in.} = 625 \text{ ft} \quad \blacktriangleleft$$



PROBLEM 4.46

For the composite beam indicated, determine the radius of curvature caused by the couple of moment 450 kip · in.

Beam of Prob. 4.42.

SOLUTION

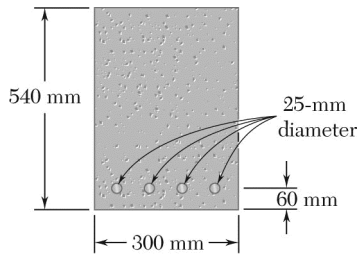
See solution to Prob. 4.42 for calculation of I .

$$I = 1874.69 \text{ in}^4 \quad E_w = 1.8 \times 10^6 \text{ psi}$$

$$M = 450 \text{ kip} \cdot \text{in} = 450 \times 10^3 \text{ lb} \cdot \text{in}$$

$$\frac{1}{\rho} = \frac{M}{EI} = \frac{450 \times 10^3}{(1.8 \times 10^6)(1874.69)} = 133.36 \times 10^{-6} \text{ in}^{-1}$$

$$\rho = 7499 \text{ in.} = 625 \text{ ft} \quad \blacktriangleleft$$



PROBLEM 4.47

The reinforced concrete beam shown is subjected to a positive bending moment of $175 \text{ kN} \cdot \text{m}$. Knowing that the modulus of elasticity is 25 GPa for the concrete and 200 GPa for the steel, determine (a) the stress in the steel, (b) the maximum stress in the concrete.

SOLUTION

$$n = \frac{E_s}{E_c} = \frac{200 \text{ GPa}}{25 \text{ GPa}} = 8.0$$

$$A_s = 4 \cdot \frac{\pi}{4} d^2 = (4) \left(\frac{\pi}{4} \right) (25)^2 = 1.9635 \times 10^3 \text{ mm}^2$$

$$nA_s = 15.708 \times 10^3 \text{ mm}^2$$

Locate the neutral axis:

$$300 \cdot x \frac{x}{2} - (15.708 \times 10^3)(480 - x) = 0$$

$$150x^2 + 15.708 \times 10^3 x - 7.5398 \times 10^6 = 0$$

Solve for x :

$$x = \frac{-15.708 \times 10^3 + \sqrt{(15.708 \times 10^3)^2 + (4)(150)(7.5398 \times 10^6)}}{(2)(150)}$$

$$x = 177.87 \text{ mm}, \quad 480 - x = 302.13 \text{ mm}$$

$$\begin{aligned} I &= \frac{1}{3}(300)x^3 + (15.708 \times 10^3)(480 - x)^2 \\ &= \frac{1}{3}(300)(177.87)^3 + (15.708 \times 10^3)(302.13)^2 \\ &= 1.9966 \times 10^9 \text{ mm}^4 = 1.9966 \times 10^{-3} \text{ m}^4 \end{aligned}$$

$$\sigma = -\frac{nMy}{I}$$

(a) Steel: $y = -302.45 \text{ mm} = -0.30245 \text{ m}$

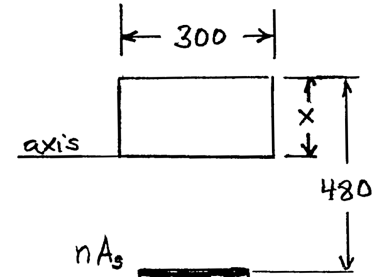
$$\sigma = -\frac{(8.0)(175 \times 10^3)(-0.30245)}{1.9966 \times 10^{-3}} = 212 \times 10^6 \text{ Pa}$$

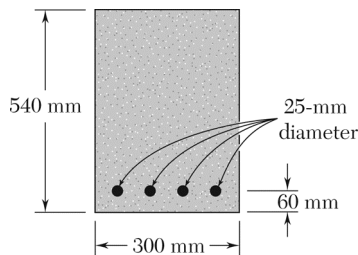
$$\sigma = 212 \text{ MPa} \quad \blacktriangleleft$$

(b) Concrete: $y = 177.87 \text{ mm} = 0.17787 \text{ m}$

$$\sigma = -\frac{(1.0)(175 \times 10^3)(0.17787)}{1.9966 \times 10^{-3}} = -15.59 \times 10^6 \text{ Pa}$$

$$\sigma = -15.59 \text{ MPa} \quad \blacktriangleleft$$





PROBLEM 4.48

Solve Prob. 4.47, assuming that the 300-mm width is increased to 350 mm.

PROBLEM 4.47 The reinforced concrete beam shown is subjected to a positive bending moment of $175 \text{ kN} \cdot \text{m}$. Knowing that the modulus of elasticity is 25 GPa for the concrete and 200 GPa for the steel, determine (a) the stress in the steel, (b) the maximum stress in the concrete.

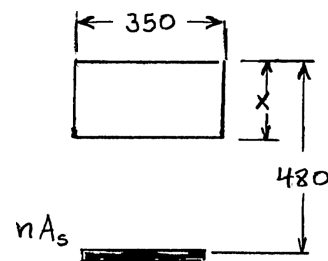
SOLUTION

$$n = \frac{E_s}{E_c} = \frac{200 \text{ GPa}}{25 \text{ GPa}} = 8.0$$

$$A_s = 4 \frac{\pi}{4} d^2 = (4) \left(\frac{\pi}{4} \right) (25)^2$$

$$= 1.9635 \times 10^3 \text{ mm}^2$$

$$nA_s = 15.708 \times 10^3 \text{ mm}^2$$



Locate the neutral axis:

$$350 x \frac{x}{2} - (15.708 \times 10^3)(480 - x) = 0$$

$$175x^2 + 15.708 \times 10^3 x - 7.5398 \times 10^6 = 0$$

Solve for x :

$$x = \frac{-15.708 \times 10^3 + \sqrt{(15.708 \times 10^3)^2 + (4)(175)(7.5398 \times 10^6)}}{(2)(175)}$$

$$x = 167.48 \text{ mm}, \quad 480 - x = 312.52 \text{ mm}$$

$$I = \frac{1}{3} (350)x^3 + (15.708 \times 10^3)(480 - x)^2$$

$$= \frac{1}{3} (350)(167.48)^3 + (15.708 \times 10^3)(312.52)^2$$

$$= 2.0823 \times 10^9 \text{ mm}^4 = 2.0823 \times 10^{-3} \text{ m}^4$$

$$\sigma = -\frac{nMy}{I}$$

(a) Steel:

$$y = -312.52 \text{ mm} = -0.31252 \text{ m}$$

$$\sigma = -\frac{(8.0)(175 \times 10^3)(-0.31252)}{2.0823 \times 10^{-3}} = 210 \times 10^6 \text{ Pa}$$

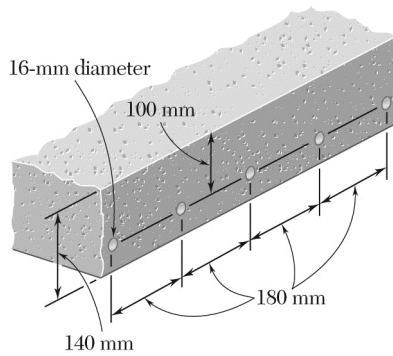
$$\sigma = 210 \text{ MPa} \quad \blacktriangleleft$$

(b) Concrete:

$$y = 167.48 \text{ mm} = 0.16748 \text{ m}$$

$$\sigma = -\frac{(1.0)(175 \times 10^3)(0.16748)}{2.0823 \times 10^{-3}} = -14.08 \times 10^6 \text{ Pa}$$

$$\sigma = -14.08 \text{ MPa} \quad \blacktriangleleft$$



PROBLEM 4.49

A concrete slab is reinforced by 16-mm-diameter steel rods placed on 180-mm centers as shown. The modulus of elasticity is 20 GPa for concrete and 200 GPa for steel. Using an allowable stress of 9 MPa for the concrete and of 120 MPa for the steel, determine the largest bending moment in a portion of slab 1 m wide.

SOLUTION

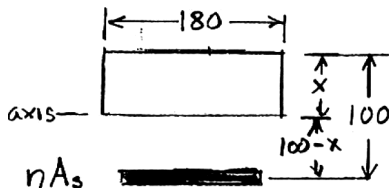
$$n = \frac{E_s}{E_c} = \frac{200 \text{ GPa}}{20 \text{ GPa}} = 10$$

Consider a section 180-mm wide with one steel rod.

$$A_s = \frac{\pi}{4} d^2 = \frac{\pi}{4} (16)^2 = 201.06 \text{ mm}^2$$

$$nA_s = 2.0106 \times 10^3 \text{ mm}^2$$

Locate the neutral axis:



$$180 x \frac{x}{2} - (100 - x)(2.0106 \times 10^3) = 0$$

$$90x^2 + 2.0106 \times 10^3 x - 201.06 \times 10^3 = 0$$

Solving for x:

$$x = \frac{-2.0106 \times 10^3 + \sqrt{(2.0106 \times 10^3)^2 + (4)(90)(201.06 \times 10^3)}}{(2)(90)}$$

$$x = 37.397 \text{ mm} \quad 100 - x = 62.603 \text{ mm}$$

$$I = \frac{1}{3} (180)x^3 + (2.0106 \times 10^3)(100 - x)^2$$

$$= \frac{1}{3} (180)(37.397)^3 + (2.0106 \times 10^3)(62.603)^2$$

$$= 11.018 \times 10^6 \text{ mm}^4 = 11.018 \times 10^{-6} \text{ m}^4$$

$$|\sigma| = \left| -\frac{nMy}{I} \right| \quad \therefore M = \frac{\sigma I}{ny}$$

Concrete:

$$n = 1, \quad y = 37.397 \text{ mm} = 0.037397 \text{ m}, \quad \sigma = 9 \times 10^6 \text{ Pa}$$

$$M = \frac{(9 \times 10^6)(11.018 \times 10^{-6})}{(1.0)(0.037397)} = 2.6516 \times 10^3 \text{ N} \cdot \text{m}$$

PROBLEM 4.49 (Continued)

Steel:

$$n = 10, \quad y = 62.603 \text{ mm} = 0.062603 \text{ m}, \quad \sigma = 120 \times 10^6 \text{ Pa}$$

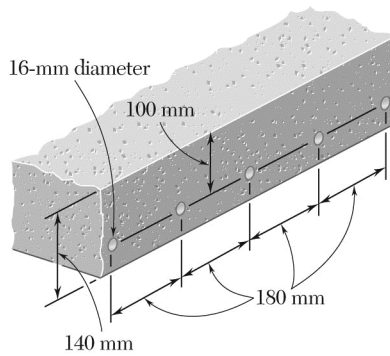
$$M = \frac{(120 \times 10^6)(11.018 \times 10^{-6})}{(10)(0.062603)} = 2.1120 \times 10^3 \text{ N} \cdot \text{m}$$

Choose the smaller value. $M = 2.1120 \times 10^3 \text{ N} \cdot \text{m}$

The above is the allowable positive moment for a 180-mm wide section. For a 1-m = 1000-mm width, multiply by $\frac{1000}{180} = 5.556$

$$M = (5.556)(2.1120 \times 10^3) = 11.73 \times 10^3 \text{ N} \cdot \text{m}$$

$$M = 11.73 \text{ kN} \cdot \text{m} \blacktriangleleft$$



PROBLEM 4.50

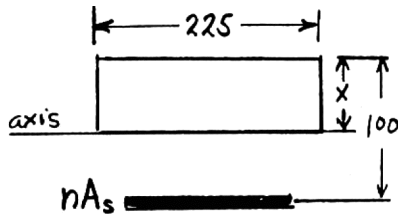
A concrete slab is reinforced by 16-mm-diameter steel rods placed on 180-mm centers as shown. The modulus of elasticity is 20 GPa for concrete and 200 GPa for steel. Using an allowable stress of 9 MPa for the concrete and of 120 MPa for the steel, determine the largest allowable positive bending moment in a portion of slab 1 m wide.

Solve Prob. 4.49, assuming that the spacing of the 16-mm-diameter rods is increased to 225 mm on centers.

SOLUTION

$$n = \frac{E_s}{E_a} = \frac{200 \text{ GPa}}{20 \text{ GPa}} = 10$$

Consider a section 225-mm wide with one steel rod.



$$A_s = \frac{\pi}{4} d^2 = \frac{\pi}{4} (16)^2 = 201.06 \text{ mm}^2$$

$$nA_s = 2.0106 \times 10^3 \text{ mm}^2$$

Locate the neutral axis:

$$225x \frac{x}{2} - (100 - x)(2.0106 \times 10^3) = 0$$

$$112.5x^2 + 2.0106x - 201.06 \times 10^3 = 0$$

$$\text{Solving for } x: \quad x = \frac{-2.0106 \times 10^3 + \sqrt{(2.0106 \times 10^3)^2 + (4)(112.5)(201.06 \times 10^3)}}{(2)(112.5)}$$

$$x = 34.273 \text{ mm} \quad 100 - x = 65.727$$

$$\begin{aligned} I &= \frac{1}{3} (225)x^3 + 2.0106 \times 10^3 (100 - x)^2 \\ &= \frac{1}{3} (225)(34.273)^3 + (2.0106 \times 10^3)(65.727)^2 \\ &= 11.705 \times 10^6 \text{ mm}^4 = 11.705 \times 10^{-6} \text{ m}^4 \end{aligned}$$

$$|\sigma| = \left| \frac{nMy}{I} \right| \quad \therefore M = \frac{\sigma I}{ny}$$

Concrete: $n = 1, \quad y = 34.273 \text{ mm} = 0.034273 \text{ m}, \quad \sigma = 9 \times 10^6 \text{ Pa}$

$$M = \frac{(9 \times 10^6)(11.705 \times 10^{-6})}{(1)(0.034273)} = 3.0738 \times 10^3 \text{ N} \cdot \text{m}$$

PROBLEM 4.50 (Continued)

Steel:

$$n = 10, \quad y = 65.727 \text{ mm} = 0.065727 \text{ m}, \quad \sigma = 120 \times 10^6 \text{ Pa}$$

$$M = \frac{(120 \times 10^6)(11.705 \times 10^{-6})}{(10)(0.065727)} = 2.1370 \times 10^3 \text{ N} \cdot \text{m}$$

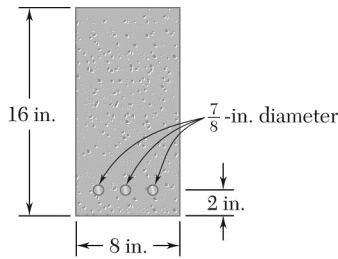
Choose the smaller value.

$$M = 2.1370 \times 10^3 \text{ N} \cdot \text{m}$$

The above is the allowable positive moment for a 225-mm-wide section. For a 1-m = 1000-mm section, multiply by $\frac{1000}{225} = 4.4444$

$$M = (4.4444)(2.1370 \times 10^3) = 9.50 \times 10^3 \text{ N} \cdot \text{m}$$

$$M = 9.50 \text{ kN} \cdot \text{m} \quad \blacktriangleleft$$



PROBLEM 4.51

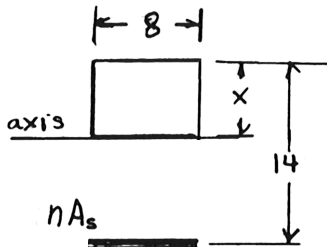
A concrete beam is reinforced by three steel rods placed as shown. The modulus of elasticity is 3×10^6 psi for the concrete and 29×10^6 psi for the steel. Using an allowable stress of 1350 psi for the concrete and 20 ksi for the steel, determine the largest allowable positive bending moment in the beam.

SOLUTION

$$n = \frac{E_s}{E_c} = \frac{29 \times 10^6}{3 \times 10^6} = 9.67$$

$$A_s = 3 \frac{\pi}{4} d^2 = (3) \left(\frac{\pi}{4} \right) \left(\frac{7}{8} \right)^2 = 1.8040 \text{ in}^2 \quad nA_s = 17.438 \text{ in}^2$$

Locate the neutral axis:



$$8x \frac{x}{2} - (17.438)(14 - x) = 0$$

$$4x^2 + 17.438x - 244.14 = 0$$

$$\text{Solve for } x. \quad x = \frac{-17.438 + \sqrt{17.438^2 + (4)(4)(244.14)}}{(2)(4)} = 5.6326 \text{ in.}$$

$$14 - x = 8.3674 \text{ in.}$$

$$I = \frac{1}{3} 8x^3 + nA_s(14 - x)^2 = \frac{1}{3} (8)(5.6326)^3 + (17.438)(8.3674)^2 = 1697.45 \text{ in}^4$$

$$|\sigma| = \left| \frac{nMy}{I} \right| \quad \therefore M = \frac{\sigma I}{ny}$$

Concrete:

$$n = 1.0, \quad |y| = 5.6326 \text{ in.}, \quad |\sigma| = 1350 \text{ psi}$$

$$M = \frac{(1350)(1697.45)}{(1.0)(5.6326)} = 406.835 \times 10^3 \text{ lb} \cdot \text{in} = 407 \text{ kip} \cdot \text{in}$$

Steel:

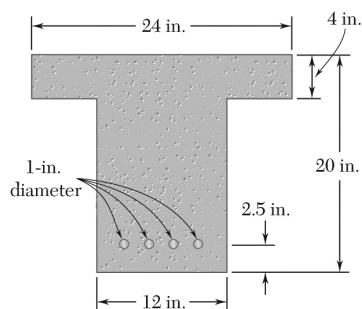
$$n = 9.67, \quad |y| = 8.3674 \text{ in.}, \quad \sigma = 20 \times 10^3 \text{ psi}$$

$$M = \frac{(20 \times 10^3)(1697.45)}{(9.67)(8.3674)} = 419.72 \text{ lb} \cdot \text{in} = 420 \text{ kip} \cdot \text{in}$$

Choose the smaller value.

$$M = 407 \text{ kip} \cdot \text{in}$$

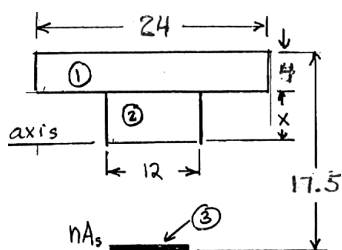
$$M = 33.9 \text{ kip} \cdot \text{ft} \quad \blacktriangleleft$$



PROBLEM 4.52

Knowing that the bending moment in the reinforced concrete beam is $+100 \text{ kip} \cdot \text{ft}$ and that the modulus of elasticity is $3.625 \times 10^6 \text{ psi}$ for the concrete and $29 \times 10^6 \text{ psi}$ for the steel, determine (a) the stress in the steel, (b) the maximum stress in the concrete.

SOLUTION



$$n = \frac{E_s}{E_c} = \frac{29 \times 10^6}{3.625 \times 10^6} = 8.0$$

$$A_s = (4) \left(\frac{\pi}{4} \right) (1)^2 = 3.1416 \text{ in}^2 \quad nA_s = 25.133 \text{ in}^2$$

Locate the neutral axis.

$$(24)(4)(x + 2) + (12x) \left(\frac{x}{2} \right) - (25.133)(17.5 - 4 - x) = 0$$

$$96x + 192 + 6x^2 - 339.3 + 25.133x = 0 \quad \text{or} \quad 6x^2 + 121.133x - 147.3 = 0$$

Solve for x .

$$x = \frac{-121.133 + \sqrt{(121.133)^2 + (4)(6)(147.3)}}{(2)(6)} = 1.150 \text{ in.}$$

$$d_3 = 17.5 - 4 - x = 12.350 \text{ in.}$$

$$I_1 = \frac{1}{12} b_1 h_1^3 + A_1 d_1^2 = \frac{1}{12} (24)(4)^3 + (24)(4)(3.150)^2 = 1080.6 \text{ in}^4$$

$$I_2 = \frac{1}{3} b_2 x^3 = \frac{1}{3} (12)(1.150)^3 = 6.1 \text{ in}^4$$

$$I_3 = nA_3 d_3^2 = (25.133)(12.350)^2 = 3833.3 \text{ in}^4$$

$$I = I_1 + I_2 + I_3 = 4920 \text{ in}^4$$

$$\sigma = -\frac{nMy}{I} \quad \text{where } M = 100 \text{ kip} \cdot \text{ft} = 1200 \text{ kip} \cdot \text{in.}$$

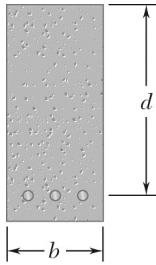
(a) Steel: $n = 8.0$ $y = -12.350 \text{ in.}$

$$\sigma_s = -\frac{(8.0)(1200)(-12.350)}{4920} \quad \sigma_s = 24.1 \text{ ksi} \quad \blacktriangleleft$$

(b) Concrete: $n = 1.0$, $y = 4 + 1.150 = 5.150 \text{ in.}$

$$\sigma_c = -\frac{(1.0)(1200)(5.150)}{4920} \quad \sigma_c = -1.256 \text{ ksi} \quad \blacktriangleleft$$

PROBLEM 4.53



The design of a reinforced concrete beam is said to be *balanced* if the maximum stresses in the steel and concrete are equal, respectively, to the allowable stresses σ_s and σ_c . Show that to achieve a balanced design the distance x from the top of the beam to the neutral axis must be

$$x = \frac{d}{1 + \frac{\sigma_s E_c}{\sigma_c E_s}}$$

where E_c and E_s are the moduli of elasticity of concrete and steel, respectively, and d is the distance from the top of the beam to the reinforcing steel.

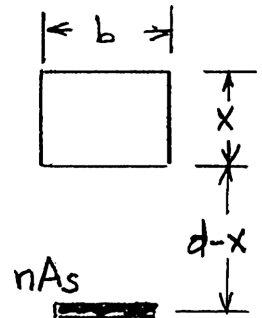
SOLUTION

$$\sigma_s = \frac{nM(d-x)}{I} \quad \sigma_c = \frac{Mx}{I}$$

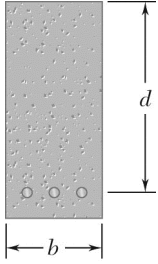
$$\frac{\sigma_s}{\sigma_c} = \frac{n(d-x)}{x} = n \frac{d}{x} - n$$

$$\frac{d}{x} = 1 + \frac{1}{n} \frac{\sigma_s}{\sigma_c} = 1 + \frac{E_c \sigma_s}{E_s \sigma_c}$$

$$x = \frac{d}{1 + \frac{E_c \sigma_s}{E_s \sigma_c}}$$



PROBLEM 4.54



For the concrete beam shown, the modulus of elasticity is 3.5×10^6 psi for the concrete and 29×10^6 psi for the steel. Knowing that $b = 8$ in. and $d = 22$ in., and using an allowable stress of 1800 psi for the concrete and 20 ksi for the steel, determine (a) the required area A_s of the steel reinforcement if the beam is to be balanced, (b) the largest allowable bending moment. (See Prob. 4.53 for definition of a balanced beam.)

The design of a reinforced concrete beam is said to be *balanced* if the maximum stresses in the steel and concrete are equal, respectively, to the allowable stresses σ_s and σ_c .

SOLUTION

$$n = \frac{E_s}{E_c} = \frac{29 \times 10^6}{3.5 \times 10^6} = 8.2857$$

$$\sigma_s = \frac{nM(d-x)}{I} \quad \sigma_c = \frac{Mx}{I}$$

$$\frac{\sigma_s}{\sigma_c} = \frac{n(d-x)}{x} = n \frac{d}{x} - n$$

$$\frac{d}{x} = 1 + \frac{1}{n} \frac{\sigma_s}{\sigma_c} = 1 + \frac{1}{8.2857} \cdot \frac{20 \times 10^3}{1800} = 2.3410$$

$$x = 0.42717d = (0.42717)(22) = 9.398 \text{ in.} \quad d - x = 22 - 9.398 = 12.602 \text{ in.}$$

Locate neutral axis: $bx \frac{x}{2} - nA_s(d-x) = 0$

$$(a) \quad A_s = \frac{bx^2}{2n(d-x)} = \frac{(8)(9.398)^2}{(2)(8.2857)(12.602)} = 3.3835 \text{ in}^2 \quad A_s = 3.38 \text{ in}^2 \blacktriangleleft$$

$$I = \frac{1}{3}bx^3 + nA_s(d-x)^2 = \frac{1}{3}(8)(9.398)^3 + (8.2857)(3.3835)(12.602)^2 = 6665.6 \text{ in}^4$$

$$\sigma = \frac{nMy}{I} \quad M = \frac{\sigma I}{ny}$$

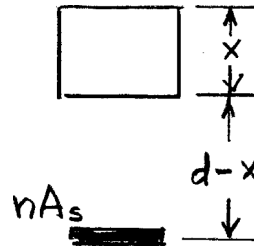
$$(b) \quad \text{Concrete:} \quad n = 1.0 \quad y = 9.398 \text{ in.} \quad \sigma_c = 1800 \text{ psi} \quad M = \frac{(1800)(6665.6)}{(1.0)(9.398)} = 1.277 \times 10^6 \text{ lb} \cdot \text{in}$$

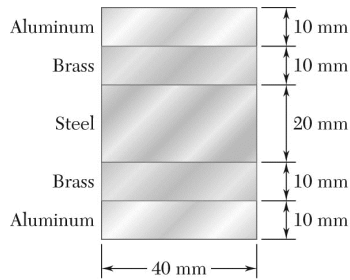
$$\text{Steel:} \quad n = 8.2857 \quad |y| = 12.602 \text{ in.} \quad \sigma_s = 20 \times 10^3 \text{ psi}$$

$$M = \frac{(20 \times 10^3)(6665.6)}{(8.2857)(12.602)} = 1.277 \times 10^6 \text{ lb} \cdot \text{in}$$

Note that both values are the same for balanced design.

$$M = 106.4 \text{ kip} \cdot \text{ft} \blacktriangleleft$$

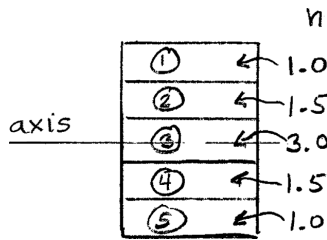




PROBLEM 4.55

Five metal strips, each 40 mm wide, are bonded together to form the composite beam shown. The modulus of elasticity is 210 GPa for the steel, 105 GPa for the brass, and 70 GPa for the aluminum. Knowing that the beam is bent about a horizontal axis by a couple of moment 1800 N · m, determine (a) the maximum stress in each of the three metals, (b) the radius of curvature of the composite beam.

SOLUTION



Use aluminum as the reference material.

$n = 1$ in aluminum.

$n = E_s / E_a = 210 / 70 = 3$ in steel.

$n = E_b / E_a = 105 / 70 = 1.5$ in brass.

Due to symmetry of both the material arrangement and the geometry, the neutral axis passes through the center of the steel portion.

For the transformed section,

$$I_1 = \frac{n_1}{12} b_1 h_1^3 + n_1 A_1 d_1^2 = \frac{1}{12} (40)(10)^3 + (40)(10)(25)^2 = 253.33 \times 10^3 \text{ mm}^4$$

$$I_2 = \frac{n_2}{12} b_2 h_2^3 + n_2 A_2 d_2^2 = \frac{1.5}{12} (40)(10)^3 + (1.5)(40)(10)(15)^2 = 140 \times 10^3 \text{ mm}^4$$

$$I_3 = \frac{n_3}{12} b_3 h_3^3 = \frac{3.0}{12} (40)(20)^3 = 80 \times 10^3 \text{ mm}^4$$

$$I_4 = I_2 = 140 \times 10^3 \text{ mm}^4 \quad I_5 = I_1 = 253.33 \times 10^3 \text{ mm}^4$$

$$I = \sum I = 866.66 \times 10^3 \text{ mm}^4 = 866.66 \times 10^{-9} \text{ m}^4$$

(a) $\sigma = -\frac{nMy}{I}$ where $M = 1800 \text{ N} \cdot \text{m}$

Aluminum: $n = 1.0$ $y = -30 \text{ mm} = 0.030 \text{ m}$

$$\sigma_a = \frac{(1.0)(1800)(0.030)}{866.66 \times 10^{-9}} = 62.3 \times 10^6 \text{ Pa} \quad \sigma_a = 62.3 \text{ MPa} \quad \blacktriangleleft$$

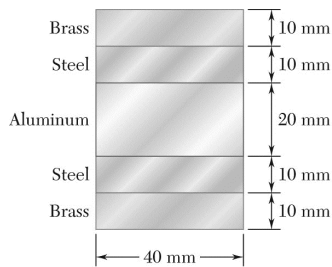
Brass: $n = 1.5$ $y = -20 \text{ mm} = -0.020 \text{ m}$

$$\sigma_b = \frac{(1.5)(1800)(0.020)}{866.66 \times 10^{-9}} = 62.3 \times 10^6 \text{ Pa} \quad \sigma_b = 62.3 \text{ MPa} \quad \blacktriangleleft$$

Steel: $n = 3.0$ $y = -10 \text{ mm} = -0.010 \text{ m}$

$$\sigma_s = \frac{(3.0)(1800)(0.010)}{866.66 \times 10^{-9}} = 62.3 \times 10^6 \text{ Pa} \quad \sigma_s = 62.3 \text{ MPa} \quad \blacktriangleleft$$

(b) Radius of curvature. $\frac{1}{\rho} = \frac{M}{E_a I} = \frac{1800}{(70 \times 10^9)(866.66 \times 10^{-9})} = 0.02967 \text{ m}^{-1} \quad \rho = 33.7 \text{ m} \quad \blacktriangleleft$



PROBLEM 4.56

Five metal strips, each of 40 mm wide, are bonded together to form the composite beam shown. The modulus of elasticity is 210 GPa for the steel, 105 GPa for the brass, and 70 GPa for the aluminum. Knowing that the beam is bent about a horizontal axis by a couple of moment 1800 N · m, determine (a) the maximum stress in each of the three metals, (b) the radius of curvature of the composite beam.

SOLUTION

Use aluminum as the reference material.

$n = 1.0$ in aluminum.

$n = E_s / E_a = 210 / 70 = 3$ in steel.

$n = E_b / E_a = 105 / 70 = 1.5$ in brass.

Due to symmetry of both the material arrangement and the geometry, the neutral axis passes through the center of the aluminum portion.

For the transformed section,

$$I_1 = \frac{n_1}{12} b_1 h_1^3 + n_1 A_1 d_1^2 = \frac{1.5}{12} (40)(10)^3 + (1.5)(40)(10)(25)^2 = 380 \times 10^3 \text{ mm}^4$$

$$I_2 = \frac{n_2}{12} b_2 h_2^3 + n_2 A_2 d_2^2 = \frac{3.0}{12} (40)(10)^3 + (3.0)(40)(10)(15)^2 = 280 \times 10^3 \text{ mm}^4$$

$$I_3 = \frac{n_3}{12} b_3 h_3^3 = \frac{1.0}{12} (40)(20)^3 = 26.67 \times 10^3 \text{ mm}^4$$

$$I_4 = I_2 = 280 \times 10^3 \text{ mm}^4 \quad I_5 = I_1 = 380 \times 10^3 \text{ mm}^4$$

$$I = \sum I = 1.3467 \times 10^6 \text{ mm}^4 = 1.3467 \times 10^{-6} \text{ m}^4$$

(a) $\sigma = -\frac{nMy}{I}$ where $M = 1800 \text{ N} \cdot \text{m}$

Aluminum: $n = 1$, $y = -10 \text{ mm} = -0.010 \text{ m}$

$$\sigma_a = \frac{(1.0)(1800)(0.010)}{1.3467 \times 10^{-6}} = 13.37 \times 10^6 \text{ Pa} \quad \sigma_a = 13.37 \text{ MPa} \quad \blacktriangleleft$$

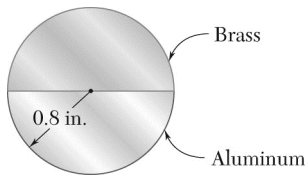
Brass: $n = 1.5$, $y = -30 \text{ mm} = -0.030 \text{ m}$

$$\sigma_b = \frac{(1.5)(1800)(0.030)}{1.3467 \times 10^{-6}} = 60.1 \times 10^6 \text{ Pa} \quad \sigma_b = 60.1 \text{ MPa} \quad \blacktriangleleft$$

Steel: $n = 3.0$, $y = -20 \text{ mm} = -0.020 \text{ m}$

$$\sigma_s = \frac{(3.0)(1800)(0.020)}{1.3467 \times 10^{-6}} = 80.1 \times 10^6 \text{ Pa} \quad \sigma_s = 80.1 \text{ MPa} \quad \blacktriangleleft$$

(b) Radius of curvature. $\frac{1}{\rho} = \frac{M}{E_a I} = \frac{1800}{(70 \times 10^9)(1.3467 \times 10^{-6})} = 0.01909 \text{ m}^{-1} \quad \rho = 52.4 \text{ m} \quad \blacktriangleleft$



PROBLEM 4.57

The composite beam shown is formed by bonding together a brass rod and an aluminum rod of semicircular cross sections. The modulus of elasticity is 15×10^6 psi for the brass and 10×10^6 psi for the aluminum. Knowing that the composite beam is bent about a horizontal axis by couples of moment $8 \text{ kip} \cdot \text{in.}$, determine the maximum stress (a) in the brass, (b) in the aluminum.

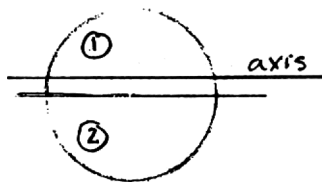
SOLUTION

For each semicircle, $r = 0.8 \text{ in.}$ $A = \frac{\pi}{2} r^2 = 1.00531 \text{ in}^2$,

$$\bar{y}_o = \frac{4r}{3\pi} = \frac{(4)(0.8)}{3\pi} = 0.33953 \text{ in.} \quad I_{\text{base}} = \frac{\pi}{8} r^4 = 0.160850 \text{ in}^4$$

$$\bar{I} = I_{\text{base}} - A\bar{y}_o^2 = 0.160850 - (1.00531)(0.33953)^2 = 0.044953 \text{ in}^4$$

Use aluminum as the reference material.



$n = 1.0$ in aluminum

$$n = \frac{E_b}{E_a} = \frac{15 \times 10^6}{10 \times 10^6} = 1.5 \text{ in brass}$$

Locate the neutral axis.

	A, in^2	nA, in^2	$\bar{y}_o, \text{in.}$	$nA\bar{y}_o, \text{in}^3$
①	1.00531	1.50796	0.33953	0.51200
②	1.00531	1.00531	-0.33953	-0.34133
Σ		2.51327		0.17067

$$\bar{Y}_o = \frac{0.17067}{2.51327} = 0.06791 \text{ in.}$$

The neutral axis lies 0.06791 in. above the material interface.

$$d_1 = 0.33953 - 0.06791 = 0.27162 \text{ in.}, \quad d_2 = 0.33953 + 0.06791 = 0.40744 \text{ in.}$$

$$I_1 = n_1 \bar{I} + n_1 A d_1^2 = (1.5)(0.044957) + (1.5)(1.00531)(0.27162)^2 = 0.17869 \text{ in}^4$$

$$I_2 = n_2 \bar{I} + n_2 A d_2^2 = (1.0)(0.044957) + (1.0)(1.00531)(0.40744)^2 = 0.21185 \text{ in}^4$$

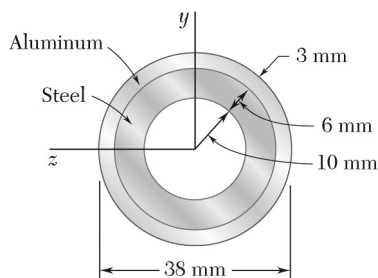
$$I = I_1 + I_2 = 0.39054 \text{ in}^4$$

(a) Brass: $n = 1.5, \quad y = 0.8 - 0.06791 = 0.73209 \text{ in.}$

$$\sigma = -\frac{nMy}{I} = -\frac{(1.5)(8)(0.73209)}{0.39054} \quad \sigma = -22.5 \text{ ksi} \quad \blacktriangleleft$$

(b) Aluminium: $n = 1.0, \quad y = -0.8 - 0.06791 = -0.86791 \text{ in.}$

$$\sigma = -\frac{nMy}{I} = -\frac{(1.0)(8)(-0.86791)}{0.39054} \quad \sigma = 17.78 \text{ ksi} \quad \blacktriangleleft$$



PROBLEM 4.58

A steel pipe and an aluminum pipe are securely bonded together to form the composite beam shown. The modulus of elasticity is 200 GPa for the steel and 70 GPa for the aluminum. Knowing that the composite beam is bent by a couple of moment 500 N · m, determine the maximum stress (a) in the aluminum, (b) in the steel.

SOLUTION

Use aluminum as the reference material.

$$n = 1.0 \text{ in aluminum}$$

$$n = E_s / E_a = 200 / 70 = 2.857 \text{ in steel}$$

For the transformed section,

$$\text{Steel:} \quad I_s = n_s \frac{\pi}{4} (r_o^4 - r_i^4) = (2.857) \left(\frac{\pi}{4} \right) (16^4 - 10^4) = 124.62 \times 10^3 \text{ mm}^4$$

$$\text{Aluminium:} \quad I_a = n_a \frac{\pi}{4} (r_o^4 - r_i^4) = (1.0) \left(\frac{\pi}{4} \right) (19^4 - 16^4) = 50.88 \times 10^3 \text{ mm}^4$$

$$I = I_s + I_a = 175.50 \times 10^3 \text{ mm}^4 = 175.5 \times 10^{-9} \text{ m}^4$$

$$(a) \text{ Aluminum:} \quad c = 19 \text{ mm} = 0.019 \text{ m}$$

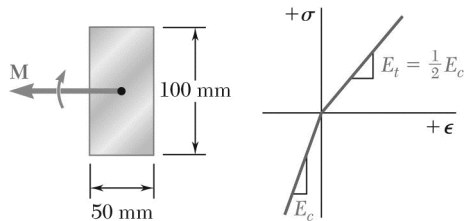
$$\sigma_a = \frac{n_a M c}{I} = \frac{(1.0)(500)(0.019)}{175.5 \times 10^{-9}} = 54.1 \times 10^6 \text{ Pa}$$

$$\sigma_a = 54.1 \text{ MPa} \quad \blacktriangleleft$$

$$(b) \text{ Steel:} \quad c = 16 \text{ mm} = 0.016 \text{ m}$$

$$\sigma_s = \frac{n_s M c}{I} = \frac{(2.857)(500)(0.016)}{175.5 \times 10^{-9}} = 130.2 \times 10^6 \text{ Pa}$$

$$\sigma_s = 130.2 \text{ MPa} \quad \blacktriangleleft$$



PROBLEM 4.59

The rectangular beam shown is made of a plastic for which the value of the modulus of elasticity in tension is one-half of its value in compression. For a bending moment $M = 600 \text{ N} \cdot \text{m}$, determine the maximum (a) tensile stress, (b) compressive stress.

SOLUTION

$$n = \frac{1}{2} \text{ on the tension side of neutral axis.}$$

$$n = 1 \text{ on the compression side.}$$

Locate neutral axis.

$$n_1 b x \frac{x}{2} - n_2 b (h - x) \frac{h - x}{2} = 0$$

$$\frac{1}{2} b x^2 - \frac{1}{4} b (h - x)^2 = 0$$

$$x^2 = \frac{1}{2} (h - x)^2 \quad x = \frac{1}{\sqrt{2}} (h - x)$$

$$x = \frac{1}{\sqrt{2} + 1} h = 0.41421 h = 41.421 \text{ mm}$$

$$h - x = 58.579 \text{ mm}$$

$$I_1 = n_1 \frac{1}{3} b x^3 = (1) \left(\frac{1}{3} \right) (50) (41.421)^3 = 1.1844 \times 10^6 \text{ mm}^4$$

$$I_2 = n_2 \frac{1}{3} b (h - x)^3 = \left(\frac{1}{2} \right) \left(\frac{1}{3} \right) (50) (58.579)^3 = 1.6751 \times 10^6 \text{ mm}^4$$

$$I = I_1 + I_2 = 2.8595 \times 10^6 \text{ mm}^4 = 2.8595 \times 10^{-6} \text{ m}^4$$

(a) Tensile stress:

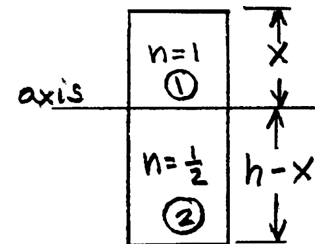
$$n = \frac{1}{2}, \quad y = -58.579 \text{ mm} = -0.058579 \text{ m}$$

$$\sigma = -\frac{nMy}{I} = -\frac{(0.5)(600)(-0.058579)}{2.8595 \times 10^{-6}} = 6.15 \times 10^6 \text{ Pa} \quad \sigma_t = 6.15 \text{ MPa} \quad \blacktriangleleft$$

(b) Compressive stress:

$$n = 1, \quad y = 41.421 \text{ mm} = 0.041421 \text{ m}$$

$$\sigma = -\frac{nMy}{I} = -\frac{(1.0)(600)(0.041421)}{2.8595 \times 10^{-6}} = -8.69 \times 10^6 \text{ Pa} \quad \sigma_c = -8.69 \text{ MPa} \quad \blacktriangleleft$$



PROBLEM 4.60*

A rectangular beam is made of material for which the modulus of elasticity is E_t in tension and E_c in compression. Show that the curvature of the beam in pure bending is

$$\frac{1}{\rho} = \frac{M}{E_r I}$$

where

$$E_r = \frac{4E_t E_c}{(\sqrt{E_t} + \sqrt{E_c})^2}$$

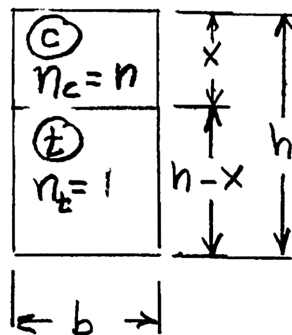
SOLUTION

Use E_t as the reference modulus.

Then $E_c = nE_t$.

Locate neutral axis.

$$\begin{aligned} nbx \frac{x}{2} - b(h-x) \frac{h-x}{2} &= 0 \\ nx^2 - (h-x)^2 &= 0 \quad \sqrt{n}x = (h-x) \\ x &= \frac{h}{\sqrt{n}+1} \quad h-x = \frac{\sqrt{n}h}{\sqrt{n}+1} \end{aligned}$$

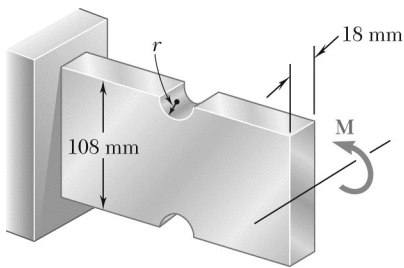


$$\begin{aligned} I_{\text{trans}} &= \frac{n}{3} bx^3 + \frac{1}{3} b(h-x)^3 = \left[\frac{h}{3} \left(\frac{1}{\sqrt{n}+1} \right)^3 + \left(\frac{\sqrt{n}}{\sqrt{n}+1} \right)^3 \right] bh^3 \\ &= \frac{1}{3} \frac{n+n^{3/2}}{(\sqrt{n}+1)^3} bh^3 = \frac{1}{3} \frac{n(1+\sqrt{n})}{(\sqrt{n}+1)^3} bh^3 = \frac{1}{3} \times \frac{n}{(\sqrt{n}+1)^2} bh^3 \end{aligned}$$

$$\frac{1}{\rho} = \frac{M}{E_t I_{\text{trans}}} = \frac{M}{E_r I} \quad \text{where} \quad I = \frac{1}{12} bh^3$$

$$E_r I = E_t I_{\text{trans}}$$

$$\begin{aligned} E_r &= \frac{E_t I_{\text{trans}}}{I} = \frac{12}{bh^3} \times E_t \times \frac{n}{3(\sqrt{n}+1)^2} bh^3 \\ &= \frac{4E_t E_c / E_t}{(\sqrt{E_c/E_t} + 1)^2} = \frac{4E_t E_c}{(\sqrt{E_c} + \sqrt{E_t})^2} \end{aligned}$$



PROBLEM 4.61

Semicircular grooves of radius r must be milled as shown in the sides of a steel member. Using an allowable stress of 60 MPa, determine the largest bending moment that can be applied to the member when (a) $r = 9$ mm, (b) $r = 18$ mm.

SOLUTION

(a) $d = D - 2r = 108 - (2)(9) = 90$ mm

$$\frac{D}{d} = \frac{108}{90} = 1.20 \quad \frac{r}{d} = \frac{9}{90} = 0.1$$

From Fig. 4.32,

$$K = 2.07$$

$$\begin{aligned} I &= \frac{1}{12}(18)(90)^3 \\ &= 1.0935 \times 10^6 \text{ mm}^4 \\ &= 1.0935 \times 10^{-6} \text{ m}^4 \end{aligned}$$

$$c = \frac{1}{2}d = 45 \text{ mm} = 0.045 \text{ m}$$

$$\sigma = \frac{KMc}{I}$$

$$M = \frac{\sigma I}{Kc} = \frac{(60 \times 10^6)(1.0935 \times 10^{-6})}{(2.07)(0.045)} = 704$$

$$M = 704 \text{ N} \cdot \text{m} \quad \blacktriangleleft$$

(b) $d = 108 - (2)(18) = 72$ mm

$$\frac{D}{d} = \frac{108}{72} = 1.5 \quad \frac{r}{d} = \frac{18}{72} = 0.25$$

$$c = \frac{1}{2}d = 36 \text{ mm} = 0.036 \text{ m}$$

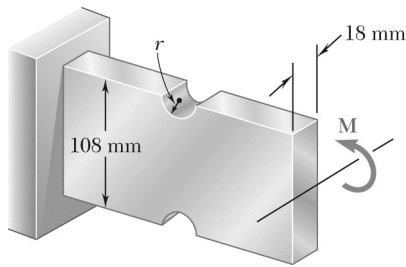
From Fig. 4.32,

$$K = 1.61$$

$$I = \frac{1}{12}(18)(72)^3 = 559.87 \times 10^3 \text{ mm}^4 = 559.87 \times 10^{-9} \text{ m}^4$$

$$M = \frac{\sigma I}{Kc} = \frac{(60 \times 10^6)(559.87 \times 10^{-9})}{(1.61)(0.036)} = 580$$

$$M = 580 \text{ N} \cdot \text{m} \quad \blacktriangleleft$$



PROBLEM 4.62

Semicircular grooves of radius r must be milled as shown in the sides of a steel member. Knowing that $M = 450 \text{ N} \cdot \text{m}$, determine the maximum stress in the member when the radius r of the semicircular grooves is

(a) $r = 9 \text{ mm}$, (b) $r = 18 \text{ mm}$.

SOLUTION

(a) $d = D - 2r = 108 - (2)(9) = 90 \text{ mm}$

$$\frac{D}{d} = \frac{108}{90} = 1.20 \quad \frac{r}{d} = \frac{9}{90} = 0.1$$

From Fig. 4.32,

$$K = 2.07$$

$$\begin{aligned} I &= \frac{1}{12}bh^3 = \frac{1}{12}(18)(90)^3 \\ &= 1.0935 \times 10^6 \text{ mm}^4 \\ &= 1.0935 \times 10^{-6} \text{ m}^4 \end{aligned}$$

$$c = \frac{1}{2}d = 45 \text{ mm} = 0.045 \text{ m}$$

$$\begin{aligned} \sigma_{\max} &= \frac{KMc}{I} = \frac{(2.07)(450)(0.045)}{1.0935 \times 10^{-6}} \\ &= 38.3 \times 10^6 \text{ Pa} \end{aligned}$$

$$\sigma_{\max} = 38.3 \text{ MPa} \quad \blacktriangleleft$$

(b) $d = D - 2r = 108 - (2)(18) = 72 \text{ mm}$

$$\frac{D}{d} = \frac{108}{72} = 1.5 \quad \frac{r}{d} = \frac{18}{72} = 0.25$$

From Fig. 4.32,

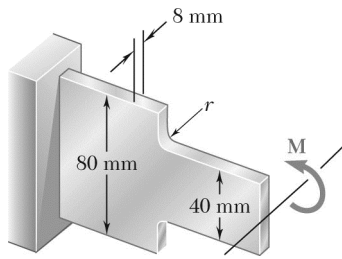
$$K = 1.61$$

$$c = \frac{1}{2}d = 36 \text{ mm} = 0.036 \text{ m}$$

$$I = \frac{1}{12}(18)(72)^3 = 559.87 \times 10^3 \text{ mm}^4 = 559.87 \times 10^{-9} \text{ m}^4$$

$$\sigma_{\max} = \frac{KMc}{I} = \frac{(1.61)(450)(0.036)}{559.87 \times 10^{-9}} = 46.6 \times 10^6 \text{ Pa}$$

$$\sigma_{\max} = 46.6 \text{ MPa} \quad \blacktriangleleft$$



PROBLEM 4.63

Knowing that the allowable stress for the beam shown is 90 MPa, determine the allowable bending moment M when the radius r of the fillets is (a) 8 mm, (b) 12 mm.

SOLUTION

$$I = \frac{1}{12}bh^3 = \frac{1}{12}(8)(40)^3 = 42.667 \times 10^3 \text{ mm}^4 = 42.667 \times 10^{-9} \text{ m}^4$$

$$c = 20 \text{ mm} = 0.020 \text{ m}$$

$$\frac{D}{d} = \frac{80 \text{ mm}}{40 \text{ mm}} = 2.00$$

$$(a) \quad \frac{r}{d} = \frac{8 \text{ mm}}{40 \text{ mm}} = 0.2$$

From Fig. 4.31, $K = 1.50$

$$\sigma_{\max} = K \frac{Mc}{I}$$

$$M = \frac{\sigma_{\max} I}{Kc} = \frac{(90 \times 10^6)(42.667 \times 10^{-9})}{(1.50)(0.020)}$$

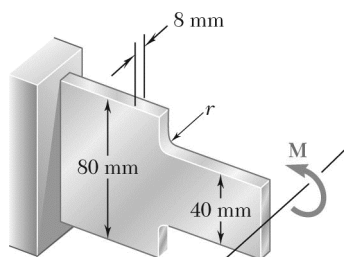
$$M = 128 \text{ N} \cdot \text{m} \quad \blacktriangleleft$$

$$(b) \quad \frac{r}{d} = \frac{12 \text{ mm}}{40 \text{ mm}} = 0.3$$

From Fig. 4.31, $K = 1.35$

$$M = \frac{(90 \times 10^6)(42.667 \times 10^{-9})}{(1.35)(0.020)}$$

$$M = 142 \text{ N} \cdot \text{m} \quad \blacktriangleleft$$



PROBLEM 4.64

Knowing that $M = 250 \text{ N} \cdot \text{m}$, determine the maximum stress in the beam shown when the radius r of the fillets is (a) 4 mm, (b) 8 mm.

SOLUTION

$$I = \frac{1}{12}bh^3 = \frac{1}{12}(8)(40)^3 = 42.667 \times 10^3 \text{ mm}^4 = 42.667 \times 10^{-9} \text{ m}^4$$

$$c = 20 \text{ mm} = 0.020 \text{ m}$$

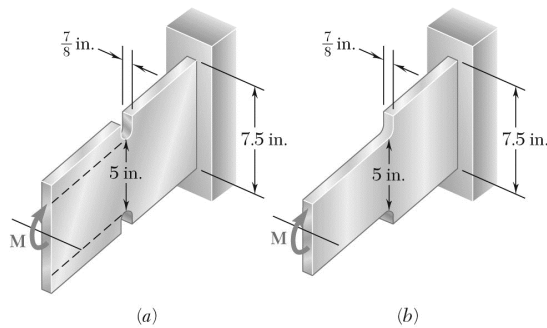
$$\frac{D}{d} = \frac{80 \text{ mm}}{40 \text{ mm}} = 2.00$$

$$(a) \quad \frac{r}{d} = \frac{4 \text{ mm}}{40 \text{ mm}} = 0.10 \quad \text{From Fig. 4.31,} \quad K = 1.87$$

$$\sigma_{\max} = K \frac{Mc}{I} = \frac{(1.87)(250)(0.020)}{42.667 \times 10^{-9}} = 219 \times 10^6 \text{ Pa} \quad \sigma_{\max} = 219 \text{ MPa} \quad \blacktriangleleft$$

$$(b) \quad \frac{r}{d} = \frac{8 \text{ mm}}{40 \text{ mm}} = 0.20 \quad \text{From Fig. 4.31,} \quad K = 1.50$$

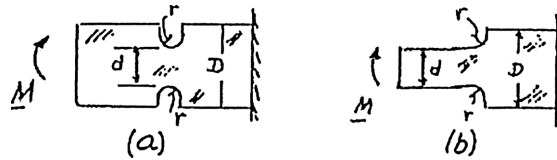
$$\sigma_{\max} = K \frac{Mc}{I} = \frac{(1.50)(250)(0.020)}{42.667 \times 10^{-9}} = 176 \times 10^6 \text{ Pa} \quad \sigma_{\max} = 176 \text{ MPa} \quad \blacktriangleleft$$



PROBLEM 4.65

The allowable stress used in the design of a steel bar is 12 ksi. Determine the largest couple M that can be applied to the bar (a) if the bar is designed with grooves having semicircular portions of radius $r = \frac{3}{4}$ in. as shown in Fig. a, (b) if the bar is redesigned by removing the material above and below the dashed lines as shown in Fig. b.

SOLUTION



Dimensions:

$$t = \frac{7}{8} \text{ in.} = 0.875 \text{ in.} \quad r = \frac{3}{4} \text{ in.} = 0.75 \text{ in.}$$

$$D = 7.5 \text{ in.} \quad d = 5 \text{ in.}$$

$$\frac{r}{d} = \frac{0.75}{5} = 0.15 \quad \frac{D}{d} = \frac{7.5}{5} = 1.5$$

Stress concentration factors: Figs. 4.32 and 4.31

Configuration (a): $K = 1.92$

Configuration (b): $K = 1.58$

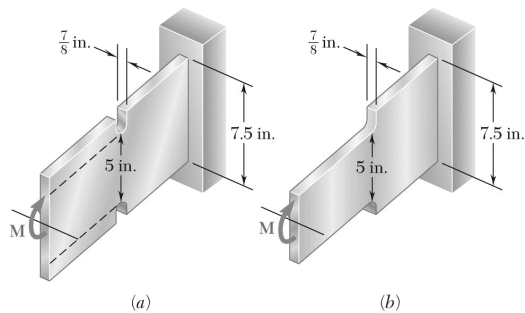
Moment of inertia: $I = \frac{1}{12}td^3 = \frac{1}{12}(0.875)(5)^3 = 9.115 \text{ in}^4$

$$c = \frac{1}{2}d = 2.5 \text{ in.} \quad \sigma_m = 12 \text{ ksi}$$

$$\sigma_m = \frac{KMc}{I} \quad M = \frac{I\sigma_m}{Kc}$$

(a) $M = \frac{(9.115)(12)}{(1.92)(2.5)} \quad M = 22.8 \text{ kip} \cdot \text{in} \quad \blacktriangleleft$

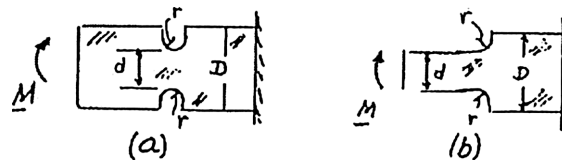
(b) $M = \frac{(9.115)(12)}{(1.58)(2.5)} \quad M = 27.7 \text{ kip} \cdot \text{in} \quad \blacktriangleleft$



PROBLEM 4.66

A couple of moment $M = 20 \text{ kip} \cdot \text{in.}$ is to be applied to the end of a steel bar. Determine the maximum stress in the bar (a) if the bar is designed with grooves having semicircular portions of radius $r = \frac{1}{2} \text{ in.}$, as shown in Fig. *a*, (b) if the bar is redesigned by removing the material above and below the dashed line as shown in Fig. *b*.

SOLUTION



Dimensions: $t = \frac{7}{8} \text{ in.} = 0.875 \text{ in.}$ $r = \frac{1}{2} \text{ in.} = 0.5 \text{ in.}$

$D = 7.5 \text{ in.}$ $d = 5 \text{ in.}$

$\frac{r}{d} = \frac{0.5}{5} = 0.10$ $\frac{D}{d} = \frac{7.5}{5} = 1.5$

Stress concentration factors: Figs. 4.32 and 4.31

Configuration (a): $K = 2.22$

Configuration (b): $K = 1.80$

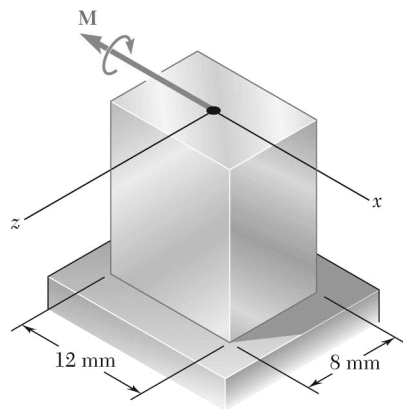
Moment of inertia: $I = \frac{1}{12}td^3 = \frac{1}{12}(0.875)(5)^3 = 9.115 \text{ in}^4$

$c = \frac{1}{2}d = 2.5 \text{ in.}$ $M = 20 \text{ kip} \cdot \text{in.}$

Maximum stress: $\sigma_m = \frac{KMc}{I}$

(a) $\sigma_m = \frac{(2.22)(20)(2.5)}{9.115}$ $\sigma_m = 12.2 \text{ ksi} \blacktriangleleft$

(b) $\sigma_m = \frac{(1.80)(20)(2.5)}{9.115}$ $\sigma_m = 9.9 \text{ ksi} \blacktriangleleft$



PROBLEM 4.67

The prismatic bar shown is made of a steel that is assumed to be elastoplastic with $\sigma_Y = 300 \text{ MPa}$ and is subjected to a couple $\mathbf{M}$ parallel to the x axis. Determine the moment M of the couple for which (a) yield first occurs, (b) the elastic core of the bar is 4 mm thick.

SOLUTION

$$(a) \quad I = \frac{1}{12}bh^3 = \frac{1}{12}(12)(8)^3 = 512 \text{ mm}^4$$

$$= 512 \times 10^{-12} \text{ m}^4$$

$$c = \frac{1}{2}h = 4 \text{ mm} = 0.004 \text{ m}$$

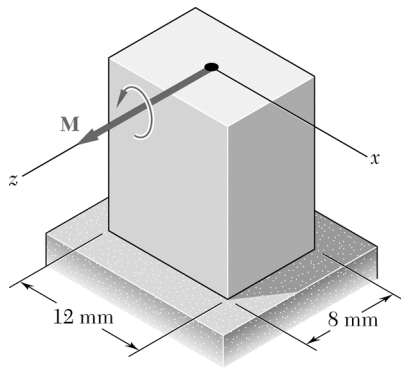
$$M_Y = \frac{\sigma_Y I}{c} = \frac{(300 \times 10^6)(512 \times 10^{-12})}{0.004} \\ = 38.4 \text{ N} \cdot \text{m}$$

$$M_Y = 38.4 \text{ N} \cdot \text{m} \quad \blacktriangleleft$$

$$(b) \quad y_Y = \frac{1}{2}(4) = 2 \text{ mm} \quad \frac{y_Y}{c} = \frac{2}{4} = 0.5$$

$$M = \frac{3}{2}M_Y \left[1 - \frac{1}{3} \left(\frac{y_Y}{c} \right)^2 \right] \\ = \frac{3}{2}(38.4) \left[1 - \frac{1}{3}(0.5)^2 \right] \\ = 52.8 \text{ N} \cdot \text{m}$$

$$M = 52.8 \text{ N} \cdot \text{m} \quad \blacktriangleleft$$



PROBLEM 4.68

Solve Prob. 4.67, assuming that the couple **M** is parallel to the *z* axis.

PROBLEM 4.67 The prismatic bar shown is made of a steel that is assumed to be elastoplastic with $\sigma_Y = 300$ MPa and is subjected to a couple **M** parallel to the *x* axis. Determine the moment *M* of the couple for which (a) yield first occurs, (b) the elastic core of the bar is 4 mm thick.

SOLUTION

$$(a) \quad I = \frac{1}{12}bh^3 = \frac{1}{12}(8)(12)^3 = 1.152 \times 10^3 \text{ mm}^4$$

$$= 1.152 \times 10^{-9} \text{ m}^4$$

$$c = \frac{1}{2}h = 6 \text{ mm} = 0.006 \text{ m}$$

$$M_Y = \frac{\sigma_Y I}{c} = \frac{(300 \times 10^6)(1.152 \times 10^{-9})}{0.006}$$

$$= 57.6 \text{ N} \cdot \text{m}$$

$$M_Y = 57.6 \text{ N} \cdot \text{m} \quad \blacktriangleleft$$

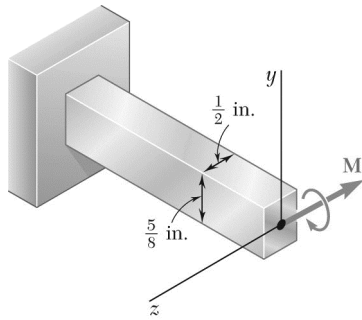
$$(b) \quad y_Y = \frac{1}{2}(4) = 2 \text{ mm} \quad \frac{y_Y}{c} = \frac{2}{6} = \frac{1}{3}$$

$$M = \frac{3}{2}M_Y \left[1 - \frac{1}{3} \left(\frac{y_Y}{c} \right)^2 \right]$$

$$= \frac{3}{2}(57.6) \left[1 - \frac{1}{3} \left(\frac{1}{3} \right)^2 \right]$$

$$= 83.2 \text{ N} \cdot \text{m}$$

$$M = 83.2 \text{ N} \cdot \text{m} \quad \blacktriangleleft$$



PROBLEM 4.69

The prismatic bar shown, made of a steel that is assumed to be elastoplastic with $E = 29 \times 10^6$ psi and $\sigma_Y = 36$ ksi, is subjected to a couple of 1350 lb·in. parallel to the z axis. Determine (a) the thickness of the elastic core, (b) the radius of curvature of the bar.

SOLUTION

$$(a) \quad I = \frac{1}{12} \left(\frac{1}{2} \right) \left(\frac{5}{8} \right)^3 = 10.1725 \times 10^{-3} \text{ in}^4$$

$$c = \frac{1}{2} \left(\frac{5}{8} \right) = 0.3125 \text{ in.}$$

$$M_Y = \frac{\sigma_Y I}{c} = \frac{(36 \times 10^3)(10.1725 \times 10^{-3})}{0.3125} = 1171.872 \text{ lb} \cdot \text{in}$$

$$M = \frac{3}{2} M_Y \left[1 - \frac{1}{3} \left(\frac{y_Y}{c} \right)^3 \right]$$

$$\frac{y_Y}{c} = \sqrt{3 - 2 \frac{M}{M_Y}} = \sqrt{3 - \frac{(2)(1350)}{1171.872}} = 0.83426$$

$$y_Y = (0.83426)(0.3125) = 0.26071 \text{ in.}$$

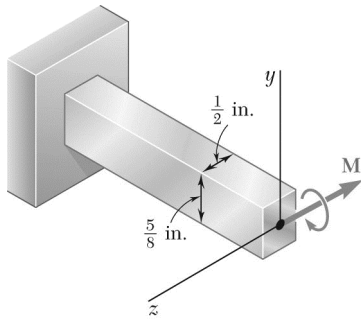
Thickness of elastic core:

$$2y_Y = 0.521 \text{ in.} \quad \blacktriangleleft$$

$$(b) \quad y_Y = \epsilon_Y \rho = \frac{\sigma_Y}{E} \rho$$

$$\rho = \frac{y_Y E}{\sigma_Y} = \frac{(0.26071)(29 \times 10^6)}{36 \times 10^3} = 210.02 \text{ in.}$$

$$\rho = 17.50 \text{ ft} \quad \blacktriangleleft$$



PROBLEM 4.70

Solve Prob. 4.69, assuming that the 1350-lb·in. couple is parallel to the y axis.

PROBLEM 4.69 The prismatic bar shown, made of a steel that is assumed to be elastoplastic with $E = 29 \times 10^6$ psi and $\sigma_Y = 36$ ksi, is subjected to a couple of 1350 lb·in parallel to the z axis. Determine (a) the thickness of the elastic core, (b) the radius of curvature of the bar.

SOLUTION

$$(a) \quad I = \frac{1}{12} \left(\frac{5}{8} \right) \left(\frac{1}{2} \right)^3 = 6.5104 \times 10^{-3} \text{ in}^4$$

$$c = \left(\frac{1}{2} \right) \left(\frac{1}{2} \right) = 0.25 \text{ in.}$$

$$M_Y = \frac{\sigma_Y I}{c} = \frac{(36 \times 10^3)(6.5104 \times 10^{-3})}{0.25} \\ = 937.5 \text{ lb} \cdot \text{in}$$

$$M = \frac{3}{2} \left[1 - \frac{1}{3} \left(\frac{y_Y}{c} \right)^2 \right]$$

$$\frac{y_Y}{c} = \sqrt{3 - 2 \frac{M}{M_Y}} = \sqrt{3 - \frac{(2)(1350)}{937.5}} = 0.34641$$

$$y_Y = (0.34641)(0.25) = 0.086603 \text{ in.}$$

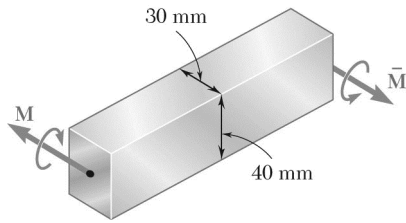
Thickness of elastic core:

$$2y_Y = 0.1732 \text{ in.} \quad \blacktriangleleft$$

$$(b) \quad y_Y = \epsilon_Y \rho = \frac{\sigma_Y}{E} \rho$$

$$\rho = \frac{y_Y E}{\sigma_Y} = \frac{(0.086603)(29 \times 10^6)}{36 \times 10^3} = 69.763 \text{ in.}$$

$$\rho = 5.81 \text{ ft} \quad \blacktriangleleft$$



PROBLEM 4.71

A bar of rectangular cross section shown is made of a steel that is assumed to be elastoplastic with $E = 200 \text{ GPa}$ and $\sigma_Y = 300 \text{ MPa}$. Determine the bending moment M for which (a) yield first occurs, (b) the plastic zones at the top and bottom of the bar are 12 mm thick.

SOLUTION

$$I = \frac{1}{12}(30)(40)^3 = 160 \times 10^3 \text{ mm}^4 = 160 \times 10^{-9} \text{ m}^4$$

$$c = \frac{1}{2}(40) = 20 \text{ mm} = 0.020 \text{ m}$$

(a) First yielding: $\sigma = \frac{Mc}{I} = \sigma_Y$

$$M_Y = \frac{I\sigma_Y}{c} = \frac{(160)(10^{-9})(300 \times 10^6)}{0.020} = 2400 \text{ N} \cdot \text{m}$$

$$M_Y = 2.40 \text{ kN} \cdot \text{m} \quad \blacktriangleleft$$

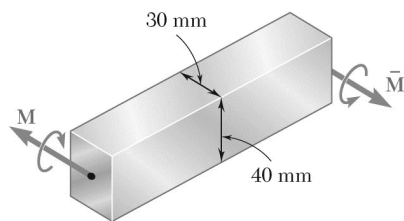
(b) Plastic zones are 12 mm thick:

$$y_Y = 20 \text{ mm} - 12 \text{ mm} = 8 \text{ mm}$$

$$\frac{y_Y}{c} = \frac{8 \text{ mm}}{20 \text{ mm}} = 0.4$$

$$\begin{aligned} M &= \frac{3}{2}M_Y \left[1 - \frac{1}{3} \left(\frac{y_Y}{c} \right)^2 \right] \\ &= \frac{3}{2}(2400) \left[1 - \frac{1}{3}(0.4)^2 \right] = 3408 \text{ N} \cdot \text{m} \end{aligned}$$

$$M = 3.41 \text{ kN} \cdot \text{m} \quad \blacktriangleleft$$



PROBLEM 4.72

Bar AB is made of a steel that is assumed to be elastoplastic with $E = 200 \text{ GPa}$ and $\sigma_Y = 240 \text{ MPa}$. Determine the bending moment M for which the radius of curvature of the bar will be (a) 18 m, (b) 9 m.

SOLUTION

$$I = \frac{1}{12}(30)(40)^3 = 160 \times 10^3 \text{ mm}^4 = 160 \times 10^{-9} \text{ m}^4$$

$$c = \frac{1}{2}(40) = 20 \text{ mm} = 0.020 \text{ m}$$

$$M_Y = \frac{I\sigma_Y}{c} = \frac{(160 \times 10^{-9})(240 \times 10^6)}{0.020} = 1920 \text{ N} \cdot \text{m}$$

$$\frac{1}{\rho_Y} = \frac{M_Y}{EI} = \frac{1920}{(200 \times 10^9)(160 \times 10^{-9})} = 0.0600 \text{ m}^{-1}$$

$$(a) \quad \rho = 18 \text{ m}: \quad \frac{1}{\rho} = 0.05556 \text{ m}^{-1} < 0.0600 \text{ m}^{-1}$$

The bar is fully elastic.

$$M = \frac{EI}{\rho} = \frac{(200 \times 10^9)(160 \times 10^{-9})}{18} = 1778 \text{ N} \cdot \text{m}$$

$$M = 1.778 \text{ kN} \cdot \text{m} \quad \blacktriangleleft$$

$$(b) \quad \rho = 9 \text{ m}: \quad \frac{1}{\rho} = 0.11111 \text{ m}^{-1} > 0.0600 \text{ m}^{-1}$$

The bar is partially plastic.

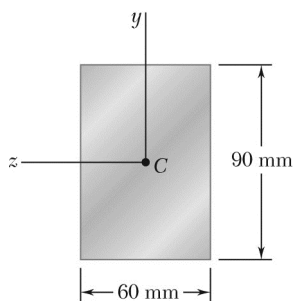
$$\sigma_Y = \frac{E y_Y}{\rho} \quad y_Y = \frac{\rho \sigma_Y}{E}$$

$$y_Y = \frac{(9)(240 \times 10^6)}{200 \times 10^9} = 0.0108 \text{ m} = 10.8 \text{ mm}$$

$$\frac{y_Y}{c} = \frac{10.8 \text{ mm}}{20 \text{ mm}} = 0.54$$

$$M = \frac{3}{2} M_Y \left[1 - \frac{1}{3} \left(\frac{y_Y}{c} \right)^2 \right] = \frac{3}{2} (1920) \left[1 - \frac{1}{3} (0.54)^2 \right] = 2600 \text{ N} \cdot \text{m}$$

$$M = 2.60 \text{ kN} \cdot \text{m} \quad \blacktriangleleft$$



PROBLEM 4.73

A beam of the cross section shown is made of a steel that is assumed to be elastoplastic with $E = 200$ GPa and $\sigma_Y = 240$ MPa. For bending about the z -axis, determine the bending moment at which (a) yield first occurs, (b) the plastic zones at the top and bottom of the bar are 30-mm thick.

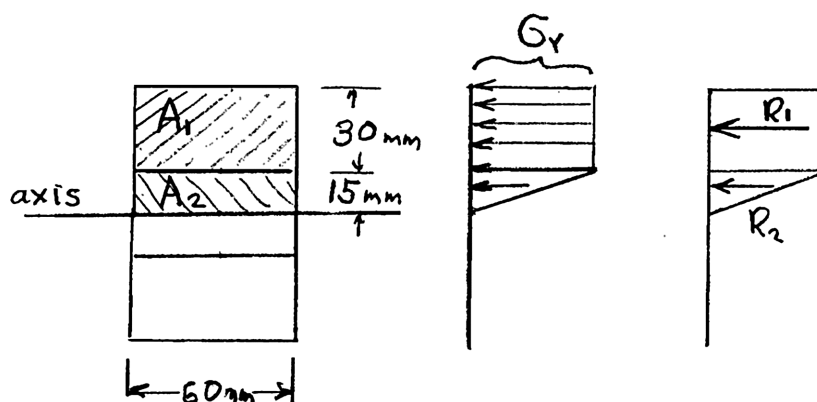
SOLUTION

$$(a) \quad I = \frac{1}{12}bh^3 = \frac{1}{12}(60)(90)^3 = 3.645 \times 10^6 \text{ mm}^4 = 3.645 \times 10^{-6} \text{ m}^4$$

$$c = \frac{1}{2}h = 45 \text{ mm} = 0.045 \text{ m}$$

$$M_Y = \frac{\sigma_Y I}{c} = \frac{(240 \times 10^6)(3.645 \times 10^{-6})}{0.045} = 19.44 \times 10 \text{ N} \cdot \text{m}$$

$$M_Y = 19.44 \text{ kN} \cdot \text{m} \quad \blacktriangleleft$$



$$R_1 = \sigma_Y A_1 = (240 \times 10^6)(0.060)(0.030) = 432 \times 10^3 \text{ N}$$

$$y_1 = 15 \text{ mm} + 15 \text{ mm} = 0.030 \text{ m}$$

$$R_2 = \frac{1}{2} \sigma_Y A_2 = \left(\frac{1}{2}\right)(240 \times 10^6)(0.060)(0.015) = 108 \times 10^3 \text{ N}$$

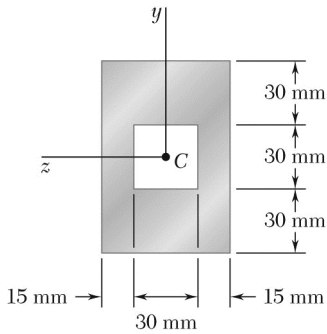
$$y_2 = \frac{2}{3}(15 \text{ mm}) = 10 \text{ mm} = 0.010 \text{ m}$$

$$(b) \quad M = 2(R_1 y_1 + R_2 y_2) = 2[(432 \times 10^3)(0.030) + (108 \times 10^3)(0.010)]$$

$$= 28.08 \times 10^3 \text{ N} \cdot \text{m}$$

$$M = 28.1 \text{ kN} \cdot \text{m} \quad \blacktriangleleft$$

PROBLEM 4.74



A beam of the cross section shown is made of a steel that is assumed to be elastoplastic with $E = 200 \text{ GPa}$ and $\sigma_Y = 240 \text{ MPa}$. For bending about the z axis, determine the bending moment at which (a) yield first occurs, (b) the plastic zones at the top and bottom of the bar are 30-mm thick.

SOLUTION

$$(a) \quad I_{\text{rect}} = \frac{1}{12}bh^3 = \frac{1}{12}(60)(90)^3 = 3.645 \times 10^6 \text{ mm}^4$$

$$I_{\text{cutout}} = \frac{1}{12}bh^3 = \frac{1}{12}(30)(30)^3 = 67.5 \times 10^3 \text{ mm}^4$$

$$I = 3.645 \times 10^6 - 67.5 \times 10^3 = 3.5775 \times 10^6 \text{ mm}^4$$

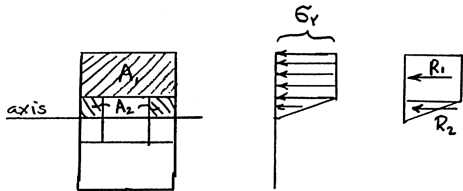
$$= 3.5775 \times 10^{-6} \text{ m}^4$$

$$c = \frac{1}{2}h = 45 \text{ mm} = 0.045 \text{ m}$$

$$M_Y = \frac{\sigma_Y I}{c} = \frac{(240 \times 10^6)(3.5775 \times 10^{-6})}{0.045}$$

$$= 19.08 \times 10^3 \text{ N} \cdot \text{m}$$

$$M_Y = 19.08 \text{ kN} \cdot \text{m} \quad \blacktriangleleft$$



$$R_1 = \sigma_Y A_1 = (240 \times 10^6)(0.060)(0.030) = 432 \times 10^3 \text{ N}$$

$$y_1 = 15 \text{ mm} + 15 \text{ mm} = 30 \text{ mm} = 0.030 \text{ m}$$

$$R_2 = \frac{1}{2} \sigma_Y A_2 = \frac{1}{2}(240 \times 10^6)(0.030)(0.015) = 54 \times 10^3 \text{ N}$$

$$y_2 = \frac{2}{3}(15 \text{ mm}) = 10 \text{ mm} = 0.010 \text{ m}$$

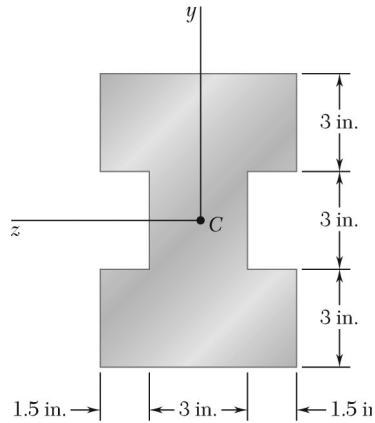
$$(b) \quad M = 2(R_1 y_1 + R_2 y_2)$$

$$= 2[(432 \times 10^3)(0.030) + (54 \times 10^3)(0.010)]$$

$$= 27.00 \times 10^3 \text{ N} \cdot \text{m}$$

$$M = 27.0 \text{ kN} \cdot \text{m} \quad \blacktriangleleft$$

PROBLEM 4.75



A beam of the cross section shown is made of a steel that is assumed to be elastoplastic with $E = 29 \times 10^6$ psi and $\sigma_y = 42$ ksi. For bending about the z axis, determine the bending moment at which (a) yield first occurs, (b) the plastic zones at the top and bottom of the bar are 3 in. thick.

SOLUTION

$$(a) \quad I_1 = \frac{1}{12} b_1 h_1^3 + A_1 d_1^2 = \frac{1}{12} (6)(3)^3 + (6)(3)(3)^2 = 175.5 \text{ in}^4$$

$$I_2 = \frac{1}{12} b_2 h_2^3 = \frac{1}{12} (3)(3)^3 = 6.75 \text{ in}^4$$

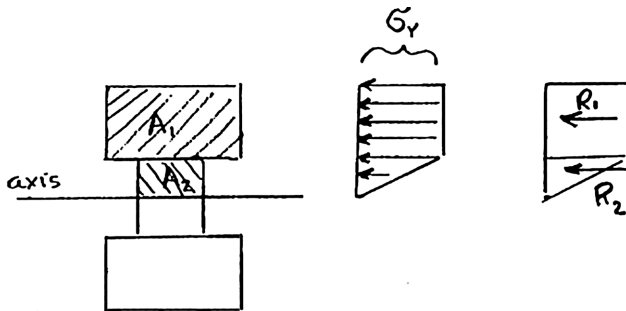
$$I_3 = I_1 = 175.5 \text{ in}^4$$

$$I = I_1 + I_2 + I_3 = 357.75 \text{ in}^4$$

$$c = 4.5 \text{ in.}$$

$$M_Y = \frac{\sigma_Y I}{c} = \frac{(42)(357.75)}{4.5}$$

$$M_Y = 3339 \text{ kip} \cdot \text{in} \quad \blacktriangleleft$$



$$R_1 = \sigma_Y A_1 = (42)(6)(3) = 756 \text{ kip}$$

$$y_1 = 1.5 + 1.5 = 3 \text{ in.}$$

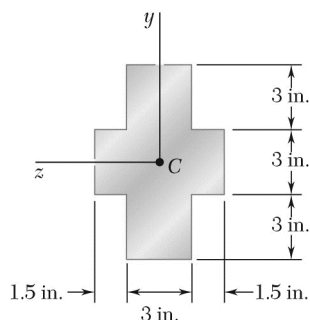
$$R_2 = \frac{1}{2} \sigma_Y A_2 = \frac{1}{2} (42)(3)(1.5) = 94.5 \text{ kip}$$

$$y_2 = \frac{2}{3} (1.5) = 1.0 \text{ in.}$$

$$(b) \quad M = 2(R_1 y_1 + R_2 y_2) = 2[(756)(3) + (94.5)(1.0)]$$

$$M = 4725 \text{ kip} \cdot \text{in} \quad \blacktriangleleft$$

PROBLEM 4.76



A beam of the cross section shown is made of a steel that is assumed to be elastoplastic with $E = 29 \times 10^6$ psi and $\sigma_Y = 42$ ksi. For bending about the z axis, determine the bending moment at which (a) yield first occurs, (b) the plastic zones at the top and bottom of the bar are 3 in. thick.

SOLUTION

$$(a) \quad I_1 = \frac{1}{12} b_1 h_1^3 + A_1 d_1^2 = \frac{1}{12} (3)(3)^3 + (3)(3)(3)^2 = 87.75 \text{ in}^4$$

$$I_2 = \frac{1}{12} b_2 h_2^3 = \frac{1}{12} (6)(3)^3 = 13.5 \text{ in}^4$$

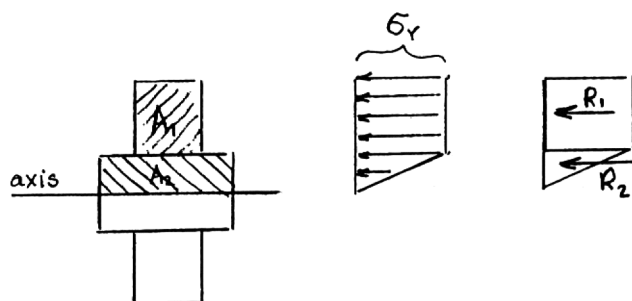
$$I_3 = I_1 = 87.75 \text{ in}^4$$

$$I = I_1 + I_2 + I_3 = 188.5 \text{ in}^4$$

$$c = 4.5 \text{ in.}$$

$$M_Y = \frac{\sigma_Y I}{c} = \frac{(42)(188.5)}{4.5}$$

$$M_Y = 1759 \text{ kip} \cdot \text{in} \quad \blacktriangleleft$$



$$R_1 = \sigma_Y A_1 = (42)(3)(3) = 378 \text{ kip}$$

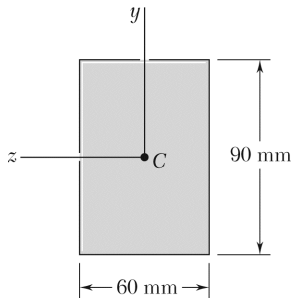
$$y_1 = 1.5 + 1.5 = 3.0 \text{ in.}$$

$$R_2 = \frac{1}{2} \sigma_Y A_2 = \frac{1}{2} (42)(6)(1.5) = 189 \text{ kip}$$

$$y_2 = \frac{2}{3} (1.5) = 1.0 \text{ in.}$$

$$(b) \quad M = 2(R_1 y_1 + R_2 y_2) = 2[(378)(3.0) + (189)(1.0)]$$

$$M = 2646 \text{ kip} \cdot \text{in} \quad \blacktriangleleft$$



PROBLEM 4.77

For the beam indicated (of Prob. 4.73), determine (a) the fully plastic moment M_p , (b) the shape factor of the cross section.

SOLUTION

From Problem 4.73,

$$E = 200 \text{ GPa} \quad \text{and} \quad \sigma_Y = 240 \text{ MPa.}$$

$$A_1 = (60)(45) = 2700 \text{ mm}^2$$

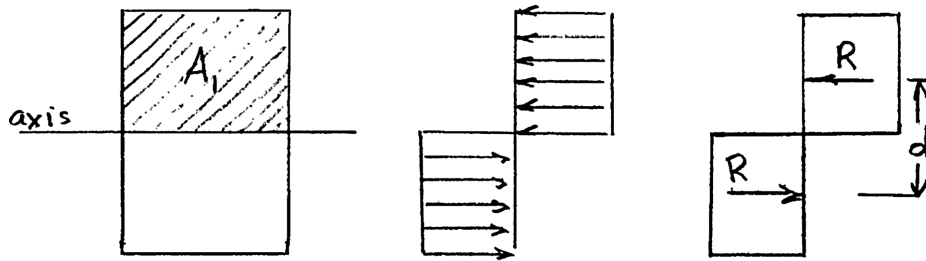
$$= 2700 \times 10^{-6} \text{ m}^2$$

$$R = \sigma_Y A_1$$

$$= (240 \times 10^6)(2700 \times 10^{-6})$$

$$= 648 \times 10^3 \text{ N}$$

$$d = 45 \text{ mm} = 0.045 \text{ m}$$



$$(a) \quad M_p = Rd = (648 \times 10^3)(0.045) = 29.16 \times 10^3 \text{ N} \cdot \text{m}$$

$$M_p = 29.2 \text{ kN} \cdot \text{m} \quad \blacktriangleleft$$

$$(b) \quad I = \frac{1}{12}bh^3 = \frac{1}{12}(60)(90)^3 = 3.645 \times 10^6 \text{ mm}^4 = 3.645 \times 10^{-6} \text{ m}^4$$

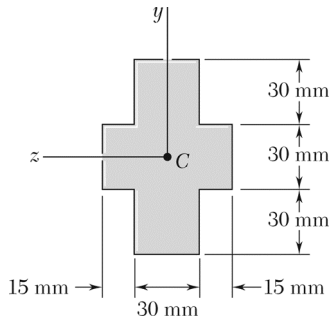
$$c = 45 \text{ mm} = 0.045 \text{ m}$$

$$M_Y = \frac{\sigma_Y I}{c} = \frac{(240 \times 10^6)(3.645 \times 10^{-6})}{0.045} = 19.44 \times 10^3 \text{ N} \cdot \text{m}$$

$$k = \frac{M_p}{M_Y} = \frac{29.16}{19.44}$$

$$k = 1.500 \quad \blacktriangleleft$$

PROBLEM 4.78



For the beam indicated (of Prob. 4.74), determine (a) the fully plastic moment M_p , (b) the shape factor of the cross section.

SOLUTION

From Problem 4.74,

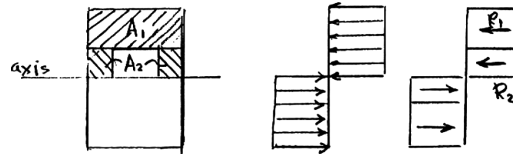
$$E = 200 \text{ GPa and } \sigma_Y = 240 \text{ MPa.}$$

$$\begin{aligned} (a) \quad R_1 &= \sigma_Y A_1 \\ &= (240 \times 10^6)(0.060)(0.030) \\ &= 432 \times 10^3 \text{ N} \\ y_1 &= 15 \text{ mm} + 15 \text{ mm} = 30 \text{ mm} \\ &= 0.030 \text{ m} \end{aligned}$$

$$\begin{aligned} R_2 &= \sigma_Y A_2 \\ &= (240 \times 10^6)(0.030)(0.015) \\ &= 108 \times 10^3 \text{ N} \\ y_2 &= \frac{1}{2}(15) = 7.5 \text{ mm} = 0.0075 \text{ m} \end{aligned}$$

$$\begin{aligned} M_p &= 2(R_1 y_1 + R_2 y_2) = 2[(432 \times 10^3)(0.030) + (108 \times 10^3)(0.0075)] \\ &= 27.54 \times 10^3 \text{ N} \cdot \text{m} \end{aligned}$$

$$M_p = 27.5 \text{ kN} \cdot \text{m} \quad \blacktriangleleft$$



$$\begin{aligned} (b) \quad I_{\text{rect}} &= \frac{1}{12}bh^3 = \frac{1}{12}(60)(90)^3 = 3.645 \times 10^6 \text{ mm}^4 \\ I_{\text{cutout}} &= \frac{1}{12}bh^3 = \frac{1}{12}(30)(30)^3 = 67.5 \times 10^3 \text{ mm}^4 \\ I &= I_{\text{rect}} - I_{\text{cutout}} = 3.645 \times 10^6 - 67.5 \times 10^3 = 3.5775 \times 10^6 \text{ mm}^4 \\ &= 3.5775 \times 10^{-9} \text{ m}^4 \end{aligned}$$

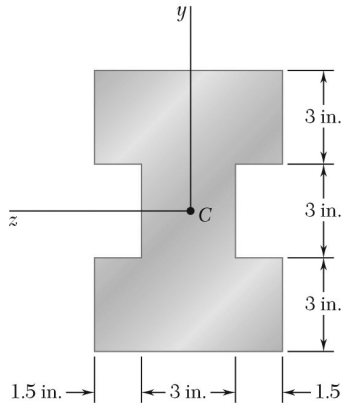
$$c = \frac{1}{2}h = 45 \text{ mm} = 0.045 \text{ m}$$

$$M_Y = \frac{\sigma_Y I}{c} = \frac{(240 \times 10^6)(3.5775 \times 10^{-9})}{0.045} = 19.08 \times 10^3 \text{ N} \cdot \text{m}$$

$$k = \frac{M_p}{M_Y} = \frac{27.54}{19.08}$$

$$k = 1.443 \quad \blacktriangleleft$$

PROBLEM 4.79



For the beam indicated (of Prob. 4.75), determine (a) the fully plastic moment M_p , (b) the shape factor of the cross section.

SOLUTION

From Problem 4.75,

$$E = 29 \times 10^6 \text{ and } \sigma_Y = 42 \text{ ksi.}$$

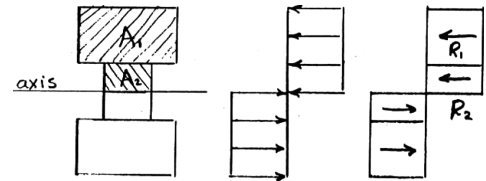
$$(a) \quad R_1 = \sigma_Y A_1 = (42)(6)(3) = 756 \text{ kip}$$

$$y_1 = 1.5 + 1.5 = 3.0 \text{ in.}$$

$$R_2 = \sigma_Y A_2 = (42)(3)(1.5) = 189 \text{ kip}$$

$$y_2 = \frac{1}{2}(1.5) = 0.75 \text{ in.}$$

$$M_p = 2(R_1 y_1 + R_2 y_2) = 2[(756)(3.0) + (189)(0.75)]$$



$$M_p = 4819.5 \text{ kip} \cdot \text{in} \quad \blacktriangleleft$$

$$(b) \quad I_1 = \frac{1}{12} b_1 h_1^3 + A_1 d_1^2 = \frac{1}{12} (6)(3)^3 + (6)(3)(3)^2 = 175.5 \text{ in}^4$$

$$I_2 = \frac{1}{12} b_2 h_2^3 = \frac{1}{12} (3)(3)^3 = 6.75 \text{ in}^4$$

$$I_3 = I_1 = 175.5 \text{ in}^4$$

$$I = I_1 + I_2 + I_3 = 357.75 \text{ in}^4$$

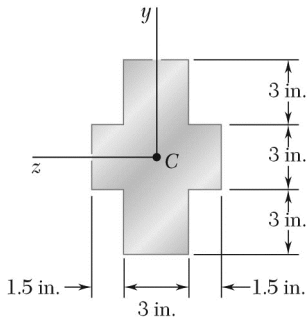
$$c = 4.5 \text{ in.}$$

$$M_Y = \frac{\sigma_Y I}{c} = \frac{(42)(357.75)}{4.5} = 3339 \text{ kip} \cdot \text{in}$$

$$k = \frac{M_p}{M_Y} = \frac{4819.5}{3339}$$

$$k = 1.443 \quad \blacktriangleleft$$

PROBLEM 4.80



For the beam indicated (of Prob. 4.76), determine (a) the fully plastic moment M_p , (b) the shape factor of the cross section.

SOLUTION

From Problem 4.76, $E = 29 \times 10^6 \text{ psi}$ and $\sigma_Y = 42 \text{ ksi}$.

(a) $R_1 = \sigma_Y A_1 = (42)(3)(3) = 378 \text{ kip}$

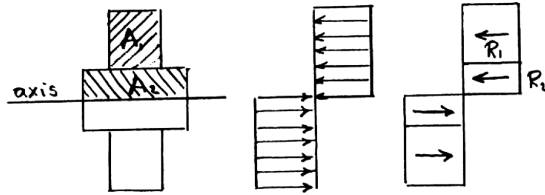
$y_1 = 1.5 + 1.5 = 3.0 \text{ in.}$

$R_2 = \sigma_Y A_2 = (42)(6)(1.5) = 378 \text{ kip}$

$y_2 = \frac{1}{2}(1.5) = 0.75 \text{ in.}$

$M_p = 2(R_1 y_1 + R_2 y_2) = 2[(378)(3.0) + (378)(0.75)]$

$M_p = 2835 \text{ kip} \cdot \text{in} \quad \blacktriangleleft$



(b) $I_1 = \frac{1}{12} b_1 h_1^3 + A_1 d_1^2 = \frac{1}{12} (3)(3)^3 + (3)(3)(3)^2 = 87.75 \text{ in}^4$

$I_2 = \frac{1}{12} b_2 h_2^3 = \frac{1}{12} (6)(3)^3 = 13.5 \text{ in}^4$

$I_3 = I_1 = 87.75 \text{ in}^4$

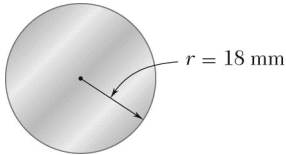
$I = I_1 + I_2 + I_3 = 188.5 \text{ in}^4$

$c = 4.5 \text{ in.}$

$M_Y = \frac{\sigma_Y I}{c} = \frac{(42)(188.5)}{4.5} = 1759.3 \text{ kip} \cdot \text{in}$

$k = \frac{M_p}{M_Y} = \frac{2835}{1759.3}$

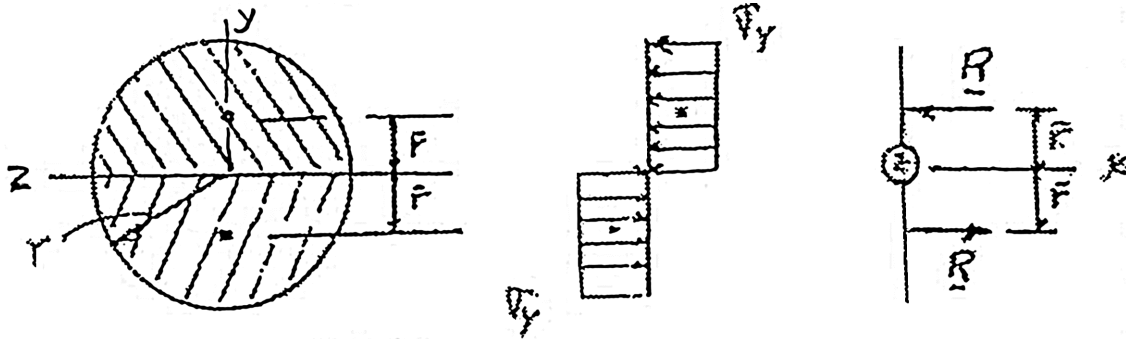
$k = 1.611 \quad \blacktriangleleft$



PROBLEM 4.81

Determine the plastic moment M_p of a steel beam of the cross section shown, assuming the steel to be elastoplastic with a yield strength of 240 MPa.

SOLUTION



For a semicircle:

$$A = \frac{\pi}{2}r^2; \quad \bar{r} = \frac{4r}{3\pi}$$

Resultant force on semicircular section:

$$R = \sigma_Y A$$

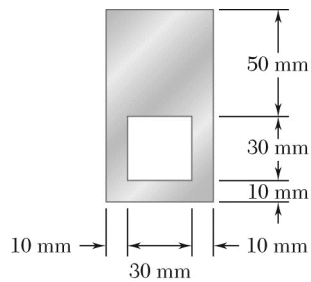
Resultant moment on entire cross section:

$$M_p = 2R\bar{r} = \frac{4}{3}\sigma_Y r^3$$

Data: $\sigma_Y = 240 \text{ MPa} = 240 \times 10^6 \text{ Pa}$, $r = 18 \text{ mm} = 0.018 \text{ m}$

$$M_p = \frac{4}{3}(240 \times 10^6)(0.018)^3 = 1866 \text{ N} \cdot \text{m}$$

$$M_p = 1.866 \text{ kN} \cdot \text{m} \quad \blacktriangleleft$$



PROBLEM 4.82

Determine the plastic moment M_p of a steel beam of the cross section shown, assuming the steel to be elastoplastic with a yield strength of 240 MPa.

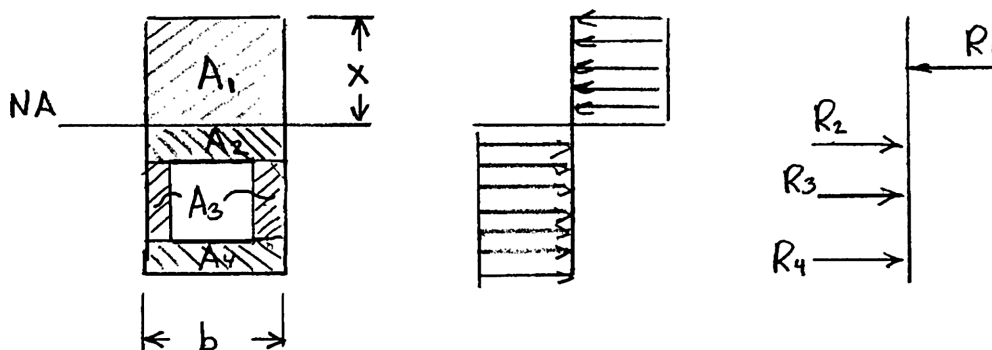
SOLUTION

Total area:

$$A = (50)(90) - (30)(30) = 3600 \text{ mm}^2$$

$$\frac{1}{2}A = 1800 \text{ mm}^2$$

$$x = \frac{\frac{1}{2}A}{b} = \frac{1800}{50} = 36 \text{ mm}$$



$$A_1 = (50)(36) = 1800 \text{ mm}^2, \quad \bar{y}_1 = 18 \text{ mm}, \quad A_1\bar{y}_1 = 32.4 \times 10^3 \text{ mm}^3$$

$$A_2 = (50)(14) = 700 \text{ mm}^2, \quad \bar{y}_2 = 7 \text{ mm}, \quad A_2\bar{y}_2 = 4.9 \times 10^3 \text{ mm}^3$$

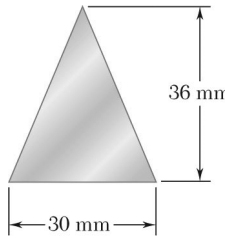
$$A_3 = (20)(30) = 600 \text{ mm}^2, \quad \bar{y}_3 = 29 \text{ mm}, \quad A_3\bar{y}_3 = 17.4 \times 10^3 \text{ mm}^3$$

$$A_4 = (50)(10) = 500 \text{ mm}^2, \quad \bar{y}_4 = 49 \text{ mm}, \quad A_4\bar{y}_4 = 24.5 \times 10^3 \text{ mm}^3$$

$$A_1\bar{y}_1 + A_2\bar{y}_2 + A_3\bar{y}_3 + A_4\bar{y}_4 = 79.2 \times 10^3 \text{ mm}^3 = 79.2 \times 10^{-6} \text{ m}^3$$

$$M_p = \sigma_Y \Sigma A_i \bar{y}_i = (240 \times 10^6)(79.2 \times 10^{-6}) = 19.008 \times 10^3 \text{ N} \cdot \text{m}$$

$$M_p = 19.01 \text{ kN} \cdot \text{m} \quad \blacktriangleleft$$



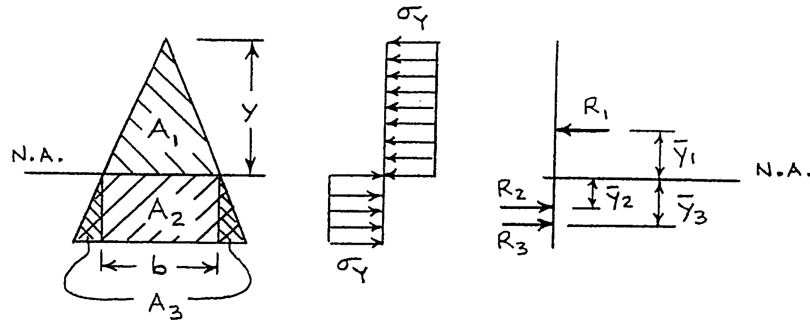
PROBLEM 4.83

Determine the plastic moment M_p of a steel beam of the cross section shown, assuming the steel to be elastoplastic with a yield strength of 240 MPa.

SOLUTION

Total area: $A = \frac{1}{2}(30)(36) = 540 \text{ mm}^2$

Half area: $\frac{1}{2}A = 270 \text{ mm}^2 = A_1$



By similar triangles, $\frac{b}{y} = \frac{30}{36} \quad b = \frac{5}{6}y$

Since $A_1 = \frac{1}{2}by = \frac{5}{12}y^2, \quad y^2 = \frac{12}{5}A_1$

$$y = \sqrt{\frac{12}{5}(270)} = 25.4558 \text{ mm}$$

$$b = 21.2132 \text{ mm}$$

$$A_1 = \frac{1}{2}(21.2132)(25.4558) = 270 \text{ mm}^2 = 270 \times 10^{-6} \text{ m}^2$$

$$A_2 = (21.2132)(36 - 25.4558) = 223.676 \text{ mm}^2 = 223.676 \times 10^{-6} \text{ m}^2$$

$$A_3 = A - A_1 - A_2 = 46.324 \text{ mm}^2 = 46.324 \times 10^{-6} \text{ m}^2$$

$$R_i = \sigma_Y A_i = 240 \times 10^6 A_i$$

$$R_1 = 64.8 \times 10^3 \text{ N}, \quad R_2 = 53.6822 \times 10^3 \text{ N}, \quad R_3 = 11.1178 \times 10^3 \text{ N}$$

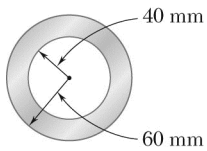
$$\bar{y}_1 = \frac{1}{3}y = 8.4853 \text{ mm} = 8.4853 \times 10^{-3} \text{ m}$$

$$\bar{y}_2 = \frac{1}{2}(36 - 25.4558) = 5.2721 \text{ mm} = 5.2721 \times 10^{-3} \text{ m}$$

$$\bar{y}_3 = \frac{2}{3}(36 - 25.4558) = 7.0295 \text{ mm} = 7.0295 \times 10^{-3} \text{ m}$$

$$M_p = R_1\bar{y}_1 + R_2\bar{y}_2 + R_3\bar{y}_3 = 911 \text{ N} \cdot \text{m}$$

$$M_p = 911 \text{ N} \cdot \text{m} \quad \blacktriangleleft$$

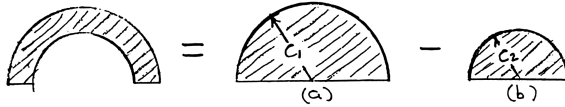
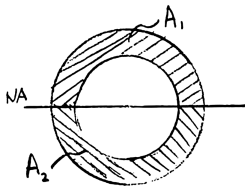
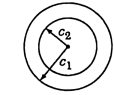


PROBLEM 4.84

Determine the plastic moment M_p of a steel beam of the cross section shown, assuming the steel to be elastoplastic with a yield strength of 240 MPa.

SOLUTION

Let c_1 be the outer radius and c_2 the inner radius.



$$\begin{aligned} A_1 \bar{y}_1 &= A_a \bar{y}_a - A_b \bar{y}_b \\ &= \left(\frac{\pi}{2} c_1^2 \right) \left(\frac{4c_1}{3\pi} \right) - \left(\frac{\pi}{2} c_2^2 \right) \left(\frac{4c_2}{3\pi} \right) \\ &= \frac{2}{3} (c_1^3 - c_2^3) \end{aligned}$$

$$A_2 \bar{y}_2 = A_1 \bar{y}_1 = \frac{2}{3} (c_1^3 - c_2^3)$$

$$M_p = \sigma_Y (A_1 \bar{y}_1 + A_2 \bar{y}_2) = \frac{4}{3} \sigma_Y (c_1^3 - c_2^3)$$

Data:

$$\sigma_Y = 240 \text{ MPa} = 240 \times 10^6 \text{ Pa}$$

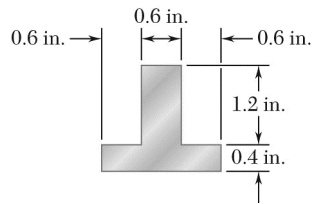
$$c_1 = 60 \text{ mm} = 0.060 \text{ m}$$

$$c_2 = 40 \text{ mm} = 0.040 \text{ m}$$

$$M_p = \frac{4}{3} (240 \times 10^6) (0.060^3 - 0.040^3)$$

$$= 48.64 \times 10^3 \text{ N} \cdot \text{m}$$

$$M_p = 48.6 \text{ kN} \cdot \text{m} \quad \blacktriangleleft$$



PROBLEM 4.85

Determine the plastic moment M_p of the cross section shown, assuming the steel to be elastoplastic with a yield strength of 36 ksi.

SOLUTION

Total area:

$$A = (1.8)(0.4) + (0.6)(1.2) = 1.44 \text{ in}^2$$

$$\frac{1}{2}A = 0.72 \text{ in}^2$$

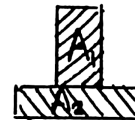
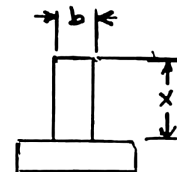
$$x = \frac{\frac{1}{2}A}{b} = \frac{0.72}{0.6} = 1.2 \text{ in.}$$

Neutral axis lies 1.2 in. below the top.

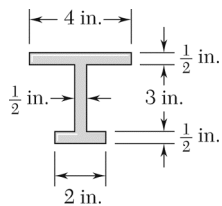
$$A_1 = \frac{1}{2}A = 0.72 \text{ in}^2, \quad \bar{y}_1 = \frac{1}{2}(1.2) = 0.6 \text{ in.}$$

$$A_2 = \frac{1}{2}A = 0.72 \text{ in}^2, \quad \bar{y}_2 = \frac{1}{2}(0.4) = 0.2 \text{ in.}$$

$$\begin{aligned} M_p &= \sigma_Y (A_1 \bar{y}_1 + A_2 \bar{y}_2) \\ &= (36)[(0.72)(0.6) + (0.72)(0.2)] = 20.7 \text{ kip} \cdot \text{in} \end{aligned}$$



$$M_p = 20.7 \text{ kip} \cdot \text{in} \quad \blacktriangleleft$$



PROBLEM 4.86

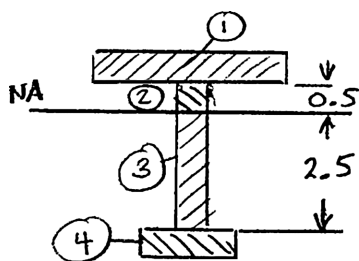
Determine the plastic moment M_p of the cross section shown, assuming the steel to be elastoplastic with a yield strength of 36 ksi.

SOLUTION

Total area:

$$A = (4)\left(\frac{1}{2}\right) + \left(\frac{1}{2}\right)(3) + (2)\left(\frac{1}{2}\right) = 4.5 \text{ in}^2$$

$$\frac{1}{2}A = 2.25 \text{ in}^2$$



$$A_1 = 2.00 \text{ in}^2, \quad \bar{y}_1 = 0.75, \quad A_1\bar{y}_1 = 1.50 \text{ in}^3$$

$$A_2 = 0.25 \text{ in}^2, \quad \bar{y}_2 = 0.25, \quad A_2\bar{y}_2 = 0.0625 \text{ in}^3$$

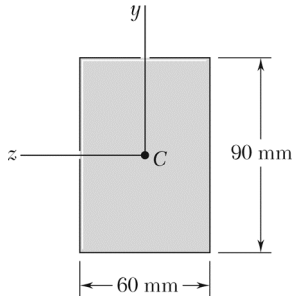
$$A_3 = 1.25 \text{ in}^2, \quad \bar{y}_3 = 1.25, \quad A_3\bar{y}_3 = 1.5625 \text{ in}^3$$

$$A_4 = 1.00 \text{ in}^2, \quad \bar{y}_4 = 2.75, \quad A_4\bar{y}_4 = 2.75 \text{ in}^3$$

$$M_p = \sigma_Y (A_1\bar{y}_1 + A_2\bar{y}_2 + A_3\bar{y}_3 + A_4\bar{y}_4)$$

$$= (36)(1.50 + 0.0625 + 1.5625 + 2.75)$$

$$M_p = 212 \text{ kip} \cdot \text{in} \quad \blacktriangleleft$$



PROBLEM 4.87

For the beam indicated (of Prob. 4.73), a couple of moment equal to the full plastic moment M_p is applied and then removed. Using a yield strength of 240 MPa, determine the residual stress at $y = 45$ mm.

SOLUTION

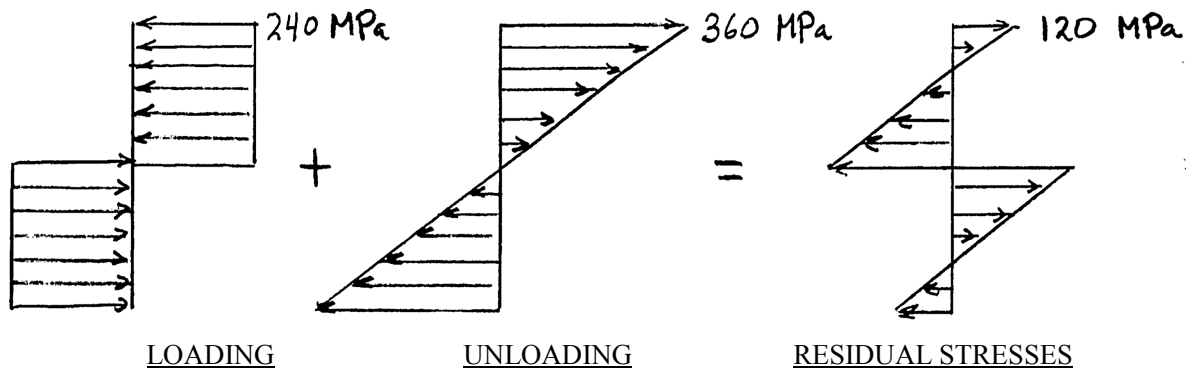
$$M_p = 29.16 \times 10^3 \text{ N} \cdot \text{m}$$

See solutions to Problems 4.73 and 4.77.

$$I = 3.645 \times 10^{-6} \text{ m}^4$$

$$c = 0.045 \text{ m}$$

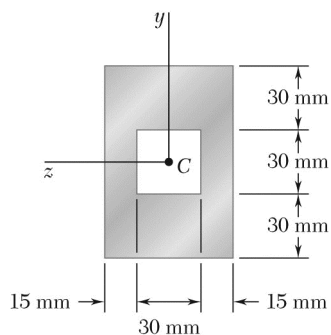
$$\sigma' = \frac{M_{\max} y}{I} = \frac{M_p c}{I} \quad \text{at } y = c = 45 \text{ mm}$$



$$\sigma' = \frac{(29.16 \times 10^3)(0.045)}{3.645 \times 10^{-6}} = 360 \times 10^6 \text{ Pa}$$

$$\begin{aligned} \sigma_{\text{res}} &= \sigma' - \sigma_Y = 360 \times 10^6 - 240 \times 10^6 \\ &= 120 \times 10^6 \text{ Pa} \end{aligned}$$

$$\sigma_{\text{res}} = 120.0 \text{ MPa} \quad \blacktriangleleft$$



PROBLEM 4.88

For the beam indicated (of Prob. 4.74), a couple of moment equal to the full plastic moment M_p is applied and then removed. Using a yield strength of 240 MPa, determine the residual stress at $y = 45$ mm.

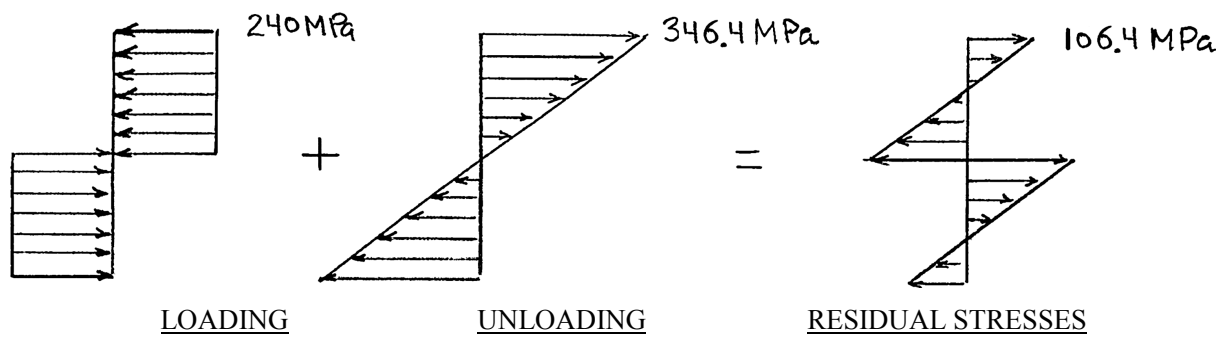
SOLUTION

$$M_p = 27.54 \times 10^3 \text{ N} \cdot \text{m} \quad (\text{See solutions to Problems 4.74 and 4.78.})$$

$$I = 3.5775 \times 10^{-6} \text{ m}^4, \quad c = 0.045 \text{ m}$$

$$\sigma' = \frac{M_{\max} y}{I} = \frac{M_p c}{I} \quad \text{at } y = c$$

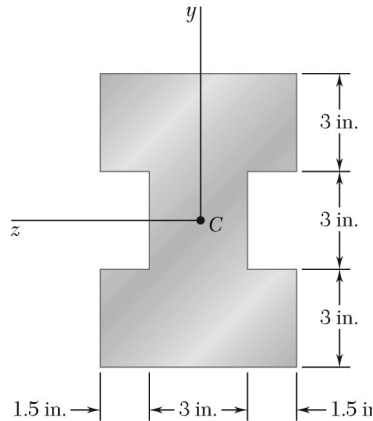
$$\sigma' = \frac{(27.54 \times 10^3)(0.045)}{3.5775 \times 10^{-6}} = 346.4 \times 10^6 \text{ Pa}$$



$$\sigma_{\text{res}} = \sigma' - \sigma_Y = 346.4 \times 10^6 - 240 \times 10^6 = 106.4 \times 10^6 \text{ Pa}$$

$$\sigma_{\text{res}} = 106.4 \text{ MPa} \quad \blacktriangleleft$$

PROBLEM 4.89



A bending couple is applied to the bar indicated, causing plastic zones 3 in. thick to develop at the top and bottom of the bar. After the couple has been removed, determine (a) the residual stress at $y = 4.5$ in., (b) the points where the residual stress is zero, (c) the radius of curvature corresponding to the permanent deformation of the bar.

SOLUTION

See solution to Problem 4.75 for bending couple and stress distribution.

$$M = 4725 \text{ kip} \cdot \text{in} \quad y_Y = 1.5 \text{ in.} \quad E = 29 \times 10^6 \text{ psi} = 29 \times 10^3 \text{ ksi}$$

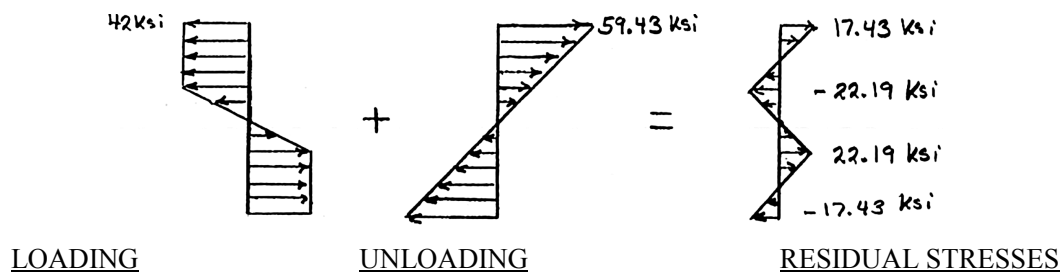
$$\sigma_Y = 42 \text{ ksi} \quad I = 357.75 \text{ in}^4 \quad c = 4.5 \text{ in.}$$

$$(a) \quad \sigma' = \frac{Mc}{I} = \frac{(4725)(4.5)}{357.75} = 59.43 \text{ ksi}$$

$$\sigma'' = \frac{My_Y}{I} = \frac{(4725)(1.5)}{357.75} = 19.81 \text{ ksi}$$

$$\text{At } y = c, \quad \sigma_{\text{res}} = \sigma' - \sigma_Y = 59.43 - 42 = 17.43 \text{ ksi}$$

$$\text{At } y = y_Y, \quad \sigma_{\text{res}} = \sigma'' - \sigma_Y = 19.81 - 42 = -22.19 \text{ ksi}$$



$$(b) \quad \sigma_{\text{res}} = 0 \quad \therefore \quad \frac{My_0}{I} - \sigma_Y = 0$$

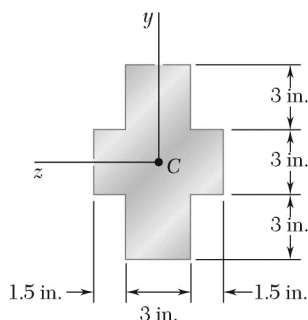
$$y_0 = \frac{I\sigma_Y}{M} = \frac{(357.75)(42)}{4725} = 3.18 \text{ in.}$$

$$\text{Answer: } y_0 = -3.18 \text{ in., } 0, \text{ } 3.18 \text{ in.} \quad \blacktriangleleft$$

$$(c) \quad \text{At } y = y_Y, \quad \sigma_{\text{res}} = -22.19 \text{ ksi}$$

$$\sigma = -\frac{Ey}{\rho} \quad \therefore \quad \rho = -\frac{Ey}{\sigma} = \frac{(29 \times 10^3)(1.5)}{22.19} = 1960 \text{ in.}$$

$$\rho = 163.4 \text{ ft.} \quad \blacktriangleleft$$



PROBLEM 4.90

A bending couple is applied to the bar indicated, causing plastic zones 3 in. thick to develop at the top and bottom of the bar. After the couple has been removed, determine (a) the residual stress at $y = 4.5$ in., (b) the points where the residual stress is zero, (c) the radius of curvature corresponding to the permanent deformation of the bar.

SOLUTION

See solution to Problem 4.76 for bending couple and stress distribution during loading.

$$M = 2646 \text{ kip} \cdot \text{in} \quad y_Y = 1.5 \text{ in.} \quad E = 29 \times 10^6 \text{ psi} = 29 \times 10^3 \text{ ksi}$$

$$\sigma_Y = 42 \text{ ksi} \quad I = 188.5 \text{ in}^4 \quad c = 4.5 \text{ in.}$$

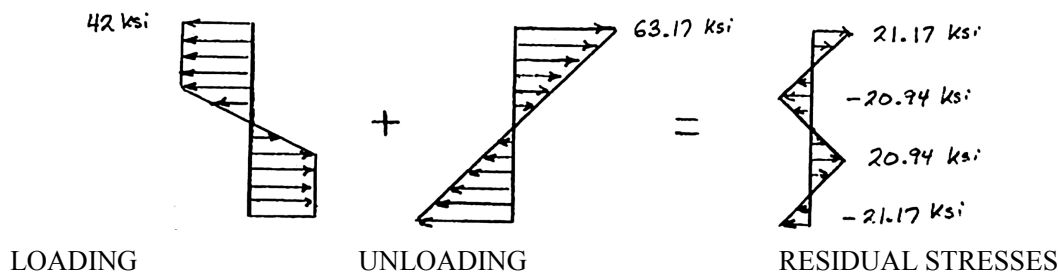
$$(a) \quad \sigma' = \frac{Mc}{I} = \frac{(2646)(4.5)}{188.5} = 63.17 \text{ ksi}$$

$$\sigma'' = \frac{My_Y}{I} = \frac{(2646)(1.5)}{188.5} = 21.06 \text{ ksi}$$

$$\text{At } y = c, \quad \sigma_{\text{res}} = \sigma' - \sigma_Y = 63.17 - 42 = 21.17 \text{ ksi}$$

$$\text{At } y = y_Y, \quad \sigma_{\text{res}} = \sigma'' - \sigma_Y = 21.06 - 42$$

$$\sigma_{\text{res}} = -20.94 \text{ ksi} \quad \blacktriangleleft$$



$$(b) \quad \sigma_{\text{res}} = 0 \quad \therefore \quad \frac{My_0}{I} = \sigma_Y$$

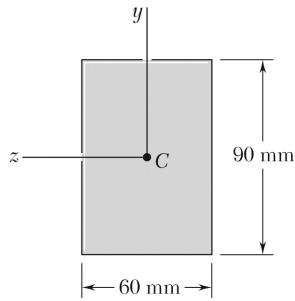
$$y_0 = \frac{I\sigma_Y}{M} = \frac{(188.5)(42)}{2646} = 2.992 \text{ in.}$$

$$\text{Answer: } y_0 = -2.992 \text{ in., } 0, \quad 2.992 \text{ in.} \quad \blacktriangleleft$$

$$(c) \quad \text{At } y = y_Y, \quad \sigma_{\text{res}} = -20.94 \text{ ksi}$$

$$\sigma = -\frac{Ey}{\rho} \quad \therefore \quad \rho = -\frac{Ey}{\sigma} = \frac{(29 \times 10^3)(1.5)}{20.94} = 2077 \text{ in.}$$

$$\rho = 173.1 \text{ ft} \quad \blacktriangleleft$$



PROBLEM 4.91

A bending couple is applied to the beam of Prob. 4.73, causing plastic zones 30 mm thick to develop at the top and bottom of the beam. After the couple has been removed, determine (a) the residual stress at $y = 45$ mm, (b) the points where the residual stress is zero, (c) the radius of curvature corresponding to the permanent deformation of the beam.

SOLUTION

See solution to Problem 4.73 for bending couple and stress distribution during loading:

$$M = 28.08 \times 10^3 \text{ N} \cdot \text{m} \quad y_Y = 15 \text{ mm} = 0.015 \text{ m} \quad E = 200 \text{ GPa}$$

$$\sigma_Y = 240 \text{ MPa} \quad I = 3.645 \times 10^{-6} \text{ m}^4 \quad c = 0.045 \text{ m}$$

$$(a) \quad \sigma' = \frac{Mc}{I} = \frac{(28.08 \times 10^3)(0.045)}{3.645 \times 10^{-6}} = 346.7 \times 10^6 \text{ Pa} = 346.7 \text{ MPa}$$

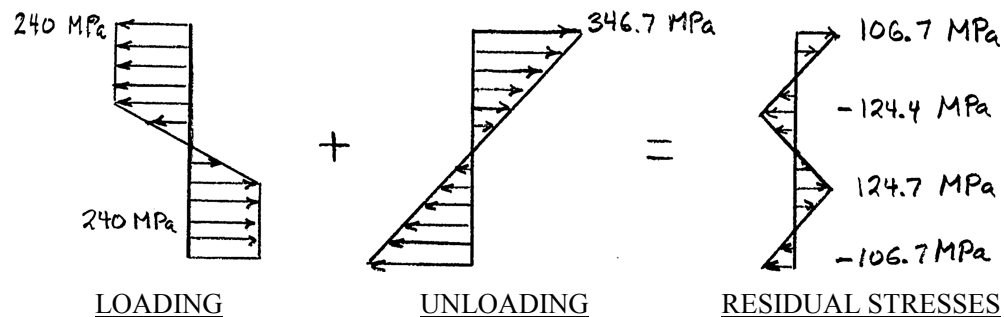
$$\sigma'' = \frac{My_Y}{I} = \frac{(28.08 \times 10^3)(0.015)}{3.645 \times 10^{-6}} = 115.6 \times 10^6 \text{ Pa} = 115.6 \text{ MPa}$$

$$\text{At } y = c, \quad \sigma_{\text{res}} = \sigma' - \sigma_Y = 346.7 - 240$$

$$\sigma_{\text{res}} = 106.7 \text{ MPa} \quad \blacktriangleleft$$

$$\text{At } y = y_Y, \quad \sigma_{\text{res}} = \sigma'' - \sigma_Y = 115.6 - 240$$

$$\sigma_{\text{res}} = -124.4 \text{ MPa}$$



$$(b) \quad \sigma_{\text{res}} = 0 \quad \therefore \quad \frac{My_0}{I} - \sigma_Y = 0$$

$$y_0 = \frac{I\sigma_Y}{M} = \frac{(3.645 \times 10^{-6})(240 \times 10^6)}{28.08 \times 10^3} = 31.15 \times 10^{-3} \text{ m} = 31.15 \text{ mm}$$

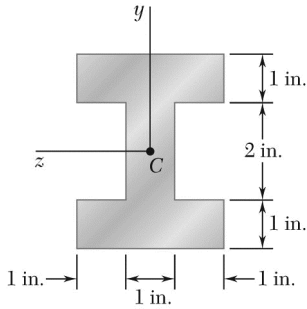
$$\text{Answer: } y_0 = -31.15 \text{ mm}, 0, 31.15 \text{ mm} \quad \blacktriangleleft$$

$$(c) \quad \text{At } y = y_Y, \quad \sigma_{\text{res}} = -124.4 \times 10^6 \text{ Pa}$$

$$\sigma = -\frac{Ey}{\rho} \quad \therefore \quad \rho = -\frac{Ey}{\sigma} = \frac{(200 \times 10^9)(0.015)}{-124.4 \times 10^6}$$

$$\rho = 24.1 \text{ m} \quad \blacktriangleleft$$

PROBLEM 4.92



A beam of the cross section shown is made of a steel that is assumed to be elastoplastic with $E = 29 \times 10^6$ psi and $\sigma_Y = 42$ ksi. A bending couple is applied to the beam about z axis, causing plastic zones 2 in. thick to develop at the top and bottom of the beam. After the couple has been removed, determine (a) the residual stress at $y = 2$ in., (b) the points where the residual stress is zero, (c) the radius of curvature corresponding to the permanent deformation of the beam.

SOLUTION

See solution to Problem 4.76 for bending couple and stress distribution during loading.

$$M = 406 \text{ kip} \cdot \text{in} \quad y_Y = 1.0 \text{ in.} \quad E = 29 \times 10^6 \text{ psi} = 29 \times 10^3 \text{ ksi}$$

$$\sigma_Y = 42 \text{ ksi} \quad I = 14.6667 \text{ in}^4 \quad c = 2 \text{ in.}$$

$$(a) \quad \sigma' = \frac{Mc}{I} = \frac{(406)(2)}{14.6667} = 55.36 \text{ ksi}$$

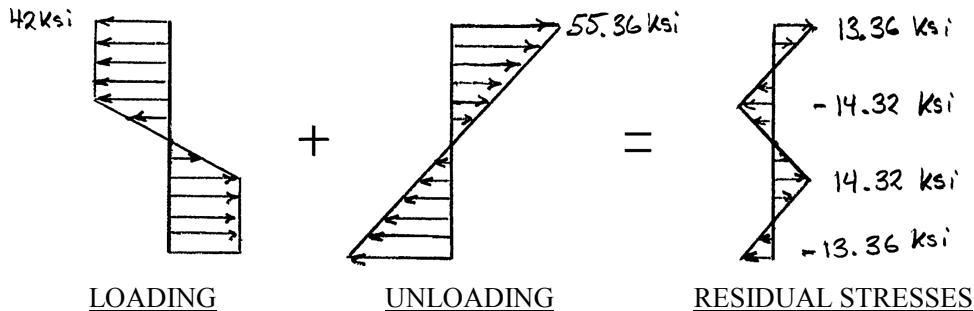
$$\sigma'' = \frac{My_Y}{I} = \frac{(406)(1.0)}{14.6667} = 27.68 \text{ ksi}$$

$$\text{At } y = c, \quad \sigma_{\text{res}} = \sigma' - \sigma_Y = 55.36 - 42$$

$$\sigma_{\text{res}} = 13.36 \text{ ksi} \quad \blacktriangleleft$$

$$\text{At } y = y_Y, \quad \sigma_{\text{res}} = \sigma'' - \sigma_Y = 27.68 - 42$$

$$\sigma_{\text{res}} = -14.32 \text{ ksi} \quad \blacktriangleleft$$



$$(b) \quad \sigma_{\text{res}} = 0 \quad \therefore \quad \frac{My_0}{I} - \sigma_Y = 0$$

$$y_0 = \frac{I\sigma_Y}{M} = \frac{(14.6667)(42)}{406} = 1.517 \text{ in.}$$

$$\text{Answer: } y_0 = -1.517 \text{ in., } 1.517 \text{ in.} \quad \blacktriangleleft$$

$$(c) \quad \text{At } y = y_Y, \quad \sigma_{\text{res}} = -14.32 \text{ ksi}$$

$$\sigma = -\frac{Ey}{\rho} \quad \therefore \quad \rho = -\frac{Ey}{\sigma} = \frac{(29 \times 10^3)(1.0)}{14.32} = 2025 \text{ in.}$$

$$\rho = 168.8 \text{ ft} \quad \blacktriangleleft$$

PROBLEM 4.93*

A rectangular bar that is straight and unstressed is bent into an arc of circle of radius by two couples of moment M . After the couples are removed, it is observed that the radius of curvature of the bar is ρ_R . Denoting by ρ_Y the radius of curvature of the bar at the onset of yield, show that the radii of curvature satisfy the following relation:

$$\frac{1}{\rho_R} = \frac{1}{\rho} \left\{ 1 - \frac{3}{2} \frac{\rho}{\rho_Y} \left[1 - \frac{1}{3} \left(\frac{\rho}{\rho_Y} \right)^2 \right] \right\}$$

SOLUTION

$$\frac{1}{\rho_Y} = \frac{M_Y}{EI}, \quad M = \frac{3}{2} M_Y \left(1 - \frac{1}{3} \frac{\rho^2}{\rho_Y^2} \right),$$

Let m denote $\frac{M}{M_Y}$:

$$m = \frac{M}{M_Y} = \frac{3}{2} \left(1 - \frac{\rho^2}{\rho_Y^2} \right) \quad \therefore \quad \frac{\rho^2}{\rho_Y^2} = 3 - 2m$$

$$\begin{aligned} \frac{1}{\rho_R} &= \frac{1}{\rho} - \frac{M}{EI} = \frac{1}{\rho} - \frac{mM_Y}{EI} = \frac{1}{\rho} - \frac{m}{\rho_Y} \\ &= \frac{1}{\rho} \left\{ 1 - \frac{\rho}{\rho_Y} m \right\} = \frac{1}{\rho} \left\{ 1 - \frac{3}{2} \frac{\rho}{\rho_Y} \left(1 - \frac{1}{3} \frac{\rho^2}{\rho_Y^2} \right) \right\} \end{aligned}$$

PROBLEM 4.94

A solid bar of rectangular cross section is made of a material that is assumed to be elastoplastic. Denoting by M_Y and ρ_Y , respectively, the bending moment and radius of curvature at the onset of yield, determine (a) the radius of curvature when a couple of moment $M = 1.25 M_Y$ is applied to the bar, (b) the radius of curvature after the couple is removed. Check the results obtained by using the relation derived in Prob. 4.93.

SOLUTION

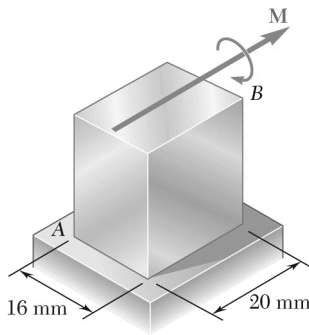
$$(a) \quad \frac{1}{\rho_Y} = \frac{M_Y}{EI}, \quad M = \frac{3}{2} M_Y \left(1 - \frac{1}{3} \frac{\rho^2}{\rho_Y^2} \right) \quad \text{Let } m = \frac{M}{M_Y} = 1.25$$

$$m = \frac{M}{M_Y} = \frac{3}{2} \left(1 - \frac{1}{3} \frac{\rho^2}{\rho_Y^2} \right) \quad \frac{\rho}{\rho_Y} = \sqrt{3 - 2m} = 0.70711$$

$$\rho = 0.70711 \rho_Y \quad \blacktriangleleft$$

$$(b) \quad \frac{1}{\rho_R} = \frac{1}{\rho} - \frac{M}{EI} = \frac{1}{\rho} - \frac{m M_Y}{EI} = \frac{1}{\rho} - \frac{m}{\rho_Y} = \frac{1}{0.70711 \rho_Y} - \frac{1.25}{\rho_Y}$$
$$= \frac{0.16421}{\rho_Y}$$

$$\rho_R = 6.09 \rho_Y \quad \blacktriangleleft$$



PROBLEM 4.95

The prismatic bar AB is made of a steel that is assumed to be elastoplastic and for which $E = 200 \text{ GPa}$. Knowing that the radius of curvature of the bar is 2.4 m when a couple of moment $M = 350 \text{ N} \cdot \text{m}$ is applied as shown, determine (a) the yield strength of the steel, (b) the thickness of the elastic core of the bar.

SOLUTION

$$\begin{aligned}
 M &= \frac{3}{2} M_Y \left(1 - \frac{1}{3} \frac{\rho^2}{\rho_Y^2} \right) \\
 &= \frac{3}{2} \frac{\sigma_Y I}{c} \left(1 - \frac{1}{3} \frac{\rho^2 \sigma_Y^2}{E^2 c^2} \right) \\
 &= \frac{3}{2} \frac{\sigma_Y b (2c)^3}{12c} \left(1 - \frac{1}{3} \frac{\rho^2 \sigma_Y^2}{E^2 c^2} \right) \\
 &= \sigma_Y b c^2 \left(1 - \frac{1}{3} \frac{\rho^2 \sigma_Y^2}{E^2 c^2} \right)
 \end{aligned}$$

$$(a) \quad b c^2 \sigma_Y \left(1 - \frac{\rho^2 \sigma_Y^2}{3 E^2 c^2} \right) = M \quad \text{Cubic equation for } \sigma_Y$$

Data:

$$E = 200 \times 10^9 \text{ Pa}$$

$$M = 420 \text{ N} \cdot \text{m}$$

$$\rho = 2.4 \text{ m}$$

$$b = 20 \text{ mm} = 0.020 \text{ m}$$

$$c = \frac{1}{2} h = 8 \text{ mm} = 0.008 \text{ m}$$

$$(1.28 \times 10^{-6}) \sigma_Y \left[1 - 750 \times 10^{-21} \sigma_Y^2 \right] = 350$$

$$\sigma_Y \left[1 - 750 \times 10^{-21} \sigma_Y^2 \right] = 273.44 \times 10^6$$

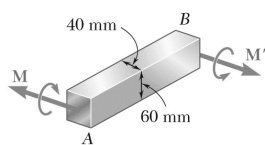
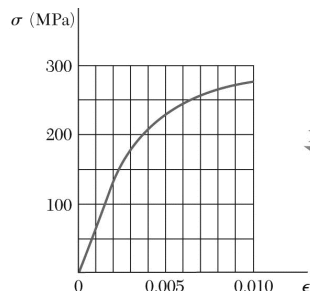
Solving by trial,

$$\sigma_Y = 292 \times 10^6 \text{ Pa}$$

$$\sigma_Y = 292 \text{ MPa} \quad \blacktriangleleft$$

$$(b) \quad y_Y = \frac{\sigma_Y \rho}{E} = \frac{(292 \times 10^6)(2.4)}{200 \times 10^9} = 3.504 \times 10^{-3} \text{ m} = 3.504 \text{ mm}$$

$$\text{thickness of elastic core} = 2y_Y = 7.01 \text{ mm} \quad \blacktriangleleft$$



PROBLEM 4.96

The prismatic bar AB is made of an aluminum alloy for which the tensile stress-strain diagram is as shown. Assuming that the σ - ϵ diagram is the same in compression as in tension, determine (a) the radius of curvature of the bar when the maximum stress is 250 MPa, (b) the corresponding value of the bending moment. (Hint: For part b, plot σ versus y and use an approximate method of integration.)

SOLUTION

(a) $\sigma_m = 250 \text{ MPa} = 250 \times 10^6 \text{ Pa}$

$\epsilon_m = 0.0064$ from curve

$c = \frac{1}{2}h = 30 \text{ mm} = 0.030 \text{ m}$

$b = 40 \text{ mm} = 0.040 \text{ m}$

$\frac{1}{\rho} = \frac{\epsilon_m}{c} = \frac{0.0064}{0.030} = 0.21333 \text{ m}^{-1}$

$\rho = 4.69 \text{ m} \quad \blacktriangleleft$

(b) Strain distribution: $\epsilon = -\epsilon_m \frac{y}{c} = -\epsilon_m u$ where $u = \frac{y}{\epsilon}$

Bending couple: $M = -\int_{-c}^c y \sigma b dy = 2b \int_0^c y |\sigma| dy = 2bc^2 \int_0^1 u |\sigma| du = 2bc^2 J$

where the integral J is given by $\int_0^1 u |\sigma| du$

Evaluate J using a method of numerical integration. If Simpson's rule is used, the integration formula is

$$J = \frac{\Delta u}{3} \sum w u |\sigma|$$

where w is a weighting factor.

Using $\Delta u = 0.25$, we get the values given in the table below:

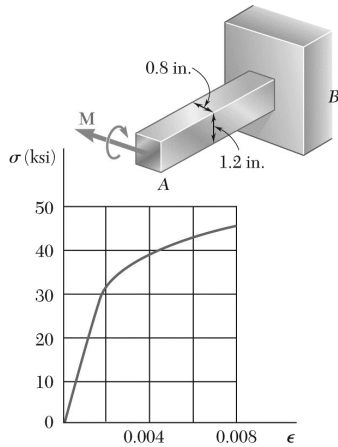
u	$ \epsilon $	$ \sigma , \text{ (MPa)}$	$u \sigma , \text{ (MPa)}$	w	$w u \sigma , \text{ (MPa)}$
0	0	0	0	1	0
0.25	0.0016	110	27.5	4	110
0.5	0.0032	180	90	2	180
0.75	0.0048	225	168.75	4	675
1.00	0.0064	250	250	1	250
					1215

$\leftarrow \sum w u |\sigma|$

$J = \frac{(0.25)(1215)}{3} = 101.25 \text{ MPa} = 101.25 \times 10^6 \text{ Pa}$

$M = (2)(0.040)(0.030)^2 (101.25 \times 10^6) = 7.29 \times 10^3 \text{ N} \cdot \text{m}$

$M = 7.29 \text{ kN} \cdot \text{m} \quad \blacktriangleleft$



PROBLEM 4.97

The prismatic bar AB is made of a bronze alloy for which the tensile stress-strain diagram is as shown. Assuming that the $\sigma - \epsilon$ diagram is the same in compression as in tension, determine (a) the maximum stress in the bar when the radius of curvature of the bar is 100 in., (b) the corresponding value of the bending moment. (See hint given in Prob. 4.96.)

SOLUTION

(a) $\rho = 100$ in., $b = 0.8$ in., $c = 0.6$ in.

$$\epsilon_m = \frac{c}{\rho} = \frac{0.6}{100} = 0.006$$

From the curve, $\sigma_m = 43$ ksi ◀

(b) Strain distribution: $\epsilon = -\epsilon_m \frac{y}{c} = -\epsilon_m u$ where $u = \frac{y}{c}$

Bending couple: $M = -\int_{-c}^c y \sigma b dy = 2b \int_0^c y |\sigma| dy = 2bc^2 \int_0^1 u |\sigma| du = 2bc^2 J$

where the integral J is given by $\int_0^1 u |\sigma| du$

Evaluate J using a method of numerical integration. If Simpson's rule is used, the integration formula is

$$J = \frac{\Delta u}{3} \sum w u |\sigma|$$

where w is a weighting factor.

Using $\Delta u = 0.25$, we get the values given the table below:

u	$ \epsilon $	$ \sigma $, ksi	$u \sigma $, ksi	w	$wu \sigma $, ksi
0	0	0	0	1	0
0.25	0.0015	25	6.25	4	25
0.5	0.003	36	18	2	36
0.75	0.0045	40	30	4	120
1.00	0.006	43	43	1	43
					224

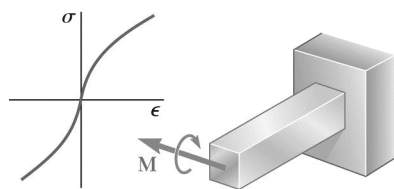
← $\sum wu |\sigma|$

$$J = \frac{(0.25)(224)}{3} = 18.67 \text{ ksi}$$

$$M = (2)(0.8)(0.6)^2(18.67)$$

$M = 10.75 \text{ kip} \cdot \text{in}$ ◀

PROBLEM 4.98



A prismatic bar of rectangular cross section is made of an alloy for which the stress-strain diagram can be represented by the relation $\epsilon = k\sigma^n$ for $\sigma > 0$ and $\epsilon = -|k\sigma^n|$ for $\sigma < 0$. If a couple $\mathbf{M}$ is applied to the bar, show that the maximum stress is

$$\sigma_m = \frac{1 + 2n}{3n} \frac{Mc}{I}$$

SOLUTION

Strain distribution: $\epsilon = -\epsilon_m \frac{y}{c} = -\epsilon_m u$ where $u = \frac{y}{c}$

Bending couple:

$$\begin{aligned} M &= -\int_{-c}^c y \sigma b dy = 2b \int_0^c y |\sigma| dy = 2bc^2 \int_0^1 \frac{y}{c} |\sigma| \frac{dy}{c} \\ &= 2bc^2 \int_0^1 u |\sigma| du \end{aligned}$$

For $\epsilon = K\sigma^n$, $\epsilon_m = K\sigma_m$

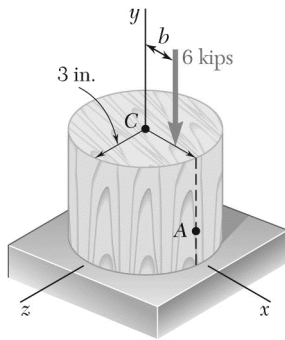
$$\frac{\epsilon}{\epsilon_m} = u = \left(\frac{\sigma}{\sigma_m} \right)^n \quad \therefore \quad |\sigma| = \sigma_m u^{\frac{1}{n}}$$

Then
$$\begin{aligned} M &= 2bc^2 \int_0^1 u \sigma_m u^{\frac{1}{n}} du = 2bc^2 \sigma_m \int_0^1 u^{1+\frac{1}{n}} du \\ &= 2bc^2 \sigma_m \frac{u^{2+\frac{1}{n}}}{2+\frac{1}{n}} \bigg|_0^1 = \frac{2n}{2n+1} bc^2 \sigma_m \end{aligned}$$

$$\sigma_m = \frac{2n+1}{2} \frac{M}{bc^2}$$

Recall: that
$$\frac{I}{c} = \frac{1}{12} \frac{b(2c)^3}{c} = \frac{2}{3} bc^2 \quad \therefore \quad \frac{1}{bc^2} = \frac{2}{3} \frac{c}{I}$$

Then
$$\sigma_m = \frac{2n+1}{3n} \frac{Mc}{I}$$



PROBLEM 4.99

A short wooden post supports a 6-kip axial load as shown. Determine the stress at point *A* when (a) $b = 0$, (b) $b = 1.5$ in., (c) $b = 3$ in.

SOLUTION

$$A = \pi r^2 = \pi(3)^2 = 28.27 \text{ in}^2$$

$$I = \frac{\pi}{4} r^4 = \frac{\pi}{4} (3)^4 = 63.62 \text{ in}^4$$

$$S = \frac{I}{c} = \frac{63.62}{3} = 21.206 \text{ in}^3$$

$$P = 6 \text{ kips} \quad M = Pb$$

(a) $b = 0 \quad M = 0$

$$\sigma = -\frac{P}{A} = -\frac{6}{28.27} = -0.212 \text{ ksi}$$

$$\sigma = -212 \text{ psi} \quad \blacktriangleleft$$

(b) $b = 1.5 \text{ in.} \quad M = (6)(1.5) = 9 \text{ kip} \cdot \text{in}$

$$\sigma = -\frac{P}{A} - \frac{M}{S} = -\frac{6}{28.27} - \frac{9}{21.206} = -0.637 \text{ ksi}$$

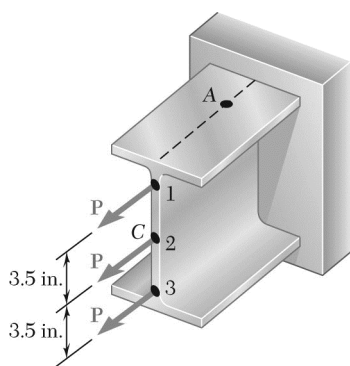
$$\sigma = -637 \text{ psi} \quad \blacktriangleleft$$

(c) $b = 3 \text{ in.} \quad M = (6)(3) = 18 \text{ kip} \cdot \text{in}$

$$\sigma = -\frac{P}{A} - \frac{M}{S} = -\frac{6}{28.27} - \frac{18}{21.206} = -1.061 \text{ ksi}$$

$$\sigma = -1061 \text{ psi} \quad \blacktriangleleft$$

PROBLEM 4.100



As many as three axial loads, each of magnitude $P = 10$ kips, can be applied to the end of a $W8 \times 21$ rolled-steel shape. Determine the stress at point A , (a) for the loading shown, (b) if loads are applied at points 1 and 2 only.

SOLUTION

For $W8 \times 21$ Appendix C gives

$$A = 6.16 \text{ in}^2, \quad d = 8.28 \text{ in.}, \quad I_x = 75.3 \text{ in}^4$$

At point A ,

$$y = \frac{1}{2}d = 4.14 \text{ in.}$$

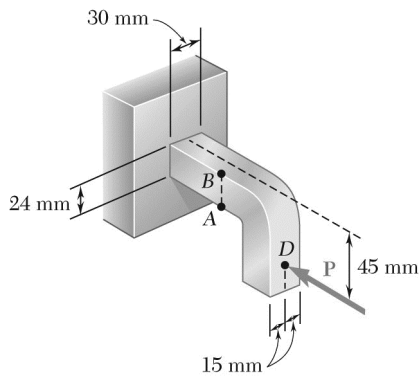
$$\sigma = \frac{F}{A} - \frac{My}{I}$$

(a) Centric loading: $F = 30$ kips, $M = 0$

$$\sigma = \frac{30}{6.16} \quad \sigma = 4.87 \text{ ksi} \quad \blacktriangleleft$$

(b) Eccentric loading: $F = 2P = 20$ kips
 $M = -(10)(3.5) = -35 \text{ kip} \cdot \text{in}$

$$\sigma = \frac{20}{6.16} - \frac{(-35)(4.14)}{75.3} \quad \sigma = 5.17 \text{ ksi} \quad \blacktriangleleft$$



PROBLEM 4.101

Knowing that the magnitude of the horizontal force **P** is 8 kN, determine the stress at (a) point *A*, (b) point *B*.

SOLUTION

$$A = (30)(24) = 720 \text{ mm}^2 = 720 \times 10^{-6} \text{ m}^2$$

$$e = 45 - 12 = 33 \text{ mm} = 0.033 \text{ m}$$

$$I = \frac{1}{12}bh^3 = \frac{1}{12}(30)(24)^3 = 34.56 \times 10^3 \text{ mm}^4 = 34.56 \times 10^{-9} \text{ m}^4$$

$$c = \frac{1}{2}(24 \text{ mm}) = 12 \text{ mm} = 0.012 \text{ m} \quad P = 8 \times 10^3 \text{ N}$$

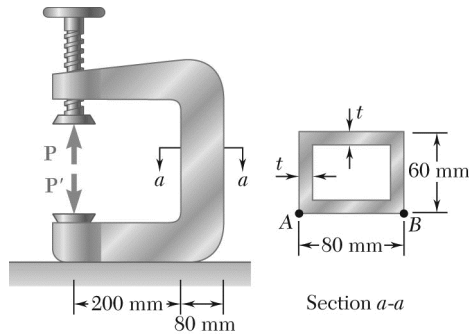
$$M = Pe = (8 \times 10^3)(0.033) = 264 \text{ N} \cdot \text{m}$$

$$(a) \quad \sigma_A = -\frac{P}{A} - \frac{Mc}{I} = -\frac{8 \times 10^3}{720 \times 10^{-6}} - \frac{(264)(0.012)}{34.56 \times 10^{-9}} = -102.8 \times 10^6 \text{ Pa}$$

$$\sigma_A = -102.8 \text{ MPa} \quad \blacktriangleleft$$

$$(b) \quad \sigma_B = -\frac{P}{A} + \frac{Mc}{I} = -\frac{8 \times 10^3}{720 \times 10^{-6}} + \frac{(264)(0.012)}{34.56 \times 10^{-9}} = 80.6 \times 10^6 \text{ Pa}$$

$$\sigma_B = 80.6 \text{ MPa} \quad \blacktriangleleft$$



PROBLEM 4.102

The vertical portion of the press shown consists of a rectangular tube of wall thickness $t = 10$ mm. Knowing that the press has been tightened on wooden planks being glued together until $P = 20$ kN, determine the stress at (a) point A , (b) point B .

SOLUTION

Rectangular cutout is $60 \text{ mm} \times 40 \text{ mm}$.

$$A = (80)(60) - (60)(40) = 2.4 \times 10^3 \text{ mm}^2 = 2.4 \times 10^{-3} \text{ m}^2$$

$$I = \frac{1}{12}(60)(80)^3 - \frac{1}{12}(40)(60)^3 = 1.84 \times 10^6 \text{ mm}^4$$

$$= 1.84 \times 10^{-6} \text{ m}^4$$

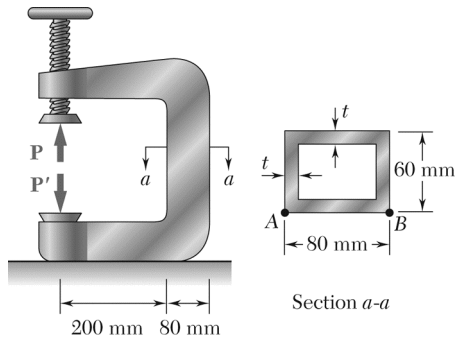
$$c = 40 \text{ mm} = 0.040 \text{ m} \quad e = 200 + 40 = 240 \text{ mm} = 0.240 \text{ m}$$

$$P = 20 \times 10^3 \text{ N}$$

$$M = Pe = (20 \times 10^3)(0.240) = 4.8 \times 10^3 \text{ N} \cdot \text{m}$$

$$(a) \quad \sigma_A = \frac{P}{A} + \frac{Mc}{I} = \frac{20 \times 10^3}{2.4 \times 10^{-3}} + \frac{(4.8 \times 10^3)(0.040)}{1.84 \times 10^{-6}} = 112.7 \times 10^6 \text{ Pa} \quad \sigma_A = 112.7 \text{ MPa} \quad \blacktriangleleft$$

$$(b) \quad \sigma_B = \frac{P}{A} - \frac{Mc}{I} = \frac{20 \times 10^3}{2.4 \times 10^{-3}} - \frac{(4.8 \times 10^3)(0.040)}{1.84 \times 10^{-6}} = -96.0 \times 10^6 \text{ Pa} \quad \sigma_B = -96.0 \text{ MPa} \quad \blacktriangleleft$$



PROBLEM 4.103

Solve Prob. 4.102, assuming that $t = 8$ mm.

PROBLEM 4.102 The vertical portion of the press shown consists of a rectangular tube of wall thickness $t = 10$ mm. Knowing that the press has been tightened on wooden planks being glued together until $P = 20$ kN, determine the stress at (a) point A, (b) point B.

SOLUTION

Rectangular cutout is $64 \text{ mm} \times 44 \text{ mm}$.

$$A = (80)(60) - (64)(44) = 1.984 \times 10^3 \text{ mm}^2$$

$$= 1.984 \times 10^{-3} \text{ m}^2$$

$$I = \frac{1}{12}(60)(80)^3 - \frac{1}{12}(44)(64)^3 = 1.59881 \times 10^6 \text{ mm}^4$$

$$= 1.59881 \times 10^{-6} \text{ m}^4$$

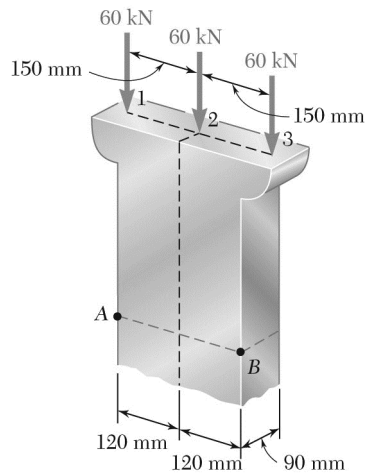
$$c = 40 \text{ mm} = 0.040 \text{ m} \quad e = 200 + 40 = 240 \text{ mm} = 0.240 \text{ m}$$

$$P = 20 \times 10^3 \text{ N}$$

$$M = Pe = (20 \times 10^3)(0.240) = 4.8 \times 10^3 \text{ N} \cdot \text{m}$$

$$(a) \quad \sigma_A = \frac{P}{A} + \frac{Mc}{I} = \frac{20 \times 10^3}{1.984 \times 10^{-3}} + \frac{(4.8 \times 10^3)(0.040)}{1.59881 \times 10^{-6}} = 130.2 \times 10^6 \text{ Pa} \quad \sigma_A = 130.2 \text{ MPa} \quad \blacktriangleleft$$

$$(b) \quad \sigma_B = \frac{P}{A} - \frac{Mc}{I} = \frac{20 \times 10^3}{1.984 \times 10^{-3}} - \frac{(4.8 \times 10^3)(0.040)}{1.59881 \times 10^{-6}} = -110.0 \times 10^6 \text{ Pa} \quad \sigma_B = -110.0 \text{ MPa} \quad \blacktriangleleft$$



PROBLEM 4.104

Determine the stress at points A and B , (a) for the loading shown, (b) if the 60-kN loads are applied at points 1 and 2 only.

SOLUTION

(a) Loading is centric.

$$P = 180 \text{ kN} = 180 \times 10^3 \text{ N}$$

$$A = (90)(240) = 21.6 \times 10^3 \text{ mm}^2 = 21.6 \times 10^{-6} \text{ m}^2$$

At A and B :

$$\sigma = -\frac{P}{A} = -\frac{180 \times 10^3}{21.6 \times 10^{-3}} = -8.33 \times 10^6 \text{ Pa}$$

$$\sigma_A = \sigma_B = -8.33 \text{ MPa} \quad \blacktriangleleft$$

(b) Eccentric loading.

$$P = 120 \text{ kN} = 120 \times 10^3 \text{ N}$$

$$M = (60 \times 10^3)(150 \times 10^{-3}) = 9.0 \times 10^3 \text{ N} \cdot \text{m}$$

$$I = \frac{1}{12}bh^3 = \frac{1}{12}(90)(240)^3 = 103.68 \times 10^6 \text{ mm}^4 = 103.68 \times 10^{-6} \text{ m}^4$$

$$c = 120 \text{ mm} = 0.120 \text{ m}$$

At A :

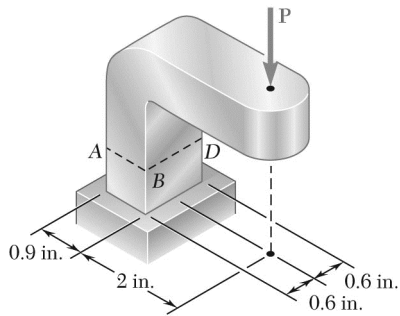
$$\sigma_A = -\frac{P}{A} - \frac{Mc}{I} = -\frac{120 \times 10^3}{21.6 \times 10^{-3}} - \frac{(9.0 \times 10^3)(0.120)}{103.68 \times 10^{-6}} = -15.97 \times 10^6 \text{ Pa}$$

At B :

$$\sigma_B = -\frac{P}{A} + \frac{Mc}{I} = -\frac{120 \times 10^3}{21.6 \times 10^{-3}} + \frac{(9.0 \times 10^3)(0.120)}{103.68 \times 10^{-6}} = 4.86 \times 10^6 \text{ Pa}$$

$$\sigma_A = -15.97 \text{ MPa} \quad \blacktriangleleft$$

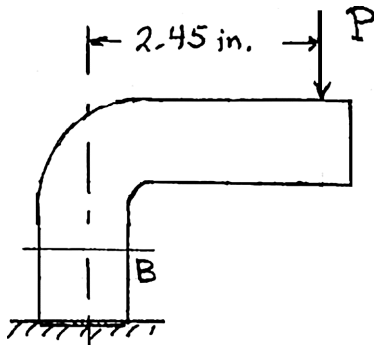
$$\sigma_B = 4.86 \text{ MPa} \quad \blacktriangleleft$$



PROBLEM 4.105

Knowing that the allowable stress in section ABD is 10 ksi, determine the largest force P that can be applied to the bracket shown.

SOLUTION



$$\text{Statics: } M = 2.45P$$

$$\text{Cross section: } A = (0.9)(1.2) = 1.08 \text{ in}^2$$

$$c = \frac{1}{2}(0.9) = 0.45 \text{ in.}$$

$$I = \frac{1}{12}(1.2)(0.9)^3 = 0.0729 \text{ in}^4$$

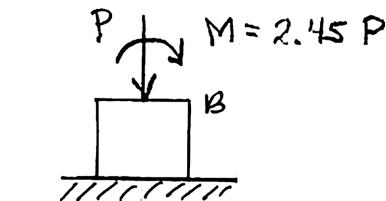
$$\text{At point } B: \sigma = -10 \text{ ksi}$$

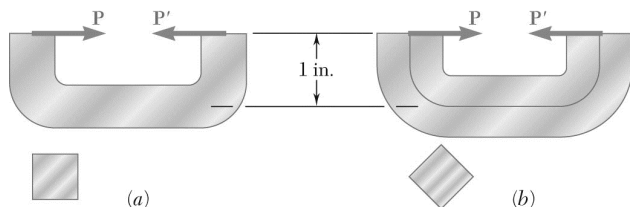
$$\sigma = -\frac{P}{A} - \frac{Mc}{I}$$

$$-10 = -\frac{P}{1.08} - \frac{(2.45P)(0.45)}{0.0729} = -16.049P$$

$$P = 0.623 \text{ kips}$$

$$P = 623 \text{ lb} \quad \blacktriangleleft$$





PROBLEM 4.106

Portions of a $\frac{1}{2} \times \frac{1}{2}$ -in. square bar have been bent to form the two machine components shown. Knowing that the allowable stress is 15 ksi, determine the maximum load that can be applied to each component.

SOLUTION

The maximum stress occurs at point B .

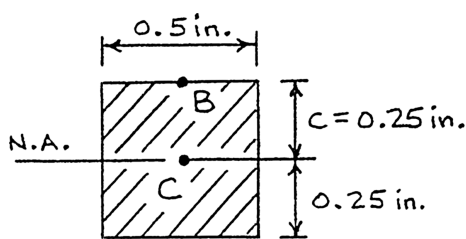
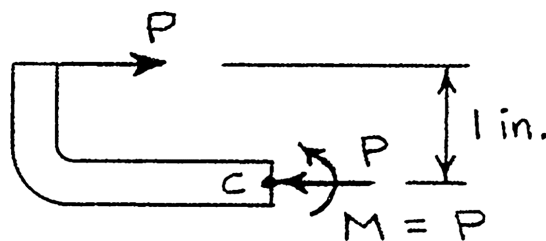
$$\sigma_B = -15 \text{ ksi} = -15 \times 10^3 \text{ psi}$$

$$\sigma_B = -\frac{P}{A} - \frac{Mc}{I} = -\frac{P}{A} - \frac{Pec}{I} = -KP$$

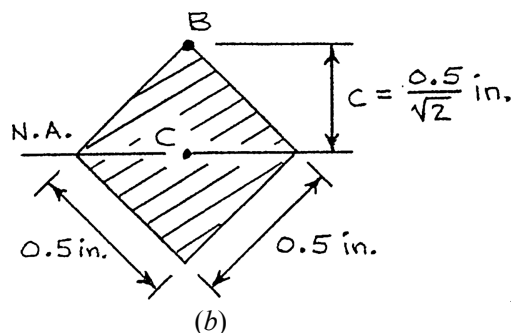
where $K = \frac{1}{A} + \frac{ec}{I}$ $e = 1.0 \text{ in.}$

$$A = (0.5)(0.5) = 0.25 \text{ in}^2$$

$$I = \frac{1}{12}(0.5)(0.5)^3 = 5.2083 \times 10^{-3} \text{ in}^4 \text{ for all centroidal axes.}$$



(a)



(b)

(a) $c = 0.25 \text{ in.}$

$$K = \frac{1}{0.25} + \frac{(1.0)(0.25)}{5.2083 \times 10^{-3}} = 52 \text{ in}^{-2}$$

$$P = -\frac{\sigma_B}{K} = -\frac{(-15 \times 10^3)}{52}$$

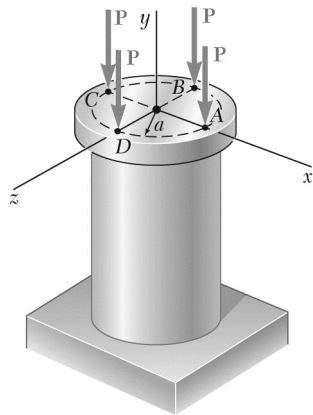
$$P = 288 \text{ lb} \quad \blacktriangleleft$$

(b) $c = \frac{0.5}{\sqrt{2}} = 0.35355 \text{ in.}$

$$K = \frac{1}{0.25} + \frac{(1.0)(0.35355)}{5.2083 \times 10^{-3}} = 71.882 \text{ in}^{-2}$$

$$P = -\frac{\sigma_B}{K} = -\frac{(-15 \times 10^3)}{71.882}$$

$$P = 209 \text{ lb} \quad \blacktriangleleft$$



PROBLEM 4.107

The four forces shown are applied to a rigid plate supported by a solid steel post of radius a . Knowing that $P = 100$ kN and $a = 40$ mm, determine the maximum stress in the post when (a) the force at D is removed, (b) the forces at C and D are removed.

SOLUTION

For a solid circular section of radius a ,

$$A = \pi a^2 \quad I = \frac{\pi}{4} a^4$$

Centric force: $F = 4P, \quad M_x = M_z = 0 \quad \sigma = -\frac{F}{A} = -\frac{4P}{\pi a^2}$

(a) Force at D is removed:

$$F = 3P, \quad M_x = -Pa, \quad M_z = 0$$

$$\sigma = -\frac{F}{A} - \frac{M_x z}{I} = -\frac{3P}{\pi a^2} - \frac{(-Pa)(-a)}{\frac{\pi}{4} a^4} = -\frac{7P}{\pi a^2}$$

(b) Forces at C and D are removed:

$$F = 2P, \quad M_x = -Pa, \quad M_z = -Pa$$

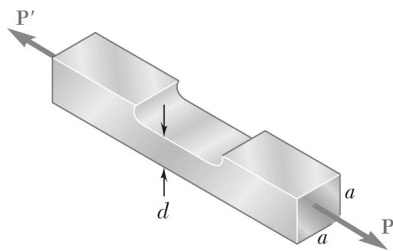
Resultant bending couple: $M = \sqrt{M_x^2 + M_z^2} = \sqrt{2} Pa$

$$\sigma = -\frac{F}{A} - \frac{Mc}{I} = -\frac{2P}{\pi a^2} - \frac{\sqrt{2} Pa a}{\frac{\pi}{4} a^4} = -\frac{2 + 4\sqrt{2}}{\pi} \frac{P}{a^2} = -2.437 P/a^2$$

Numerical data: $P = 100 \times 10^3$ N, $a = 0.040$ m

Answers: (a) $\sigma = -\frac{(7)(100 \times 10^3)}{\pi(0.040)^2} = -139.3 \times 10^6$ Pa $\sigma = -139.3$ MPa ◀

(b) $\sigma = -\frac{(2.437)(100 \times 10^3)}{(0.040)^2} = -152.3 \times 10^6$ Pa $\sigma = -152.3$ MPa ◀



PROBLEM 4.108

A milling operation was used to remove a portion of a solid bar of square cross section. Knowing that $a = 30$ mm, $d = 20$ mm, and $\sigma_{\text{all}} = 60$ MPa, determine the magnitude P of the largest forces that can be safely applied at the centers of the ends of the bar.

SOLUTION

$$A = ad, \quad I = \frac{1}{12}ad^3, \quad c = \frac{1}{2}d$$

$$e = \frac{a}{2} - \frac{d}{2}$$

$$\sigma = \frac{P}{A} + \frac{Mc}{I} = \frac{P}{ad} + \frac{6Ped}{ad^3}$$

$$\sigma = \frac{P}{ad} + \frac{3P(a-d)}{ad^2} = KP \quad \text{where} \quad K = \frac{1}{ad} + \frac{3(a-d)}{ad^2}$$

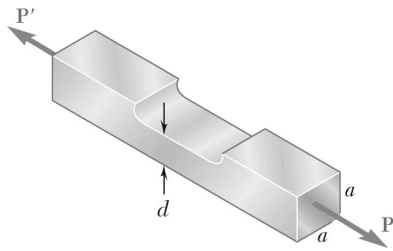
Data:

$$a = 30 \text{ mm} = 0.030 \text{ m} \quad d = 20 \text{ mm} = 0.020 \text{ m}$$

$$K = \frac{1}{(0.030)(0.020)} + \frac{(3)(0.010)}{(0.030)(0.020)^2} = 4.1667 \times 10^3 \text{ m}^{-2}$$

$$P = \frac{\sigma}{K} = \frac{60 \times 10^6}{4.1667 \times 10^3} = 14.40 \times 10^3 \text{ N}$$

$$P = 14.40 \text{ kN} \quad \blacktriangleleft$$



PROBLEM 4.109

A milling operation was used to remove a portion of a solid bar of square cross section. Forces of magnitude $P = 18 \text{ kN}$ are applied at the centers of the ends of the bar. Knowing that $a = 30 \text{ mm}$ and $\sigma_{\text{all}} = 135 \text{ MPa}$, determine the smallest allowable depth d of the milled portion of the bar.

SOLUTION

$$A = ad, \quad I = \frac{1}{12}ad^3, \quad c = \frac{1}{2}d$$

$$e = \frac{a}{2} - \frac{d}{2}$$

$$\sigma = \frac{P}{A} + \frac{Mc}{I} = \frac{P}{ad} + \frac{Pec}{I} = \frac{P}{ad} + \frac{P \frac{1}{2}(a-d) \frac{1}{2}d}{\frac{1}{12}ad^3} = \frac{P}{ad} + \frac{3P(a-d)}{ad^2}$$

$$\sigma = \frac{3P}{d^2} - \frac{2P}{ad} \quad \text{or} \quad \sigma d^2 + \frac{2P}{a}d - 3P = 0$$

Solving for d ,

$$d = \frac{1}{2\sigma} \left\{ \sqrt{\left(\frac{2P}{a} \right)^2 + 12P\sigma} - \frac{2P}{a} \right\}$$

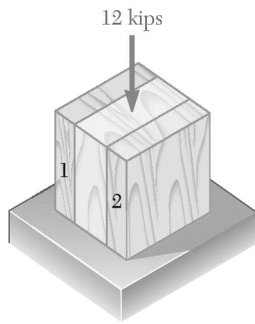
Data:

$$a = 0.030 \text{ m}, \quad P = 18 \times 10^3 \text{ N}, \quad \sigma = 135 \times 10^6 \text{ Pa}$$

$$d = \frac{1}{(2)(135 \times 10^6)} \left\{ \sqrt{\left[\frac{(2)(18 \times 10^3)}{0.030} \right]^2 + 12(18 \times 10^3)(135 \times 10^6)} - \frac{(2)(18 \times 10^3)}{0.030} \right\}$$

$$= 16.04 \times 10^{-3} \text{ m}$$

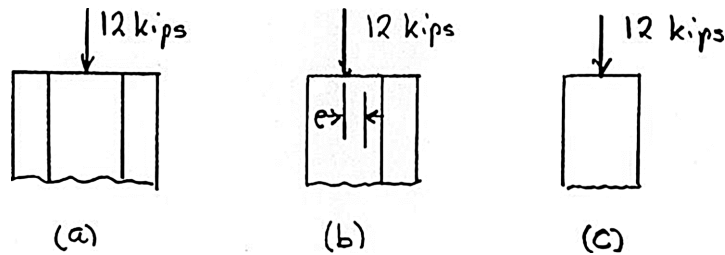
$$d = 16.04 \text{ mm} \quad \blacktriangleleft$$



PROBLEM 4.110

A short column is made by nailing two 1×4-in. planks to a 2×4-in. timber. Determine the largest compressive stress created in the column by a 12-kip load applied as shown in the center of the top section of the timber if (a) the column is as described, (b) plank 1 is removed, (c) both planks are removed.

SOLUTION



- (a) Centric loading: 4 in. × 4 in. cross section $A = (4)(4) = 16 \text{ in}^2$

$$\sigma = -\frac{P}{A} = -\frac{12}{16} \quad \sigma = -0.75 \text{ ksi} \blacktriangleleft$$

- (b) Eccentric loading: 4 in. × 3 in. cross section $A = (4)(3) = 12 \text{ in}^2$

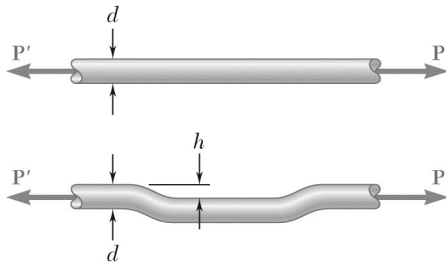
$$c = \left(\frac{1}{2}\right)(3) = 1.5 \text{ in.} \quad e = 1.5 - 1.0 = 0.5 \text{ in.}$$

$$I = \frac{1}{12}bh^3 = \frac{1}{12}(4)(3)^3 = 9 \text{ in}^4$$

$$\sigma = -\frac{P}{A} - \frac{Pec}{I} = -\frac{12}{12} - \frac{(12)(0.5)(1.5)}{9} \quad \sigma = -2.00 \text{ ksi} \blacktriangleleft$$

- (c) Centric loading: 4 in. × 2 in. cross section $A = (4)(2) = 8 \text{ in}^2$

$$\sigma = -\frac{P}{A} = -\frac{12}{8} \quad \sigma = -1.50 \text{ ksi} \blacktriangleleft$$



PROBLEM 4.111

An offset h must be introduced into a solid circular rod of diameter d . Knowing that the maximum stress after the offset is introduced must not exceed 5 times the stress in the rod when it is straight, determine the largest offset that can be used.

SOLUTION

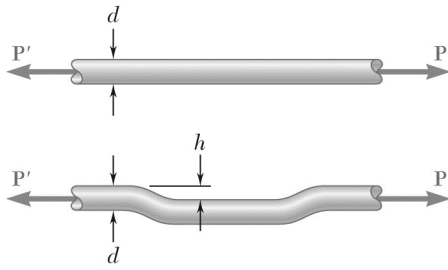
For centric loading, $\sigma_c = \frac{P}{A}$

For eccentric loading, $\sigma_e = \frac{P}{A} + \frac{Phc}{I}$

Given $\sigma_e = 5 \sigma_c$

$$\frac{P}{A} + \frac{Phc}{I} = 5 \frac{P}{A}$$

$$\frac{Phc}{I} = 4 \frac{P}{A} \quad \therefore \quad h = \frac{4I}{cA} = \frac{(4) \left(\frac{\pi}{64} d^4 \right)}{\left(\frac{d}{2} \right) \left(\frac{\pi}{4} d^2 \right)} = \frac{1}{2} d \quad h = 0.500 d \quad \blacktriangleleft$$



PROBLEM 4.112

An offset h must be introduced into a metal tube of 0.75-in. outer diameter and 0.08-in. wall thickness. Knowing that the maximum stress after the offset is introduced must not exceed 4 times the stress in the tube when it is straight, determine the largest offset that can be used.

SOLUTION

$$c = \frac{1}{2}d = 0.375 \text{ in.}$$

$$c_1 = c - t = 0.375 - 0.08 = 0.295 \text{ in.}$$

$$A = \pi(c^2 - c_1^2) = \pi(0.375^2 - 0.295^2) \\ = 0.168389 \text{ in}^2$$

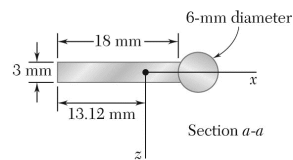
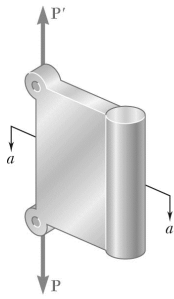
$$I = \frac{\pi}{4}(c^4 - c_1^4) = \frac{\pi}{4}(0.375^4 - 0.295^4) \\ = 9.5835 \times 10^{-3} \text{ in}^4$$

For centric loading, $\sigma_{\text{cen}} = \frac{P}{A}$

For eccentric loading, $\sigma_{\text{ecc}} = \frac{P}{A} + \frac{Phc}{I}$

$$\sigma_{\text{ecc}} = 4\sigma_{\text{cen}} \quad \text{or} \quad \frac{P}{A} + \frac{Phc}{I} = 4\frac{P}{A}$$

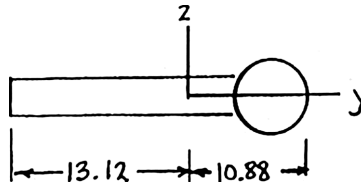
$$\frac{hc}{I} = \frac{3}{A} \quad h = \frac{3I}{Ac} = \frac{(3)(9.5835 \times 10^{-3})}{(0.168389)(0.375)} \quad h = 0.455 \text{ in.} \quad \blacktriangleleft$$



PROBLEM 4.113

A steel rod is welded to a steel plate to form the machine element shown. Knowing that the allowable stress is 135 MPa, determine (a) the largest force **P** that can be applied to the element, (b) the corresponding location of the neutral axis. *Given:* The centroid of the cross section is at *C* and $I_z = 4195 \text{ mm}^4$.

SOLUTION



$$(a) \quad A = (3)(18) + \frac{\pi}{4}(6)^2 = 82.27 \text{ mm}^2 = 82.27 \times 10^{-6} \text{ m}^2$$

$$I = 4195 \text{ mm}^4 = 4195 \times 10^{-12} \text{ m}^4$$

$$e = 13.12 \text{ mm} = 0.01312 \text{ m}$$

Based on tensile stress at $y = -13.12 \text{ mm} = -0.01312 \text{ m}$

$$\sigma = \frac{P}{A} + \frac{Pec}{I} = \left(\frac{1}{A} + \frac{ec}{I} \right) P = KP$$

$$K = \frac{1}{A} + \frac{ec}{I} = \frac{1}{82.27 \times 10^{-6}} + \frac{(0.01312)(0.01312)}{4195 \times 10^{-12}} = 53.188 \times 10^3 \text{ m}^{-2}$$

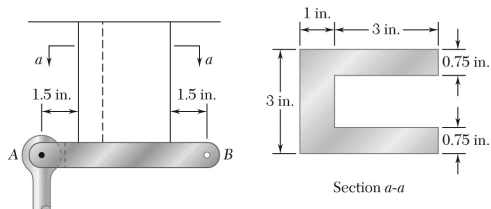
$$P = \frac{\sigma}{K} = \frac{135 \times 10^6}{53.188 \times 10^3} = 2.538 \times 10^3 \text{ N} \quad P = 2.54 \text{ kN} \quad \blacktriangleleft$$

(b) Location of neutral axis. $\sigma = 0$

$$\sigma = \frac{P}{A} - \frac{My}{I} = \frac{P}{A} - \frac{Pey}{I} = 0 \quad \frac{ey}{I} = \frac{1}{A}$$

$$y = \frac{I}{Ae} = \frac{4195 \times 10^{-12}}{(82.27 \times 10^{-6})(0.01312)} = 3.89 \times 10^{-3} \text{ m} \quad y = 3.89 \text{ mm} \quad \blacktriangleleft$$

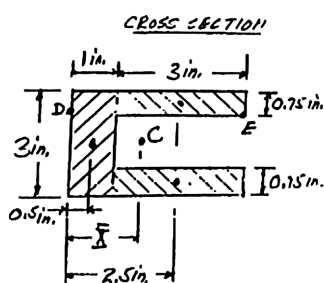
The neutral axis lies 3.89 mm to the right of the centroid or 17.01 mm to the right of the line of action of the loads.



PROBLEM 4.114

A vertical rod is attached at point *A* to the cast iron hanger shown. Knowing that the allowable stresses in the hanger are $\sigma_{\text{all}} = +5$ ksi and $\sigma_{\text{all}} = -12$ ksi, determine the largest downward force and the largest upward force that can be exerted by the rod.

SOLUTION



$$\bar{X} = \frac{\sum A\bar{y}}{\sum A} = \frac{(1 \times 3)(0.5) + 2(3 \times 0.75)(2.5)}{(1 \times 3) + 2(3 \times 0.75)}$$

$$\bar{X} = \frac{12.75 \text{ in}^3}{7.5 \text{ in}^2} = 1.700 \text{ in.}$$

$$A = 7.5 \text{ in}^2$$

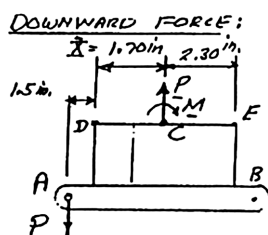
$$\sigma_{\text{all}} = +5 \text{ ksi} \quad \sigma_{\text{all}} = -12 \text{ ksi}$$

$$I_c = \sum \left(\frac{1}{12} bh^3 + Ad^2 \right)$$

$$= \frac{1}{12} (3)(1)^3 + (3 \times 1)(1.70 - 0.5)^2 + \frac{1}{12} (1.5)(3)^3 + (1.5 \times 3)(2.5 - 1.70)^2$$

$$I_c = 10.825 \text{ in}^4$$

Downward Force.



$$M = P(1.5 \text{ in.} + 1.70 \text{ in.}) = (3.20 \text{ in.})P$$

$$\text{At } D: \sigma_D = +\frac{P}{A} + \frac{Mc}{I}$$

$$+ 5 \text{ ksi} = \frac{P}{7.5} + \frac{(3.20)P(1.70)}{10.825}$$

$$+ 5 = P(+0.6359)$$

$$P = 7.86 \text{ kips} \downarrow$$

$$\text{At } E: \sigma_E = +\frac{P}{A} - \frac{Mc}{I}$$

$$- 12 \text{ ksi} = \frac{P}{7.5} - \frac{(3.20)P(2.30)}{10.825}$$

$$- 12 = P(-0.5466)$$

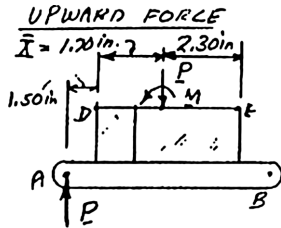
$$P = 21.95 \text{ kips} \downarrow$$

$$P = 7.96 \text{ kips} \downarrow \blacktriangleleft$$

We choose the smaller value.

PROBLEM 4.114 (Continued)

Upward Force.



$$M = P(1.5 \text{ in.} + 1.70 \text{ in.}) = (3.20 \text{ in.})P$$

$$\text{At } D: \sigma_D = +\frac{P}{A} - \frac{Mc}{I}$$

$$-12 \text{ ksi} = -\frac{P}{7.5} - \frac{(3.20)P(1.70)}{10.825}$$

$$-12 = P(-0.6359)$$

$$P = 18.87 \text{ kips } \uparrow$$

$$\text{At } E: \sigma_E = +\frac{P}{A} + \frac{Mc}{I}$$

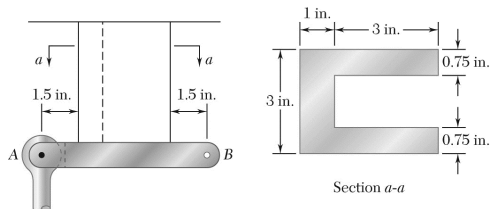
$$+5 \text{ ksi} = \frac{P}{7.5} + \frac{(3.20)P(2.30)}{10.825}$$

$$+5 = P(+0.5466)$$

$$P = 9.15 \text{ kips } \uparrow$$

We choose the smaller value.

$$P = 9.15 \text{ kips } \uparrow \blacktriangleleft$$

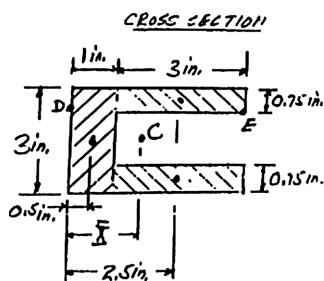


PROBLEM 4.115

Solve Prob. 4.114, assuming that the vertical rod is attached at point *B* instead of point *A*.

PROBLEM 4.114 A vertical rod is attached at point *A* to the cast iron hanger shown. Knowing that the allowable stresses in the hanger are $\sigma_{\text{all}} = +5$ ksi and $\sigma_{\text{all}} = -12$ ksi, determine the largest downward force and the largest upward force that can be exerted by the rod.

SOLUTION



$$\bar{X} = \frac{\sum A\bar{y}}{\sum A} = \frac{(1 \times 3)(0.5) + 2(3 \times 0.25)(2.5)}{(1 \times 3) + 2(3 \times 0.75)}$$

$$\bar{X} = \frac{12.75 \text{ in}^3}{7.5 \text{ in}^2} = 1.700 \text{ in.}$$

$$A = 7.5 \text{ in}^2$$

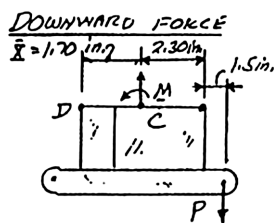
$$\sigma_{\text{all}} = +5 \text{ ksi} \quad \sigma_{\text{all}} = -12 \text{ ksi}$$

$$I_c = \sum \left(\frac{1}{12} bh^3 + Ad^2 \right)$$

$$= \frac{1}{12} (3)(1)^3 + (3 \times 1)(1.70 - 0.5)^2 + \frac{1}{12} (1.5)(3)^3 + (1.5 \times 3)(2.5 - 1.70)^2$$

$$I_c = 10.825 \text{ in}^4$$

Downward Force.



$$\sigma_{\text{all}} = +5 \text{ ksi} \quad \sigma_{\text{all}} = -12 \text{ ksi}$$

$$M = (2.30 \text{ in.} + 1.5 \text{ in.})P = (3.80 \text{ in.})P$$

$$\text{At } D: \sigma_D = +\frac{P}{A} - \frac{Mc}{I}$$

$$-12 \text{ ksi} = +\frac{P}{7.5} - \frac{(3.80)P(1.70)}{10.825}$$

$$-12 = P(-0.4634)$$

$$P = 25.9 \text{ kips} \downarrow$$

$$\text{At } E: \sigma_E = +\frac{P}{A} + \frac{Mc}{I}$$

$$+5 \text{ ksi} = +\frac{P}{7.5} + \frac{(3.80)P(2.30)}{10.825}$$

$$+5 = P(+0.9407)$$

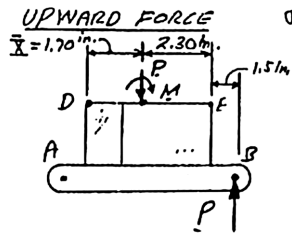
$$P = 5.32 \text{ kips} \downarrow$$

We choose the smaller value.

$$P = 5.32 \text{ kips} \downarrow \blacktriangleleft$$

PROBLEM 4.115 (Continued)

Upward Force.



$$\sigma_{\text{all}} = +5 \text{ ksi} \quad \sigma_{\text{all}} = -12 \text{ ksi}$$

$$M = (2.30 \text{ in.} + 1.5 \text{ in.})P = (3.80 \text{ in.})P$$

$$\text{At } D: \sigma_D = -\frac{P}{A} + \frac{Mc}{I}$$

$$5 \text{ ksi} = -\frac{P}{7.5} + \frac{(3.80)P(1.70)}{10.825}$$

$$5 = P(+0.4634)$$

$$P = 10.79 \text{ kips } \uparrow$$

$$\text{At } E: \sigma_E = -\frac{P}{A} - \frac{Mc}{I}$$

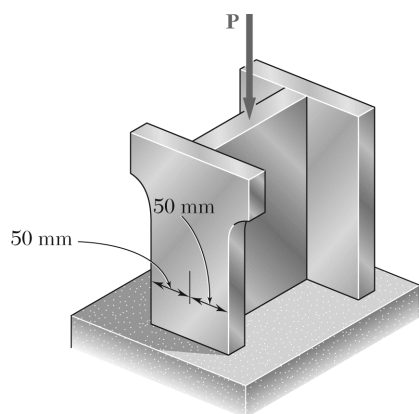
$$-12 \text{ ksi} = -\frac{P}{7.5} - \frac{(3.80)P(2.30)}{10.825}$$

$$-12 = P(-0.9407)$$

$$P = 12.76 \text{ kips } \uparrow$$

We choose the smaller value.

$$P = 10.79 \text{ kips } \uparrow \blacktriangleleft$$



PROBLEM 4.116

Three steel plates, each of 25×150 -mm cross section, are welded together to form a short H-shaped column. Later, for architectural reasons, a 25-mm strip is removed from each side of one of the flanges. Knowing that the load remains centric with respect to the original cross section, and that the allowable stress is 100 MPa, determine the largest force **P** (a) that could be applied to the original column, (b) that can be applied to the modified column.

SOLUTION

(a) Centric loading: $\sigma = -\frac{P}{A}$

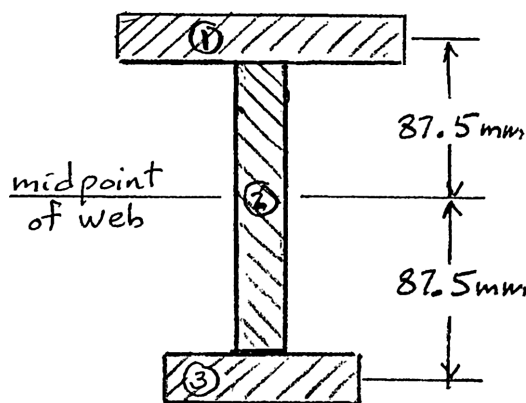
$$A = (3)(150)(25) = 11.25 \times 10^3 \text{ mm}^2 = 11.25 \times 10^{-3} \text{ m}^2$$

$$P = -\sigma A = -(-100 \times 10^6)(11.25 \times 10^{-3})$$

$$= 1.125 \times 10^6 \text{ N}$$

$$P = 1125 \text{ kN} \quad \blacktriangleleft$$

(b) Eccentric loading (reduced cross section):



	$A, 10^3 \text{ mm}^2$	$\bar{y}, \text{ mm}$	$A\bar{y} (10^3 \text{ mm}^3)$	$d, \text{ mm}$
①	3.75	87.5	328.125	76.5625
②	3.75	0	0	10.9375
③	2.50	-87.5	-218.75	98.4375
Σ	10.00		109.375	

$$\bar{Y} = \frac{\Sigma A\bar{y}}{\Sigma A} = \frac{109.375 \times 10^3}{10.00 \times 10^3} = 10.9375 \text{ mm}$$

PROBLEM 4.116 (Continued)

The centroid lies 10.9375 mm from the midpoint of the web.

$$I_1 = \frac{1}{12}b_1h_1^3 + A_1d_1^2 = \frac{1}{12}(150)(25)^3 + (3.75 \times 10^3)(76.5625)^2 = 22.177 \times 10^6 \text{ mm}^4$$

$$I_2 = \frac{1}{12}b_2h_2^3 + A_2d_2^2 = \frac{1}{12}(25)(150)^3 + (3.75 \times 10^3)(10.9375)^2 = 7.480 \times 10^6 \text{ mm}^4$$

$$I_3 = \frac{1}{12}b_3h_3^3 + A_3d_3^2 = \frac{1}{12}(100)(25)^3 + (2.50 \times 10^3)(98.4375)^2 = 24.355 \times 10^6 \text{ mm}^4$$

$$I = I_1 + I_2 + I_3 = 54.012 \times 10^6 \text{ mm}^4 = 54.012 \times 10^{-6} \text{ m}^4$$

$$c = 10.9375 + 75 + 25 = 110.9375 \text{ mm} = 0.1109375 \text{ m}$$

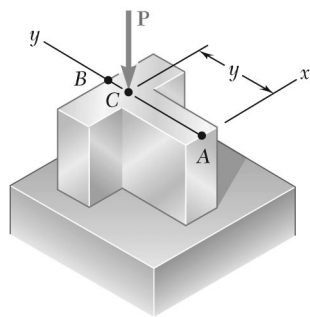
$$M = Pe \quad \text{where} \quad e = 10.4375 \text{ mm} = 10.4375 \times 10^{-3} \text{ m}$$

$$\sigma = -\frac{P}{A} - \frac{Mc}{I} = -\frac{P}{A} - \frac{Pec}{I} = -KP \quad A = 10.00 \times 10^{-3} \text{ m}^2$$

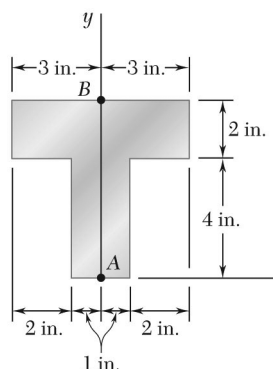
$$K = \frac{1}{A} + \frac{ec}{I} = \frac{1}{10.00 \times 10^{-3}} + \frac{(101.9375 \times 10^{-3})(0.1109375)}{54.012 \times 10^{-6}} = 122.465 \text{ m}^{-2}$$

$$P = -\frac{\sigma}{K} = -\frac{(-100 \times 10^6)}{122.465} = 817 \times 10^3 \text{ N}$$

$$P = 817 \text{ kN} \quad \blacktriangleleft$$



(a)



(b)

PROBLEM 4.117

A vertical force **P** of magnitude 20 kips is applied at point **C** located on the axis of symmetry of the cross section of a short column. Knowing that $y = 5$ in., determine (a) the stress at point **A**, (b) the stress at point **B**, (c) the location of the neutral axis.

SOLUTION

Locate centroid.

Part	A, in^2	$\bar{y}, \text{in.}$	$A\bar{y}, \text{in}^3$
①	12	5	60
②	8	2	16
Σ	20		76

$$\bar{y} = \frac{\Sigma A\bar{y}}{\Sigma A} = \frac{76}{20} = 3.8 \text{ in.}$$

Eccentricity of load: $e = 5 - 3.8 = 1.2 \text{ in.}$

$$I_1 = \frac{1}{12}(6)(2)^3 + (12)(1.2)^2 = 21.28 \text{ in}^4 \quad I_2 = \frac{1}{12}(2)(4)^3 + (8)(1.8)^2 = 36.587 \text{ in}^4$$

$$I = I_1 + I_2 = 57.867 \text{ in}^4$$

(a) Stress at **A**: $c_A = 3.8 \text{ in.}$

$$\sigma_A = -\frac{P}{A} + \frac{Pec_A}{I} = -\frac{20}{20} + \frac{20(1.2)(3.8)}{57.867} \quad \sigma_A = 0.576 \text{ ksi} \quad \blacktriangleleft$$

(b) Stress at **B**: $c_B = 6 - 3.8 = 2.2 \text{ in.}$

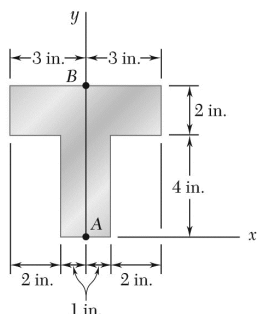
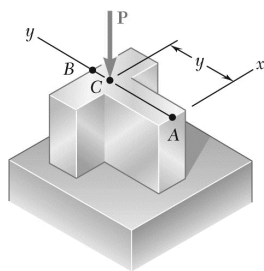
$$\sigma_B = -\frac{P}{A} + \frac{Pec_B}{I} = -\frac{20}{20} - \frac{20(1.2)(2.2)}{57.867} \quad \sigma_B = -1.912 \text{ ksi} \quad \blacktriangleleft$$

(c) Location of neutral axis: $\sigma = 0 \quad \sigma = -\frac{P}{A} + \frac{Pea}{I} = 0 \quad \therefore \frac{ea}{I} = \frac{1}{A}$

$$a = \frac{I}{Ae} = \frac{57.867}{(20)(1.2)} = 2.411 \text{ in.}$$

Neutral axis lies 2.411 in. below centroid or $3.8 - 2.411 = 1.389 \text{ in.}$ above point **A**.

Answer: 1.389 in. from point **A**. $\blacktriangleleft$



PROBLEM 4.118

A vertical force P is applied at point C located on the axis of symmetry of the cross section of a short column. Determine the range of values of y for which tensile stresses do not occur in the column.

SOLUTION

Locate centroid.

	A, in^2	$\bar{y}, \text{in.}$	$A\bar{y}, \text{in}^3$
①	12	5	60
②	8	2	16
Σ	20		76

$$\bar{y} = \frac{\Sigma A_i \bar{y}_i}{\Sigma A_i} = \frac{76}{20} = 3.8 \text{ in.}$$

Eccentricity of load: $e = y - 3.8 \text{ in.}$ $y = e + 3.8 \text{ in.}$

$$I_1 = \frac{1}{12}(6)(2)^3 + (12)(1.2)^2 = 21.28 \text{ in}^4 \quad I_2 = \frac{1}{12}(2)(4)^3 + (8)(1.8)^2 = 36.587 \text{ in}^4$$

$$I = I_1 + I_2 = 57.867 \text{ in}^4$$

If stress at A equals zero, $c_A = 3.8 \text{ in.}$

$$\sigma_A = -\frac{P}{A} + \frac{Pec_A}{I} = 0 \quad \therefore \quad \frac{ec_A}{I} = \frac{1}{A}$$

$$e = \frac{I}{Ac_A} = \frac{57.867}{(20)(3.8)} = 0.761 \text{ in.} \quad y = 0.761 + 3.8 = 4.561 \text{ in.}$$

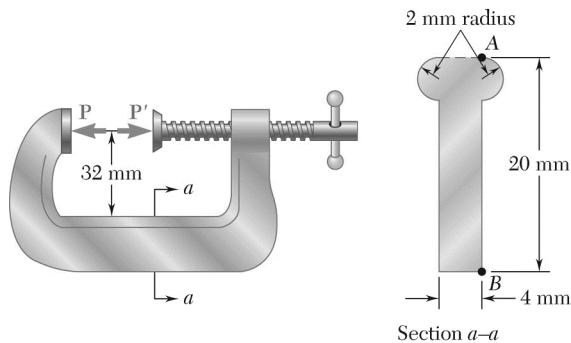
If stress at B equals zero. $c_B = 6 - 3.8 = 2.2 \text{ in.}$

$$\sigma_B = -\frac{P}{A} - \frac{Pec_B}{I} = 0 \quad \therefore \quad \frac{ec_B}{I} = -\frac{1}{A}$$

$$e = -\frac{I}{Ac_B} = -\frac{57.867}{(20)(2.2)} = -1.315 \text{ in.}$$

$$y = -1.315 + 3.8 = 2.485 \text{ in.}$$

Answer: $2.485 \text{ in.} < y < 4.561 \text{ in.}$ ◀

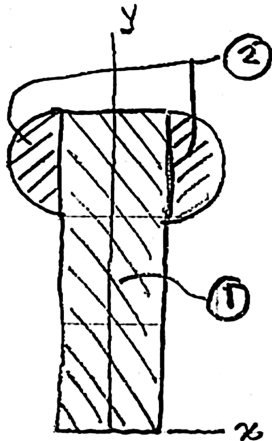


PROBLEM 4.119

Knowing that the clamp shown has been tightened until $P = 400 \text{ N}$, determine (a) the stress at point A , (b) the stress at point B , (c) the location of the neutral axis of section $a - a$.

SOLUTION

Cross section: Rectangle ① + Circle ②



$$A_1 = (20 \text{ mm})(4 \text{ mm}) = 80 \text{ mm}^2$$

$$\bar{y}_1 = \frac{1}{2}(20 \text{ mm}) = 10 \text{ mm}$$

$$A_2 = \pi(2 \text{ mm})^2 = 4\pi \text{ mm}^2$$

$$\bar{y}_2 = 20 - 2 = 18 \text{ mm}$$

$$c_B = \bar{y} = \frac{\sum A\bar{y}}{\sum A} = \frac{(80)(10) + (4\pi)(18)}{80 + 4\pi} = 11.086 \text{ mm}$$

$$c_A = 20 - \bar{y} = 8.914 \text{ mm}$$

$$d_1 = 11.086 - 10 = 1.086 \text{ mm}$$

$$d_2 = 18 - 11.086 = 6.914 \text{ mm}$$

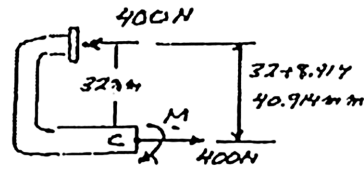
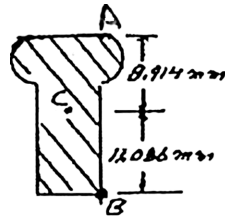
$$I_1 = \bar{I}_1 + A_1 d_1^2 = \frac{1}{12}(4)(20)^3 + (80)(1.086)^2 = 2.761 \times 10^3 \text{ mm}^4$$

$$I_2 = \bar{I}_2 + A_2 d_2^2 = \frac{\pi}{4}(2)^4 + (4\pi)(6.914)^2 = 0.613 \times 10^3 \text{ mm}^4$$

$$I = I_1 + I_2 = 3.374 \times 10^3 \text{ mm}^4 = 3.374 \times 10^{-9} \text{ m}^4$$

$$A = A_1 + A_2 = 92.566 \text{ mm}^2 = 92.566 \times 10^{-6} \text{ m}^2$$

PROBLEM 4.119 (Continued)



$$e = 32 + 8.914 = 40.914 \text{ mm} = 0.040914 \text{ m}$$

$$M = Pe = (400 \text{ N})(0.040914 \text{ m}) = 16.3656 \text{ N} \cdot \text{m}$$

(a) Point A:

$$\begin{aligned} \sigma_A &= \frac{P}{A} + \frac{Mc}{I} = \frac{400}{92.566 \times 10^{-6}} + \frac{(16.3656)(8.914 \times 10^{-3})}{3.374 \times 10^{-9}} \\ &= 4.321 \times 10^6 + 43.23 \times 10^6 = 47.55 \times 10^6 \text{ Pa} \end{aligned}$$

$$\sigma_A = 47.6 \text{ MPa} \quad \square$$

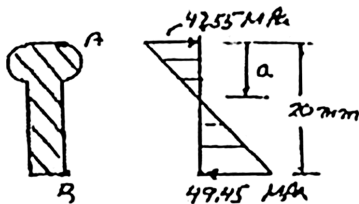
(b) Point B:

$$\begin{aligned} \sigma_B &= \frac{P}{A} - \frac{Mc}{I} = \frac{400}{92.566 \times 10^{-6}} - \frac{(16.3656)(11.086)}{3.374 \times 10^{-9}} \\ &= 4.321 \times 10^6 - 53.72 \times 10^6 = -49.45 \times 10^6 \text{ Pa} \end{aligned}$$

$$\sigma_B = -49.4 \text{ MPa} \quad \square$$

(c) Neutral axis:

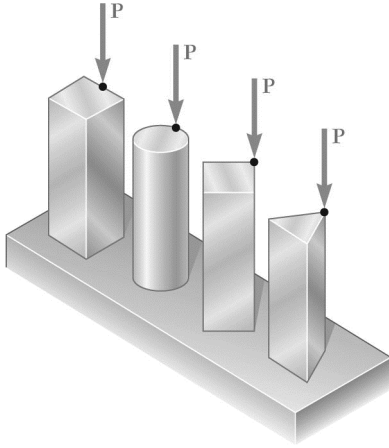
By proportions,



$$\begin{aligned} \frac{a}{47.55} &= \frac{20}{47.55 + 49.45} \\ a &= 9.80 \text{ mm} \end{aligned}$$

$$9.80 \text{ mm below top of section} \quad \square$$

PROBLEM 4.120



The four bars shown have the same cross-sectional area. For the given loadings, show that (a) the maximum compressive stresses are in the ratio 4:5:7:9, (b) the maximum tensile stresses are in the ratio 2:3:5:3. (Note: the cross section of the triangular bar is an equilateral triangle.)

SOLUTION

Stresses:

At A,

$$\sigma_A = -\frac{P}{A} - \frac{Pec_A}{I} = -\frac{P}{A} \left(1 + \frac{Aec_A}{I} \right)$$

At B,

$$\sigma_B = -\frac{P}{A} + \frac{Pec_B}{I} = \frac{P}{A} \left(\frac{Aec_B}{I} - 1 \right)$$

$$\left\{ \begin{aligned} A_1 &= a^2, \quad I_1 = \frac{1}{12}a^4, \quad c_A = c_B = \frac{1}{2}a, \quad e = \frac{1}{2}a \\ \sigma_A &= -\frac{P}{A} \left(1 + \frac{(a^2) \left(\frac{1}{2}a \right) \left(\frac{1}{2}a \right)}{\frac{1}{12}a^4} \right) \\ \sigma_B &= \frac{P}{A} \left(\frac{(a^2) \left(\frac{1}{2}a \right) \left(\frac{1}{2}a \right)}{\frac{1}{12}a^4} - 1 \right) \end{aligned} \right.$$

$$\sigma_A = -4 \frac{P}{A_1} \blacktriangleleft$$

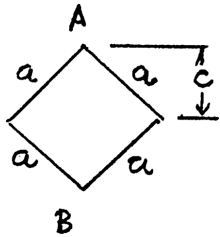
$$\sigma_B = 2 \frac{P}{A_1} \blacktriangleleft$$

$$\left\{ \begin{aligned} A_2 &= \pi c^2 = a^2 \quad \therefore c = \frac{a}{\sqrt{\pi}}, \quad I_2 = \frac{\pi}{4}c^4, \quad e = c \\ \sigma_A &= -\frac{P}{A_2} \left(1 + \frac{(\pi c^2)(c)(c)}{\frac{\pi}{4}c^4} \right) \\ \sigma_B &= \frac{P}{A_2} \left(\frac{(\pi c^2)(c)(c)}{\frac{\pi}{4}c^4} - 1 \right) \end{aligned} \right.$$

$$\sigma_A = -5 \frac{P}{A_2} \blacktriangleleft$$

$$\sigma_B = 3 \frac{P}{A_2} \blacktriangleleft$$

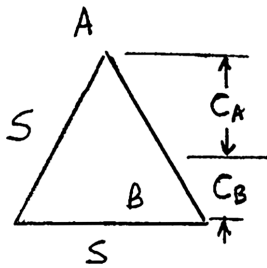
PROBLEM 4.120 (Continued)



$$\left\{ \begin{array}{l} A_3 = a^2 \quad c = \frac{\sqrt{2}}{2}a \quad I_3 = \frac{1}{12}a^4 \quad e = c \\ \sigma_A = -\frac{P}{A_3} \left(1 + \frac{(a^2) \left(\frac{\sqrt{2}}{2}a \right) \left(\frac{\sqrt{2}}{2}a \right)}{\frac{1}{12}a^4} \right) \\ \sigma_B = \frac{P}{A_3} \left(\frac{(a^2) \left(\frac{\sqrt{2}}{2}a \right) \left(\frac{\sqrt{2}}{2}a \right)}{\frac{1}{12}a^4} - 1 \right) \end{array} \right.$$

$$\sigma_A = -7 \frac{P}{A_3} \quad \blacktriangleleft$$

$$\sigma_B = 5 \frac{P}{A_3} \quad \blacktriangleleft$$



$$A_4 = \frac{1}{2}(s) \left(\frac{\sqrt{3}}{2}s \right) = \frac{\sqrt{3}}{4}s^2$$

$$I_4 = \frac{1}{36}s \left(\frac{\sqrt{3}}{2}s \right)^3 = \frac{\sqrt{3}}{96}s^4$$

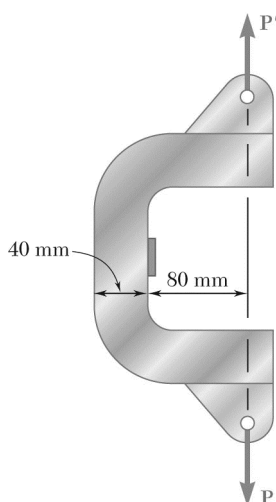
$$c_A = \frac{2}{3} \frac{\sqrt{3}}{2}s = \frac{s}{\sqrt{3}} = e \quad c_B = s$$

$$\sigma_A = -\frac{P}{A_4} \left(1 + \frac{\left(\frac{\sqrt{3}}{4}s^2 \right) \left(\frac{s}{\sqrt{3}} \right) \left(\frac{s}{\sqrt{3}} \right)}{\frac{\sqrt{3}}{96}s^4} \right)$$

$$\sigma_A = -9 \frac{P}{A_4} \quad \blacktriangleleft$$

$$\sigma_B = \frac{P}{A_4} \left(\frac{\left(\frac{\sqrt{3}}{4}s^2 \right) \left(\frac{s}{\sqrt{3}} \right) \left(\frac{s}{2\sqrt{3}} \right)}{\frac{\sqrt{3}}{96}s^4} - 1 \right)$$

$$\sigma_B = 3 \frac{P}{A_4} \quad \blacktriangleleft$$



PROBLEM 4.121

The C-shaped steel bar is used as a dynamometer to determine the magnitude P of the forces shown. Knowing that the cross section of the bar is a square of side 40 mm and that strain on the inner edge was measured and found to be 450μ , determine the magnitude P of the forces. Use $E = 200 \text{ GPa}$.

SOLUTION

At the strain gage location,

$$\sigma = E\epsilon = (200 \times 10^9)(450 \times 10^{-6}) = 90 \times 10^6 \text{ Pa}$$

$$A = (40)(40) = 1600 \text{ mm}^2 = 1600 \times 10^{-6} \text{ m}^2$$

$$I = \frac{1}{12}(40)(40)^3 = 213.33 \times 10^3 \text{ mm}^4 = 213.33 \times 10^{-9} \text{ m}^4$$

$$e = 80 + 20 = 100 \text{ mm} = 0.100 \text{ m}$$

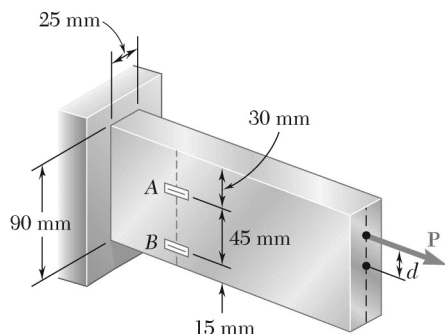
$$c = 20 \text{ mm} = 0.020 \text{ m}$$

$$\sigma = \frac{P}{A} + \frac{Mc}{I} = \frac{P}{A} + \frac{Pec}{I} = KP$$

$$K = \frac{1}{A} + \frac{ec}{I} = \frac{1}{1600 \times 10^{-6}} + \frac{(0.100)(0.020)}{213.33 \times 10^{-9}} = 10.00 \times 10^3 \text{ m}^{-2}$$

$$P = \frac{\sigma}{K} = \frac{90 \times 10^6}{10.00 \times 10^3} = 9.00 \times 10^3 \text{ N}$$

$$P = 9.00 \text{ kN} \quad \blacktriangleleft$$



PROBLEM 4.122

An eccentric force $\mathbf{P}$ is applied as shown to a steel bar of 25×90 -mm cross section. The strains at A and B have been measured and found to be

$$\epsilon_A = +350 \mu \quad \epsilon_B = -70 \mu$$

Knowing that $E = 200$ GPa, determine (a) the distance d , (b) the magnitude of the force $\mathbf{P}$.

SOLUTION

$$h = 15 + 45 + 30 = 90 \text{ mm} \quad b = 25 \text{ mm} \quad c = \frac{1}{2}h = 45 \text{ mm} = 0.045 \text{ m}$$

$$A = bh - (25)(90) = 2.25 \times 10^3 \text{ mm}^2 = 2.25 \times 10^{-3} \text{ m}^2$$

$$I = \frac{1}{12}bh^3 = \frac{1}{12}(25)(90)^3 = 1.51875 \times 10^6 \text{ mm}^4 = 1.51875 \times 10^{-6} \text{ m}^4$$

$$y_A = 60 - 45 = 15 \text{ mm} = 0.015 \text{ m} \quad y_B = 15 - 45 = -30 \text{ mm} = -0.030 \text{ m}$$

Stresses from strain gages at A and B : $\sigma_A = E\epsilon_A = (200 \times 10^9)(350 \times 10^{-6}) = 70 \times 10^6 \text{ Pa}$

$$\sigma_B = E\epsilon_B = (200 \times 10^9)(-70 \times 10^{-6}) = -14 \times 10^6 \text{ Pa}$$

$$\sigma_A = \frac{P}{A} - \frac{My_A}{I} \quad (1)$$

$$\sigma_B = \frac{P}{A} - \frac{My_B}{I} \quad (2)$$

Subtracting, $\sigma_A - \sigma_B = -\frac{M(y_A - y_B)}{I}$

$$M = -\frac{I(\sigma_A - \sigma_B)}{y_A - y_B} = -\frac{(1.51875 \times 10^{-6})(84 \times 10^6)}{0.045} = -2835 \text{ N} \cdot \text{m}$$

Multiplying (2) by y_A and (1) by y_B and subtracting, $y_A\sigma_B - y_B\sigma_A = (y_A - y_B)\frac{P}{A}$

$$P = \frac{A(y_A\sigma_B - y_B\sigma_A)}{y_A - y_B} = \frac{(2.25 \times 10^{-3})[(0.015)(-14 \times 10^6) - (-0.030)(70 \times 10^6)]}{0.045} = 94.5 \times 10^3 \text{ N}$$

(a) $M = -Pd \quad \therefore \quad d = -\frac{M}{P} = -\frac{-2835}{94.5 \times 10^3} = 0.030 \text{ m} \quad d = 30.0 \text{ mm} \quad \blacktriangleleft$

(b) $P = 94.5 \text{ kN} \quad \blacktriangleleft$

PROBLEM 4.123

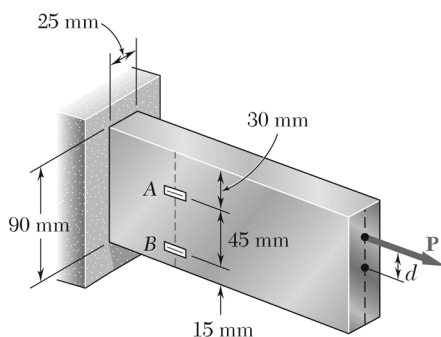
Solve Prob. 4.122, assuming that the measured strains are

$$\epsilon_A = +600 \mu \quad \epsilon_B = +420 \mu$$

PROBLEM 4.122 An eccentric force **P** is applied as shown to a steel bar of 25×90-mm cross section. The strains at *A* and *B* have been measured and found to be

$$\epsilon_A = +350 \mu \quad \epsilon_B = -70 \mu$$

Knowing that $E = 200$ GPa, determine (a) the distance d , (b) the magnitude of the force **P**.



SOLUTION

$$h = 15 + 45 + 30 = 90 \text{ mm} \quad b = 25 \text{ mm} \quad c = \frac{1}{2}h = 45 \text{ mm} = 0.045 \text{ m}$$

$$A = bh = (25)(90) = 2.25 \times 10^3 \text{ mm}^2 = 2.25 \times 10^{-3} \text{ m}^2$$

$$I = \frac{1}{12}bh^3 = \frac{1}{12}(25)(90)^3 = 1.51875 \times 10^6 \text{ mm}^4 = 1.51875 \times 10^{-6} \text{ m}^4$$

$$y_A = 60 - 45 = 15 \text{ mm} = 0.015 \text{ m} \quad y_B = 15 - 45 = -30 \text{ mm} = -0.030 \text{ m}$$

Stresses from strain gages at *A* and *B*: $\sigma_A = E\epsilon_A = (200 \times 10^9)(600 \times 10^{-6}) = 120 \times 10^6 \text{ Pa}$

$$\sigma_B = E\epsilon_B = (200 \times 10^9)(420 \times 10^{-6}) = 84 \times 10^6 \text{ Pa}$$

$$\sigma_A = \frac{P}{A} - \frac{My_A}{I} \quad (1)$$

$$\sigma_B = \frac{P}{A} - \frac{My_B}{I} \quad (2)$$

Subtracting, $\sigma_A - \sigma_B = -\frac{M(y_A - y_B)}{I}$

$$M = -\frac{I(\sigma_A - \sigma_B)}{y_A - y_B} = -\frac{(1.51875 \times 10^{-6})(36 \times 10^6)}{0.045} = -1215 \text{ N} \cdot \text{m}$$

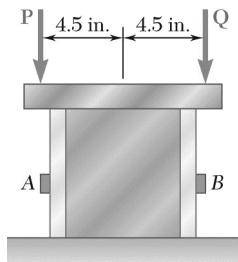
Multiplying (2) by y_A and (1) by y_B and subtracting, $y_A\sigma_B - y_B\sigma_A = (y_A - y_B)\frac{P}{A}$

$$P = \frac{A(y_A\sigma_B - y_B\sigma_A)}{y_A - y_B} = \frac{(2.25 \times 10^{-3})[(0.015)(84 \times 10^6) - (-0.030)(120 \times 10^6)]}{0.045} = 243 \times 10^3 \text{ N}$$

$$M = -Pd$$

$$(a) \quad \therefore d = -\frac{M}{P} = -\frac{-1215}{243 \times 10^3} = 5 \times 10^{-3} \text{ m} \quad d = 5.00 \text{ mm} \quad \blacktriangleleft$$

$$(b) \quad P = 243 \text{ kN} \quad \blacktriangleleft$$



PROBLEM 4.124

A short length of a W8 × 31 rolled-steel shape supports a rigid plate on which two loads **P** and **Q** are applied as shown. The strains at two points **A** and **B** on the centerline of the outer faces of the flanges have been measured and found to be

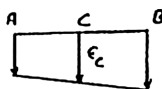
$$\epsilon_A = -550 \times 10^{-6} \text{ in./in.} \quad \epsilon_B = -680 \times 10^{-6} \text{ in./in.}$$

Knowing that $E = 29 \times 10^6$ psi, determine the magnitude of each load.

SOLUTION

Strains:

$$\epsilon_A = -550 \times 10^{-6} \text{ in./in.}$$



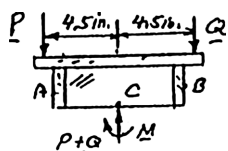
$$\epsilon_B = -680 \times 10^{-6} \text{ in./in.}$$

$$\epsilon_C = \frac{1}{2}(\epsilon_A + \epsilon_B) = \frac{1}{2}(-550 - 680)10^{-6} = -615 \times 10^{-6} \text{ in./in.}$$

Stresses:

$$\sigma_A = E\epsilon_A = (29 \times 10^6 \text{ psi})(-550 \times 10^{-6} \text{ in./in.}) = -15.95 \text{ ksi}$$

$$\sigma_C = E\epsilon_C = (29 \times 10^6 \text{ psi})(-615 \times 10^{-6} \text{ in./in.}) = -17.935 \text{ ksi}$$



W8 × 31

$$A = 9.13 \text{ in}^2$$

$$S = 27.5 \text{ in}^3$$

$$M = (4.5 \text{ in.})(P - Q)$$

At point C:

$$\sigma_C = -\frac{P + Q}{A}; \quad -17.835 \text{ ksi} = -\frac{P + Q}{9.13 \text{ in}^2}$$

$$P + Q = 162.83 \text{ kips} \quad (1)$$

At point A:

$$\sigma_A = -\frac{P + Q}{A} - \frac{M}{S}$$

$$-15.95 \text{ ksi} = -17.835 \text{ ksi} - \frac{(4.5 \text{ in.})(P - Q)}{27.5 \text{ in}^3};$$

$$P - Q = -11.52 \text{ kips} \quad (2)$$

Solve simultaneously,

$$P = 25.7 \text{ kips} \quad Q = 87.2 \text{ kips}$$

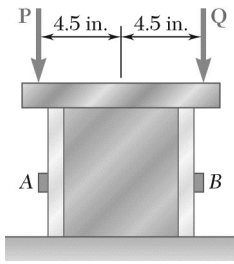
$$P = 25.7 \text{ kips} \downarrow \blacktriangleleft$$

$$Q = 87.2 \text{ kips} \downarrow \blacktriangleleft$$

PROBLEM 4.125

Solve Prob. 4.124, assuming that the measured strains are

$$\epsilon_A = +35 \times 10^{-6} \text{ in./in.} \quad \epsilon_B = -450 \times 10^{-6} \text{ in./in.}$$



PROBLEM 4.124 A short length of a W8 × 31 rolled-steel shape supports a rigid plate on which two loads **P** and **Q** are applied as shown. The strains at two points *A* and *B* on the centerline of the outer faces of the flanges have been measured and found to be

$$\epsilon_A = -550 \times 10^{-6} \text{ in./in.} \quad \epsilon_B = -680 \times 10^{-6} \text{ in./in.}$$

Knowing that $E = 29 \times 10^6$ psi, determine the magnitude of each load.

SOLUTION

See solution and figures of Prob. 4.124.

$$\epsilon_A = +35 \times 10^{-6} \text{ in./in.}; \quad \epsilon_B = -450 \times 10^{-6} \text{ in./in.}$$

$$\epsilon_C = \frac{1}{2}(\epsilon_A + \epsilon_B) = \frac{1}{2}(35 - 450)10^{-6} \text{ in./in.} = -207.5 \times 10^{-6} \text{ in./in.}$$

Stresses:

$$\sigma_A = E\epsilon_A = (29 \times 10^6 \text{ psi})(+35 \times 10^{-6} \text{ in./in.}) = +1.015 \text{ ksi}$$

$$\sigma_C = E\epsilon_C = (29 \times 10^6 \text{ psi})(-207.5 \times 10^{-6} \text{ in./in.}) = -6.0175 \text{ ksi}$$

At point C:
$$\sigma_C = -\frac{P + Q}{A}; \quad -6.0175 \text{ ksi} = -\frac{P + Q}{9.13 \text{ in}^2}$$

$$P + Q = 54.94 \text{ kips} \quad (1)$$

At point A:
$$\sigma_A = -\frac{P + Q}{A} - \frac{M}{S}$$

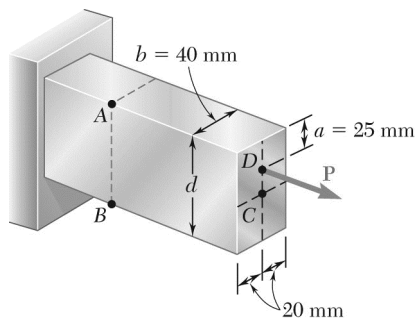
$$+1.015 \text{ ksi} = -6.0175 - \frac{(4.5 \text{ in.})(P - Q)}{27.5 \text{ in}^3}$$

$$P - Q = -42.98 \text{ kips} \quad (2)$$

Solve simultaneously, $P = 5.98 \text{ kips} \quad Q = 49.0 \text{ kips}$

$$P = 5.98 \text{ kips} \downarrow \blacktriangleleft$$

$$Q = 49.0 \text{ kips} \downarrow \blacktriangleleft$$



PROBLEM 4.126

The eccentric axial force $\mathbf{P}$ acts at point D , which must be located 25 mm below the top surface of the steel bar shown. For $P = 60$ kN, determine (a) the depth d of the bar for which the tensile stress at point A is maximum, (b) the corresponding stress at point A .

SOLUTION

$$A = bd \quad I = \frac{1}{12}bd^3$$

$$c = \frac{1}{2}d \quad e = \frac{1}{2}d - a$$

$$\sigma_A = \frac{P}{A} + \frac{Pec}{I}$$

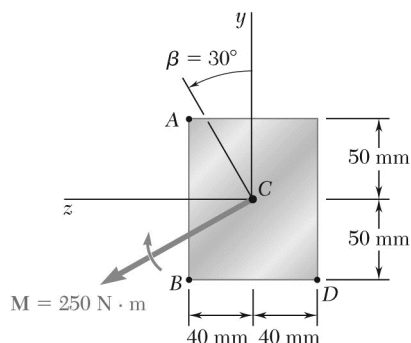
$$\sigma_A = \frac{P}{b} \left\{ \frac{1}{d} + \frac{12 \left(\frac{1}{2}d - a \right) \left(\frac{1}{2}d \right)}{d^3} \right\} = \frac{P}{b} \left\{ \frac{4}{d} - \frac{6a}{d^2} \right\}$$

(a) Depth d for maximum σ_A : Differentiate with respect to d .

$$\frac{d\sigma_A}{dd} = \frac{P}{b} \left\{ -\frac{4}{d^2} + \frac{12a}{d^3} \right\} = 0 \quad d = 3a \quad d = 75 \text{ mm} \blacktriangleleft$$

$$(b) \quad \sigma_A = \frac{60 \times 10^3}{40 \times 10^{-3}} \left\{ \frac{4}{75 \times 10^{-3}} - \frac{(6)(25 \times 10^{-3})}{(75 \times 10^{-3})^2} \right\} = 40 \times 10^6 \text{ Pa} \quad \sigma_A = 40 \text{ MPa} \blacktriangleleft$$

PROBLEM 4.127



The couple $\mathbf{M}$ is applied to a beam of the cross section shown in a plane forming an angle β with the vertical. Determine the stress at (a) point A , (b) point B , (c) point D .

SOLUTION

$$I_z = \frac{1}{12}(80)(100)^3 = 6.6667 \times 10^6 \text{ mm}^4 = 6.6667 \times 10^{-6} \text{ m}^4$$

$$I_y = \frac{1}{12}(100)(80)^3 = 4.2667 \times 10^6 \text{ mm}^4 = 4.2667 \times 10^{-6} \text{ m}^4$$

$$y_A = -y_B = -y_D = 50 \text{ mm}$$

$$z_A = z_B = -z_D = 40 \text{ mm}$$

$$M_y = -250 \sin 30^\circ = -125 \text{ N} \cdot \text{m}$$

$$M_z = 250 \cos 30^\circ = 216.51 \text{ N} \cdot \text{m}$$

$$\begin{aligned} (a) \quad \sigma_A &= -\frac{M_z y_A}{I_z} + \frac{M_y z_A}{I_y} = -\frac{(216.51)(0.050)}{6.6667 \times 10^{-6}} + \frac{(-125)(0.040)}{4.2667 \times 10^{-6}} \\ &= -2.80 \times 10^6 \text{ Pa} \end{aligned}$$

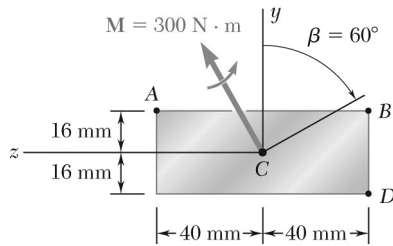
$$\sigma_A = -2.80 \text{ MPa} \quad \blacktriangleleft$$

$$\begin{aligned} (b) \quad \sigma_B &= -\frac{M_z y_B}{I_z} + \frac{M_y z_B}{I_y} = -\frac{(216.51)(-0.050)}{6.6667 \times 10^{-6}} + \frac{(-125)(0.040)}{4.2667 \times 10^{-6}} \\ &= 0.452 \times 10^6 \text{ Pa} \end{aligned}$$

$$\sigma_B = 0.452 \text{ MPa} \quad \blacktriangleleft$$

$$\begin{aligned} (c) \quad \sigma_D &= -\frac{M_z y_D}{I_z} + \frac{M_y z_D}{I_y} = -\frac{(216.51)(-0.050)}{6.6667 \times 10^{-6}} + \frac{(-125)(-0.040)}{4.2667 \times 10^{-6}} \\ &= 2.80 \times 10^6 \text{ Pa} \end{aligned}$$

$$\sigma_D = 2.80 \text{ MPa} \quad \blacktriangleleft$$



PROBLEM 4.128

The couple $\mathbf{M}$ is applied to a beam of the cross section shown in a plane forming an angle β with the vertical. Determine the stress at (a) point A , (b) point B , (c) point D .

SOLUTION

$$I_z = \frac{1}{12}(80)(32)^3 = 218.45 \times 10^3 \text{ mm}^4 = 218.45 \times 10^{-9} \text{ m}^4$$

$$I_y = \frac{1}{12}(32)(80)^3 = 1.36533 \times 10^6 \text{ mm}^4 = 1.36533 \times 10^{-6} \text{ m}^4$$

$$y_A = y_B = -y_D = 16 \text{ mm}$$

$$z_A = -z_B = -z_D = 40 \text{ mm}$$

$$M_y = 300 \cos 30^\circ = 259.81 \text{ N} \cdot \text{m} \quad M_z = 300 \sin 30^\circ = 150 \text{ N} \cdot \text{m}$$

$$(a) \quad \sigma_A = -\frac{M_z y_A}{I_z} + \frac{M_y z_A}{I_y} = -\frac{(150)(16 \times 10^{-3})}{218.45 \times 10^{-9}} + \frac{(259.81)(40 \times 10^{-3})}{1.36533 \times 10^{-6}}$$

$$= -3.37 \times 10^6 \text{ Pa}$$

$$\sigma_A = -3.37 \text{ MPa} \quad \blacktriangleleft$$

$$(b) \quad \sigma_B = -\frac{M_z y_B}{I_z} + \frac{M_y z_B}{I_y} = -\frac{(150)(16 \times 10^{-3})}{218.45 \times 10^{-9}} + \frac{(259.81)(-40 \times 10^{-3})}{1.36533 \times 10^{-6}}$$

$$= -18.60 \times 10^6 \text{ Pa}$$

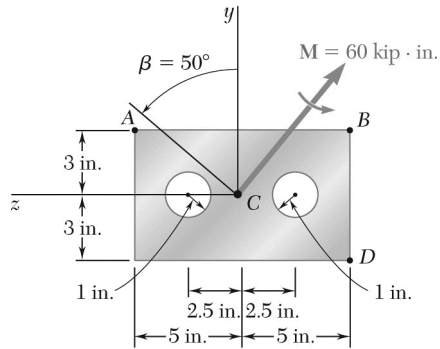
$$\sigma_B = -18.60 \text{ MPa} \quad \blacktriangleleft$$

$$(c) \quad \sigma_D = -\frac{M_z y_D}{I_z} + \frac{M_y z_D}{I_y} = -\frac{(150)(-16 \times 10^{-3})}{218.45 \times 10^{-9}} + \frac{(259.81)(-40 \times 10^{-3})}{1.36533 \times 10^{-6}}$$

$$= 3.37 \times 10^6 \text{ Pa}$$

$$\sigma_D = 3.37 \text{ MPa} \quad \blacktriangleleft$$

PROBLEM 4.129



The couple $\mathbf{M}$ is applied to a beam of the cross section shown in a plane forming an angle β with the vertical. Determine the stress at (a) point A, (b) point B, (c) point D.

SOLUTION

$$M_z = -60 \sin 40^\circ = -38.567 \text{ kip} \cdot \text{in}$$

$$M_y = 60 \cos 40^\circ = 45.963 \text{ kip} \cdot \text{in}$$

$$y_A = y_B = -y_D = 3 \text{ in.}$$

$$z_A = -z_B = -z_D = 5 \text{ in.}$$

$$I_z = \frac{1}{12}(10)(6)^3 - 2 \left[\frac{\pi}{4}(1)^2 \right] = 178.429 \text{ in}^4$$

$$I_y = \frac{1}{12}(6)(10)^3 - 2 \left[\frac{\pi}{4}(1)^4 + \pi(1)^2(2.5)^2 \right] = 459.16 \text{ in}^4$$

$$(a) \quad \sigma_A = -\frac{M_z y_A}{I_z} + \frac{M_y z_A}{I_y} = -\frac{(-38.567)(3)}{178.429} + \frac{(45.963)(5)}{459.16}$$

$$= 1.149 \text{ ksi}$$

$$\sigma_A = 1.149 \text{ ksi} \quad \blacktriangleleft$$

$$(b) \quad \sigma_B = -\frac{M_z y_B}{I_z} + \frac{M_y z_B}{I_y} = -\frac{(-38.567)(3)}{178.429} + \frac{(45.963)(-5)}{459.16}$$

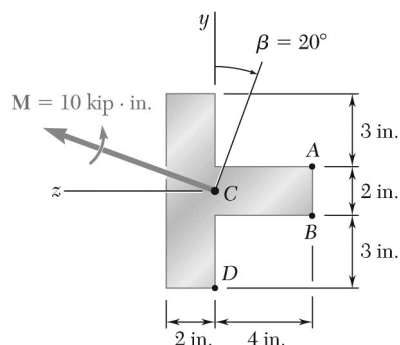
$$= 0.1479$$

$$\sigma_B = 0.1479 \text{ ksi} \quad \blacktriangleleft$$

$$(c) \quad \sigma_D = -\frac{M_z y_D}{I_z} + \frac{M_y z_D}{I_y} = -\frac{(-38.567)(-3)}{178.429} + \frac{(45.963)(-5)}{459.16}$$

$$= -1.149 \text{ ksi}$$

$$\sigma_D = -1.149 \text{ ksi} \quad \blacktriangleleft$$



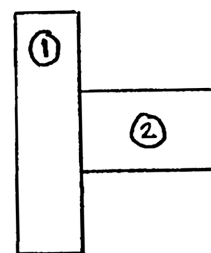
PROBLEM 4.130

The couple **M** is applied to a beam of the cross section shown in a plane forming an angle β with the vertical. Determine the stress at (a) point A, (b) point B, (c) point D.

SOLUTION

Locate centroid.

	$A, \text{ in}^2$	$\bar{z}, \text{ in.}$	$A\bar{z}, \text{ in}^3$
①	16	-1	-16
②	8	2	16
Σ	24		0



The centroid lies at point C.

$$I_z = \frac{1}{12}(2)(8)^3 + \frac{1}{12}(4)(2)^3 = 88 \text{ in}^4$$

$$I_y = \frac{1}{3}(8)(2)^3 + \frac{1}{3}(2)(4)^3 = 64 \text{ in}^4$$

$$y_A = -y_B = 1 \text{ in.}, \quad y_D = -4 \text{ in.}$$

$$z_A = z_B = -4 \text{ in.}, \quad z_D = 0$$

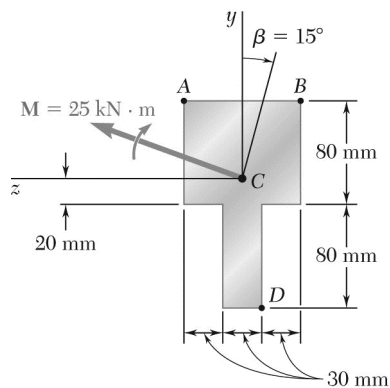
$$M_z = 10 \cos 20^\circ = 9.3969 \text{ kip} \cdot \text{in}$$

$$M_y = 10 \sin 20^\circ = 3.4202 \text{ kip} \cdot \text{in}$$

$$(a) \quad \sigma_A = -\frac{M_z y_A}{I_z} + \frac{M_y z_A}{I_y} = -\frac{(9.3969)(1)}{88} + \frac{(3.4202)(-4)}{64} \quad \sigma_A = 0.321 \text{ ksi} \quad \blacktriangleleft$$

$$(b) \quad \sigma_B = -\frac{M_z y_B}{I_z} + \frac{M_y z_B}{I_y} = -\frac{(9.3969)(-1)}{88} + \frac{(3.4202)(-4)}{64} \quad \sigma_B = -0.107 \text{ ksi} \quad \blacktriangleleft$$

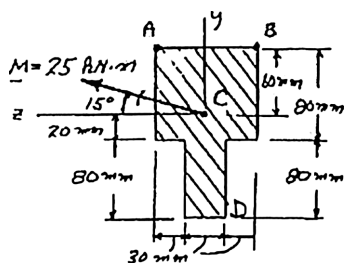
$$(c) \quad \sigma_D = -\frac{M_z y_D}{I_z} + \frac{M_y z_D}{I_y} = -\frac{(9.3969)(-4)}{88} + \frac{(3.4202)(0)}{64} \quad \sigma_D = 0.427 \text{ ksi} \quad \blacktriangleleft$$



PROBLEM 4.131

The couple $\mathbf{M}$ is applied to a beam of the cross section shown in a plane forming an angle β with the vertical. Determine the stress at (a) point A, (b) point B, (c) point D.

SOLUTION



$$M_y = 25 \sin 15^\circ = 6.4705 \text{ kN} \cdot \text{m}$$

$$M_z = 25 \cos 15^\circ = 24.148 \text{ kN} \cdot \text{m}$$

$$I_y = \frac{1}{12}(80)(90)^3 + \frac{1}{12}(80)(30)^3 = 5.04 \times 10^6 \text{ mm}^4$$

$$I_y = 5.04 \times 10^{-6} \text{ m}^4$$

$$I_z = \frac{1}{3}(90)(60)^3 + \frac{1}{3}(60)(20)^3 + \frac{1}{3}(30)(100)^3 = 16.64 \times 10^6 \text{ mm}^4 = 16.64 \times 10^{-6} \text{ m}^4$$

$$\text{Stress: } \sigma = \frac{M_y z}{I_y} - \frac{M_z y}{I_z}$$

$$(a) \quad \sigma_A = \frac{(6.4705 \text{ kN} \cdot \text{m})(0.045 \text{ m})}{5.04 \times 10^{-6} \text{ m}^4} - \frac{(24.148 \text{ kN} \cdot \text{m})(0.060 \text{ m})}{16.64 \times 10^{-6} \text{ m}^4}$$

$$= 57.772 \text{ MPa} - 87.072 \text{ MPa}$$

$$\sigma_A = -29.3 \text{ MPa} \quad \blacktriangleleft$$

$$(b) \quad \sigma_B = \frac{(6.4705 \text{ kN} \cdot \text{m})(-0.045 \text{ m})}{5.04 \times 10^{-6} \text{ m}^4} - \frac{(24.148 \text{ kN} \cdot \text{m})(0.060 \text{ m})}{16.64 \times 10^{-6} \text{ m}^4}$$

$$= -57.772 \text{ MPa} - 87.072 \text{ MPa}$$

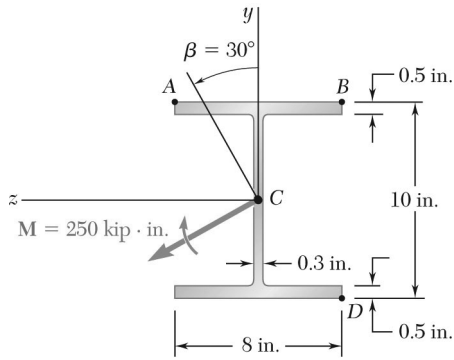
$$\sigma_B = -144.8 \text{ MPa} \quad \blacktriangleleft$$

$$(c) \quad \sigma_D = \frac{(6.4705 \text{ kN} \cdot \text{m})(-0.015 \text{ m})}{5.04 \times 10^{-6} \text{ m}^4} - \frac{(24.148 \text{ kN} \cdot \text{m})(-0.100 \text{ m})}{16.64 \times 10^{-6} \text{ m}^4}$$

$$= -19.257 \text{ MPa} + 145.12 \text{ MPa}$$

$$\sigma_D = -125.9 \text{ MPa} \quad \blacktriangleleft$$

PROBLEM 4.132



The couple M is applied to a beam of the cross section shown in a plane forming an angle β with the vertical. Determine the stress at (a) point A, (b) point B, (c) point D.

SOLUTION

Flange:

$$I_z = \frac{1}{12}(8)(0.5)^3 + (8)(0.5)(4.75)^2$$

$$= 90.333 \text{ in}^4$$

$$I_y = \frac{1}{12}(0.5)(8)^3 = 21.333 \text{ in}^4$$

Web:

$$I_z = \frac{1}{12}(0.3)(9)^3 = 18.225 \text{ in}^4$$

$$I_y = \frac{1}{12}(9)(0.3)^3 = 0.02025 \text{ in}^4$$

Total:

$$I_z = (2)(90.333) + 18.225 = 198.89 \text{ in}^4$$

$$I_y = (2)(21.333) + 0.02025 = 42.687 \text{ in}^4$$

$$y_A = y_B = -y_D = 5 \text{ in.}$$

$$z_A = -z_B = -z_C = 4 \text{ in.}$$

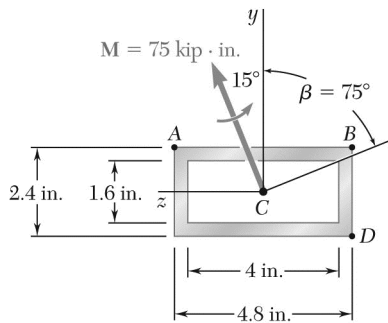
$$M_z = 250 \cos 30^\circ = 216.51 \text{ kip} \cdot \text{in}$$

$$M_y = -250 \sin 30^\circ = -125 \text{ kip} \cdot \text{in}$$

$$(a) \quad \sigma_A = -\frac{M_z y_A}{I_z} + \frac{M_y z_A}{I_y} = -\frac{(216.51)(5)}{198.89} + \frac{(-125)(4)}{42.687} \quad \sigma_A = -17.16 \text{ ksi} \quad \blacktriangleleft$$

$$(b) \quad \sigma_B = -\frac{M_z y_B}{I_z} + \frac{M_y z_B}{I_y} = -\frac{(216.51)(5)}{198.89} + \frac{(-125)(-4)}{42.687} \quad \sigma_B = 6.27 \text{ ksi} \quad \blacktriangleleft$$

$$(c) \quad \sigma_D = -\frac{M_z y_D}{I_z} + \frac{M_y z_D}{I_y} = -\frac{(216.51)(-5)}{198.89} + \frac{(-125)(-4)}{42.687} \quad \sigma_D = 17.16 \text{ ksi} \quad \blacktriangleleft$$



PROBLEM 4.133

The couple $\mathbf{M}$ is applied to a beam of the cross section shown in a plane forming an angle β with the vertical. Determine the stress at (a) point A, (b) point B, (c) point D.

SOLUTION

$$I_z = \frac{1}{12}(4.8)(2.4)^3 - \frac{1}{12}(4)(1.6)^3 = 4.1643 \text{ in}^4$$

$$I_y = \frac{1}{12}(2.4)(4.8)^3 - \frac{1}{12}(1.6)(4)^3 = 13.5851 \text{ in}^4$$

$$y_A = y_B = -y_D = 1.2 \text{ in.}$$

$$z_A = -z_B = -z_D = 2.4 \text{ in.}$$

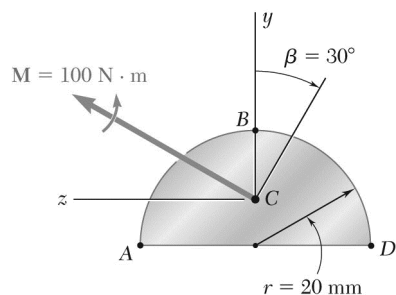
$$M_z = 75 \sin 15^\circ = 19.4114 \text{ kip} \cdot \text{in}$$

$$M_y = 75 \cos 15^\circ = 72.444 \text{ kip} \cdot \text{in}$$

$$(a) \quad \sigma_A = -\frac{M_z y_A}{I_z} + \frac{M_y z_A}{I_y} = -\frac{(19.4114)(1.2)}{4.1643} + \frac{(72.444)(2.4)}{13.5851} \quad \sigma_A = 7.20 \text{ ksi} \quad \blacktriangleleft$$

$$(b) \quad \sigma_B = -\frac{M_z y_B}{I_z} + \frac{M_y z_B}{I_y} = -\frac{(19.4114)(1.2)}{4.1643} + \frac{(72.444)(-2.4)}{13.5851} \quad \sigma_B = -18.39 \text{ ksi} \quad \blacktriangleleft$$

$$(c) \quad \sigma_D = -\frac{M_z y_D}{I_z} + \frac{M_y z_D}{I_y} = -\frac{(19.4114)(-1.2)}{4.1643} + \frac{(72.444)(-2.4)}{13.5851} \quad \sigma_D = -7.20 \text{ ksi} \quad \blacktriangleleft$$



PROBLEM 4.134

The couple $\mathbf{M}$ is applied to a beam of the cross section shown in a plane forming an angle β with the vertical. Determine the stress at (a) point A, (b) point B, (c) point D.

SOLUTION

$$I_z = \frac{\pi}{8} r^4 - \left(\frac{\pi}{2} r^2 \right) \left(\frac{4r}{3\pi} \right)^2 = \left(\frac{\pi}{8} - \frac{8}{9\pi} \right) r^4$$

$$= (0.109757)(20)^4 = 17.5611 \times 10^{-3} \text{ mm}^4 = 17.5611 \times 10^{-9} \text{ m}^4$$

$$I_y = \frac{\pi}{8} r^4 = \frac{\pi(20)^4}{8} = 62.832 \times 10^{-3} \text{ mm}^4 = 62.832 \times 10^{-9} \text{ m}^4$$

$$y_A = y_D = -\frac{4r}{3\pi} = -\frac{(4)(20)}{3\pi} = -8.4883 \text{ mm}$$

$$y_B = 20 - 8.4883 = 11.5117 \text{ mm}$$

$$z_A = -z_D = 20 \text{ mm} \quad z_B = 0$$

$$M_z = 100 \cos 30^\circ = 86.603 \text{ N} \cdot \text{m}$$

$$M_y = 100 \sin 30^\circ = 50 \text{ N} \cdot \text{m}$$

$$(a) \quad \sigma_A = -\frac{M_z y_A}{I_z} + \frac{M_y z_A}{I_y} = -\frac{(86.603)(-8.4883 \times 10^{-3})}{17.5611 \times 10^{-9}} + \frac{(50)(20 \times 10^{-3})}{62.832 \times 10^{-9}}$$

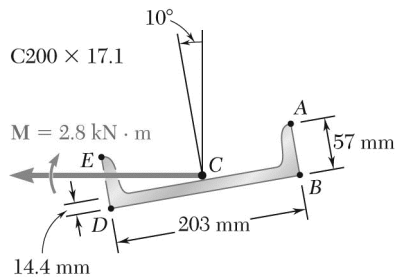
$$= 57.8 \times 10^6 \text{ Pa} \quad \sigma_A = 57.8 \text{ MPa} \quad \blacktriangleleft$$

$$(b) \quad \sigma_B = -\frac{M_z y_B}{I_z} + \frac{M_y z_B}{I_y} = -\frac{(86.603)(11.5117 \times 10^{-3})}{17.5611 \times 10^{-9}} + \frac{(50)(0)}{62.832 \times 10^{-9}}$$

$$= -56.8 \times 10^6 \text{ Pa} \quad \sigma_B = -56.8 \text{ MPa} \quad \blacktriangleleft$$

$$(c) \quad \sigma_D = -\frac{M_z y_D}{I_z} + \frac{M_y z_D}{I_y} = -\frac{(86.603)(-8.4883 \times 10^{-3})}{17.5611 \times 10^{-9}} + \frac{(50)(-20 \times 10^{-3})}{62.832 \times 10^{-9}}$$

$$= 25.9 \times 10^6 \text{ Pa} \quad \sigma_D = 25.9 \text{ MPa} \quad \blacktriangleleft$$



PROBLEM 4.135

The couple $\mathbf{M}$ acts in a vertical plane and is applied to a beam oriented as shown. Determine (a) the angle that the neutral axis forms with the horizontal, (b) the maximum tensile stress in the beam.

SOLUTION

For C 200×17.1 rolled steel shape, $I_z = 0.538 \times 10^6 \text{ mm}^4 = 0.538 \times 10^{-6} \text{ m}^4$
 $I_y = 13.4 \times 10^6 \text{ mm}^4 = 13.4 \times 10^{-6} \text{ m}^4$

$$z_E = z_D = -z_A = -z_B = \frac{1}{2}(203) = 101.5 \text{ mm}$$

$$y_D = y_B = -14.4 \text{ mm} \quad y_E = y_A = 57 - 14.4 = 42.6 \text{ mm}$$

$$M_z = (2.8 \times 10^3) \cos 10^\circ = 2.7575 \times 10^3 \text{ N} \cdot \text{m}$$

$$M_y = (2.8 \times 10^3) \sin 10^\circ = 486.21 \text{ N} \cdot \text{m}$$

(a) Angle of neutral axis.

$$\tan \phi = \frac{I_z}{I_y} \tan \theta = \frac{0.538}{13.4} \tan 10^\circ = 0.007079 \quad \phi = 0.4056^\circ$$

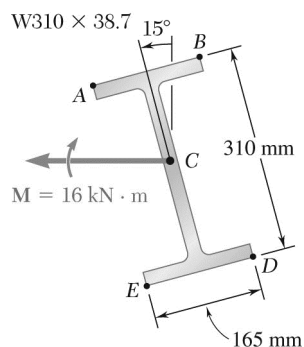
$$\alpha = 10^\circ - 0.4056^\circ$$

$$\alpha = 9.59^\circ \quad \blacktriangleleft$$

(b) Maximum tensile stress occurs at point D.

$$\sigma_D = -\frac{M_z y_D}{I_z} + \frac{M_y z_D}{I_y} = -\frac{(2.7575 \times 10^3)(-14.4 \times 10^{-3})}{0.538 \times 10^{-6}} + \frac{(486.21)(0.1015)}{13.4 \times 10^{-6}}$$

$$= 73.807 \times 10^6 + 3.682 \times 10^6 = 77.5 \times 10^6 \text{ Pa} \quad \sigma_D = 77.5 \text{ MPa} \quad \blacktriangleleft$$



PROBLEM 4.136

The couple $\mathbf{M}$ acts in a vertical plane and is applied to a beam oriented as shown. Determine (a) the angle that the neutral axis forms with the horizontal, (b) the maximum tensile stress in the beam.

SOLUTION

For W 310×38.7 rolled steel shape,

$$I_z = 85.1 \times 10^6 \text{ mm}^4 = 85.1 \times 10^{-6} \text{ m}^4$$

$$I_y = 7.27 \times 10^6 \text{ mm}^4 = 7.27 \times 10^{-6} \text{ m}^4$$

$$y_A = y_B = -y_D = -y_E = \left(\frac{1}{2}\right)(310) = 155 \text{ mm}$$

$$z_A = z_E = -z_B = -z_D = \left(\frac{1}{2}\right)(165) = 82.5 \text{ mm}$$

$$M_z = (16 \times 10^3) \cos 15^\circ = 15.455 \times 10^3 \text{ N} \cdot \text{m}$$

$$M_y = (16 \times 10^3) \sin 15^\circ = 4.1411 \times 10^3 \text{ N} \cdot \text{m}$$

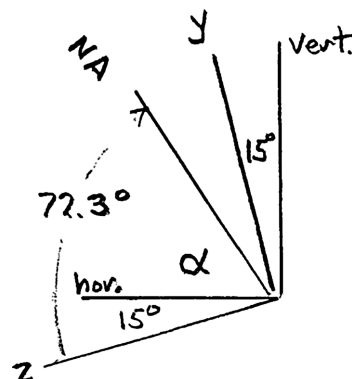
(a) Angle of neutral axis.

$$\tan \varphi = \frac{I_z}{I_y} \tan \theta = \frac{85.1 \times 10^{-6}}{7.27 \times 10^{-6}} \tan 15^\circ = 3.1365$$

$$\varphi = 72.3^\circ$$

$$\alpha = 72.3^\circ - 15^\circ$$

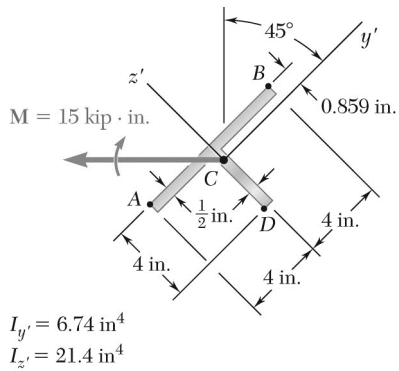
$$\alpha = 57.3^\circ \quad \blacktriangleleft$$



(b) Maximum tensile stress occurs at point E.

$$\begin{aligned} \sigma_E &= -\frac{M_z y_E}{I_z} + \frac{M_y z_E}{I_y} \\ &= -\frac{(15.455 \times 10^3)(-155 \times 10^{-3})}{85.1 \times 10^{-6}} + \frac{(4.1411 \times 10^3)(82.5 \times 10^{-3})}{7.27 \times 10^{-6}} \\ &= 75.1 \times 10^6 \text{ Pa} \end{aligned}$$

$$\sigma_E = 75.1 \text{ MPa} \quad \blacktriangleleft$$



PROBLEM 4.137

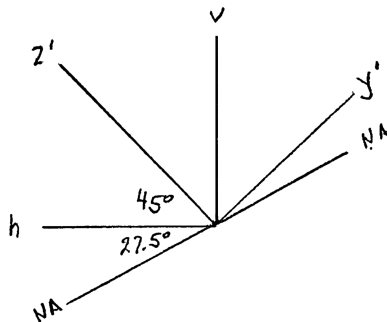
The couple $\mathbf{M}$ acts in a vertical plane and is applied to a beam oriented as shown. Determine (a) the angle that the neutral axis forms with the horizontal, (b) the maximum tensile stress in the beam.

SOLUTION

$$\begin{aligned}
 I_{z'} &= 21.4 \text{ in}^4 & I_{y'} &= 6.74 \text{ in}^4 \\
 z'_A &= z'_B = 0.859 \text{ in.} & z'_D &= -4 + 0.859 \text{ in.} = -3.141 \text{ in.} \\
 y_A &= -4 \text{ in.} & y_{B'} &= 4 \text{ in.} & y_{D'} &= -0.25 \text{ in.} \\
 M_{y'} &= -15 \sin 45^\circ = -10.6066 \text{ kip} \cdot \text{in} \\
 M_{z'} &= 15 \cos 45^\circ = 10.6066 \text{ kip} \cdot \text{in}
 \end{aligned}$$

(a) Angle of neutral axis.

$$\tan \varphi = \frac{I_{z'}}{I_{y'}} \tan \theta = \frac{21.4}{6.74} \tan (-45^\circ) = 3.1751$$



$$\begin{aligned}
 \varphi &= -72.5^\circ \\
 \alpha &= 72.5^\circ - 45^\circ
 \end{aligned}$$

$$\alpha = 27.5^\circ \blacktriangleleft$$

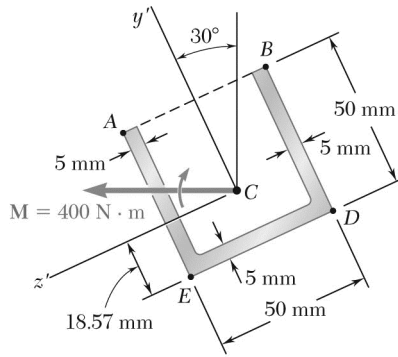
(b) The maximum tensile stress occurs at point D.

$$\begin{aligned}
 \sigma_D &= -\frac{M_z y_D}{I_z} + \frac{M_y z_D}{I_y} = -\frac{(10.6066)(-0.25)}{21.4} + \frac{(-10.6066)(-3.141)}{6.74} \\
 &= 0.12391 + 4.9429
 \end{aligned}$$

$$\sigma_D = 5.07 \text{ ksi} \blacktriangleleft$$

PROBLEM 4.138

The couple $\mathbf{M}$ acts in a vertical plane and is applied to a beam oriented as shown. Determine (a) the angle that the neutral axis forms with the horizontal, (b) the maximum tensile stress in the beam.



$$I_{y'} = 281 \times 10^3 \text{ mm}^4$$

$$I_{z'} = 176.9 \times 10^3 \text{ mm}^4$$

SOLUTION

$$I_{z'} = 176.9 \times 10^3 \text{ mm}^4 = 176.9 \times 10^{-9} \text{ m}^4$$

$$I_{y'} = 281 \times 10^3 \text{ mm}^4 = 281 \times 10^{-9} \text{ m}^4$$

$$y'_E = -18.57 \text{ mm}, \quad z_E = 25 \text{ mm}$$

$$M_{z'} = 400 \cos 30^\circ = 346.41 \text{ N} \cdot \text{m}$$

$$M_{y'} = 400 \sin 30^\circ = 200 \text{ N} \cdot \text{m}$$

$$(a) \quad \tan \phi = \frac{I_{z'}}{I_{y'}} \tan \theta = \frac{176.9 \times 10^{-9}}{281 \times 10^{-9}} \cdot \tan 30^\circ = 0.36346$$

$$\phi = 19.97^\circ$$

$$\alpha = 30^\circ - 19.97^\circ$$

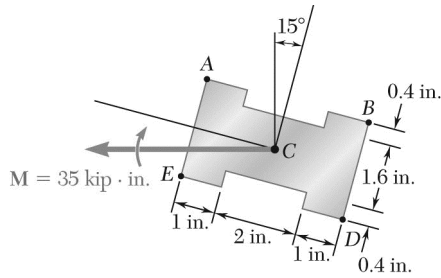
$$\alpha = 10.03^\circ \quad \blacktriangleleft$$

(b) Maximum tensile stress occurs at point E.

$$\begin{aligned} \sigma_E &= -\frac{M_{z'} y'_E}{I_{z'}} + \frac{M_{y'} z'_E}{I_{y'}} = -\frac{(346.41)(-18.57 \times 10^{-3})}{176.9 \times 10^{-9}} + \frac{(200)(25 \times 10^{-3})}{281 \times 10^{-9}} \\ &= 36.36 \times 10^6 + 17.79 \times 10^6 = 54.2 \times 10^6 \text{ Pa} \end{aligned}$$

$$\sigma_E = 54.2 \text{ MPa} \quad \blacktriangleleft$$

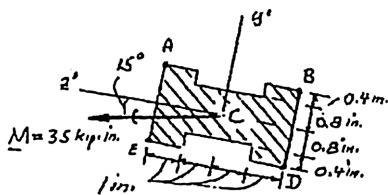
PROBLEM 4.139



The couple $\mathbf{M}$ acts in a vertical plane and is applied to a beam oriented as shown. Determine (a) the angle that the neutral axis forms with the horizontal, (b) the maximum tensile stress in the beam.

SOLUTION

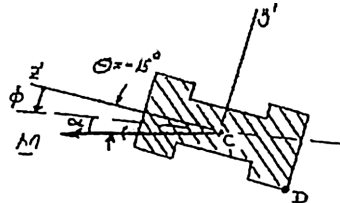
Bending moments:



$$M_{y'} = -35 \sin 15^\circ = -9.059 \text{ kip} \cdot \text{in}$$

$$M_{z'} = -35 \cos 15^\circ = 33.807 \text{ kip} \cdot \text{in}$$

Moments of inertia:



$$I_{y'} = \frac{1}{12} (2.4)(4)^3 - \frac{1}{12} (2 \times 0.4)(2)^3 = 12.267 \text{ in}^4$$

$$I_{z'} = \frac{1}{12} (2)(2.4)^3 + \frac{1}{12} (2)(1.6)^3 = 2.987 \text{ in}^4$$

(a) Neutral axis:

$$\tan \phi = \frac{I_{z'}}{I_{y'}} \tan \theta = \frac{2.987 \text{ in}^4}{12.267 \text{ in}^4} \tan(-15^\circ) = -0.06525$$

$$\phi = 3.73^\circ \nearrow$$

$$\alpha = 15^\circ - \phi = 15^\circ - 3.73^\circ = 11.27^\circ \searrow$$

$$\alpha = 11.3^\circ \searrow \blacktriangleleft$$

(b) Maximum tensile stress at D:

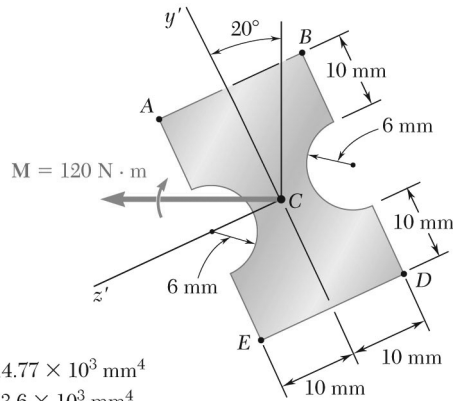
$$y'_D = -1.2 \text{ in.} \quad z'_D = -2 \text{ in.}$$

$$\begin{aligned} \sigma_D &= \frac{M_{y'} z'_D}{I_{y'}} - \frac{M_{z'} y'_D}{I_{z'}} = \frac{(-9.059 \text{ kip} \cdot \text{in})(-2 \text{ in.})}{12.267 \text{ in}^4} - \frac{(33.807 \text{ kip} \cdot \text{in})(-1.2 \text{ in.})}{2.987 \text{ in}^4} \\ &= 1.477 \text{ ksi} + 13.582 \text{ ksi} = 15.059 \text{ ksi} \end{aligned}$$

$$\sigma_D = 15.06 \text{ ksi} \blacktriangleleft$$

PROBLEM 4.140

The couple $\mathbf{M}$ acts in a vertical plane and is applied to a beam oriented as shown. Determine (a) the angle that the neutral axis forms with the horizontal, (b) the maximum tensile stress in the beam.



SOLUTION

$$I_{z'} = 53.6 \times 10^3 \text{ mm}^4 = 53.6 \times 10^{-9} \text{ m}^4$$

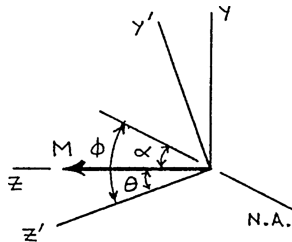
$$I_{y'} = 14.77 \times 10^3 \text{ mm}^4 = 14.77 \times 10^{-9} \text{ m}^4$$

$$M_{z'} = 120 \sin 70^\circ = 112.763 \text{ N} \cdot \text{m}$$

$$M_{y'} = 120 \cos 70^\circ = 41.042 \text{ N} \cdot \text{m}$$

(a) Angle of neutral axis.

$$\theta = 20^\circ.$$



$$\tan \phi = \frac{I_{z'}}{I_{y'}} \quad \tan \theta = \frac{53.6 \times 10^{-9}}{14.77 \times 10^{-9}} \tan 20^\circ = 1.32084$$

$$\phi = 52.871^\circ$$

$$\alpha = 52.871^\circ - 20^\circ$$

$$\alpha = 32.9^\circ \quad \blacktriangleleft$$

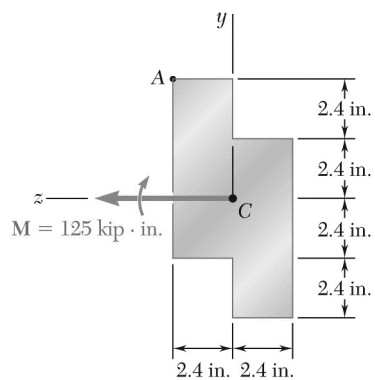
(b) The maximum tensile stress occurs at point E.

$$y'_E = -16 \text{ mm} = -0.016 \text{ m}$$

$$z_E = 10 \text{ mm} = 0.010 \text{ m}$$

$$\begin{aligned} \sigma_E &= -\frac{M_{z'} y'_E}{I_{z'}} + \frac{M_{y'} z'_E}{I_{y'}} \\ &= -\frac{(112.763)(-0.016)}{53.6 \times 10^{-9}} + \frac{(41.042)(0.010)}{14.77 \times 10^{-9}} \\ &= 61.448 \times 10^6 \text{ Pa} \end{aligned}$$

$$\sigma_E = 61.4 \text{ MPa} \quad \blacktriangleleft$$



PROBLEM 4.141

The couple M acts in a vertical plane and is applied to a beam oriented as shown. Determine the stress at point A .

SOLUTION

$$I_y = 2 \left\{ \frac{1}{3} (7.2)(2.4)^3 \right\} = 66.355 \text{ in}^4$$

$$I_z = 2 \left\{ \frac{1}{12} (2.4)(7.2)^3 + (2.4)(7.2)(1.2)^2 \right\} = 199.066 \text{ in}^4$$

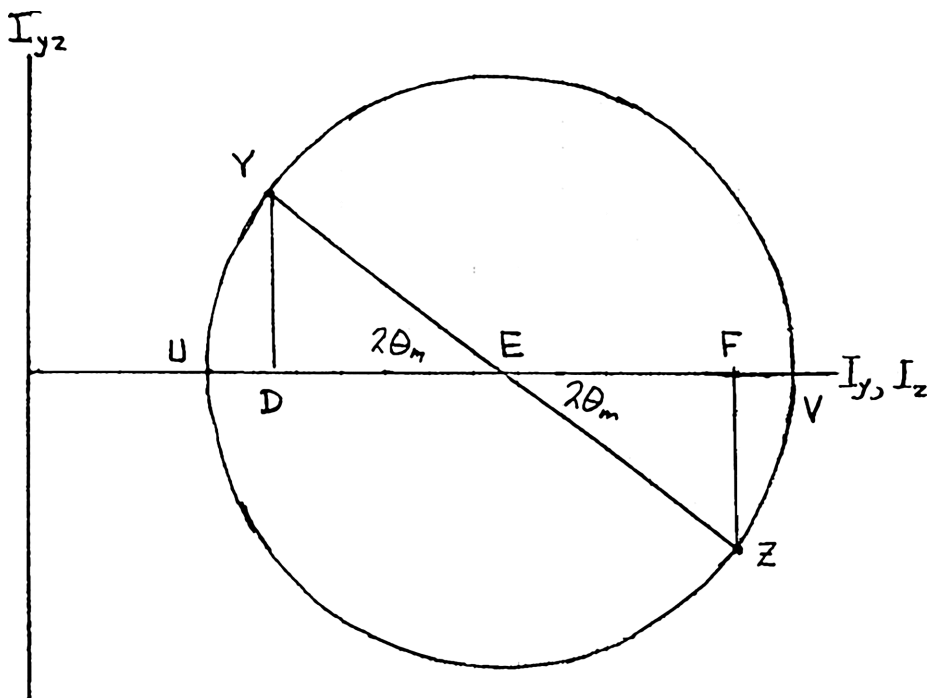
$$I_{yz} = 2 \{ (2.4)(7.2)(1.2)(1.2) \} = 49.766 \text{ in}^4$$

Using Mohr's circle determine the principal axes and principal moments of inertia.

$$Y: (66.355 \text{ in}^4, 49.766 \text{ in}^4)$$

$$Z: (199.066 \text{ in}^4, -49.766 \text{ in}^4)$$

$$E: (132.710 \text{ in}^4, 0)$$



PROBLEM 4.141 (Continued)

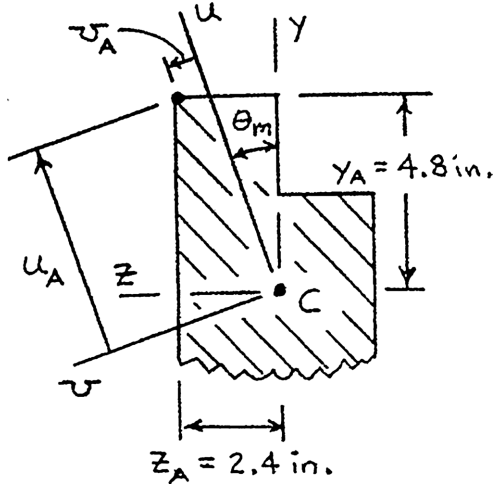
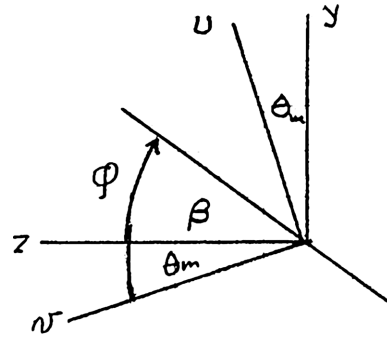
$$\tan 2\theta_m = \frac{DY}{DE} = \frac{49.766}{66.355}$$

$$2\theta_m = 36.87^\circ \quad \theta_m = 18.435^\circ$$

$$R = \sqrt{DE^2 + DY^2} = 82.944 \text{ in}^4$$

$$I_u = 132.710 - 82.944 = 49.766 \text{ in}^4$$

$$I_v = 132.710 + 82.944 = 215.654 \text{ in}^4$$



$$M_u = 125 \sin 18.435^\circ = 39.529 \text{ kip} \cdot \text{in}$$

$$M_v = 125 \cos 18.435^\circ = 118.585 \text{ kip} \cdot \text{in}$$

$$u_A = 4.8 \cos 18.435^\circ + 2.4 \sin 18.435^\circ = 5.3126 \text{ in.}$$

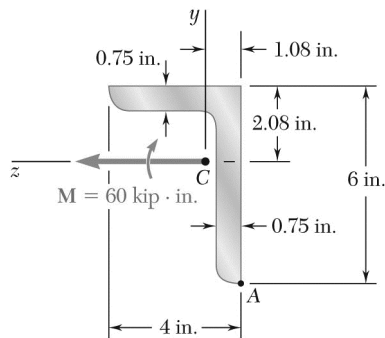
$$v_A = -4.8 \sin 18.435^\circ + 2.4 \cos 18.435^\circ = 0.7589 \text{ in.}$$

$$\sigma_A = -\frac{M_v u_A}{I_v} + \frac{M_u v_A}{I_u}$$

$$= -\frac{(118.585)(5.3126)}{215.654} + \frac{(39.529)(0.7589)}{49.766}$$

$$= -2.32 \text{ ksi}$$

$$\sigma_A = -2.32 \text{ ksi} \quad \blacktriangleleft$$



$$I_y = 8.7 \text{ in}^4$$

$$I_z = 24.5 \text{ in}^4$$

$$I_{yz} = +8.3 \text{ in}^4$$

PROBLEM 4.142

The couple $\mathbf{M}$ acts in a vertical plane and is applied to a beam oriented as shown. Determine the stress at point A .

SOLUTION

Using Mohr's circle determine the principal axes and principal moments of inertia.

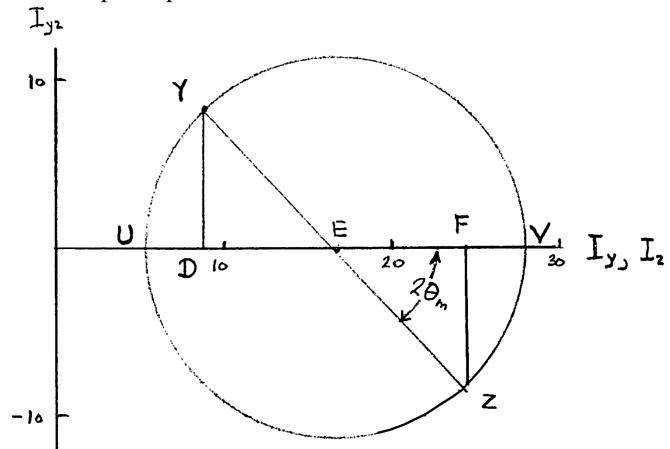
$$Y: (8.7, 8.3) \text{ in}^4$$

$$Z: (24.5, -8.3) \text{ in}^4$$

$$E: (16.6, 0) \text{ in}^4$$

$$EF = 7.9 \text{ in}^4$$

$$FZ = 8.3 \text{ in}^4$$



$$R = \sqrt{7.9^2 + 8.3^2} = 11.46 \text{ in}^4 \quad \tan 2\theta_m = \frac{FZ}{EF} = \frac{8.3}{7.9} = 1.0506$$

$$\theta_m = 23.2^\circ \quad I_u = 16.6 - 11.46 = 5.14 \text{ in}^4 \quad I_v = 16.6 + 11.46 = 28.06 \text{ in}^4$$

$$M_u = M \sin \theta_m = (60) \sin 23.2^\circ = 23.64 \text{ kip} \cdot \text{in}$$

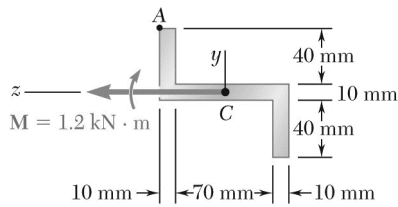
$$M_v = M \cos \theta_m = (60) \cos 23.2^\circ = 55.15 \text{ kip} \cdot \text{in}$$

$$u_A = y_A \cos \theta_m + z_A \sin \theta_m = -3.92 \cos 23.2^\circ - 1.08 \sin 23.2^\circ = -4.03 \text{ in.}$$

$$v_A = z_A \cos \theta_m - y_A \sin \theta_m = -1.08 \cos 23.2^\circ + 3.92 \sin 23.2^\circ = 0.552 \text{ in.}$$

$$\sigma_A = -\frac{M_v u_A}{I_v} + \frac{M_u v_A}{I_u} = -\frac{(55.15)(-4.03)}{28.06} + \frac{(23.64)(0.552)}{5.14}$$

$$\sigma_A = 10.46 \text{ ksi} \quad \blacktriangleleft$$



PROBLEM 4.143

The couple M acts in a vertical plane and is applied to a beam oriented as shown. Determine the stress at point A .

$$I_y = 1.894 \times 10^6 \text{ mm}^4$$

$$I_z = 0.614 \times 10^6 \text{ mm}^4$$

$$I_{yz} = +0.800 \times 10^6 \text{ mm}^4$$

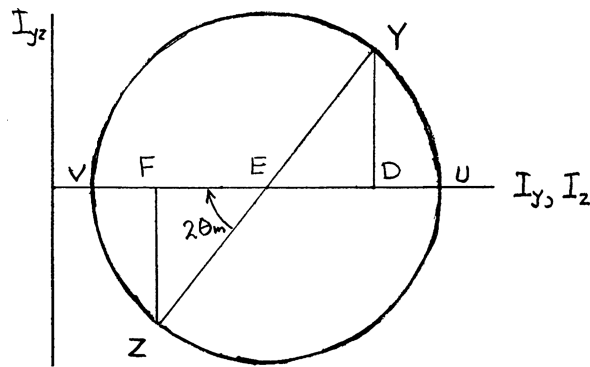
SOLUTION

Using Mohr's circle determine the principal axes and principal moments of inertia.

$$Y: (1.894, 0.800) \times 10^6 \text{ mm}^4$$

$$Z: (0.614, 0.800) \times 10^6 \text{ mm}^4$$

$$E: (1.254, 0) \times 10^6 \text{ mm}^4$$



$$R = \sqrt{EF^2 + FZ^2} = \sqrt{0.640^2 + 0.800^2} \times 10^{-6} = 1.0245 \times 10^6 \text{ mm}^4$$

$$I_v = (1.254 - 1.0245) \times 10^6 \text{ mm}^4 = 0.2295 \times 10^6 \text{ mm}^4 = 0.2295 \times 10^{-6} \text{ m}^4$$

$$I_u = (1.254 + 1.0245) \times 10^6 \text{ mm}^4 = 2.2785 \times 10^6 \text{ mm}^4 = 2.2785 \times 10^{-6} \text{ m}^4$$

$$\tan 2\theta_m = \frac{FZ}{FE} = \frac{0.800 \times 10^6}{0.640 \times 10^6} = 1.25 \quad \theta_m = 25.67^\circ$$

$$M_v = M \cos \theta_m = (1.2 \times 10^3) \cos 25.67^\circ = 1.0816 \times 10^3 \text{ N} \cdot \text{m}$$

$$M_u = -M \sin \theta_m = -(1.2 \times 10^3) \sin 25.67^\circ = -0.5198 \times 10^3 \text{ N} \cdot \text{m}$$

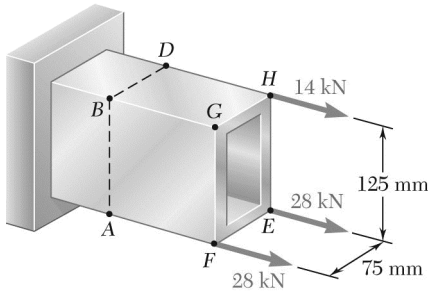
$$u_A = y_A \cos \theta_m - z_A \sin \theta_m = 45 \cos 25.67^\circ - 45 \sin 25.67^\circ = 21.07 \text{ mm}$$

$$v_A = z_A \cos \theta_m + y_A \sin \theta_m = 45 \cos 25.67^\circ + 45 \sin 25.67^\circ = 60.05 \text{ mm}$$

$$\sigma_A = -\frac{M_v u_A}{I_v} + \frac{M_u v_A}{I_u} = -\frac{(1.0816 \times 10^3)(21.07 \times 10^{-3})}{0.2295 \times 10^{-6}} + \frac{(-0.5198 \times 10^3)(60.05 \times 10^{-3})}{2.2785 \times 10^{-6}}$$

$$= 113.0 \times 10^6 \text{ Pa}$$

$$\sigma_A = 113.0 \text{ MPa} \quad \blacktriangleleft$$



PROBLEM 4.144

The tube shown has a uniform wall thickness of 12 mm. For the loading given, determine (a) the stress at points A and B, (b) the point where the neutral axis intersects line ABD.

SOLUTION

Add y- and z-axes as shown. Cross section is a 75 mm $\times$ 125 mm rectangle with a 51 mm $\times$ 101 mm rectangular cutout.

$$I_z = \frac{1}{12}(75)(125)^3 - \frac{1}{12}(51)(101)^3 = 7.8283 \times 10^6 \text{ mm}^4 = 7.8283 \times 10^{-6} \text{ m}^4$$

$$I_y = \frac{1}{12}(125)(75)^3 - \frac{1}{12}(101)(51)^3 = 3.2781 \times 10^3 \text{ mm}^4 = 3.2781 \times 10^{-6} \text{ m}^4$$

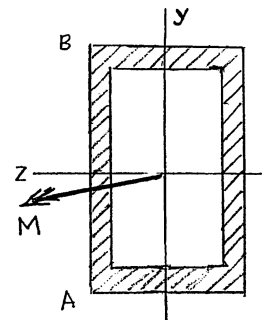
$$A = (75)(125) - (51)(101) = 4.224 \times 10^3 \text{ mm}^2 = 4.224 \times 10^{-3} \text{ m}^2$$

Resultant force and bending couples:

$$P = 14 + 28 + 28 = 70 \text{ kN} = 70 \times 10^3 \text{ N}$$

$$M_z = -(62.5 \text{ mm})(14 \text{ kN}) + (62.5 \text{ mm})(28 \text{ kN}) + (62.5 \text{ mm})(28 \text{ kN}) = 2625 \text{ N} \cdot \text{m}$$

$$M_y = -(37.5 \text{ mm})(14 \text{ kN}) + (37.5 \text{ mm})(28 \text{ kN}) + (37.5 \text{ mm})(28 \text{ kN}) = -525 \text{ N} \cdot \text{m}$$



$$(a) \quad \sigma_A = \frac{P}{A} - \frac{M_z y_A}{I_z} + \frac{M_y z_A}{I_y} = \frac{70 \times 10^3}{4.224 \times 10^{-3}} - \frac{(2625)(-0.0625)}{7.8283 \times 10^{-6}} + \frac{(-525)(0.0375)}{3.2781 \times 10^{-6}}$$

$$= 31.524 \times 10^6 \text{ Pa}$$

$$\sigma_A = 31.5 \text{ MPa} \quad \blacktriangleleft$$

$$\sigma_B = \frac{P}{A} - \frac{M_z y_B}{I_z} + \frac{M_y z_B}{I_y} = \frac{70 \times 10^3}{4.224 \times 10^{-3}} - \frac{(2625)(0.0625)}{7.8283 \times 10^{-6}} + \frac{(-525)(0.0375)}{3.2781 \times 10^{-6}}$$

$$= -10.39 \times 10^6 \text{ Pa}$$

$$\sigma_B = -10.39 \text{ MPa} \quad \blacktriangleleft$$

(b) Let point H be the point where the neutral axis intersects AB.

$$z_H = 0.0375 \text{ m}, \quad y_H = ?, \quad \sigma_H = 0$$

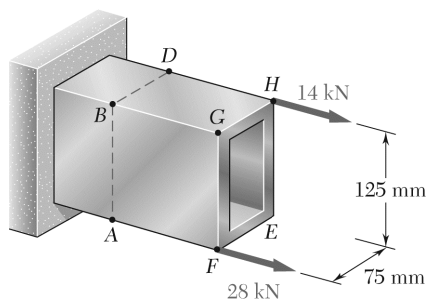
$$0 = \frac{P}{A} - \frac{M_z y_H}{I_z} + \frac{M_y z_H}{I_y}$$

$$y_H = \frac{I_z}{M_z} \left(\frac{P}{A} + \frac{M_y z_H}{I_y} \right) = \frac{7.8283 \times 10^{-6}}{2625} \left[\frac{70 \times 10^3}{4.224 \times 10^{-3}} + \frac{(-525)(0.0375)}{3.2781 \times 10^{-6}} \right]$$

$$= 0.03151 \text{ m} = 31.51 \text{ mm}$$

$$31.51 + 62.5 = 94.0 \text{ mm}$$

$$\text{Answer: } 94.0 \text{ mm above point A.} \quad \blacktriangleleft$$



PROBLEM 4.145

Solve Prob. 4.144, assuming that the 28-kN force at point *E* is removed.

PROBLEM 4.144 The tube shown has a uniform wall thickness of 12 mm. For the loading given, determine (a) the stress at points *A* and *B*, (b) the point where the neutral axis intersects line *ABD*.

SOLUTION

Add *y*- and *z*-axes as shown. Cross section is a 75 mm × 125 mm rectangle with a 51 mm × 101 mm rectangular cutout.

$$I_z = \frac{1}{12}(75)(125)^3 - \frac{1}{12}(51)(101)^3 = 7.8283 \times 10^6 \text{ mm}^4 = 7.8283 \times 10^{-6} \text{ m}^4$$

$$I_y = \frac{1}{12}(125)(75)^3 - \frac{1}{12}(101)(51)^3 = 3.2781 \times 10^6 \text{ mm}^4 = 3.2781 \times 10^{-6} \text{ m}^4$$

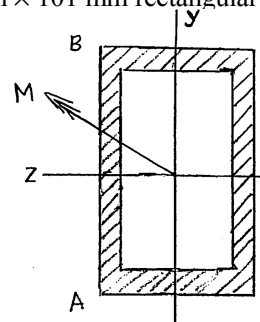
$$A = (75)(125) - (51)(101) = 4.224 \times 10^3 \text{ mm}^2 = 4.224 \times 10^{-3} \text{ m}^2$$

Resultant force and bending couples:

$$P = 14 + 28 = 42 \text{ kN} = 42 \times 10^3 \text{ N}$$

$$M_z = -(62.5 \text{ mm})(14 \text{ kN}) + (62.5 \text{ mm})(28 \text{ kN}) = 875 \text{ N} \cdot \text{m}$$

$$M_y = -(37.5 \text{ mm})(14 \text{ kN}) + (37.5 \text{ mm})(28 \text{ kN}) = 525 \text{ N} \cdot \text{m}$$



$$(a) \quad \sigma_A = \frac{P}{A} - \frac{M_z y_A}{I_z} + \frac{M_y z_A}{I_y} = \frac{42 \times 10^3}{4.224 \times 10^{-3}} - \frac{(875)(-0.0625)}{7.8283 \times 10^{-6}} + \frac{(525)(0.0375)}{3.2781 \times 10^{-6}}$$

$$= 22.935 \times 10^6 \text{ Pa} \quad \sigma_A = 22.9 \text{ MPa} \quad \blacktriangleleft$$

$$\sigma_B = \frac{P}{A} - \frac{M_z y_B}{I_z} + \frac{M_y z_B}{I_y} = \frac{42 \times 10^3}{4.224 \times 10^{-3}} - \frac{(875)(0.0625)}{7.8283 \times 10^{-6}} + \frac{(525)(0.0375)}{3.2781 \times 10^{-6}}$$

$$= 8.9631 \times 10^6 \text{ Pa} \quad \sigma_B = 8.96 \text{ MPa} \quad \blacktriangleleft$$

(b) Let point *K* be the point where the neutral axis intersects *BD*.

$$z_K = ?, \quad y_K = 0.0625 \text{ m}, \quad \sigma_H = 0$$

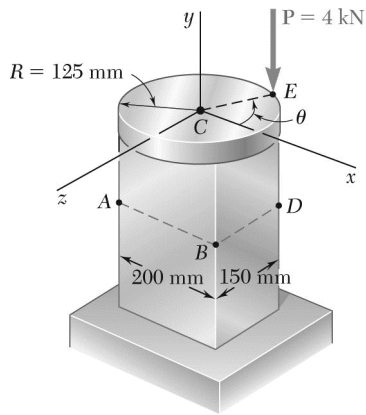
$$0 = \frac{P}{A} - \frac{M_z y_H}{I_z} + \frac{M_y z_H}{I_y}$$

$$z_H = \frac{I_y}{M_y} \left(\frac{M_z y_H}{I_z} - \frac{P}{A} \right) = \frac{3.2781 \times 10^{-6}}{525} \left[\frac{(875)(0.0625)}{7.8283 \times 10^{-6}} - \frac{42 \times 10^3}{4.224 \times 10^{-3}} \right]$$

$$= -0.018465 \text{ m} = -18.465 \text{ mm}$$

$$37.5 + 18.465 = 56.0 \text{ mm}$$

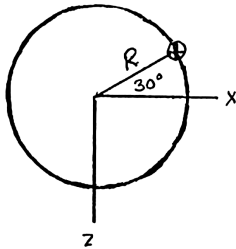
Answer: 56.0 mm to the right of point *B*. ◀



PROBLEM 4.146

A rigid circular plate of 125-mm radius is attached to a solid 150×200 -mm rectangular post, with the center of the plate directly above the center of the post. If a 4-kN force $\mathbf{P}$ is applied at E with $\theta = 30^\circ$, determine (a) the stress at point A , (b) the stress at point B , (c) the point where the neutral axis intersects line ABD .

SOLUTION



$$P = 4 \times 10^3 \text{ N (compression)}$$

$$M_x = -PR \sin 30^\circ = -(4 \times 10^3)(125 \times 10^{-3}) \sin 30^\circ = -250 \text{ N} \cdot \text{m}$$

$$M_z = -PR \cos 30^\circ = -(4 \times 10^3)(125 \times 10^{-3}) \cos 30^\circ = -433 \text{ N} \cdot \text{m}$$

$$I_x = \frac{1}{12}(200)(150)^3 = 56.25 \times 10^6 \text{ mm}^4 = 56.25 \times 10^{-6} \text{ m}^4$$

$$I_z = \frac{1}{12}(150)(200)^3 = 100 \times 10^6 \text{ mm}^4 = 100 \times 10^{-6} \text{ m}^4$$

$$-x_A = x_B = 100 \text{ mm} \quad z_A = z_B = 75 \text{ mm}$$

$$A = (200)(150) = 30 \times 10^3 \text{ mm}^2 = 30 \times 10^{-3} \text{ m}^2$$

$$(a) \quad \sigma_A = -\frac{P}{A} - \frac{M_x z_A}{I_x} + \frac{M_z x_A}{I_z} = -\frac{4 \times 10^3}{30 \times 10^{-3}} - \frac{(-250)(75 \times 10^{-3})}{56.25 \times 10^{-6}} + \frac{(-433)(-100 \times 10^{-3})}{100 \times 10^{-6}}$$

$$\sigma_A = 633 \times 10^3 \text{ Pa} = 633 \text{ kPa} \quad \blacktriangleleft$$

$$(b) \quad \sigma_B = -\frac{P}{A} - \frac{M_x z_B}{I_x} + \frac{M_z x_B}{I_z} = -\frac{4 \times 10^3}{30 \times 10^{-3}} - \frac{(-250)(75 \times 10^{-3})}{56.25 \times 10^{-6}} + \frac{(-433)(100 \times 10^{-3})}{100 \times 10^{-6}}$$

$$\sigma_B = -233 \times 10^3 \text{ Pa} = -233 \text{ kPa} \quad \blacktriangleleft$$

(c) Let G be the point on AB where the neutral axis intersects.

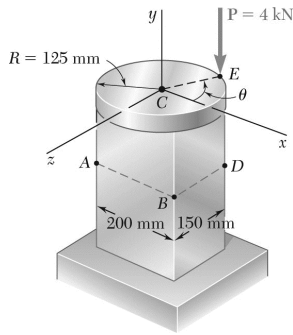
$$\sigma_G = 0 \quad z_G = 75 \text{ mm} \quad x_G = ?$$

$$\sigma_G = -\frac{P}{A} - \frac{M_x z_G}{I_x} + \frac{M_z x_G}{I_z} = 0$$

$$x_G = \frac{I_z}{M_z} \left\{ \frac{P}{A} + \frac{M_x z_G}{I_x} \right\} = \frac{100 \times 10^{-6}}{-433} \left\{ \frac{4 \times 10^3}{30 \times 10^{-3}} + \frac{(-250)(75 \times 10^{-3})}{56.25 \times 10^{-6}} \right\}$$

$$= 46.2 \times 10^{-3} \text{ m} = 46.2 \text{ mm}$$

Point G lies 146.2 mm from point A $\blacktriangleleft$

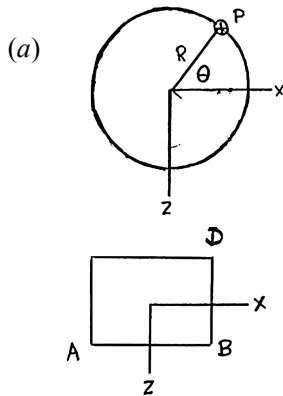


PROBLEM 4.147

4.147 In Prob. 4.146, determine (a) the value of θ for which the stress at D reaches its largest value, (b) the corresponding values of the stress, at A , B , C , and D .

4.146 A rigid circular plate of 125-mm radius is attached to a solid 150×200 -mm rectangular post, with the center of the plate directly above the center of the post. If a 4-kN force $\mathbf{P}$ is applied at E with $\theta = 30^\circ$, determine (a) the stress at point A , (b) the stress at point B , (c) the point where the neutral axis intersects line ABD .

SOLUTION



$$P = 4 \times 10^3 \text{ N} \quad PR = (4 \times 10^3)(125 \times 10^{-3}) = 500 \text{ N} \cdot \text{m}$$

$$M_x = -PR \sin \theta = -500 \sin \theta \quad M_z = -PR \cos \theta = -500 \cos \theta$$

$$I_x = \frac{1}{2}(200)(150)^3 = 56.25 \times 10^6 \text{ mm}^4 = 56.25 \times 10^{-6} \text{ m}^4$$

$$I_z = \frac{1}{2}(150)(200)^3 = 100 \times 10^6 \text{ mm}^4 = 100 \times 10^{-6} \text{ m}^4$$

$$x_D = 100 \text{ mm} \quad z_D = -75 \text{ mm}$$

$$A = (200)(150) = 30 \times 10^3 \text{ mm}^2 = 30 \times 10^{-3} \text{ m}^2$$

$$\sigma = -\frac{P}{A} - \frac{M_x z}{I_x} + \frac{M_z x}{I_z} = -P \left\{ \frac{1}{A} - \frac{Rz \sin \theta}{I_x} + \frac{Rx \cos \theta}{I_z} \right\}$$

For σ to be a maximum, $\frac{d\sigma}{d\theta} = 0$ with $z = z_D$, $x = x_D$

$$\frac{d\sigma_D}{d\theta} = -P \left\{ 0 + \frac{Rz_D \cos \theta}{I_x} + \frac{Rx_D \sin \theta}{I_z} \right\} = 0$$

$$\frac{\sin \theta}{\cos \theta} = \tan \theta = -\frac{I_z z_D}{I_x x_D} = -\frac{(100 \times 10^{-6})(-75 \times 10^{-3})}{(56.25 \times 10^{-6})(100 \times 10^{-3})} = \frac{4}{3}$$

$$\sin \theta = 0.8, \quad \cos \theta = 0.6, \quad \theta = 53.1^\circ \quad \blacktriangleleft$$

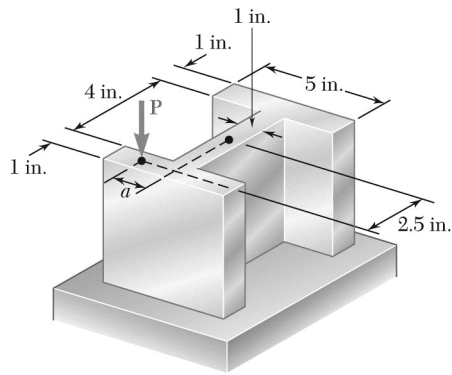
$$(b) \quad \sigma_A = -\frac{P}{A} - \frac{M_x z_A}{I_x} + \frac{M_z x_A}{I_z} = -\frac{4 \times 10^3}{30 \times 10^{-3}} + \frac{(500)(0.8)(75 \times 10^{-3})}{56.25 \times 10^{-6}} - \frac{(500)(0.6)(-100 \times 10^{-3})}{100 \times 10^{-6}}$$

$$= (-0.13333 + 0.53333 + 0.300) \times 10^6 \text{ Pa} = 0.700 \times 10^6 \text{ Pa} \quad \sigma_A = 700 \text{ kPa} \quad \blacktriangleleft$$

$$\sigma_B = (-0.13333 + 0.53333 - 0.300) \times 10^6 \text{ Pa} = 0.100 \times 10^6 \text{ Pa} \quad \sigma_B = 100 \text{ kPa} \quad \blacktriangleleft$$

$$\sigma_C = (-0.13333 + 0 + 0) \times 10^6 \text{ Pa} \quad \sigma_C = -133.3 \text{ kPa} \quad \blacktriangleleft$$

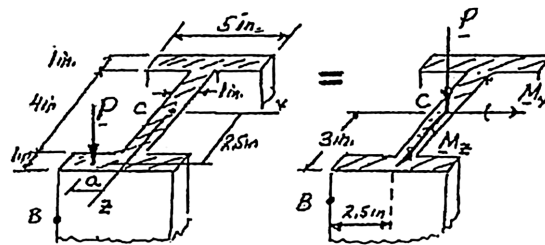
$$\sigma_D = (-0.13333 - 0.53333 - 0.300) \times 10^6 \text{ Pa} = -0.967 \times 10^6 \text{ Pa} \quad \sigma_D = -967 \text{ kPa} \quad \blacktriangleleft$$



PROBLEM 4.148

Knowing that $P = 90$ kips, determine the largest distance a for which the maximum compressive stress does not exceed 18 ksi.

SOLUTION



$$A = (5 \text{ in.})(6 \text{ in.}) - 2(2 \text{ in.})(4 \text{ in.}) = 14 \text{ in}^2$$

$$I_x = \frac{1}{12}(5 \text{ in.})(6 \text{ in.})^3 - 2\frac{1}{12}(2 \text{ in.})(4 \text{ in.})^3 = 68.67 \text{ in}^4$$

$$I_z = 2\frac{1}{12}(1 \text{ in.})(5 \text{ in.})^3 + \frac{1}{12}(4 \text{ in.})(1 \text{ in.})^3 = 21.17 \text{ in}^4$$

Force-couple system at C: $P = P \quad M_x = P(2.5 \text{ in.}) \quad M_z = Pa$

For $P = 90$ kips: $P = 90 \text{ kips} \quad M_x = (90 \text{ kips})(2.5 \text{ in.}) = 225 \text{ kip} \cdot \text{in.} \quad M_z = (90 \text{ kips})a$

Maximum compressive stress at B: $\sigma_B = -18 \text{ ksi}$

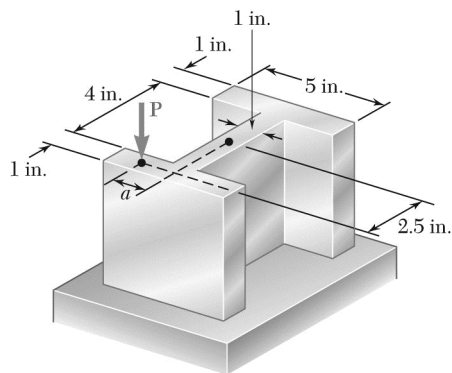
$$\sigma_B = -\frac{P}{A} - \frac{M_x(3 \text{ in.})}{I_x} - \frac{M_z(2.5 \text{ in.})}{I_z}$$

$$-18 \text{ ksi} = -\frac{90 \text{ kips}}{14 \text{ in}^2} - \frac{(225 \text{ kip} \cdot \text{in.})(3 \text{ in.})}{68.67 \text{ in}^4} - \frac{(90 \text{ kips})a(2.5 \text{ in.})}{21.17 \text{ in}^4}$$

$$-18 = -6.429 - 9.830 - 10.628a$$

$$-1.741 = -10.628a$$

$$a = 0.1638 \text{ in.} \quad \blacktriangleleft$$



PROBLEM 4.149

Knowing that $a = 1.25$ in., determine the largest value of P that can be applied without exceeding either of the following allowable stresses:

$$\sigma_{\text{ten}} = 10 \text{ ksi}$$

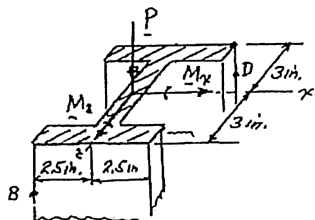
$$\sigma_{\text{comp}} = 18 \text{ ksi}$$

SOLUTION

$$A = (5 \text{ in.})(1 \text{ in.}) - (2 \text{ in.})(4 \text{ in.}) = 14 \text{ in}^2$$

$$I_x = \frac{1}{12}(5 \text{ in.})(1 \text{ in.})^3 - 2\frac{1}{12}(2 \text{ in.})(4 \text{ in.})^3 = 68.67 \text{ in}^4$$

$$I_z = 2\frac{1}{12}(1 \text{ in.})(5 \text{ in.})^3 + \frac{1}{12}(4 \text{ in.})(1 \text{ in.})^3 = 21.17 \text{ in}^4$$



Force-couple system at C: For $a = 1.25$ in.,

$$P = P \quad M_x = P(2.5 \text{ in.})$$

$$M_y = Pa = (1.25 \text{ in.})$$

Maximum compressive stress at B: $\sigma_B = -18 \text{ ksi}$

$$\sigma_B = -\frac{P}{A} - \frac{M_x(3 \text{ in.})}{I_x} - \frac{M_z(2.5 \text{ in.})}{I_z}$$

$$-18 \text{ ksi} = -\frac{P}{14 \text{ in}^2} - \frac{P(2.5 \text{ in.})(3 \text{ in.})}{68.67 \text{ in}^4} - \frac{P(1.25 \text{ in.})(2.5 \text{ in.})}{21.17 \text{ in}^4}$$

$$-18 = -0.0714P - 0.1092P - 0.1476P$$

$$-18 = 0.3282P \quad P = 54.8 \text{ kips}$$

Maximum tensile stress at D: $\sigma_D = +10 \text{ ksi}$

$$\sigma_D = -\frac{P}{A} + \frac{M_x(3 \text{ in.})}{I_x} + \frac{M_z(2.5 \text{ in.})}{I_z}$$

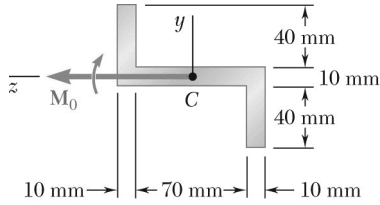
$$+10 \text{ ksi} = -0.0714P + 0.1092P + 0.1476P$$

$$10 = 0.1854P \quad P = 53.9 \text{ kips}$$

The smaller value of P is the largest allowable value.

$$P = 53.9 \text{ kips} \quad \blacktriangleleft$$

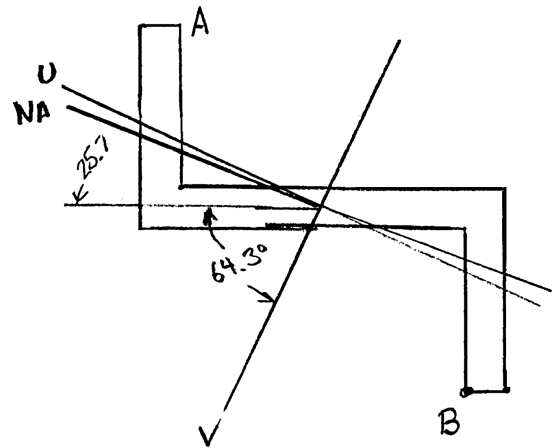
PROBLEM 4.150



The Z section shown is subjected to a couple M_0 acting in a vertical plane. Determine the largest permissible value of the moment M_0 of the couple if the maximum stress is not to exceed 80 MPa. *Given:* $I_{\max} = 2.28 \times 10^{-6} \text{ mm}^4$, $I_{\min} = 0.23 \times 10^{-6} \text{ mm}^4$, principal axes 25.7° and 64.3° .

SOLUTION

$$\begin{aligned}
 I_v &= I_{\max} = 2.28 \times 10^6 \text{ mm}^4 = 2.28 \times 10^{-6} \text{ m}^4 \\
 I_u &= I_{\min} = 0.23 \times 10^6 \text{ mm}^4 = 0.23 \times 10^{-6} \text{ m}^4 \\
 M_v &= M_0 \cos 64.3^\circ \\
 M_u &= M_0 \sin 64.3^\circ \\
 \theta &= 64.3^\circ \\
 \tan \phi &= \frac{I_v}{I_u} \tan \theta \\
 &= \frac{2.28 \times 10^{-6}}{0.23 \times 10^{-6}} \tan 64.3^\circ \\
 &= 20.597 \\
 \phi &= 87.22^\circ
 \end{aligned}$$

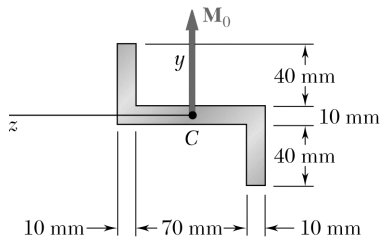


Points A and B are farthest from the neutral axis.

$$\begin{aligned}
 u_B &= y_B \cos 64.3^\circ + z_B \sin 64.3^\circ = (-45) \cos 64.3^\circ + (-35) \sin 64.3^\circ \\
 &= -51.05 \text{ mm} \\
 v_B &= z_B \cos 64.3^\circ - y_B \sin 64.3^\circ = (-35) \cos 64.3^\circ - (-45) \sin 64.3^\circ \\
 &= +25.37 \text{ mm} \\
 \sigma_B &= -\frac{M_v u_B}{I_v} + \frac{M_u v_B}{I_u} \\
 80 \times 10^6 &= -\frac{(M_0 \cos 64.3^\circ)(-51.05 \times 10^{-3})}{2.28 \times 10^{-6}} + \frac{(M_0 \sin 64.3^\circ)(25.37 \times 10^{-3})}{0.23 \times 10^{-6}} \\
 &= 109.1 \times 10^3 M_0 \\
 M_0 &= \frac{80 \times 10^6}{109.1 \times 10^3}
 \end{aligned}$$

$$M_0 = 733 \text{ N} \cdot \text{m} \quad \blacktriangleleft$$

PROBLEM 4.151



Solve Prob. 4.150 assuming that the couple M_0 acts in a horizontal plane.

PROBLEM 4.150 The Z section shown is subjected to a couple M_0 acting in a vertical plane. Determine the largest permissible value of the moment M_0 of the couple if the maximum stress is not to exceed 80 MPa. *Given:* $I_{\max} = 2.28 \times 10^{-6} \text{ mm}^4$, $I_{\min} = 0.23 \times 10^{-6} \text{ mm}^4$, principal axes 25.7° and 64.3° .

SOLUTION

$$I_v = I_{\min} = 0.23 \times 10^6 \text{ mm}^4 = 0.23 \times 10^6 \text{ m}^4$$

$$I_u = I_{\max} = 2.28 \times 10^6 \text{ mm}^4 = 2.28 \times 10^6 \text{ m}^4$$

$$M_v = M_0 \cos 64.3^\circ$$

$$M_u = M_0 \sin 64.3^\circ$$

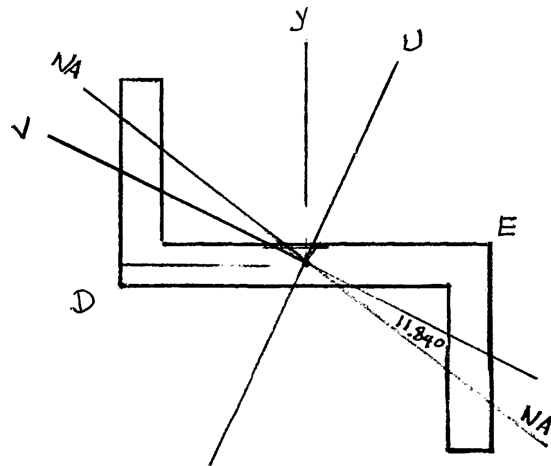
$$\theta = 64.3^\circ$$

$$\tan \varphi = \frac{I_v}{I_u} \tan \theta$$

$$= \frac{0.23 \times 10^{-6}}{2.28 \times 10^{-6}} \tan 64.3^\circ$$

$$= 0.20961$$

$$\varphi = 11.84^\circ$$



Points D and E are farthest from the neutral axis.

$$u_D = y_D \cos 25.7^\circ - z_D \sin 25.7^\circ = (-5) \cos 25.7^\circ - 45 \sin 25.7^\circ$$

$$= -24.02 \text{ mm}$$

$$v_D = z_D \cos 25.7^\circ + y_D \sin 25.7^\circ = 45 \cos 25.7^\circ + (-5) \sin 25.7^\circ$$

$$= 38.38 \text{ mm}$$

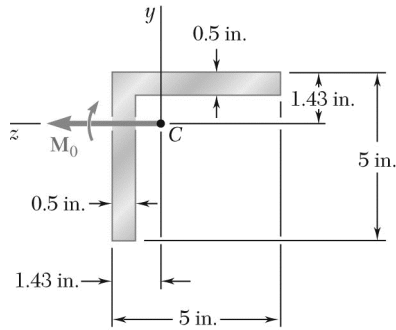
$$\sigma_D = -\frac{M_v u_D}{I_v} + \frac{M_u v_D}{I_u} = -\frac{(M_0 \cos 64.3^\circ)(-24.02 \times 10^{-3})}{0.23 \times 10^{-6}}$$

$$+ \frac{(M_0 \sin 64.3^\circ)(38.38 \times 10^{-3})}{2.28 \times 10^{-6}}$$

$$80 \times 10^6 = 60.48 \times 10^3 M_0$$

$$M_0 = 1.323 \times 10^3 \text{ N} \cdot \text{m}$$

$$M_0 = 1.323 \text{ kN} \cdot \text{m} \quad \blacktriangleleft$$



PROBLEM 4.152

A beam having the cross section shown is subjected to a couple $\mathbf{M}_0$ that acts in a vertical plane. Determine the largest permissible value of the moment M_0 of the couple if the maximum stress in the beam is not to exceed 12 ksi. *Given:* $I_y = I_z = 11.3 \text{ in.}^4$, $A = 4.75 \text{ in.}^2$, $k_{\min} = 0.983 \text{ in.}$ (*Hint:* By reason of symmetry, the principal axes form an angle of 45° with the coordinate axes. Use the relations $I_{\min} = Ak_{\min}^2$ and $I_{\min} + I_{\max} = I_y + I_z$)

SOLUTION

$$M_u = M_0 \sin 45^\circ = 0.70711 M_0$$

$$M_v = M_0 \cos 45^\circ = 0.7071 M_0$$

$$I_{\min} = Ak_{\min}^2 = (4.75)(0.983)^2 = 4.59 \text{ in.}^4$$

$$I_{\max} = I_y + I_z - I_{\min} = 11.3 + 11.3 - 4.59 = 18.01 \text{ in.}^4$$

$$u_B = y_B \cos 45^\circ + z_B \sin 45^\circ = -3.57 \cos 45^\circ + 0.93 \sin 45^\circ = -1.866 \text{ in.}$$

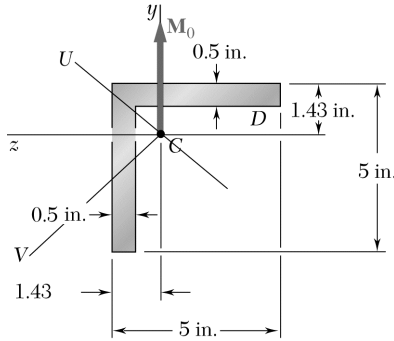
$$v_B = z_B \cos 45^\circ - y_B \sin 45^\circ = 0.93 \cos 45^\circ - (-3.57) \sin 45^\circ = 3.182 \text{ in.}$$

$$\begin{aligned} \sigma_B &= -\frac{M_v u_B}{I_v} + \frac{M_u v_B}{I_u} = -0.70711 M_0 \left[-\frac{u_B}{I_{\min}} + \frac{v_B}{I_{\max}} \right] \\ &= 0.70711 M_0 \left[-\frac{(-1.866)}{4.59} + \frac{3.182}{18.01} \right] = 0.4124 M_0 \end{aligned}$$

$$M_0 = \frac{\sigma_B}{0.4124} = \frac{12}{0.4124}$$

$$M_0 = 29.1 \text{ kip} \cdot \text{in} \quad \blacktriangleleft$$

PROBLEM 4.153



Solve Prob. 4.152, assuming that the couple $\mathbf{M}_0$ acts in a horizontal plane.

PROBLEM 4.152 A beam having the cross section shown is subjected to a couple $\mathbf{M}_0$ that acts in a vertical plane. Determine the largest permissible value of the moment M_0 of the couple if the maximum stress in the beam is not to exceed 12 ksi. *Given:* $I_y = I_z = 11.3 \text{ in.}^4$, $A = 4.75 \text{ in.}^2$, $k_{\min} = 0.983 \text{ in.}$ (*Hint:* By reason of symmetry, the principal axes form an angle of 45° with the coordinate axes. Use the relations $I_{\min} = Ak_{\min}^2$ and $I_{\min} + I_{\max} = I_y + I_z$)

SOLUTION

$$M_u = M_0 \cos 45^\circ = 0.70711 M_0$$

$$M_v = -M_0 \sin 45^\circ = -0.70711 M_0$$

$$I_{\min} = Ak_{\min}^2 = (4.75)(0.983)^2 = 4.59 \text{ in.}^4$$

$$I_{\max} = I_y + I_z - I_{\min} = 11.3 + 11.3 - 4.59 = 18.01 \text{ in.}^4$$

$$u_D = y_D \cos 45^\circ + z_D \sin 45^\circ = -0.93 \cos 45^\circ + (-3.57 \sin 45^\circ) = -1.866 \text{ in.}$$

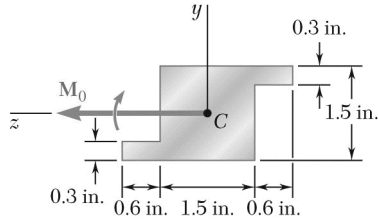
$$v_D = z_D \cos 45^\circ - y_D \sin 45^\circ = (-3.57) \cos 45^\circ - (0.93) \sin 45^\circ = 3.182 \text{ in.}$$

$$\begin{aligned} \sigma_D &= -\frac{M_v u_D}{I_v} + \frac{M_u v_D}{I_u} = -0.70711 M_0 \left[-\frac{u_D}{I_{\min}} + \frac{v_D}{I_{\max}} \right] \\ &= 0.70711 M_0 \left[-\frac{(-1.866)}{4.59} + \frac{3.182}{18.01} \right] = 0.4124 M_0 \end{aligned}$$

$$M_0 = \frac{\sigma_D}{0.4124} = \frac{12}{0.4124}$$

$$M_0 = 29.1 \text{ kip} \cdot \text{in} \quad \blacktriangleleft$$

PROBLEM 4.154



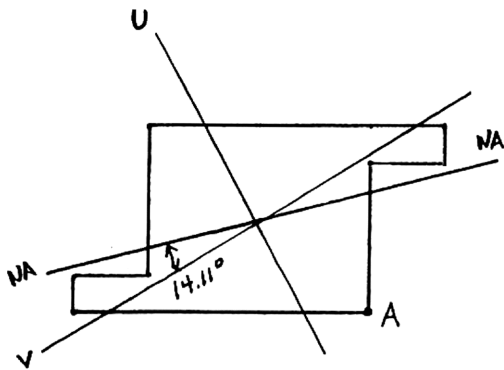
An extruded aluminum member having the cross section shown is subjected to a couple acting in a vertical plane. Determine the largest permissible value of the moment M_0 of the couple if the maximum stress is not to exceed 12 ksi. *Given:* $I_{\max} = 0.957 \text{ in}^4$, $I_{\min} = 0.427 \text{ in}^4$, principal axes 29.4° and 60.6° .

SOLUTION

$$I_u = I_{\max} = 0.957 \text{ in}^4$$

$$I_v = I_{\min} = 0.427 \text{ in}^4$$

$$M_u = M_0 \sin 29.4^\circ, \quad M_v = M_0 \cos 29.4^\circ$$



$$\theta = 29.4^\circ$$

$$\begin{aligned} \tan \phi &= \frac{I_v}{I_u} \tan \theta = \frac{0.427}{0.957} \tan 29.4^\circ \\ &= 0.2514 \quad \phi = 14.11^\circ \end{aligned}$$

Point A is farthest from the neutral axis.

$$y_A = -0.75 \text{ in.}, \quad z_A = -0.75 \text{ in.}$$

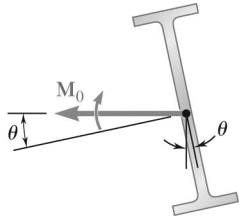
$$u_A = y_A \cos 29.4^\circ + z_A \sin 29.4^\circ = -1.0216 \text{ in.}$$

$$v_A = z_A \cos 29.4^\circ - y_A \sin 29.4^\circ = -0.2852 \text{ in.}$$

$$\begin{aligned} \sigma_A &= -\frac{M_v u_A}{I_v} + \frac{M_u v_A}{I_u} = -\frac{(M_0 \cos 29.4^\circ)(-1.0216)}{0.427} + \frac{(M_0 \sin 29.4^\circ)(-0.2852)}{0.957} \\ &= 1.9381 M_0 \end{aligned}$$

$$M_0 = \frac{\sigma_A}{1.9381} = \frac{12}{1.9381}$$

$$M_0 = 6.19 \text{ kip} \cdot \text{in} \quad \blacktriangleleft$$



PROBLEM 4.155

A couple M_0 acting in a vertical plane is applied to a W12 $\times$ 16 rolled-steel beam, whose web forms an angle θ with the vertical. Denoting by σ_0 the maximum stress in the beam when $\theta = 0$, determine the angle of inclination θ of the beam for which the maximum stress is $2\sigma_0$.

SOLUTION

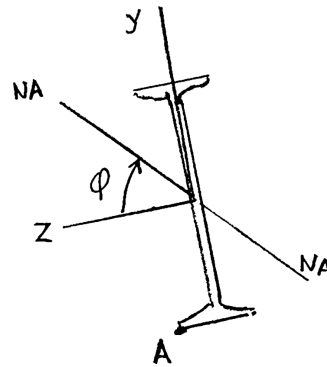
For W12 $\times$ 16 rolled steel section,

$$I_z = 103 \text{ in}^4 \quad I_y = 2.82 \text{ in}^4$$

$$d = 11.99 \text{ in.} \quad b_f = 3.990 \text{ in.}$$

$$y_A = -\frac{d}{2} \quad z_A = \frac{b_f}{2}$$

$$\tan \phi = \frac{I_z}{I_y} \tan \theta = \frac{103}{2.82} \tan \theta = 36.52 \tan \theta$$



Point A is farthest from the neutral axis.

$$M_y = M_0 \sin \theta \quad M_z = M_0 \cos \theta$$

$$\sigma_A = -\frac{M_z y_A}{I_z} + \frac{M_y z_A}{I_y} = \frac{M_0 d}{2I_z} \cos \theta + \frac{M_0 b_f}{2I_y} \sin \theta = \frac{M_0 d}{2I_z} \left(1 + \frac{I_z b_f}{I_y d} \tan \theta \right) \cos \theta$$

$$\text{For } \theta = 0, \quad \sigma_0 = \frac{M_0 d}{2I_z}$$

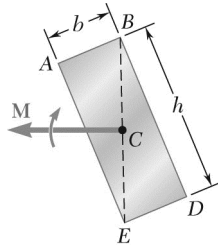
$$\sigma_A = \sigma_0 \left(1 + \frac{I_z b_f}{I_y d} \tan \theta \right) \cos \theta = 2\sigma_0$$

$$\frac{I_z b_f}{I_y d} \tan \theta = \frac{(103)(3.990)}{(2.82)(11.99)} \left(\frac{2}{\cos \theta} - 1 \right)$$

$$\tan \theta = 0.082273 \left(\frac{2}{\cos \theta} - 1 \right)$$

Assuming $\cos \theta \approx 1$, we get

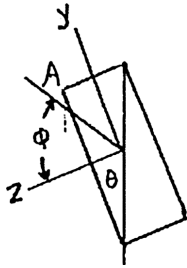
$$\theta = 4.70^\circ \quad \blacktriangleleft$$



PROBLEM 4.156

Show that, if a solid rectangular beam is bent by a couple applied in a plane containing one diagonal of a rectangular cross section, the neutral axis will lie along the other diagonal.

SOLUTION



$$\tan \theta = \frac{b}{h}$$

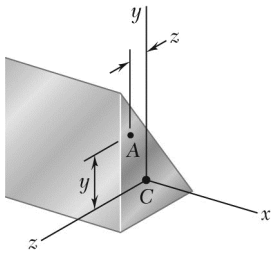
$$M_z = M \cos \theta, \quad M_z = M \sin \theta$$

$$I_z = \frac{1}{12} b h^3 \quad I_y = \frac{1}{12} h b^3$$

$$\tan \phi = \frac{I_z}{I_y} \tan \theta = \frac{\frac{1}{12} b h^3}{\frac{1}{12} h b^3} \cdot \frac{b}{h} = \frac{h}{b}$$

The neutral axis passes through corner *A* of the diagonal *AD*. ◀

PROBLEM 4.157



A beam of unsymmetric cross section is subjected to a couple $\mathbf{M}_0$ acting in the horizontal plane xz . Show that the stress at point A , of coordinates y and z , is

$$\sigma_A = \frac{zI_z - yI_{yz}}{I_yI_z - I_{yz}^2} M_y$$

where I_y , I_z , and I_{yz} denote the moments and product of inertia of the cross section with respect to the coordinate axes, and M_y the moment of the couple.

SOLUTION

The stress σ_A varies linearly with the coordinates y and z . Since the axial force is zero, the y - and z -axes are centroidal axes:

$$\sigma_A = C_1 y + C_2 z$$

where C_1 and C_2 are constants.

$$\begin{aligned} M_z &= -\int y \sigma_A dA = -C_1 \int y^2 dA - C_2 \int yz dA \\ &= -I_z C_1 - I_{yz} C_2 = 0 \end{aligned}$$

$$C_1 = -\frac{I_{yz}}{I_z} C_2$$

$$\begin{aligned} M_y &= \int z \sigma_A dA = C_1 \int yz dA + C_2 \int z^2 dA \\ &= I_{yz} C_1 + I_y C_2 \\ &\quad - I_{yz} \frac{I_{yz}}{I_z} C_2 + I_y C_2 \end{aligned}$$

$$I_z M_y = (I_y I_z - I_{yz}^2) C_2$$

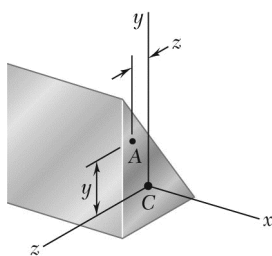
$$C_2 = \frac{I_z M_y}{I_y I_z - I_{yz}^2}$$

$$C_1 = -\frac{I_{yz} M_y}{I_y I_z - I_{yz}^2}$$

$$\sigma_A = \frac{I_z z - I_{yz} y}{I_y I_z - I_{yz}^2} M_y$$



PROBLEM 4.158



A beam of unsymmetric cross section is subjected to a couple $\mathbf{M}_0$ acting in the vertical plane xy . Show that the stress at point A , of coordinates y and z , is

$$\sigma_A = \frac{yI_y - zI_{yz}}{I_yI_z - I_{yz}^2} M_z$$

where I_y , I_z , and I_{yz} denote the moments and product of inertia of the cross section with respect to the coordinate axes, and M_z the moment of the couple.

SOLUTION

The stress σ_A varies linearly with the coordinates y and z . Since the axial force is zero, the y - and z -axes are centroidal axes:

$$\sigma_A = C_1 y + C_2 z$$

where C_1 and C_2 are constants.

$$\begin{aligned} M_y = \int z \sigma_A dA &= C_1 \int yz dA + C_2 \int z^2 dA \\ &= I_{yz} C_1 + I_y C_2 = 0 \end{aligned}$$

$$C_2 = -\frac{I_{yz}}{I_y} C_1$$

$$\begin{aligned} M_z = -\int y \sigma_A dz &= -C_1 \int y^2 dA + C_2 \int yz dA \\ &= -I_z C_1 - I_{yz} \frac{I_{yz}}{I_y} C_1 \end{aligned}$$

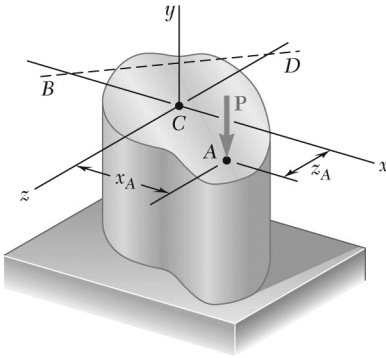
$$I_y M_z = -(I_y I_z - I_{yz}^2) C_1$$

$$C_1 = -\frac{I_y M_z}{I_y I_z - I_{yz}^2}$$

$$C_2 = +\frac{I_{yz} M_z}{I_y I_z - I_{yz}^2}$$

$$\sigma_A = -\frac{I_y y - I_{yz} z}{I_y I_z - I_{yz}^2} M_z$$

PROBLEM 4.159



(a) Show that, if a vertical force $\mathbf{P}$ is applied at point A of the section shown, the equation of the neutral axis BD is

$$\left(\frac{x_A}{r_z^2} \right) x + \left(\frac{z_A}{r_x^2} \right) z = -1$$

where r_z and r_x denote the radius of gyration of the cross section with respect to the z axis and the x axis, respectively. (b) Further show that, if a vertical force $\mathbf{Q}$ is applied at any point located on line BD , the stress at point A will be zero.

SOLUTION

Definitions:

$$r_x^2 = \frac{I_x}{A}, \quad r_z^2 = \frac{I_z}{A}$$

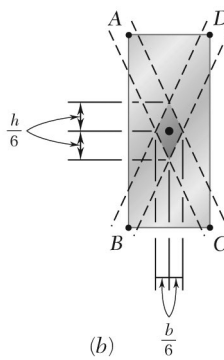
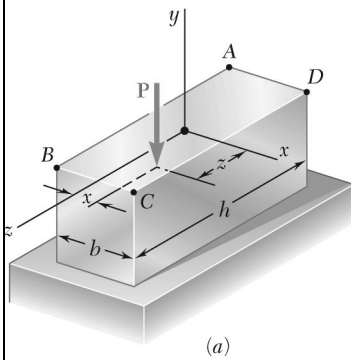
$$\begin{aligned} (a) \quad M_x &= Pz_A \quad M_z = -Px_A \\ \sigma_E &= -\frac{P}{A} + \frac{M_z x_E}{I_z} - \frac{M_x z_E}{I_x} = -\frac{P}{A} - \frac{Px_A x_E}{Ar_z^2} - \frac{Pz_A z_E}{Ar_x^2} \\ &= -\frac{P}{A} \left[1 + \left(\frac{x_A}{r_z^2} \right) x_E + \left(\frac{z_A}{r_x^2} \right) z_E \right] = 0 \end{aligned}$$

if E lies on neutral axis.

$$1 + \left(\frac{x_A}{r_z^2} \right) x + \left(\frac{z_A}{r_x^2} \right) z = 0, \quad \left(\frac{x_A}{r_z^2} \right) x + \left(\frac{z_A}{r_x^2} \right) z = -1$$

$$\begin{aligned} (b) \quad M_x &= Pz_E \quad M_z = -Px_E \\ \sigma_A &= -\frac{P}{A} + \frac{M_z x_A}{I_z} - \frac{M_x z_A}{I_x} = -\frac{P}{A} - \frac{Px_E x_A}{Ar_z^2} - \frac{Pz_E z_A}{Ar_x^2} \\ &= 0 \text{ by equation from part (a).} \end{aligned}$$

PROBLEM 4.160



(a) Show that the stress at corner A of the prismatic member shown in part a of the figure will be zero if the vertical force $\mathbf{P}$ is applied at a point located on the line

$$\frac{x}{b/6} + \frac{z}{h/6} = 1$$

(b) Further show that, if no tensile stress is to occur in the member, the force $\mathbf{P}$ must be applied at a point located within the area bounded by the line found in part a and three similar lines corresponding to the condition of zero stress at B , C , and D , respectively. This area, shown in part b of the figure, is known as the *kern* of the cross section.

SOLUTION

$$I_z = \frac{1}{12}hb^3 \quad I_x = \frac{1}{12}bh^3 \quad A = bh$$

$$z_A = -\frac{h}{2} \quad x_A = -\frac{b}{2}$$

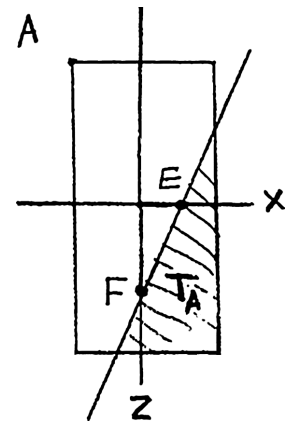
Let P be the load point.

$$M_z = -Px_P \quad M_x = Pz_P$$

$$\sigma_A = -\frac{P}{A} + \frac{M_z x_A}{I_z} - \frac{M_x z_A}{I_x}$$

$$= -\frac{P}{bh} + \frac{(-Px_P)\left(-\frac{b}{2}\right)}{\frac{1}{12}hb^3} - \frac{(Pz_P - \frac{h}{2})}{\frac{1}{12}bh^3}$$

$$= -\frac{P}{bh} \left[1 - \frac{x_P}{b/6} - \frac{z_P}{h/6} \right]$$



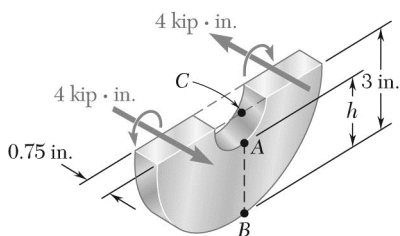
(a) For $\sigma_A = 0$, $1 - \frac{x}{b/6} - \frac{z}{h/6} = 0 \quad \frac{x}{b/6} + \frac{z}{h/6} = 1$

(b) At point E : $z = 0 \quad \therefore x_E = b/6$

At point F : $x = 0 \quad \therefore z_F = h/6$

If the line of action (x_P, z_P) lies within the portion marked T_A , a tensile stress will occur at corner A .

By considering $\sigma_B = 0$, $\sigma_C = 0$, and $\sigma_D = 0$, the other portions producing tensile stresses are identified.



PROBLEM 4.161

For the machine component and loading shown, determine the stress at point A when (a) $h = 2$ in., (b) $h = 2.6$ in.

SOLUTION

$$M = -4 \text{ kip} \cdot \text{in}$$

Rectangular cross section: $A = bh$ $r_2 = 3 \text{ in.}$ $r_1 = r_2 - h$

$$\bar{r} = \frac{1}{2}(r_1 + r_2), \quad R = \frac{h}{\ln \frac{r_2}{r_1}}, \quad e = \bar{r} - R$$

(a) $h = 2 \text{ in.}$ $A = (0.75)(2) = 1.5 \text{ in}^2$

$$r_1 = 3 - 2 = 1 \text{ in.} \quad \bar{r} = \frac{1}{2}(3 + 1) = 2 \text{ in.}$$

$$R = \frac{2}{\ln \frac{3}{1}} = 1.8205 \text{ in.} \quad e = 2 - 1.8205 = 0.1795 \text{ in.}$$

At point A : $r = r_1 = 1 \text{ in.}$

$$\sigma_A = \frac{M(r - R)}{Aer} = \frac{(-4)(1 - 1.8205)}{(1.5)(0.1795)(1)} = 12.19 \text{ ksi}$$

$$\sigma_A = 12.19 \text{ ksi} \quad \blacktriangleleft$$

(b) $h = 2.6 \text{ in.}$ $A = (0.75)(2.6) = 1.95 \text{ in}^2$

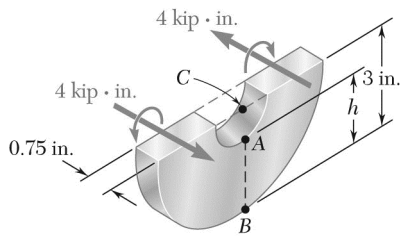
$$r_1 = 3 - 2.6 = 0.4 \text{ in.} \quad \bar{r} = \frac{1}{2}(3 + 0.4) = 1.7 \text{ in.}$$

$$R = \frac{2.6}{\ln \frac{3}{0.4}} = 1.2904 \text{ in.} \quad e = 1.7 - 1.2904 = 0.4096 \text{ in.}$$

At point A : $r = r_1 = 0.4 \text{ in.}$

$$\sigma_A = \frac{M(r - R)}{Aer} = \frac{(-4)(0.4 - 1.2904)}{(1.95)(0.4096)(0.4)} = 11.15 \text{ ksi}$$

$$\sigma_A = 11.15 \text{ ksi} \quad \blacktriangleleft$$



PROBLEM 4.162

For the machine component and loading shown, determine the stress at points *A* and *B* when $h = 2.5$ in.

SOLUTION

$$M = -4 \text{ kip} \cdot \text{in}$$

Rectangular cross section: $h = 2.5 \text{ in.}$ $b = 0.75 \text{ in.}$ $A = 1.875 \text{ in.}^2$

$$r_2 = 3 \text{ in.} \quad r_1 = r_2 - h = 0.5 \text{ in.}$$

$$\bar{r} = \frac{1}{2}(r_1 + r_2) = \frac{1}{2}(0.5 + 3.0) = 1.75 \text{ in.}$$

$$R = \frac{h}{\ln \frac{r_2}{r_1}} = \frac{2.5}{\ln \frac{3.0}{0.5}} = 1.3953 \text{ in.}$$

$$e = \bar{r} - R = 1.75 - 1.3953 = 0.3547 \text{ in.}$$

At point *A*:

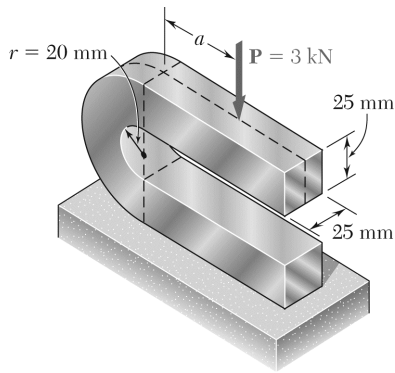
$$r = r_1 = 0.5 \text{ in.}$$

$$\sigma_A = \frac{M(r - R)}{Aer} = \frac{(-4 \text{ kip} \cdot \text{in})(0.5 \text{ in.} - 1.3953 \text{ in.})}{(0.75 \text{ in.})(2.5 \text{ in.})(0.3547 \text{ in.})(0.5 \text{ in.})} \quad \sigma_A = 10.77 \text{ ksi} \quad \blacktriangleleft$$

At point *B*:

$$r = r_2 = 3 \text{ in.}$$

$$\sigma_B = \frac{M(r - R)}{Aer} = \frac{(-4 \text{ kip} \cdot \text{in})(3 \text{ in.} - 1.3953 \text{ in.})}{(0.75 \text{ in.} \times 2.5 \text{ in.})(0.3547 \text{ in.})(3 \text{ in.})} \quad \sigma_B = -3.22 \text{ ksi} \quad \blacktriangleleft$$



PROBLEM 4.163

The curved portion of the bar shown has an inner radius of 20 mm. Knowing that the allowable stress in the bar is 150 MPa, determine the largest permissible distance a from the line of action of the 3-kN force to the vertical plane containing the center of curvature of the bar.

SOLUTION

Reduce the internal forces transmitted across section AB to a force-couple system at the centroid of the section. The bending couple is

$$M = P(a + \bar{r})$$

For the rectangular section, the neutral axis for bending couple only lies at

$$R = \frac{h}{\ln \frac{r_2}{r_1}}$$

$$\text{Also } e = \bar{r} - R$$

The maximum compressive stress occurs at point A . It is given by

$$\sigma_A = -\frac{P}{A} - \frac{My_A}{Aer_1} = -\frac{P}{A} - \frac{P(a + \bar{r})y_A}{Aer_1} = -K \frac{P}{A}$$

with

$$y_A = R - r_1$$

Thus,

$$K = 1 + \frac{(a + \bar{r})(R - r_1)}{er_1}$$

Data:

$$h = 25 \text{ mm}, \quad r_1 = 20 \text{ mm}, \quad r_2 = 45 \text{ mm}, \quad \bar{r} = 32.5 \text{ mm}$$

$$R = \frac{25}{\ln \frac{45}{20}} = 30.8288 \text{ mm}, \quad e = 32.5 - 30.8288 = 1.6712 \text{ mm}$$

$$b = 25 \text{ mm}, \quad A = bh = (25)(25) = 625 \text{ mm}^2 = 625 \times 10^{-6} \text{ m}^2 \quad R - r_1 = 10.8288 \text{ mm}$$

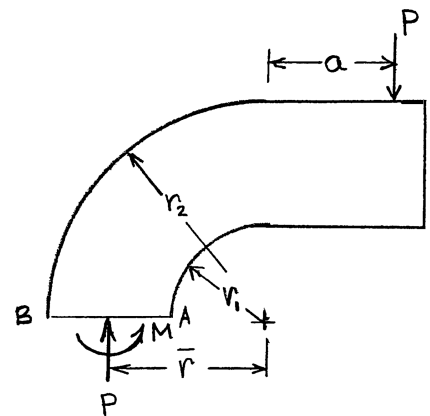
$$P = 3 \times 10^3 \text{ N} \cdot \text{m}, \quad \sigma_A = -150 \times 10^6 \text{ Pa}$$

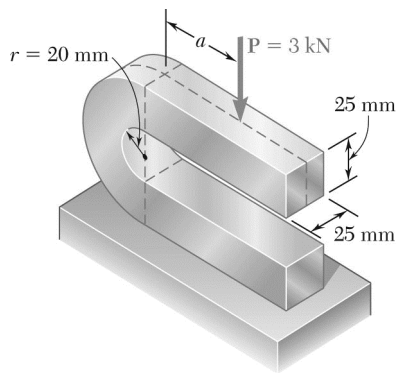
$$K = -\frac{\sigma_A A}{P} = -\frac{(-150 \times 10^6)(625 \times 10^{-6})}{3 \times 10^3} = 31.25$$

$$a + \bar{r} = \frac{(K - 1)er_1}{R - r_1} = \frac{(30.25)(1.6712)(20)}{10.8288} = 93.37 \text{ mm}$$

$$a = 93.37 - 32.5$$

$$a = 60.9 \text{ mm} \quad \blacktriangleleft$$





PROBLEM 4.164

The curved portion of the bar shown has an inner radius of 20 mm. Knowing that the line of action of the 3-kN force is located at a distance $a = 60$ mm from the vertical plane containing the center of curvature of the bar, determine the largest compressive stress in the bar.

SOLUTION

Reduce the internal forces transmitted across section AB to a force-couple system at the centroid of the section. The bending couple is

$$M = P(a + \bar{r})$$

For the rectangular section, the neutral axis for bending couple only lies at

$$R = \frac{h}{\ln \frac{r_2}{r_1}}$$

$$\text{Also } e = \bar{r} - R$$

The maximum compressive stress occurs at point A . It is given by

$$\sigma_A = -\frac{P}{A} - \frac{My_A}{Aer_1} = -\frac{P}{A} - \frac{P(a + \bar{r})y_A}{Aer_1} = -K \frac{P}{A}$$

$$\text{with } y_A = R - r_1$$

$$\text{Thus, } K = 1 + \frac{(a + \bar{r})(R - r_1)}{er_1}$$

$$\text{Data: } h = 25 \text{ mm, } r_1 = 20 \text{ mm, } r_2 = 45 \text{ mm, } \bar{r} = 32.5 \text{ mm}$$

$$R = \frac{25}{\ln \frac{45}{20}} = 30.8288 \text{ mm, } e = 32.5 - 30.8288 = 1.6712 \text{ mm}$$

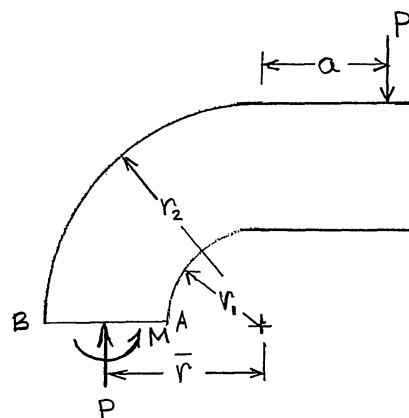
$$b = 25 \text{ mm, } A = bh = (25)(25) = 625 \text{ mm}^2 = 625 \times 10^{-6} \text{ m}^2$$

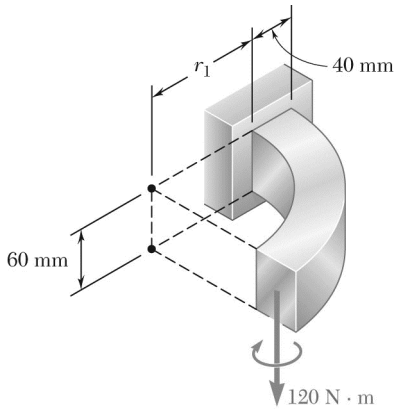
$$a = 60 \text{ mm, } a + \bar{r} = 92.5 \text{ mm, } R - r_1 = 10.8288 \text{ mm}$$

$$K = 1 + \frac{(92.5)(10.8288)}{(1.6712)(20)} = 30.968 \quad P = 30 \times 10^3 \text{ N}$$

$$\sigma_A = \frac{KP}{A} = -\frac{(30.968)(3 \times 10^3)}{625 \times 10^{-6}} = -148.6 \times 10^6 \text{ Pa}$$

$$\sigma_A = -148.6 \text{ MPa} \quad \blacktriangleleft$$





PROBLEM 4.165

The curved bar shown has a cross section of 40×60 mm and an inner radius $r_1 = 15$ mm. For the loading shown, determine the largest tensile and compressive stresses.

SOLUTION

$$h = 40 \text{ mm}, \quad r_1 = 15 \text{ mm}, \quad r_2 = 55 \text{ mm}$$

$$A = (60)(40) = 2400 \text{ mm}^2 = 2400 \times 10^{-6} \text{ m}^2$$

$$R = \frac{h}{\ln \frac{r_2}{r_1}} = \frac{40}{\ln \frac{55}{15}} = 30.786 \text{ mm}$$

$$\bar{r} = \frac{1}{2}(r_1 + r_2) = 35 \text{ mm}$$

$$e = \bar{r} - R = 4.214 \text{ mm} \qquad \sigma = -\frac{My}{Aer}$$

$$\text{At } r = 15 \text{ mm}, \quad y = 30.786 - 15 = 15.756 \text{ mm}$$

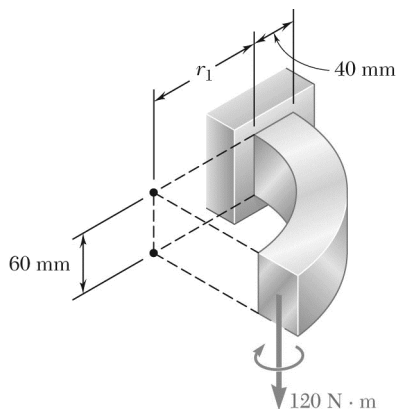
$$\sigma = -\frac{(120)(15.786 \times 10^{-3})}{(2400 \times 10^{-6})(4.214 \times 10^{-3})(15 \times 10^{-3})} = -12.49 \times 10^6 \text{ Pa}$$

$$\sigma = -12.49 \text{ MPa} \quad \blacktriangleleft$$

$$\text{At } r = 55 \text{ mm}, \quad y = 30.786 - 55 = -24.214 \text{ mm}$$

$$\sigma = -\frac{(120)(-24.214 \times 10^{-3})}{(2400 \times 10^{-6})(4.214 \times 10^{-3})(55 \times 10^{-3})} = 5.22 \times 10^6 \text{ Pa}$$

$$\sigma = 5.22 \text{ MPa} \quad \blacktriangleleft$$



PROBLEM 4.166

For the curved bar and loading shown, determine the percent error introduced in the computation of the maximum stress by assuming that the bar is straight. Consider the case when (a) $r_1 = 20$ mm, (b) $r_1 = 200$ mm, (c) $r_1 = 2$ m.

SOLUTION

$$h = 40 \text{ mm}, A = (60)(40) = 2400 \text{ mm}^2 = 2400 \times 10^{-6} \text{ m}^2, M = 120 \text{ N} \cdot \text{m}$$

$$I = \frac{1}{12}bh^3 = \frac{1}{12}(60)(40)^3 = 0.32 \times 10^6 \text{ mm}^4 = 0.32 \times 10^{-6} \text{ m}^4, c = \frac{1}{2}h = 20 \text{ mm}$$

Assuming that the bar is straight

$$\sigma_s = -\frac{Mc}{I} = -\frac{(120)(20 \times 10^{-3})}{(0.32 \times 10^{-6})} = 7.5 \times 10^6 \text{ Pa} = 7.5 \text{ MPa}$$

(a) $r_1 = 20 \text{ mm}$ $r_2 = 60 \text{ mm}$

$$R = \frac{h}{\ln \frac{r_2}{r_1}} = \frac{40}{\ln \frac{60}{20}} = 36.4096 \text{ mm} \quad r_1 - R = -16.4096 \text{ mm}$$

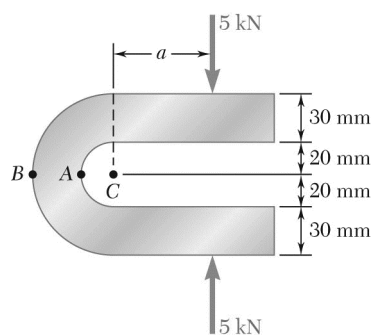
$$\bar{r} = \frac{1}{2}(r_1 + r_2) = 40 \text{ mm} \quad e = \bar{r} - R = 3.5904 \text{ mm}$$

$$\sigma_a = \frac{M(r_1 - R)}{Aer} = \frac{(120)(-16.4096 \times 10^{-3})}{(2400 \times 10^{-6})(3.5904 \times 10^{-3})(20 \times 10^{-3})} = -11.426 \times 10^6 \text{ Pa} = -11.426 \text{ MPa}$$

$$\% \text{ error} = \frac{-11.426 - (-7.5)}{-11.426} \times 100\% = -34.4\%$$

For parts (b) and (c), we get the values in the table below:

	r_1 , mm	r_2 , mm	R , mm	$\bar{r}$, mm	e , mm	σ , MPa	% error
(a)	20	60	36.4096	40	3.5904	-11.426	-34.4 %
(b)	200	240	219.3926	220	0.6074	-7.982	6.0 %
(c)	2000	2040	2019.9340	2020	0.0660	-7.546	0.6 %



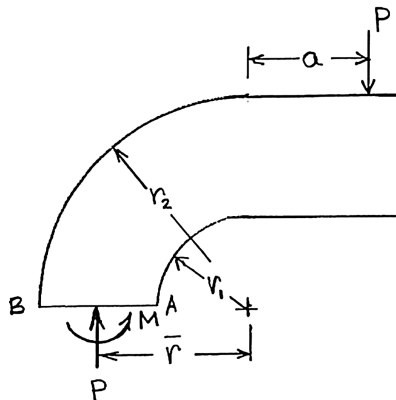
PROBLEM 4.167

The curved bar shown has a cross section of 30×30 mm. Knowing that $a = 60$ mm, determine the stress at (a) point A, (b) point B.

SOLUTION

Reduce the internal forces transmitted across section AB to a force-couple system at the centroid of the section. The bending couple is

$$M = P(a + \bar{r})$$



For the rectangular section, the neutral axis for bending couple only lies at

$$R = \frac{h}{\ln \frac{r_2}{r_1}}$$

$$\text{Also } e = \bar{r} - R$$

The maximum compressive stress occurs at point A. It is given by

$$\sigma_A = -\frac{P}{A} - \frac{My_A}{Aer_1} = -\frac{P}{A} - \frac{P(a + \bar{r})y_A}{Aer_1}$$

$$= -K \frac{P}{A} \quad \text{with} \quad y_A = R - r_1$$

$$\text{Thus,} \quad K = 1 + \frac{(a + \bar{r})(R - r_1)}{er_1}$$

$$\text{Data:} \quad h = 30 \text{ mm}, \quad r_1 = 20 \text{ mm}, \quad r_2 = 50 \text{ mm}, \quad \bar{r} = 35 \text{ mm}$$

$$R = \frac{30}{\ln \frac{50}{20}} = 32.7407 \text{ mm}, \quad e = 35 - 32.7407 = 2.2593 \text{ mm}$$

$$b = 30 \text{ mm}, \quad A = bh = (30)(30) = 900 \text{ mm}^2 = 900 \times 10^{-6} \text{ m}^2$$

$$a = 60 \text{ mm}, \quad a + \bar{r} = 95 \text{ mm}, \quad R - r_1 = 12.7407 \text{ mm}$$

$$K = 1 + \frac{(95)(12.7407)}{(2.2593)(20)} = 27.786$$

$$P = 5 \times 10^3 \text{ N}$$

PROBLEM 4.167 (Continued)

$$(a) \quad \sigma_A = -\frac{KP}{A} = -\frac{(27.786)(5 \times 10^3)}{900 \times 10^{-6}} = -154.4 \times 10^6 \text{ Pa}$$

$$\sigma_A = -154.4 \text{ MPa} \quad \blacktriangleleft$$

$$(b) \quad \text{At point B,} \quad y_B = r_2 - R = 50 - 32.7407 = 17.2953 \text{ mm}$$

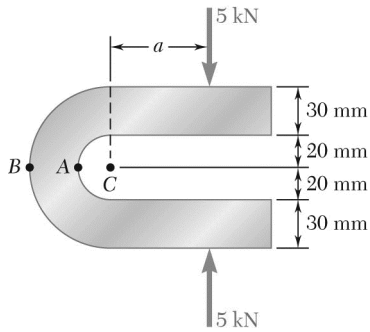
$$\sigma_B = -\frac{P}{A} + \frac{My_B}{Aer_2} = \frac{P}{A} \left[-1 + \frac{(a + \bar{r})y_B}{er_2} \right]$$

$$= \frac{K'P}{A} \quad \text{where} \quad K' = \frac{(a + \bar{r})y_B}{er_2} - 1$$

$$K' = \frac{(95)(17.2953)}{(2.2593)(50)} - 1 = 13.545$$

$$\sigma_B = \frac{(13.545)(5 \times 10^3)}{900 \times 10^{-6}} = 75.2 \times 10^6 \text{ Pa}$$

$$\sigma_B = 75.2 \text{ MPa} \quad \blacktriangleleft$$



PROBLEM 4.168

The curved bar shown has a cross section of 30×30 mm. Knowing that the allowable compressive stress is 175 MPa, determine the largest allowable distance a .

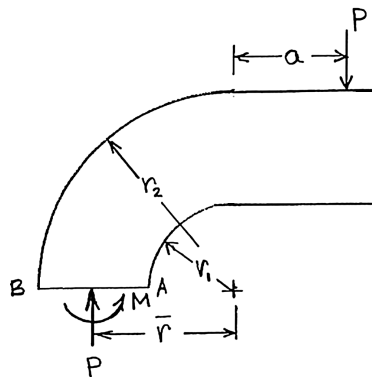
SOLUTION

Reduce the internal forces transmitted across section AB to a force-couple system at the centroid of the section. The bending couple is

$$M = P(a + \bar{r})$$

For the rectangular section, the neutral axis for bending couple only lies at

$$R = \frac{h}{\ln \frac{r_2}{r_1}}$$



$$\text{Also } e = \bar{r} - R$$

The maximum compressive stress occurs at point A . It is given by

$$\begin{aligned} \sigma_A &= -\frac{P}{A} - \frac{My_A}{Aer_1} = -\frac{P}{A} - \frac{P(a + \bar{r})y_A}{Aer_1} \\ &= -K \frac{P}{A} \quad \text{with } y_A = R - r_1 \end{aligned}$$

$$\text{Thus, } K = 1 + \frac{(a + \bar{r})(R - r_1)}{er_1} \quad (1)$$

$$\text{Data: } h = 30 \text{ mm, } r_1 = 20 \text{ mm, } r_2 = 50 \text{ mm, } \bar{r} = 35 \text{ mm, } R = \frac{30}{\ln \frac{50}{20}} = 32.7407 \text{ mm}$$

$$e = 35 - 32.7407 = 2.2593 \text{ mm, } b = 30 \text{ mm, } R - r_1 = 12.7407 \text{ mm, } a = ?$$

$$\sigma_A = -175 \text{ MPa} = -175 \times 10^6 \text{ Pa, } P = 5 \text{ kN} = 5 \times 10^3 \text{ N}$$

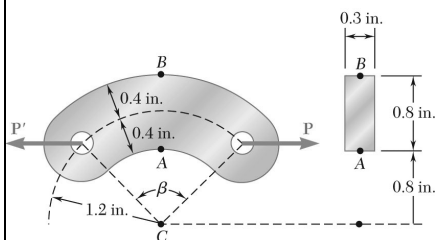
$$K = -\frac{A\sigma_A}{P} = -\frac{(900 \times 10^{-6})(-175 \times 10^6)}{5 \times 10^3} = 31.5$$

$$\text{Solving (1) for } a + \bar{r}, \quad a + \bar{r} = \frac{(K - 1)er_1}{R - r_1}$$

$$a + \bar{r} = \frac{(30.5)(2.2593)(20)}{12.7407} = 108.17 \text{ mm}$$

$$a = 108.17 \text{ mm} - 35 \text{ mm}$$

$$a = 73.2 \text{ mm} \quad \blacktriangleleft$$



PROBLEM 4.169

Steel links having the cross section shown are available with different central angles β . Knowing that the allowable stress is 12 ksi, determine the largest force $\mathbf{P}$ that can be applied to a link for which $\beta = 90^\circ$.

SOLUTION

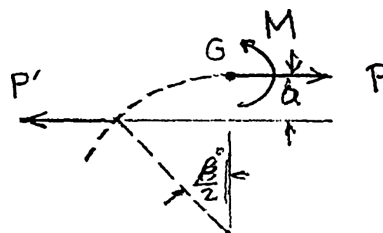
Reduce section force to a force-couple system at G ; the centroid of the cross section AB .

$$a = \bar{r} \left(1 - \cos \frac{\beta}{2} \right)$$

The bending couple is $M = -Pa$.

For the rectangular section, the neutral axis for bending couple only lies at

$$R = \frac{h}{\ln \frac{r_2}{r_1}}$$



Also $e = \bar{r} - R$

At point A the tensile stress is
$$\sigma_A = \frac{P}{A} - \frac{My_A}{Aer_1} = \frac{P}{A} + \frac{Pay_A}{Aer_1} = \frac{P}{A} \left(1 + \frac{ay_A}{er_1} \right) = K \frac{P}{A}$$

where
$$K = 1 + \frac{ay_A}{er_1} \quad \text{and} \quad y_A = R - r_1$$

$$P = \frac{A\sigma_A}{K}$$

Data: $\bar{r} = 1.2 \text{ in.}, \quad r_1 = 0.8 \text{ in.}, \quad r_2 = 1.6 \text{ in.}, \quad h = 0.8 \text{ in.}, \quad b = 0.3 \text{ in.}$

$$A = (0.3)(0.8) = 0.24 \text{ in}^2 \quad R = \frac{0.8}{\ln \frac{1.6}{0.8}} = 1.154156 \text{ in.}$$

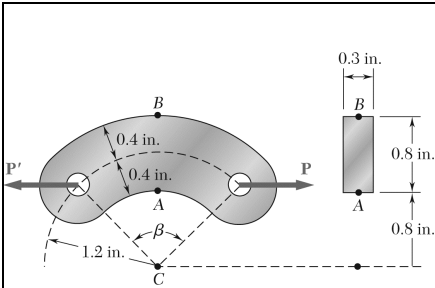
$$e = 1.2 - 1.154156 = 0.045844 \text{ in.}, \quad y_A = 1.154156 - 0.8 = 0.35416 \text{ in.}$$

$$a = 1.2(1 - \cos 45^\circ) = 0.35147 \text{ in.}$$

$$K = 1 + \frac{(0.35147)(0.35416)}{(0.045844)(0.8)} = 4.3940$$

$$P = \frac{(0.24)(12)}{4.3940} = 0.65544 \text{ kips}$$

$$P = 655 \text{ lb} \quad \blacktriangleleft$$



PROBLEM 4.170

Solve Prob. 4.169, assuming that $\beta = 60^\circ$.

PROBLEM 4.169 Steel links having the cross section shown are available with different central angles β . Knowing that the allowable stress is 12 ksi, determine the largest force P that can be applied to a link for which $\beta = 90^\circ$.

SOLUTION

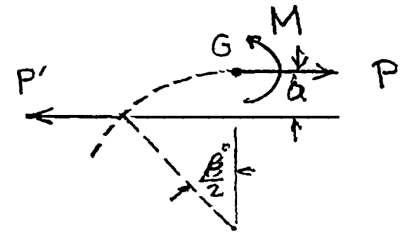
Reduce section force to a force-couple system at G , the centroid of the cross section AB .

$$a = \bar{r} \left(1 - \cos \frac{\beta}{2} \right)$$

The bending couple is $M = -Pa$.

For the rectangular section, the neutral axis for bending couple only lies at

$$R = \frac{h}{\ln \frac{r_2}{r_1}}$$



Also $e = \bar{r} - R$

At point A , the tensile stress is
$$\sigma_A = \frac{P}{A} - \frac{My_A}{Aer_1} = \frac{P}{A} + \frac{Pay_A}{Aer_1} = \frac{P}{A} \left(1 + \frac{ay_A}{er_1} \right) = K \frac{P}{A}$$

where
$$K = 1 + \frac{ay_A}{er_1} \quad \text{and} \quad y_A = R - r_1$$

$$P = \frac{A\sigma_A}{K}$$

Data: $\bar{r} = 1.2 \text{ in.}, \quad r_1 = 0.8 \text{ in.}, \quad r_2 = 1.6 \text{ in.}, \quad h = 0.8 \text{ in.}, \quad b = 0.3 \text{ in.}$

$$A = (0.3)(0.8) = 0.24 \text{ in}^2 \quad R = \frac{0.8}{\ln \frac{1.6}{0.8}} = 1.154156 \text{ in.}$$

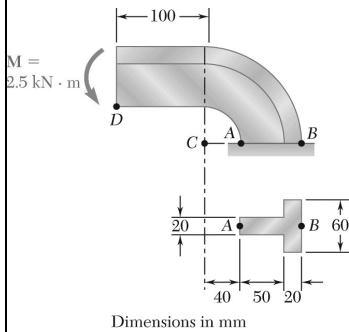
$$e = 1.2 - 1.154156 = 0.045844 \text{ in.} \quad y_A = 1.154156 - 0.8 = 0.35416 \text{ in.}$$

$$a = (1.2)(1 - \cos 30^\circ) = 0.160770 \text{ in.}$$

$$K = 1 + \frac{(0.160770)(0.35416)}{(0.045844)(0.8)} = 2.5525$$

$$P = \frac{(0.24)(12)}{2.5525} = 1.128 \text{ kips}$$

$$P = 1128 \text{ lb} \quad \blacktriangleleft$$



PROBLEM 4.171

A machine component has a T-shaped cross section that is orientated as shown. Knowing that $M = 2500 \text{ N} \cdot \text{m}$, determine the stress at (a) point A, (b) point B.

SOLUTION

Properties of the cross section.

$$R = \frac{\sum A}{\sum \int \frac{1}{r} dA} = \frac{\sum b_i h_i}{\sum b_i \ln \frac{r_{i+1}}{r_i}} = \frac{\sum A_i}{\sum b_i \ln \frac{r_{i+1}}{r_i}} \quad \bar{r} = \frac{\sum A_i \bar{r}_i}{\sum A_i}$$

r , mm	Part	b , mm	h , mm	A , mm ²	$b_i \ln \frac{r_{i+1}}{r_i}$, mm	$\bar{r}_i$, mm	$A_i \bar{r}_i$, mm ³
40	1	20	50	1000	16.2186	65	65,000
90	2	60	20	1200	12.0402	100	120,000
110	Σ			2200	28.2588		185,000

$$R = \frac{2200}{28.2588} = 77.852 \text{ mm}, \quad \bar{r} = \frac{185000}{2200} = 84.091 \text{ mm}, \quad e = \bar{r} - R = 6.239 \text{ mm}$$

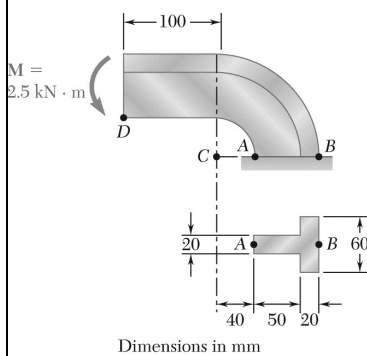
Stresses.

(a) Point A: $r = r_A = 40 \text{ mm}$

$$\sigma_A = \frac{M(r - R)}{Aer} = \frac{(2.5 \text{ kN} \cdot \text{m})(0.040 \text{ m} - 0.0778517 \text{ m})}{(2.2 \times 10^{-3} \text{ m}^2)(6.2392 \times 10^{-3} \text{ m})(0.040 \text{ m})} \quad \sigma_A = -172.4 \text{ MPa} \blacktriangleleft$$

(b) Point B: $r = r_B = 110 \text{ mm}$

$$\sigma_B = \frac{M(r - R)}{Aer} = \frac{(2.5 \text{ kN} \cdot \text{m})(0.110 \text{ m} - 0.0778517 \text{ m})}{(2.2 \times 10^{-3} \text{ m}^2)(6.2392 \times 10^{-3} \text{ m})(0.110 \text{ m})} \quad \sigma_B = 53.2 \text{ MPa} \blacktriangleleft$$



PROBLEM 4.172

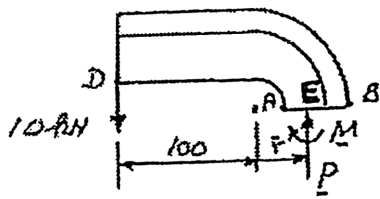
Assuming that the couple shown is replaced by a vertical 10-kN force attached at point *D* and acting downward, determine the stress at (a) point *A*, (b) point *B*.

SOLUTION

Properties of the cross section. $R = \frac{\sum A}{\sum \int \frac{1}{r} dA} = \frac{\sum b_i h_i}{\sum b_i \ln \frac{r_{i+1}}{r_i}} = \frac{\sum A_i}{\sum b_i \ln \frac{r_{i+1}}{r_i}} \quad \bar{r} = \frac{\sum A_i \bar{r}_i}{\sum A_i}$

r , mm	Part	b , mm	h , mm	A , mm ²	$b_i \ln \frac{r_{i+1}}{r_i}$, mm	$\bar{r}_i$, mm	$A_i \bar{r}_i$, mm ³
40	1	20	50	1000	16.2186	65	65,000
90	2	60	20	1200	12.0402	100	120,000
110	Σ			2200	28.2588		185,000

$$R = \frac{2200}{28.2588} = 77.852 \text{ mm}, \quad \bar{r} = \frac{185000}{2200} = 84.091 \text{ mm}, \quad e = \bar{r} - R = 6.239 \text{ mm}$$



Force-couple system at the centroid *E*.

$$P = 10 \text{ kN}$$

$$M = (10 \text{ kN})(100 \text{ mm} + 84.0909 \text{ mm}) = 1840.9 \text{ N} \cdot \text{m}$$

PROBLEM 4.172 (Continued)

Stresses.

(a) Point A: $r = r_A = 40 \text{ mm}$

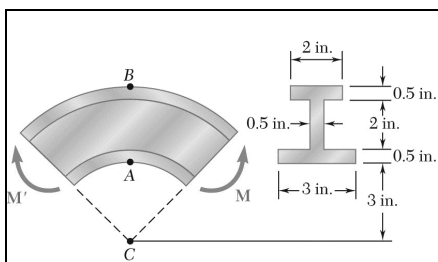
$$\begin{aligned}\sigma_A &= -\frac{P}{A} + \frac{M(r - R)}{Aer} = -\frac{10 \text{ kN}}{2.2 \times 10^{-3} \text{ m}^2} + \frac{(1840.9 \text{ N} \cdot \text{m})(0.040 \text{ m} - 0.0778517 \text{ m})}{(2.2 \times 10^{-3} \text{ m}^2)(6.2392 \times 10^{-3} \text{ m})(0.040 \text{ m})} \\ &= -4.545 \text{ MPa} - 126.913 \text{ MPa}\end{aligned}$$

$$\sigma_A = -131.5 \text{ MPa} \quad \blacktriangleleft$$

(b) Point B: $r = r_B = 110 \text{ mm}$

$$\begin{aligned}\sigma_B &= -\frac{P}{A} + \frac{M(r - R)}{Aer} = -\frac{10 \text{ kN}}{2.2 \times 10^{-3} \text{ m}^2} + \frac{(1840.9 \text{ N} \cdot \text{m})(0.110 \text{ m} - 0.0778517 \text{ m})}{(2.2 \times 10^{-3} \text{ m}^2)(6.2392 \times 10^{-3} \text{ m})(0.110 \text{ m})} \\ &= -4.545 \text{ MPa} + 39.196 \text{ MPa}\end{aligned}$$

$$\sigma_B = 34.7 \text{ MPa} \quad \blacktriangleleft$$



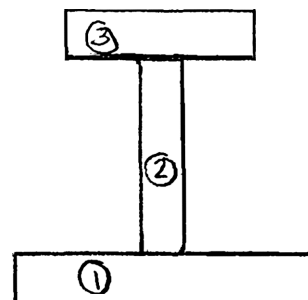
PROBLEM 4.173

Three plates are welded together to form the curved beam shown. For the given loading, determine the distance e between the neutral axis and the centroid of the cross section.

SOLUTION

$$R = \frac{\Sigma A}{\Sigma \int \frac{1}{r} dA} = \frac{\Sigma b_i h_i}{\Sigma b_i \ln \frac{r_{i+1}}{r_i}} = \frac{\Sigma A}{\Sigma b_i \ln \frac{r_{i+1}}{r_i}}$$

$$\bar{r} = \frac{\Sigma A \bar{r}_i}{\Sigma A}$$

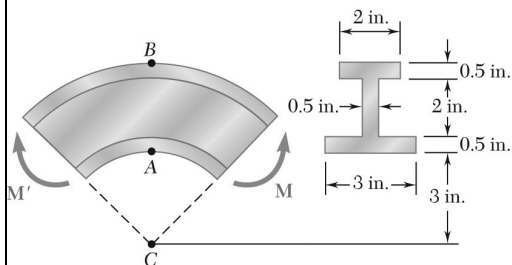


r	Part	b	h	A	$b \ln \frac{r_{i+1}}{r_i}$	$\bar{r}$	$A \bar{r}$
3	①	3	0.5	1.5	0.462452	3.25	4.875
3.5	②	0.5	2	1.0	0.225993	4.5	4.5
5.5	③	2	0.5	1.0	0.174023	5.75	5.75
6	Σ			3.5	0.862468		15.125

$$R = \frac{3.5}{0.862468} = 4.05812 \text{ in.}, \quad \bar{r} = \frac{15.125}{3.5} = 4.32143 \text{ in.}$$

$$e = \bar{r} - R = 0.26331 \text{ in.}$$

$$e = 0.263 \text{ in.} \quad \blacktriangleleft$$



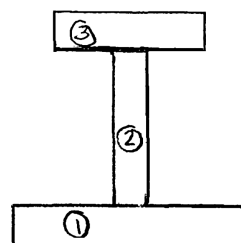
PROBLEM 4.174

Three plates are welded together to form the curved beam shown. For $M = 8 \text{ kip} \cdot \text{in.}$, determine the stress at (a) point A, (b) point B, (c) the centroid of the cross section.

SOLUTION

$$R = \frac{\Sigma A}{\Sigma \int \frac{1}{r} dA} = \frac{\Sigma b_i h_i}{\Sigma b_i \ln \frac{r_{i+1}}{r_i}} = \frac{\Sigma A}{\Sigma b_i \ln \frac{r_{i+1}}{r_i}}$$

$$\bar{r} = \frac{\Sigma A \bar{r}_i}{\Sigma A}$$



r	Part	b	h	A	$b \ln \frac{r_{i+1}}{r_i}$	$\bar{r}$	$A\bar{r}$
3	①	3	0.5	1.5	0.462452	3.25	4.875
3.5	②	0.5	2	1.0	0.225993	4.5	4.5
5.5	③	2	0.5	1.0	0.174023	5.75	5.75
6	Σ			3.5	0.862468		15.125

$$R = \frac{3.5}{0.862468} = 4.05812 \text{ in.}, \quad \bar{r} = \frac{15.125}{3.5} = 4.32143 \text{ in.}$$

$$e = \bar{r} - R = 0.26331 \text{ in.} \quad M = -8 \text{ kip} \cdot \text{in.}$$

(a) $y_A = R - r_1 = 4.05812 - 3 = 1.05812 \text{ in.}$

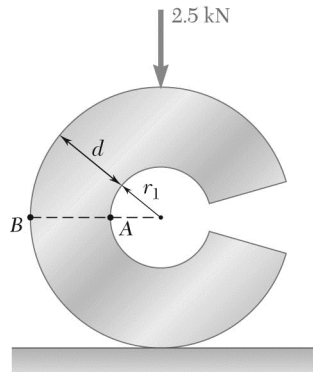
$$\sigma_A = -\frac{My_A}{Aer_1} = -\frac{(-8)(1.05812)}{(3.5)(0.26331)(3)} \quad \sigma_A = 3.06 \text{ ksi} \blacktriangleleft$$

(b) $y_B = R - r_2 = 4.05812 - 6 = -1.94188 \text{ in.}$

$$\sigma_B = -\frac{My_B}{Aer_2} = -\frac{(-8)(-1.94188)}{(3.5)(0.26331)(6)} \quad \sigma_B = -2.81 \text{ ksi} \blacktriangleleft$$

(c) $y_C = R - \bar{r} = -e$

$$\sigma_C = -\frac{My_C}{Ae\bar{r}} = -\frac{Me}{Ae\bar{r}} = -\frac{M}{A\bar{r}} = -\frac{-8}{(3.5)(4.32143)} \quad \sigma_C = 0.529 \text{ ksi} \blacktriangleleft$$



PROBLEM 4.175

The split ring shown has an inner radius $r_1 = 20$ mm and a circular cross section of diameter $d = 32$ mm. For the loading shown, determine the stress at (a) point A, (b) point B.

SOLUTION

$$c = \frac{1}{2}d = 16 \text{ mm}, \quad r_1 = 20 \text{ mm}, \quad r_2 = r_1 + d = 52 \text{ mm}$$

$$\bar{r} = r_1 + c = 36 \text{ mm}$$

$$R = \frac{1}{2} \left[\bar{r} + \sqrt{\bar{r}^2 - c^2} \right] = \frac{1}{2} \left[36 + \sqrt{36^2 - 16^2} \right]$$

$$= 34.1245 \text{ mm}$$

$$e = \bar{r} - R = 1.8755 \text{ mm}$$

$$A = \frac{\pi}{4}d^2 = \frac{\pi}{4}(32)^2 = 804.25 \text{ mm}^2 = 804.25 \times 10^{-6} \text{ m}^2$$

$$P = 2.5 \times 10^3 \text{ N} \quad M = P\bar{r} = (2.5 \times 10^3)(36 \times 10^{-3}) = 90 \text{ N} \cdot \text{m}$$

(a) Point A: $y_A = R - r_1 = 34.1245 - 20 = 14.125 \text{ mm}$

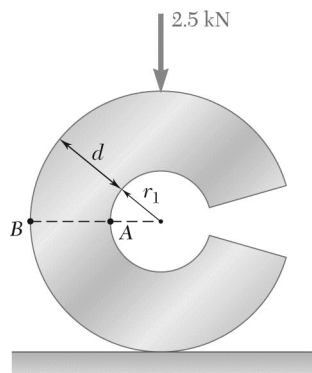
$$\sigma_A = -\frac{P}{A} - \frac{My_A}{Aer_1} = -\frac{2.5 \times 10^3}{804.25 \times 10^{-6}} - \frac{(90)(14.1245 \times 10^{-3})}{(804.25 \times 10^{-6})(1.8755 \times 10^{-3})(20 \times 10^{-3})}$$

$$= -45.2 \times 10^6 \text{ Pa} \quad \sigma_A = -45.2 \text{ MPa} \quad \blacktriangleleft$$

(b) Point B: $y_B = R - r_2 = 34.1245 - 52 = -17.8755 \text{ mm}$

$$\sigma_B = -\frac{P}{A} - \frac{My_B}{Aer_2} = -\frac{2.5 \times 10^3}{804.25 \times 10^{-6}} - \frac{(90)(-17.8755 \times 10^{-3})}{(804.25 \times 10^{-6})(1.8755 \times 10^{-3})(52 \times 10^{-3})}$$

$$= 17.40 \times 10^6 \text{ Pa} \quad \sigma_B = 17.40 \text{ MPa} \quad \blacktriangleleft$$



PROBLEM 4.176

The split ring shown has an inner radius $r_1 = 16$ mm and a circular cross section of diameter $d = 32$ mm. For the loading shown, determine the stress at (a) point A, (b) point B.

SOLUTION

$$c = \frac{1}{2}d = 16 \text{ mm}, \quad r_1 = 16 \text{ mm}, \quad r_2 = r_1 + d = 48 \text{ mm}$$

$$\bar{r} = r_1 + c = 32 \text{ mm}$$

$$R = \frac{1}{2} \left[\bar{r} + \sqrt{\bar{r}^2 - c^2} \right] = \frac{1}{2} \left[32 + \sqrt{32^2 - 16^2} \right]$$

$$= 29.8564 \text{ mm}$$

$$e = \bar{r} - R = 2.1436 \text{ mm}$$

$$A = \frac{\pi}{4} d^2 = \frac{\pi}{4} (32)^2 = 804.25 \text{ mm}^2 = 804.25 \times 10^{-6} \text{ m}^2$$

$$P = 2.5 \times 10^3 \text{ N} \quad M = P\bar{r} = (2.5 \times 10^3)(32 \times 10^{-3}) = 80 \text{ N} \cdot \text{m}$$

(a) Point A: $y_A = R - r_1 = 29.8564 - 16 = 13.8564 \text{ mm}$

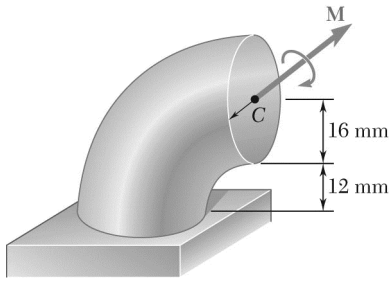
$$\sigma_A = -\frac{P}{A} - \frac{My_A}{Aer_1} = -\frac{2.5 \times 10^3}{804.25 \times 10^{-6}} - \frac{(80)(13.8564 \times 10^{-3})}{(804.25 \times 10^{-6})(2.1436 \times 10^{-3})(16 \times 10^{-3})}$$

$$= -43.3 \times 10^6 \text{ Pa} \quad \sigma_A = -43.3 \text{ MPa} \quad \blacktriangleleft$$

(b) Point B: $y_B = R - r_2 = 29.8564 - 48 = -18.1436 \text{ mm}$

$$\sigma_B = -\frac{P}{A} - \frac{My_B}{Aer_2} = -\frac{2.5 \times 10^3}{804.25 \times 10^{-6}} - \frac{(80)(-18.1436 \times 10^{-3})}{(804.25 \times 10^{-6})(2.1436 \times 10^{-3})(48 \times 10^{-3})}$$

$$= 14.43 \times 10^6 \text{ Pa} \quad \sigma_B = 14.43 \text{ MPa} \quad \blacktriangleleft$$



PROBLEM 4.177

The curved bar shown has a circular cross section of 32-mm diameter. Determine the largest couple **M** that can be applied to the bar about a horizontal axis if the maximum stress is not to exceed 60 MPa.

SOLUTION

$$c = 16 \text{ mm} \quad \bar{r} = 12 + 16 = 28 \text{ mm}$$

$$R = \frac{1}{2} \left[\bar{r} + \sqrt{\bar{r}^2 - c^2} \right]$$

$$= \frac{1}{2} \left[28 + \sqrt{28^2 - 16^2} \right] = 25.4891 \text{ mm}$$

$$e = \bar{r} - R = 28 - 25.4891 = 2.5109 \text{ mm}$$

$\sigma_{\max}$ occurs at *A*, which lies at the inner radius.

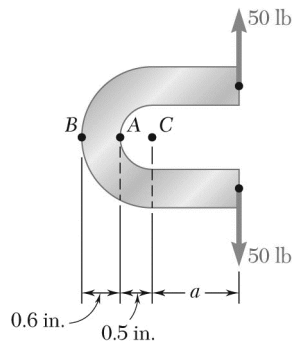
It is given by $|\sigma_{\max}| = \left| \frac{My_A}{Aer_1} \right|$ from which $M = \frac{Aer_1|\sigma_{\max}|}{y_A}$.

Also, $A = \pi c^2 = \pi(16)^2 = 804.25 \text{ mm}^2$

Data: $y_A = R - r_1 = 25.4891 - 12 = 13.4891 \text{ mm}$

$$M = \frac{(804.25 \times 10^{-6})(2.5109 \times 10^{-3})(12 \times 10^{-3})(60 \times 10^6)}{13.4891 \times 10^{-3}}$$

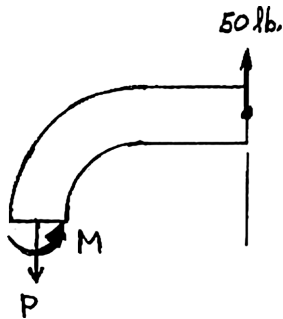
$$M = 107.8 \text{ N} \cdot \text{m} \quad \blacktriangleleft$$



PROBLEM 4.178

The bar shown has a circular cross section of 0.6 in. diameter. Knowing that $a = 1.2$ in., determine the stress at (a) point A, (b) point B.

SOLUTION



$$c = \frac{1}{2}d = 0.3 \text{ in.} \quad \bar{r} = 0.5 + 0.3 = 0.8 \text{ in.}$$

$$R = \frac{1}{2} \left[\bar{r} + \sqrt{\bar{r}^2 - c^2} \right] = \frac{1}{2} \left[0.8 + \sqrt{0.8^2 - 0.3^2} \right] \\ = 0.77081 \text{ in.}$$

$$e = \bar{r} - R = 0.02919 \text{ in.}$$

$$A = \pi c^2 = \pi(0.3)^2 = 0.28274 \text{ in}^2$$

$$P = 50 \text{ lb}$$

$$M = -P(a + \bar{r}) = -50(1.2 + 0.8) = -100 \text{ lb} \cdot \text{in}$$

$$y_A = R - r_1 = 0.77081 - 0.5 = 0.27081 \text{ in.}$$

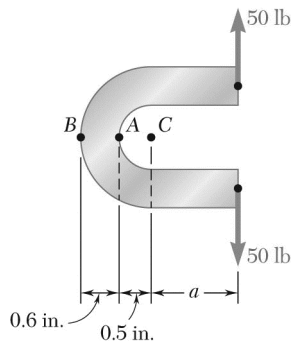
$$y_B = R - r_2 = 0.77081 - 1.1 = -0.32919 \text{ in.}$$

$$(a) \quad \sigma_A = \frac{P}{A} - \frac{My_A}{Aer_1} = \frac{50}{0.28274} - \frac{(-100)(0.27081)}{(0.28274)(0.02919)(0.5)} = 6.74 \times 10^3 \text{ psi}$$

$$\sigma_A = 6.74 \text{ ksi} \quad \blacktriangleleft$$

$$(b) \quad \sigma_B = \frac{P}{A} - \frac{My_B}{Aer_2} = \frac{50}{0.28274} - \frac{(-100)(-0.32919)}{(0.28274)(0.02919)(1.1)} = -3.45 \times 10^3 \text{ psi}$$

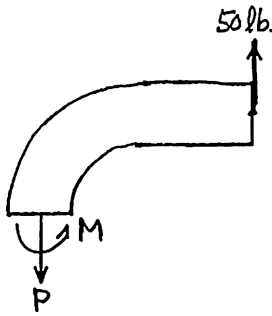
$$\sigma_B = -3.45 \text{ ksi} \quad \blacktriangleleft$$



PROBLEM 4.179

The bar shown has a circular cross section of 0.6 in. diameter. Knowing that the allowable stress is 8 ksi, determine the largest permissible distance a from the line of action of the 50-lb forces to the plane containing the center of curvature of the bar.

SOLUTION



$$c = \frac{1}{2}d = 0.3 \text{ in.}, \quad \bar{r} = 0.5 + 0.3 = 0.8 \text{ in.}$$

$$R = \frac{1}{2} \left[\bar{r} + \sqrt{\bar{r}^2 - c^2} \right] = \frac{1}{2} \left[0.8 + \sqrt{0.8^2 - 0.3^2} \right]$$

$$= 0.77081 \text{ in.} \quad e = \bar{r} - R = 0.02919 \text{ in.}$$

$$A = \pi c^2 = \pi(0.3)^2 = 0.28274 \text{ in}^2$$

$$M = -P(a + \bar{r})$$

$$y_A = R - r_1 = 0.77081 - 0.5 = 0.27081 \text{ in.}$$

$$\sigma_A = \frac{P}{A} - \frac{My_A}{Aer_1} = \frac{P}{A} + \frac{P(a + \bar{r})y_A}{Aer_1} = \frac{P}{A} \left[1 + \frac{(a + \bar{r})y_A}{er_1} \right]$$

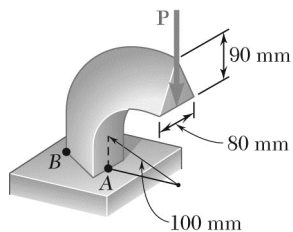
$$= \frac{KP}{A} \quad \text{where} \quad K = 1 + \frac{(a + \bar{r})y_A}{er_1}$$

$$K = \frac{\sigma_A A}{P} = \frac{(8 \times 10^3)(0.28274)}{50} = 45.238$$

$$a + \bar{r} = \frac{(K - 1)er_1}{y_A} = \frac{(44.238)(0.02919)(0.5)}{0.27081} = 2.384 \text{ in.}$$

$$a = 2.384 - 0.8$$

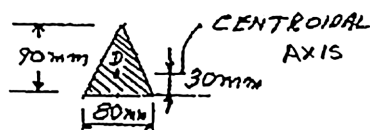
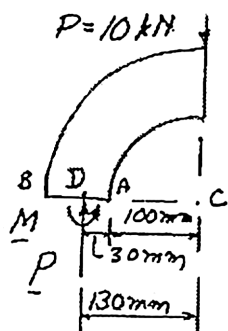
$$a = 1.584 \text{ in.} \quad \blacktriangleleft$$



PROBLEM 4.180

Knowing that $P = 10 \text{ kN}$, determine the stress at (a) point A, (b) point B.

SOLUTION



Locate the centroid D of the cross section.

$$\bar{r} = 100 \text{ mm} + \frac{90 \text{ mm}}{3} = 130 \text{ mm}$$

Force-couple system at D .

$$P = 10 \text{ kN}$$

$$M = P\bar{r} = (10 \text{ kN})(130 \text{ mm}) = 1300 \text{ N} \cdot \text{m}$$

Triangular cross section.

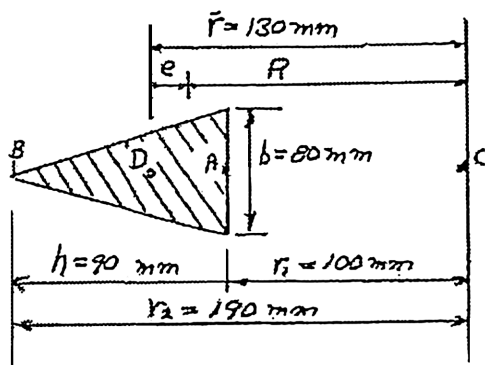
$$A = \frac{1}{2}bh = \frac{1}{2}(90 \text{ mm})(80 \text{ mm})$$

$$= 3600 \text{ mm}^2 = 3600 \times 10^{-6} \text{ m}^2$$

$$R = \frac{\frac{1}{2}h}{\frac{r_2 \ln \frac{r_2}{r_1} - 1}} = \frac{\frac{1}{2}(90)}{\frac{190 \ln \frac{190}{100} - 1}} = \frac{45 \text{ mm}}{0.355025}$$

$$R = 126.752 \text{ mm}$$

$$e = \bar{r} - R = 130 \text{ mm} - 126.752 \text{ mm} = 3.248 \text{ mm}$$



(a) Point A: $r_A = 100 \text{ mm} = 0.100 \text{ m}$

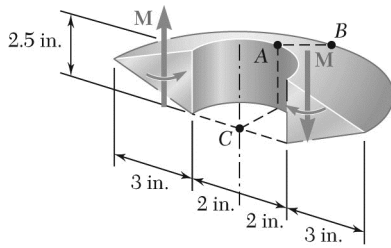
$$\sigma_A = -\frac{P}{A} + \frac{M(r_A - R)}{Aer_A} = -\frac{10 \text{ kN}}{3600 \times 10^{-6} \text{ m}^2} + \frac{(1300 \text{ N} \cdot \text{m})(0.100 \text{ m} - 0.126752 \text{ m})}{(3600 \times 10^{-6} \text{ m}^2)(3.248 \times 10^{-3} \text{ m})(0.100 \text{ m})}$$

$$= -2.778 \text{ MPa} - 29.743 \text{ MPa} \quad \sigma_A = -32.5 \text{ MPa} \quad \blacktriangleleft$$

(b) Point B: $r_B = 190 \text{ mm} = 0.190 \text{ m}$

$$\sigma_B = -\frac{P}{A} + \frac{M(r_B - R)}{Aer_B} = -\frac{10 \text{ kN}}{3600 \times 10^{-6} \text{ m}^2} + \frac{(1300 \text{ N} \cdot \text{m})(0.190 \text{ m} - 0.126752 \text{ m})}{(3600 \times 10^{-6} \text{ m}^2)(3.248 \times 10^{-3} \text{ m})(0.190 \text{ m})}$$

$$= -2.778 \text{ MPa} + 37.01 \text{ MPa} \quad \sigma_B = +34.2 \text{ MPa} \quad \blacktriangleleft$$



PROBLEM 4.181

Knowing that $M = 5 \text{ kip} \cdot \text{in.}$, determine the stress at (a) point A, (b) point B.

SOLUTION

$$A = \frac{1}{2}bh = \frac{1}{2}(2.5)(3) = 3.75 \text{ in}^2$$

$$\bar{r} = 2 + 1 = 3.00000 \text{ in.}$$

$$b_1 = 2.5 \text{ in.}, \quad r_1 = 2 \text{ in.}, \quad b_2 = 0, \quad r_2 = 5 \text{ in.}$$

Use formula for trapezoid with $b_2 = 0$.

$$R = \frac{\frac{1}{2}h^2(b_1 + b_2)}{(b_1r_2 - b_2r_1)\ln\frac{r_2}{r_1} - h(b_1 - b_2)}$$

$$= \frac{(0.5)(3)^2(2.5 + 0)}{[(2.5)(5) - (0)(2)]\ln\frac{5}{2} - (3)(2.5 - 0)} = 2.84548 \text{ in.}$$

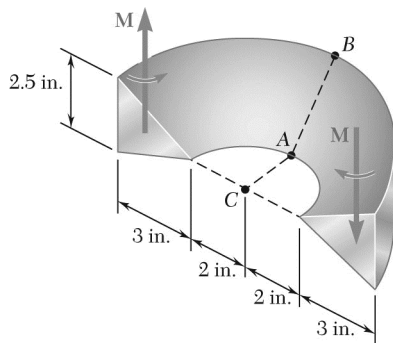
$$e = \bar{r} - R = 0.15452 \text{ in.} \quad M = 5 \text{ kip} \cdot \text{in.}$$

$$(a) \quad y_A = R - r_1 = 0.84548 \text{ in.}$$

$$\sigma_A = -\frac{My_A}{Aer_1} = -\frac{(5)(0.84548)}{(3.75)(0.15452)(2)} \quad \sigma_A = -3.65 \text{ ksi} \quad \blacktriangleleft$$

$$(b) \quad y_B = R - r_2 = -2.15452 \text{ in.}$$

$$\sigma_B = -\frac{My_B}{Aer_2} = -\frac{(5)(-2.15452)}{(3.75)(0.15452)(5)} \quad \sigma_B = 3.72 \text{ ksi} \quad \blacktriangleleft$$



PROBLEM 4.182

Knowing that $M = 5 \text{ kip} \cdot \text{in}$, determine the stress at (a) point A, (b) point B.

SOLUTION

$$A = \frac{1}{2}(2.5)(3) = 3.75 \text{ in}^2$$

$$\bar{r} = 2 + 2 = 4.00000 \text{ in.}$$

$$b_1 = 0, \quad r_1 = 2 \text{ in.}, \quad b_2 = 2.5 \text{ in.}, \quad r_2 = 5 \text{ in.}$$

Use formula for trapezoid with $b_1 = 0$.

$$R = \frac{\frac{1}{2}h^2(b_1 + b_2)}{(b_1r_2 - b_2r_1) \ln \frac{r_2}{r_1} - h(b_1 - b_2)}$$

$$= \frac{(0.5)(3)^2(0 + 2.5)}{[(0)(5) - (2.5)(2)] \ln \frac{5}{2} - (3)(0 - 2.5)} = 3.85466 \text{ in.}$$

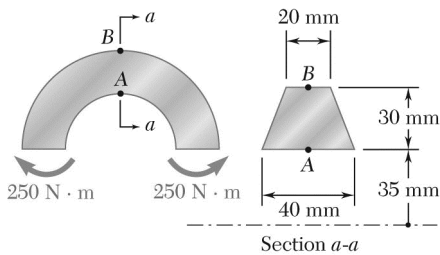
$$e = \bar{r} - R = 0.14534 \text{ in.} \quad M = 5 \text{ kip} \cdot \text{in}$$

$$(a) \quad y_A = R - r_1 = 1.85466 \text{ in.}$$

$$\sigma_A = -\frac{My_A}{Aer_1} = -\frac{(5)(1.85466)}{(3.75)(0.14534)(2)} \quad \sigma_A = -8.51 \text{ ksi} \quad \blacktriangleleft$$

$$(b) \quad y_B = R - r_2 = -1.14534 \text{ in.}$$

$$\sigma_B = -\frac{My_B}{Aer_2} = -\frac{(5)(-1.14534)}{(3.75)(0.14534)(5)} \quad \sigma_B = 2.10 \text{ ksi} \quad \blacktriangleleft$$



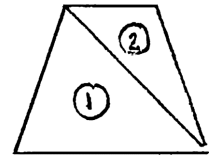
PROBLEM 4.183

For the curved beam and loading shown, determine the stress at (a) point A, (b) point B.

SOLUTION

Locate centroid.

	$A, \text{ mm}^2$	$\bar{r}, \text{ mm}$	$A\bar{r}, \text{ mm}^3$
①	600	45	27×10^3
②	300	55	16.5×10^3
Σ	900		43.5×10^3



$$\bar{r} = \frac{43.5 \times 10^3}{900} = 48.333 \text{ mm}$$

$$R = \frac{\frac{1}{2} h^2 (b_1 + b_2)}{(b_1 r_2 - b_2 r_1) \ln \frac{r_2}{r_1} - h(b_2 - b_1)}$$

$$= \frac{(0.5)(30)^2(40 + 20)}{[(40)(65) - (20)(35)] \ln \frac{65}{35} - (30)(40 - 20)} = 46.8608 \text{ mm}$$

$$e = \bar{r} - R = 1.4725 \text{ mm} \quad M = -250 \text{ N} \cdot \text{m}$$

(a) $y_A = R - r_1 = 11.8608 \text{ mm}$

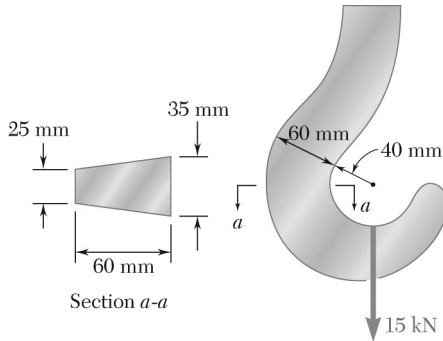
$$\sigma_A = -\frac{My_A}{Aer_1} = -\frac{(-250)(11.8608 \times 10^{-3})}{(900 \times 10^{-6})(1.4725 \times 10^{-3})(35 \times 10^{-3})} = 63.9 \times 10^6 \text{ Pa} \quad \sigma_A = 63.9 \text{ MPa} \quad \blacktriangleleft$$

(b) $y_B = R - r_2 = -18.1392 \text{ mm}$

$$\sigma_B = -\frac{My_B}{Aer_2} = -\frac{(-250)(-18.1392 \times 10^{-3})}{(900 \times 10^{-6})(1.4725 \times 10^{-3})(65 \times 10^{-3})} = -52.6 \times 10^6 \text{ Pa} \quad \sigma_B = -52.6 \text{ MPa} \quad \blacktriangleleft$$

PROBLEM 4.184

For the crane hook shown, determine the largest tensile stress in section $a-a$.



SOLUTION

Locate centroid.

	A, mm^2	$\bar{r}, \text{mm}$	$A\bar{r}, \text{mm}^3$
①	1050	60	63×10^3
②	750	80	60×10^3
Σ	1800		103×10^3

$$\bar{r} = \frac{103 \times 10^3}{1800} = 63.333 \text{ mm}$$

Force-couple system at centroid:

$$P = 15 \times 10^3 \text{ N}$$

$$M = -P\bar{r} = -(15 \times 10^3)(63.333 \times 10^{-3}) = -1.025 \times 10^3 \text{ N} \cdot \text{m}$$

$$R = \frac{\frac{1}{2}h^2(b_1 + b_2)}{(b_1r_2 - b_2r_1)\ln \frac{r_2}{r_1} - h(b_1 - b_2)}$$

$$= \frac{(0.5)(60)^2(35 + 25)}{[(35)(100) - (25)(40)]\ln \frac{100}{40} - (60)(35 + 25)} = 63.878 \text{ mm}$$

$$e = \bar{r} - R = 4.452 \text{ mm}$$

Maximum tensile stress occurs at point A .

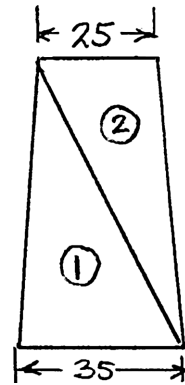
$$y_A = R - r_1 = 23.878 \text{ mm}$$

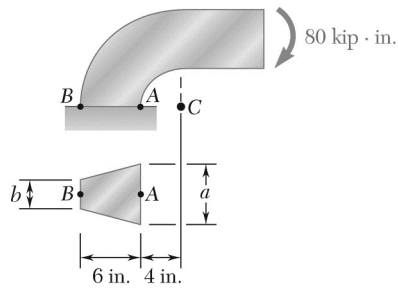
$$\sigma_A = \frac{P}{A} - \frac{My_A}{Aer_1}$$

$$= \frac{15 \times 10^3}{1800 \times 10^{-6}} - \frac{-(1.025 \times 10^3)(23.878 \times 10^{-3})}{(1800 \times 10^{-6})(4.452 \times 10^{-3})(40 \times 10^{-3})}$$

$$= 84.7 \times 10^6 \text{ Pa}$$

$$\sigma_A = 84.7 \text{ MPa} \quad \blacktriangleleft$$





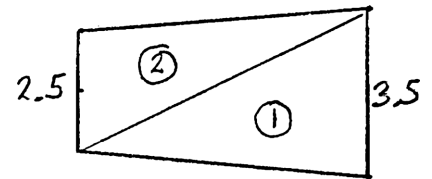
PROBLEM 4.185

Knowing that the machine component shown has a trapezoidal cross section with $a = 3.5$ in. and $b = 2.5$ in., determine the stress at (a) point A, (b) point B.

SOLUTION

Locate centroid.

	A, in^2	$\bar{r}, \text{in.}$	$A\bar{r}, \text{in}^3$
①	10.5	6	63
②	7.5	8	60
Σ	18		123



$$\bar{r} = \frac{123}{18} = 6.8333 \text{ in.}$$

$$R = \frac{\frac{1}{2}h^2(b_1 + b_2)}{(b_1r_2 - b_2r_1)\ln \frac{r_2}{r_1} - h(b_1 - b_2)}$$

$$= \frac{(0.5)(6)^2(3.5 + 2.5)}{[(3.5)(10) - (2.5)(4)]\ln \frac{10}{4} - (6)(3.5 - 2.5)} = 6.3878 \text{ in.}$$

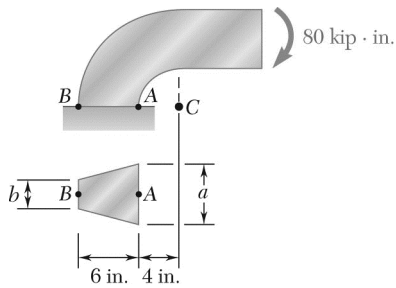
$$e = \bar{r} - R = 0.4452 \text{ in.} \quad M = 80 \text{ kip} \cdot \text{in}$$

(a) $y_A = R - r_1 = 6.3878 - 4 = 2.3878 \text{ in.}$

$$\sigma_A = -\frac{My_A}{Aer_1} = -\frac{(80)(2.3878)}{(18)(0.4452)(4)} \quad \sigma_A = -5.96 \text{ ksi} \quad \blacktriangleleft$$

(b) $y_B = R - r_2 = 6.3878 - 10 = -3.6122 \text{ in.}$

$$\sigma_B = -\frac{My_B}{Aer_2} = -\frac{(80)(-3.6122)}{(18)(0.4452)(10)} \quad \sigma_B = 3.61 \text{ ksi} \quad \blacktriangleleft$$



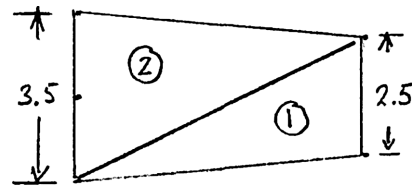
PROBLEM 4.186

Knowing that the machine component shown has a trapezoidal cross section with $a = 2.5$ in. and $b = 3.5$ in., determine the stress at (a) point A, (b) point B.

SOLUTION

Locate centroid.

	A, in^2	$\bar{r}, \text{in.}$	$A\bar{r}, \text{in}^3$
①	7.5	6	45
②	10.5	8	84
Σ	18		129



$$\bar{r} = \frac{129}{18} = 7.1667 \text{ in.}$$

$$R = \frac{\frac{1}{2}h^2(b_1 + b_2)}{(b_1r_2 - b_2r_1)\ln\frac{r_2}{r_1} - h(b_1 - b_2)}$$

$$= \frac{(0.5)(6)^2(2.5 + 3.5)}{[(2.5)(10) - (3.5)(4)]\ln\frac{10}{4} - (6)(2.5 - 3.5)} = 6.7168 \text{ in.}$$

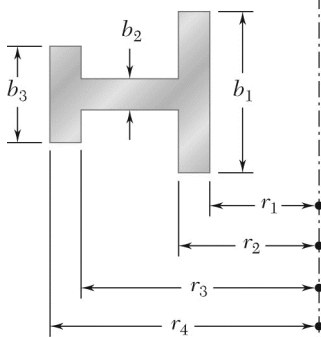
$$e = \bar{r} - R = 0.4499 \text{ in.} \quad M = 80 \text{ kip} \cdot \text{in}$$

(a) $y_A = R - r_1 = 2.7168 \text{ in.}$

$$\sigma_A = -\frac{My_A}{Aer_1} = -\frac{(80)(2.7168)}{(18)(0.4499)(4)} \quad \sigma_A = -6.71 \text{ ksi} \quad \blacktriangleleft$$

(b) $y_B = R - r_2 = -3.2832 \text{ in.}$

$$\sigma_B = -\frac{My_B}{Aer_2} = -\frac{(80)(-3.2832)}{(18)(0.4499)(10)} \quad \sigma_B = 3.24 \text{ ksi} \quad \blacktriangleleft$$



PROBLEM 4.187

Show that if the cross section of a curved beam consists of two or more rectangles, the radius R of the neutral surface can be expressed as

$$R = \frac{A}{\ln \left[\left(\frac{r_2}{r_1} \right)^{b_1} \left(\frac{r_3}{r_2} \right)^{b_2} \left(\frac{r_4}{r_3} \right)^{b_3} \right]}$$

where A is the total area of the cross section.

SOLUTION

$$\begin{aligned} R &= \frac{\Sigma A}{\Sigma \int \frac{1}{r} dA} = \frac{A}{\Sigma b_i \ln \frac{r_{i+1}}{r_i}} \\ &= \frac{A}{\Sigma \ln \left(\frac{r_{i+1}}{r_i} \right)^{b_i}} = \frac{A}{\ln \left[\left(\frac{r_2}{r_1} \right)^{b_1} \left(\frac{r_3}{r_2} \right)^{b_2} \left(\frac{r_4}{r_3} \right)^{b_3} \right]} \end{aligned}$$

Note that for each rectangle,

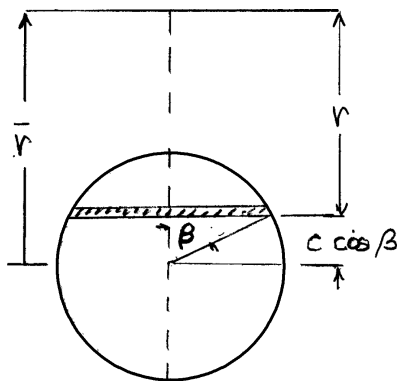
$$\begin{aligned} \int \frac{1}{r} dA &= \int_{r_i}^{r_{i+1}} b_i \frac{dr}{r} \\ &= b_i \int_{r_i}^{r_{i+1}} \frac{dr}{r} = b_i \ln \frac{r_{i+1}}{r_i} \end{aligned}$$

PROBLEM 4.188

Using Eq. (4.66), derive the expression for R given in Fig. 4.73 for a circular cross section.

SOLUTION

Use polar coordinate β as shown. Let w be the width as a function of β



$$w = 2c \sin \beta$$

$$r = \bar{r} - c \cos \beta$$

$$dr = c \sin \beta d\beta$$

$$dA = w dr = 2c^2 \sin^2 \beta d\beta$$

$$\int \frac{dA}{r} = \int_0^\pi \frac{2c^2 \sin^2 \beta}{\bar{r} - c \cos \beta} d\beta$$

$$\int \frac{dA}{r} = \int_0^\pi \frac{c^2 (1 - \cos^2 \beta)}{\bar{r} - c \cos \beta} d\beta$$

$$= 2 \int_0^\pi \frac{\bar{r}^2 - c^2 \cos^2 \beta - (\bar{r}^2 - c^2)}{\bar{r} - c \cos \beta} d\beta$$

$$= 2 \int_0^\pi (\bar{r} + c \cos \beta) d\beta - 2(\bar{r}^2 - c^2) \int_0^\pi \frac{dr}{\bar{r} - c \cos \beta}$$

$$= 2\bar{r} \beta \Big|_0^\pi + 2c \sin \beta \Big|_0^\pi$$

$$- 2(\bar{r}^2 - c^2) \frac{2}{\sqrt{\bar{r}^2 - c^2}} \tan^{-1} \frac{\sqrt{\bar{r}^2 - c^2} \tan\left(\frac{1}{2}\beta\right)}{\bar{r} + c} \Big|_0^\pi$$

$$= 2\bar{r}(\pi - 0) + 2c(0 - 0) - 4\sqrt{\bar{r}^2 - c^2} \cdot \left(\frac{\pi}{2} - 0\right)$$

$$= 2\pi\bar{r} - 2\pi\sqrt{\bar{r}^2 - c^2}$$

$$A = \pi c^2$$

$$R = \frac{A}{\int \frac{dA}{r}} = \frac{\pi c^2}{2\pi\bar{r} - 2\pi\sqrt{\bar{r}^2 - c^2}} = \frac{1}{2} \frac{c^2}{\bar{r} - \sqrt{\bar{r}^2 - c^2}} \times \frac{\bar{r} + \sqrt{\bar{r}^2 - c^2}}{\bar{r} + \sqrt{\bar{r}^2 - c^2}}$$

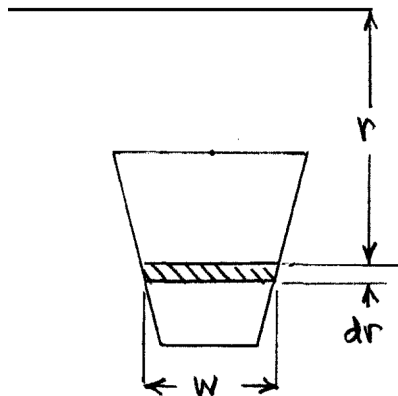
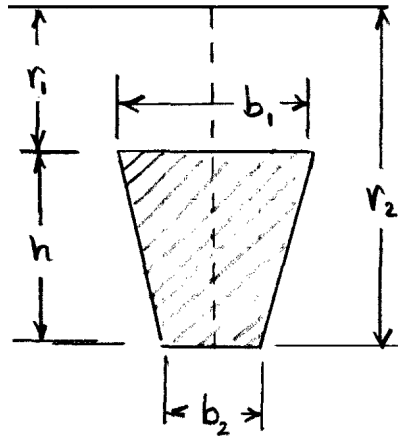
$$= \frac{1}{2} \frac{c^2 (\bar{r} + \sqrt{\bar{r}^2 - c^2})}{\bar{r}^2 - (\bar{r}^2 - c^2)} = \frac{1}{2} \frac{c^2 (\bar{r} + \sqrt{\bar{r}^2 - c^2})}{c^2} = \frac{1}{2} (\bar{r} + \sqrt{\bar{r}^2 - c^2})$$

PROBLEM 4.189

Using Eq. (4.66), derive the expression for R given in Fig. 4.73 for a trapezoidal cross section.

SOLUTION

The section width w varies linearly with r .



$$\begin{aligned} w &= c_0 + c_1 r \\ w &= b_1 \text{ at } r = r_1 \text{ and } w = b_2 \text{ at } r = r_2 \\ b_1 &= c_0 + c_1 r_1 \\ b_2 &= c_0 + c_1 r_2 \\ b_1 - b_2 &= c_1 (r_1 - r_2) = -c_1 h \end{aligned}$$

$$c_1 = -\frac{b_1 - b_2}{h}$$

$$r_2 b_1 - r_1 b_2 = (r_2 - r_1) c_0 = h c_0$$

$$c_0 = \frac{r_2 b_1 - r_1 b_2}{h}$$

$$\int \frac{dA}{r} = \int_{r_1}^{r_2} \frac{w}{r} dr = \int_{r_1}^{r_2} \frac{c_0 + c_1 r}{r} dr$$

$$= c_0 \ln r \Big|_{r_1}^{r_2} + c_1 r \Big|_{r_1}^{r_2}$$

$$= c_0 \ln \frac{r_2}{r_1} + c_1 (r_2 - r_1)$$

$$= \frac{r_2 b_1 - r_1 b_2}{h} \ln \frac{r_2}{r_1} - \frac{b_1 - b_2}{h} h$$

$$= \frac{r_2 b_1 - r_1 b_2}{h} \ln \frac{r_2}{r_1} - (b_1 - b_2)$$

$$A = \frac{1}{2} (b_1 + b_2) h$$

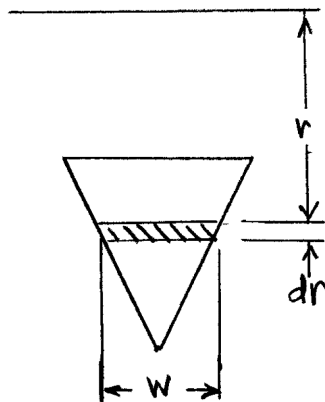
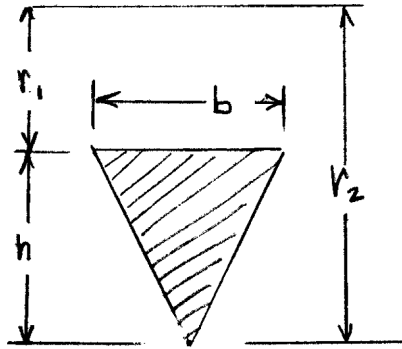
$$R = \frac{A}{\int \frac{dA}{r}} = \frac{\frac{1}{2} h^2 (b_1 + b_2)}{(r_2 b_1 - r_1 b_2) \ln \frac{r_2}{r_1} - h(b_1 - b_2)}$$

PROBLEM 4.190

Using Equation (4.66), derive the expression for R given in Fig. 4.73 for a triangular cross section.

SOLUTION

The section width w varies linearly with r .



$$w = c_0 + c_1 r$$

$$w = b \text{ at } r = r_1 \quad \text{and} \quad w = 0 \text{ at } r = r_2$$

$$b = c_0 + c_1 r_1$$

$$0 = c_0 + c_1 r_2$$

$$b = c_1 (r_1 - r_2) = -c_1 h$$

$$c_1 = -\frac{b}{h} \quad \text{and} \quad c_0 = -c_1 r_2 = \frac{b r_2}{h}$$

$$\int \frac{dA}{r} = \int_{r_1}^{r_2} \frac{w}{r} dr = \int_{r_1}^{r_2} \frac{c_0 + c_1 r}{r} dr$$

$$= c_0 \ln r \Big|_{r_1}^{r_2} + c_1 r \Big|_{r_1}^{r_2}$$

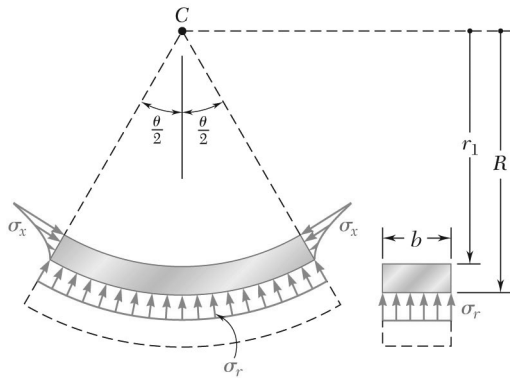
$$= c_0 \ln \frac{r_2}{r_1} + c_1 (r_2 - r_1)$$

$$= \frac{b r_2}{h} \ln \frac{r_2}{r_1} - \frac{b}{h} h$$

$$= \frac{b r_2}{h} \ln \frac{r_2}{r_1} - b = b \left(\frac{r_2}{h} \ln \frac{r_2}{r_1} - 1 \right)$$

$$A = \frac{1}{2} b h$$

$$R = \frac{A}{\int \frac{dA}{r}} = \frac{\frac{1}{2} b h}{b \left(\frac{r_2}{h} \ln \frac{r_2}{r_1} - 1 \right)} = \frac{\frac{1}{2} h}{\frac{r_2}{h} \ln \frac{r_2}{r_1} - 1}$$



PROBLEM 4.191

For a curved bar of rectangular cross section subjected to a bending couple M , show that the radial stress at the neutral surface is

$$\sigma_r = \frac{M}{Ae} \left(1 - \frac{r_1}{R} - \ln \frac{R}{r_1} \right)$$

and compute the value of σ_r for the curved bar of Examples 4.10 and 4.11. (Hint: consider the free-body diagram of the portion of the beam located above the neutral surface.)

SOLUTION

At radial distance r ,

$$\sigma_r = \frac{M(r - R)}{Aer} = \frac{M}{Ae} - \frac{MR}{Aer}$$

For portion above the neutral axis, the resultant force is

$$\begin{aligned} H &= \int \sigma_r dA = \int_{r_1}^R \sigma_r b dr \\ &= \frac{Mb}{Ae} \int_{r_1}^R dr - \frac{MRb}{Ae} \int_{r_1}^R \frac{dr}{r} \\ &= \frac{Mb}{Ae} (R - r_1) - \frac{MRb}{Ae} \ln \frac{R}{r_1} = \frac{MbR}{Ae} \left(1 - \frac{r_1}{R} - \ln \frac{R}{r_1} \right) \end{aligned}$$

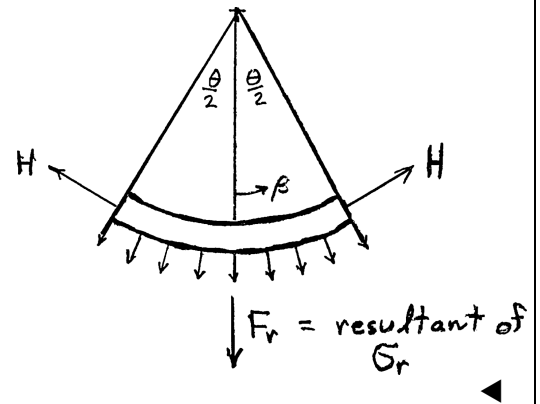
Resultant of σ_θ :

$$\begin{aligned} F_r &= \int \sigma_\theta \cos \beta dA \\ &= \int_{-\theta/2}^{\theta/2} \sigma_r \cos \beta (bR d\beta) = \sigma_r bR \int_{-\theta/2}^{\theta/2} \cos \beta d\beta \\ &= \sigma_r bR \sin \beta \Big|_{-\theta/2}^{\theta/2} = 2\sigma_r bR \sin \frac{\theta}{2} \end{aligned}$$

For equilibrium: $F_r - 2H \sin \frac{\theta}{2} = 0$

$$2\sigma_r bR \sin \frac{\theta}{2} - 2 \frac{MbR}{Ae} \left(1 - \frac{r_1}{R} - \ln \frac{R}{r_1} \right) \sin \frac{\theta}{2} = 0$$

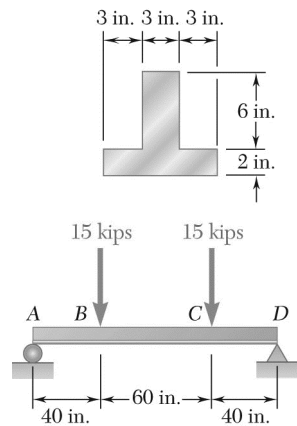
$$\sigma_r = \frac{M}{Ae} \left(1 - \frac{r_1}{R} - \ln \frac{R}{r_1} \right)$$



Using results of Examples 4.10 and 4.11 as data,

$$M = 8 \text{ kip} \cdot \text{in}, \quad A = 3.75 \text{ in}^2, \quad R = 5.9686 \text{ in}, \quad e = 0.0314 \text{ in}, \quad r_1 = 5.25 \text{ in}.$$

$$\sigma_r = \frac{8}{(3.75)(0.0314)} \left[1 - \frac{5.25}{5.9686} - \ln \frac{5.9686}{5.25} \right] \quad \sigma_r = -0.54 \text{ ksi} \quad \blacktriangleleft$$



PROBLEM 4.192

Two vertical forces are applied to a beam of the cross section shown. Determine the maximum tensile and compressive stresses in portion *BC* of the beam.

SOLUTION

	<i>A</i>	$\bar{y}_0$	<i>A</i> $\bar{y}_0$
①	18	5	90
②	18	1	18
Σ	36		108

$$\bar{Y}_0 = \frac{108}{36} = 3 \text{ in.}$$

Neutral axis lies 3 in. above the base.

$$I_1 = \frac{1}{12} b_1 h_1^3 + A_1 d_1^2 = \frac{1}{12} (3)(6)^3 + (18)(2)^2 = 126 \text{ in}^4$$

$$I_2 = \frac{1}{12} b_2 h_2^3 + A_2 d_2^2 = \frac{1}{12} (9)(2)^3 + (18)(2)^2 = 78 \text{ in}^4$$

$$I = I_1 + I_2 = 126 + 78 = 204 \text{ in}^4$$

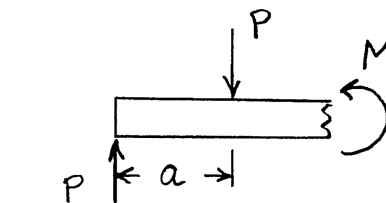
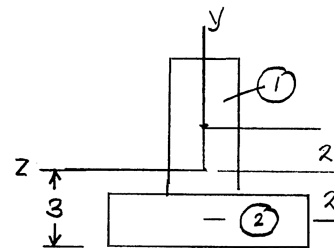
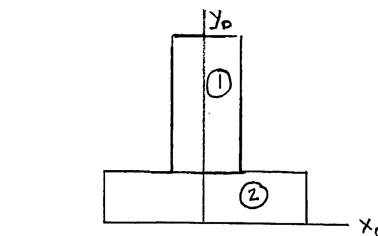
$$y_{\text{top}} = 5 \text{ in.} \quad y_{\text{bot}} = -3 \text{ in.}$$

$$M - Pa = 0$$

$$M = Pa = (15)(40) = 600 \text{ kip} \cdot \text{in}$$

$$\sigma_{\text{top}} = -\frac{M y_{\text{top}}}{I} = -\frac{(600)(5)}{204}$$

$$\sigma_{\text{bot}} = -\frac{M y_{\text{bot}}}{I} = -\frac{(600)(-3)}{204}$$



$$\sigma_{\text{top}} = -14.71 \text{ ksi (compression)} \quad \blacktriangleleft$$

$$\sigma_{\text{bot}} = 8.82 \text{ ksi (tension)} \quad \blacktriangleleft$$



PROBLEM 4.193

Straight rods of 6-mm diameter and 30-m length are stored by coiling the rods inside a drum of 1.25-m inside diameter. Assuming that the yield strength is not exceeded, determine (a) the maximum stress in a coiled rod, (b) the corresponding bending moment in the rod. Use $E = 200$ GPa.

SOLUTION

Let $D =$ inside diameter of the drum.

$d =$ diameter of rod, $c = \frac{1}{2}d$,

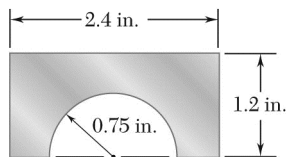
$\rho =$ radius of curvature of centerline of rods when bent.

$$\rho = \frac{1}{2}D - \frac{1}{2}d = \frac{1}{2}(1.25) - \frac{1}{2}(6 \times 10^{-3}) = 0.622 \text{ m}$$

$$I = \frac{\pi}{4}c^4 = \frac{\pi}{4}(0.003)^4 = 63.617 \times 10^{-12} \text{ m}^4$$

$$(a) \quad \sigma_{\max} = \frac{Ec}{\rho} = \frac{(200 \times 10^9)(0.003)}{0.622} = 965 \times 10^6 \text{ Pa} \quad \sigma = 965 \text{ MPa} \quad \blacktriangleleft$$

$$(b) \quad M = \frac{EI}{\rho} = \frac{(200 \times 10^9)(63.617 \times 10^{-12})}{0.622} = 20.5 \text{ N} \cdot \text{m} \quad M = 20.5 \text{ N} \cdot \text{m} \quad \blacktriangleleft$$



PROBLEM 4.194

Knowing that for the beam shown the allowable stress is 12 ksi in tension and 16 ksi in compression, determine the largest couple **M** that can be applied.

SOLUTION

① = rectangle

② = semi-circular cutout

$$A_1 = (2.4)(1.2) = 2.88 \text{ in}^2$$

$$A_2 = \frac{\pi}{2}(0.75)^2 = 0.8836 \text{ in}^2$$

$$A = 2.88 - 0.8836 = 1.9964 \text{ in}^2$$

$$\bar{y}_1 = 0.6 \text{ in.}$$

$$\bar{y}_2 = \frac{4r}{3\pi} = \frac{(4)(0.75)}{3\pi} = 0.3183 \text{ in.}$$

$$\bar{Y} = \frac{\Sigma A\bar{y}}{\Sigma A} = \frac{(2.88)(0.6) - (0.8836)(0.3183)}{1.9964} = 0.7247 \text{ in.}$$

Neutral axis lies 0.7247 in. above the bottom.

Moment of inertia about the base:

$$I_b = \frac{1}{3}bh^3 - \frac{\pi}{8}r^4 = \frac{1}{3}(2.4)(1.2)^3 - \frac{\pi}{8}(0.75)^4 = 1.25815 \text{ in}^4$$

Centroidal moment of inertia:

$$\bar{I} = I_b - A\bar{Y}^2 = 1.25815 - (1.9964)(0.7247)^2 = 0.2097 \text{ in}^4$$

$$y_{\text{top}} = 1.2 - 0.7247 = 0.4753 \text{ in.},$$

$$y_{\text{bot}} = -0.7247 \text{ in.}$$

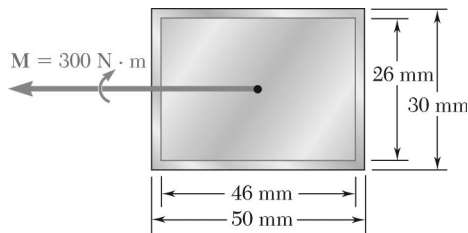
$$|\sigma| = \left| \frac{My}{I} \right| \quad M = \left| \frac{\sigma I}{y} \right|$$

Top: (tension side) $M = \frac{(12)(0.2097)}{0.4753} = 5.29 \text{ kip} \cdot \text{in}$

Bottom: (compression) $M = \frac{(16)(0.2097)}{0.7247} = 4.63 \text{ kip} \cdot \text{in}$

Choose the smaller value.

$$M = 4.63 \text{ kip} \cdot \text{in} \quad \blacktriangleleft$$



PROBLEM 4.195

In order to increase corrosion resistance, a 2-mm-thick cladding of aluminum has been added to a steel bar as shown. The modulus of elasticity is 200 GPa for steel and 70 GPa for aluminum. For a bending moment of 300 N · m, determine (a) the maximum stress in the steel, (b) the maximum stress in the aluminum, (c) the radius of curvature of the bar.

SOLUTION

Use aluminum as the reference material.

$$n = 1 \text{ in aluminum.} \quad n = \frac{E_s}{E_a} = \frac{200}{70} = 2.857 \text{ in steel.}$$

Cross section geometry:

$$\text{Steel: } A_s = (46 \text{ mm})(26 \text{ mm}) = 1196 \text{ mm}^2 \quad I_s = \frac{1}{12}(46 \text{ mm})(26 \text{ mm})^3 = 67,375 \text{ mm}^4$$

$$\text{Aluminum: } A_a = (50 \text{ mm})(30 \text{ mm}) - 1196 \text{ mm}^2 = 304 \text{ mm}^2$$

$$I_a = \frac{1}{12}(50 \text{ mm})(30 \text{ mm})^3 - 67,375 \text{ mm}^4 = 45,125 \text{ mm}^4$$

Transformed section.

$$I = n_a I_a + n_s I_s = (1)(45,125) + (2.857)(67,375) = 237,615 \text{ mm}^4 = 237.615 \times 10^{-9} \text{ m}^4$$

Bending moment. $M = 300 \text{ N} \cdot \text{m}$

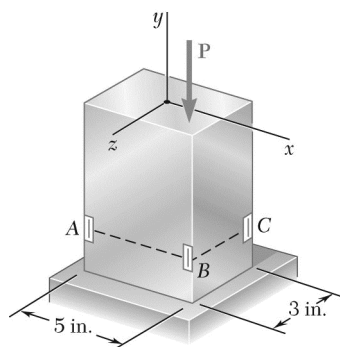
$$(a) \quad \text{Maximum stress in steel:} \quad n_s = 2.857 \quad y_s = 13 \text{ mm} = 0.013 \text{ m}$$

$$\sigma_s = \frac{n_s M y_s}{I} = \frac{(2.857)(300)(0.013)}{237.615 \times 10^{-9}} = 46.9 \times 10^6 \text{ Pa} \quad \sigma_s = 46.9 \text{ MPa} \quad \blacktriangleleft$$

$$(b) \quad \text{Maximum stress in aluminum:} \quad n_a = 1, \quad y_a = 15 \text{ mm} = 0.015 \text{ m}$$

$$\sigma_a = \frac{n_a M y_a}{I} = \frac{(1)(300)(0.015)}{237.615 \times 10^{-9}} = 18.94 \times 10^6 \text{ Pa} \quad \sigma_a = 18.94 \text{ MPa} \quad \blacktriangleleft$$

$$(c) \quad \text{Radius of curvature: } \rho = \frac{EI}{M} \quad \rho = \frac{(70 \times 10^9)(237.615 \times 10^{-9})}{300} \quad \rho = 55.4 \text{ m} \quad \blacktriangleleft$$



PROBLEM 4.196

A single vertical force $\mathbf{P}$ is applied to a short steel post as shown. Gages located at A , B , and C indicate the following strains:

$$\epsilon_A = -500\mu \quad \epsilon_B = -1000\mu \quad \epsilon_C = -200\mu$$

Knowing that $E = 29 \times 10^6$ psi, determine (a) the magnitude of $\mathbf{P}$, (b) the line of action of $\mathbf{P}$, (c) the corresponding strain at the hidden edge of the post, where $x = -2.5$ in. and $z = -1.5$ in.

SOLUTION

$$I_x = \frac{1}{12}(5)(3)^3 = 11.25 \text{ in}^4 \quad I_z = \frac{1}{12}(3)(5)^3 = 31.25 \text{ in}^4 \quad A = (5)(3) = 15 \text{ in}^2$$

$$M_x = Pz \quad M_z = -Px$$

$$x_A = -2.5 \text{ in.}, \quad x_B = 2.5 \text{ in.}, \quad x_C = 2.5 \text{ in.}, \quad x_D = -2.5 \text{ in.}$$

$$z_A = 1.5 \text{ in.}, \quad z_B = 1.5 \text{ in.}, \quad z_C = -1.5 \text{ in.}, \quad z_D = -1.5 \text{ in.}$$

$$\sigma_A = E\epsilon_A = (29 \times 10^6)(-500 \times 10^{-6}) = -14,500 \text{ psi} = -14.5 \text{ ksi}$$

$$\sigma_B = E\epsilon_B = (29 \times 10^6)(-1000 \times 10^{-6}) = -29,000 \text{ psi} = -29 \text{ ksi}$$

$$\sigma_C = E\epsilon_C = (29 \times 10^6)(-200 \times 10^{-6}) = -5800 \text{ psi} = -5.8 \text{ ksi}$$

$$\sigma_A = -\frac{P}{A} - \frac{M_x z_A}{I_x} + \frac{M_z x_A}{I_z} = -0.06667P - 0.13333M_x - 0.08M_z \quad (1)$$

$$\sigma_B = -\frac{P}{A} - \frac{M_x z_B}{I_x} + \frac{M_z x_B}{I_z} = -0.06667P - 0.13333M_x + 0.08M_z \quad (2)$$

$$\sigma_C = -\frac{P}{A} - \frac{M_x z_C}{I_x} + \frac{M_z x_C}{I_z} = -0.06667P + 0.13333M_x + 0.08M_z \quad (3)$$

Substituting the values for σ_A , σ_B , and σ_C into (1), (2), and (3) and solving the simultaneous equations gives

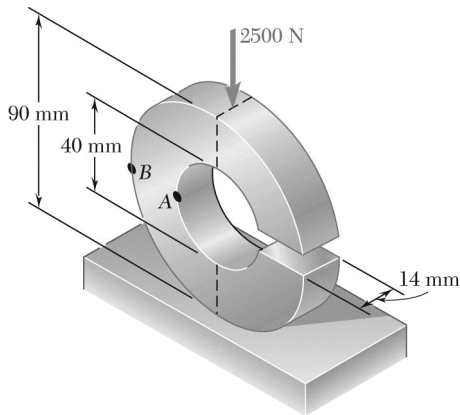
$$M_x = 87 \text{ kip} \cdot \text{in.}, \quad M_z = -90.625 \text{ kip} \cdot \text{in.}, \quad (a) \quad P = 152.25 \text{ kips} \quad \blacktriangleleft$$

$$x = -\frac{M_z}{P} = -\frac{-90.625}{152.25} \quad (b) \quad x = 0.595 \text{ in.} \quad \blacktriangleleft$$

$$z = \frac{M_x}{P} = \frac{87}{152.25} \quad z = 0.571 \text{ in.} \quad \blacktriangleleft$$

$$\begin{aligned} \sigma_D &= -\frac{P}{A} - \frac{M_x z_D}{I_x} + \frac{M_z x_D}{I_z} = -0.06667P + 0.13333M_x - 0.08M_z \\ &= -(0.06667)(152.25) + (0.13333)(87) - (0.08)(-90.625) = 8.70 \text{ ksi} \end{aligned}$$

$$(c) \quad \text{Strain at hidden edge:} \quad \epsilon = \frac{\sigma_D}{E} = \frac{8.70 \times 10^3}{29 \times 10^6} \quad \epsilon = 300 \mu \quad \blacktriangleleft$$



PROBLEM 4.197

For the split ring shown, determine the stress at (a) point A, (b) point B.

SOLUTION

$$r_1 = \frac{1}{2}(40) = 20 \text{ mm}, \quad r_2 = \frac{1}{2}(90) = 45 \text{ mm} \quad h = r_2 - r_1 = 25 \text{ mm}$$

$$A = (14)(25) = 350 \text{ mm}^2 \quad R = \frac{h}{\ln \frac{r_2}{r_1}} = \frac{25}{\ln \frac{45}{20}} = 30.8288 \text{ mm}$$

$$\bar{r} = \frac{1}{2}(r_1 + r_2) = 32.5 \text{ mm} \quad e = \bar{r} - R = 1.6712 \text{ mm}$$

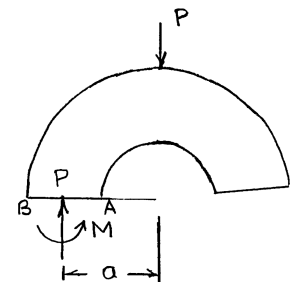
Reduce the internal forces transmitted across section AB to a force-couple system at the centroid of the cross section. The bending couple is

$$M = Pa = P\bar{r} = (2500)(32.5 \times 10^{-3}) = 81.25 \text{ N} \cdot \text{m}$$

(a) Point A: $r_A = 20 \text{ mm}$ $y_A = 30.8288 - 20 = 10.8288 \text{ mm}$

$$\sigma_A = -\frac{P}{A} - \frac{My_A}{AeR} = -\frac{2500}{350 \times 10^{-6}} - \frac{(81.25)(10.8288 \times 10^{-3})}{(350 \times 10^{-6})(1.6712 \times 10^{-3})(20 \times 10^{-3})}$$

$$= -82.4 \times 10^6 \text{ Pa}$$



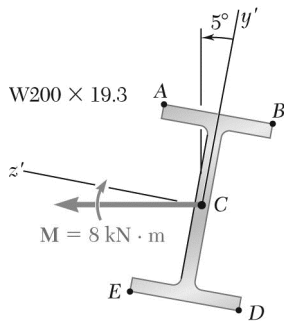
$$\sigma_A = -82.4 \text{ MPa} \quad \blacktriangleleft$$

(b) Point B: $r_B = 45 \text{ mm}$ $y_B = 30.8288 - 45 = -14.1712 \text{ mm}$

$$\sigma_B = -\frac{P}{A} - \frac{My_B}{AeR} = -\frac{2500}{350 \times 10^{-6}} - \frac{(81.25)(-14.1712 \times 10^{-3})}{(350 \times 10^{-6})(1.6712 \times 10^{-3})(45 \times 10^{-3})}$$

$$= 36.6 \times 10^6 \text{ Pa}$$

$$\sigma_B = 36.6 \text{ MPa} \quad \blacktriangleleft$$



PROBLEM 4.198

A couple $\mathbf{M}$ of moment $8 \text{ kN} \cdot \text{m}$ acting in a vertical plane is applied to a $W200 \times 19.3$ rolled-steel beam as shown. Determine
 (a) the angle that the neutral axis forms with the horizontal plane,
 (b) the maximum stress in the beam.

SOLUTION

For $W200 \times 19.3$ rolled steel sector,

$$I_{z'} = 16.6 \times 10^6 \text{ mm}^4 = 16.6 \times 10^{-6} \text{ m}^4$$

$$I_{y'} = 1.15 \times 10^6 \text{ mm}^4 = 1.15 \times 10^{-6} \text{ m}^4$$

$$y_A = y_B = -y_D = -y_E = \frac{203}{2} = 101.5 \text{ mm}$$

$$z_A = -z_B = -z_D = z_E = \frac{102}{2} = 51 \text{ mm}$$

$$M_z = (8 \times 10^3) \cos 5^\circ = 7.9696 \times 10^3 \text{ N} \cdot \text{m}$$

$$M_y = -(8 \times 10^3) \sin 5^\circ = -0.6972 \times 10^3 \text{ N} \cdot \text{m}$$

$$(a) \quad \tan \varphi = \frac{I_z}{I_y} \tan \theta = \frac{16.6 \times 10^{-6}}{1.15 \times 10^{-6}} \tan(-5^\circ) = -1.2629$$

$$\varphi = -51.6^\circ$$

$$\alpha = 51.6^\circ - 5^\circ$$

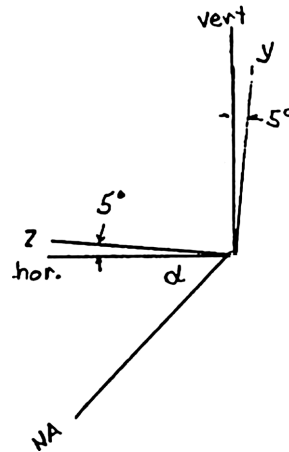
$$\alpha = 46.6^\circ \quad \blacktriangleleft$$

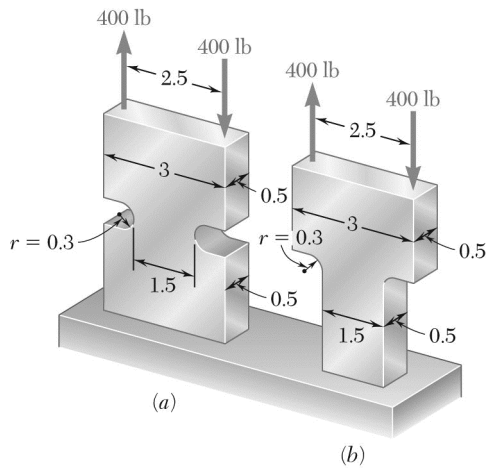
(b) Maximum tensile stress occurs at point D.

$$\sigma_D = -\frac{M_z y_D}{I_z} + \frac{M_y z_D}{I_y} = -\frac{(7.9696 \times 10^3)(-101.5 \times 10^{-3})}{16.6 \times 10^{-6}} + \frac{(-0.6972 \times 10^3)(-51 \times 10^{-3})}{1.15 \times 10^{-6}}$$

$$= 79.6 \times 10^6 \text{ Pa}$$

$$\sigma_D = 79.6 \text{ MPa} \quad \blacktriangleleft$$





PROBLEM 4.199

Determine the maximum stress in each of the two machine elements shown.

SOLUTION

For each case, $M = (400)(2.5) = 1000 \text{ lb} \cdot \text{in}$

At the minimum section,

$$I = \frac{1}{12}(0.5)(1.5)^3 = 0.140625 \text{ in}^4$$

$$c = 0.75 \text{ in.}$$

$$(a) \quad D/d = 3/1.5 = 2$$

$$r/d = 0.3/1.5 = 0.2$$

From Fig 4.32, $K = 1.75$

$$\sigma_{\max} = \frac{KMc}{I} = \frac{(1.75)(1000)(0.75)}{0.140625} = 9.33 \times 10^3 \text{ psi}$$

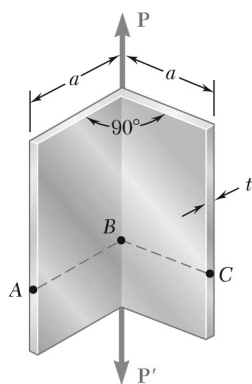
$$\sigma_{\max} = 9.33 \text{ ksi} \quad \blacktriangleleft$$

$$(b) \quad D/d = 3/1.5 = 2 \quad r/d = 0.3/1.5 = 0.2$$

From Fig. 4.31, $K = 1.50$

$$\sigma_{\max} = \frac{KMc}{I} = \frac{(1.50)(1000)(0.75)}{0.140625} = 8.00 \times 10^3 \text{ psi}$$

$$\sigma_{\max} = 8.00 \text{ ksi} \quad \blacktriangleleft$$



PROBLEM 4.200

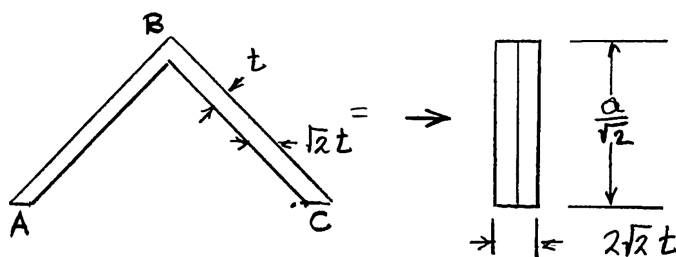
The shape shown was formed by bending a thin steel plate. Assuming that the thickness t is small compared to the length a of a side of the shape, determine the stress (a) at A , (b) at B , (c) at C .

SOLUTION

Moment of inertia about centroid:

$$I = \frac{1}{12} (2\sqrt{2}t) \left(\frac{a}{\sqrt{2}} \right)^3$$

$$= \frac{1}{12} ta^3$$



Area: $A = (2\sqrt{2}t) \left(\frac{a}{\sqrt{2}} \right) = 2at, \quad c = \frac{a}{2\sqrt{2}}$

$$(a) \quad \sigma_A = \frac{P}{A} - \frac{Pec}{I} = \frac{P}{2at} - \frac{P \left(\frac{a}{2\sqrt{2}} \right) \left(\frac{a}{2\sqrt{2}} \right)}{\frac{1}{12} ta^3}$$

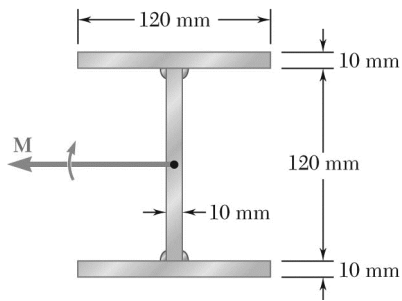
$$\sigma_A = -\frac{P}{2at} \quad \blacktriangleleft$$

$$(b) \quad \sigma_B = \frac{P}{A} + \frac{Pec}{I} = \frac{P}{2at} + \frac{P \left(\frac{a}{\sqrt{2}} \right) \left(\frac{a}{\sqrt{2}} \right)}{\frac{1}{12} ta^3}$$

$$\sigma_B = -\frac{2P}{at} \quad \blacktriangleleft$$

$$(c) \quad \sigma_C = \sigma_A$$

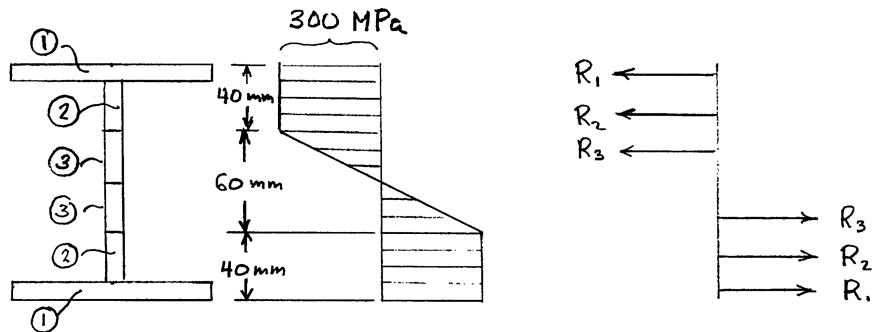
$$\sigma_C = -\frac{P}{2at} \quad \blacktriangleleft$$



PROBLEM 4.201

Three 120×10 -mm steel plates have been welded together to form the beam shown. Assuming that the steel is elastoplastic with $E = 200$ GPa and $\sigma_Y = 300$ MPa, determine (a) the bending moment for which the plastic zones at the top and bottom of the beam are 40 mm thick, (b) the corresponding radius of curvature of the beam.

SOLUTION



$$A_1 = (120)(10) = 1200 \text{ mm}^2$$

$$R_1 = \sigma_Y A_1 = (300 \times 10^6)(1200 \times 10^{-6}) = 360 \times 10^3 \text{ N}$$

$$A_2 = (30)(10) = 300 \text{ mm}^2$$

$$R_2 = \sigma_Y A_2 = (300 \times 10^6)(300 \times 10^{-6}) = 90 \times 10^3 \text{ N}$$

$$A_3 = (30)(10) = 300 \text{ mm}^2$$

$$R_3 = \frac{1}{2} \sigma_Y A_2 = \frac{1}{2} (300 \times 10^6)(300 \times 10^{-6}) = 45 \times 10^3 \text{ N}$$

$$y_1 = 65 \text{ mm} = 65 \times 10^{-3} \text{ m} \quad y_2 = 45 \text{ mm} = 45 \times 10^{-3} \text{ m} \quad y_3 = 20 \text{ mm} = 20 \times 10^{-3} \text{ m}$$

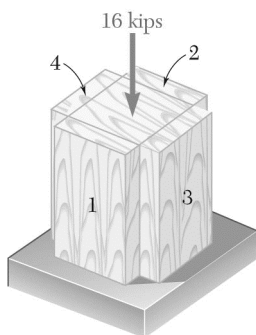
$$(a) \quad M = 2(R_1 y_1 + R_2 y_2 + R_3 y_3) = 2\{(360)(65) + (90)(45) + (45)(20)\}$$

$$= 56.7 \times 10^3 \text{ N} \cdot \text{m}$$

$$M = 56.7 \text{ kN} \cdot \text{m} \quad \blacktriangleleft$$

$$(b) \quad \frac{y_Y}{\rho} = \frac{\sigma_Y}{E} \quad \rho = \frac{E y_Y}{\sigma_Y} = \frac{(200 \times 10^9)(30 \times 10^{-3})}{300 \times 10^6}$$

$$\rho = 20 \text{ m} \quad \blacktriangleleft$$



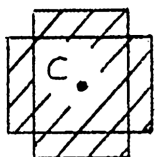
PROBLEM 4.202

A short column is made by nailing four 1×4-in. planks to a 4×4-in. timber. Determine the largest compressive stress created in the column by a 16-kip load applied as shown in the center of the top section of the timber if (a) the column is as described, (b) plank 1 is removed, (c) planks 1 and 2 are removed, (d) planks 1, 2, and 3 are removed, (e) all planks are removed.

SOLUTION

(a) Centric loading:

$$M = 0 \quad \sigma = -\frac{P}{A}$$



$$A = (4)(4) + (4)(1)(4) = 32 \text{ in}^2$$

$$\sigma = -\frac{16 \times 10^3}{32}$$

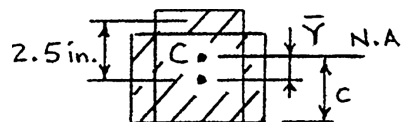
$$\sigma = -500 \text{ psi} \quad \blacktriangleleft$$

(b) Eccentric loading:

$$M = Pe \quad \sigma = -\frac{P}{A} - \frac{Pec}{I}$$

$$A = (4)(4) + (3)(1)(3) = 28 \text{ in}^2 \quad e = \bar{y}$$

$$\bar{y} = \frac{\Sigma A\bar{y}}{A} = \frac{(1)(4)(2.5)}{28} = 0.35714 \text{ in.}$$



$$I = \Sigma(\bar{I} + Ad^2) = \frac{1}{12}(6)(4)^3 + (6)(4)(0.35714)^2$$

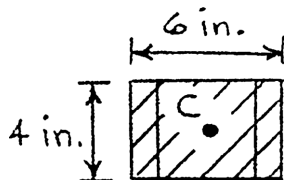
$$+ \frac{1}{12}(4)(1)^3 + (4)(1)(2.14286)^2 = 53.762 \text{ in}^4$$

$$\sigma = -\frac{16 \times 10^3}{28} - \frac{(16 \times 10^3)(0.35714)(2.35714)}{53.762}$$

$$\sigma = -822 \text{ psi} \quad \blacktriangleleft$$

(c) Centric loading:

$$M = 0 \quad \sigma = -\frac{P}{A}$$



$$A = (6)(4) = 24 \text{ in}^2$$

$$\sigma = -\frac{16 \times 10^3}{24}$$

$$\sigma = -667 \text{ psi} \quad \blacktriangleleft$$

PROBLEM 4.202 (Continued)

(d) Eccentric loading:

$$M = Pe \quad \sigma = -\frac{P}{A} - \frac{Pec}{I}$$

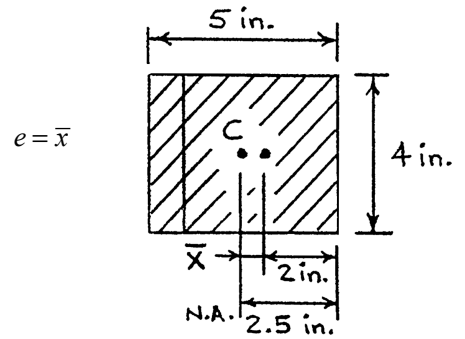
$$A = (4)(1) = (1)(4)(1) = 20 \text{ in}^2$$

$$\bar{x} = 2.5 - 2 = 0.5 \text{ in.}$$

$$I = \frac{1}{12}(4)(5)^3 = 41.667 \text{ in}^4$$

$$\sigma = -\frac{16 \times 10^3}{20} - \frac{(16 \times 10^3)(0.5)(2.5)}{41.667}$$

$$\sigma = -1280 \text{ psi} \quad \blacktriangleleft$$



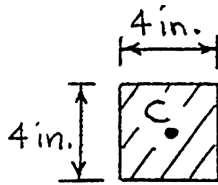
(e) Centric loading:

$$M = 0 \quad \sigma = -\frac{P}{A}$$

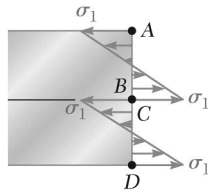
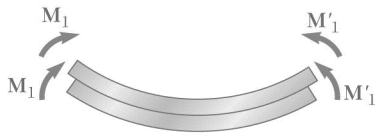
$$A = (4)(4) = 16 \text{ in}^2$$

$$\sigma = -\frac{16 \times 10^3}{16}$$

$$\sigma = -1000 \text{ psi} \quad \blacktriangleleft$$



PROBLEM 4.203



Two thin strips of the same material and same cross section are bent by couples of the same magnitude and glued together. After the two surfaces of contact have been securely bonded, the couples are removed. Denoting by σ_1 the maximum stress and by ρ_1 the radius of curvature of each strip while the couples were applied, determine (a) the final stresses at points A , B , C , and D , (b) the final radius of curvature.

SOLUTION

Let b = width and t = thickness of one strip.

Loading one strip, $M = M_1$

$$I_1 = \frac{1}{12}bt^3, \quad c = \frac{1}{2}t$$

$$\sigma_1 = \frac{M_1 c}{I} = \frac{\sigma M_1}{bt^2}$$

$$\frac{1}{\rho_1} = \frac{M_1}{EI_1} = \frac{12 M_1}{Et^3}$$

After M_1 is applied to each of the strips, the stresses are those given in the sketch above. They are

$$\sigma_A = -\sigma_1, \quad \sigma_B = \sigma_1, \quad \sigma_C = -\sigma_1, \quad \sigma_D = \sigma_1$$

The total bending couple is $2M_1$.

After the strips are glued together, this couple is removed.

$$M' = 2M_1, \quad I' = \frac{1}{12}b(2t)^3 = \frac{2}{3}bt^3 \quad c = t$$

The stresses removed are

$$\sigma' = -\frac{M'y}{I} = -\frac{2M_1 y}{\frac{2}{3}bt^3} = -\frac{3M_1 y}{bt^2}$$

$$\sigma'_A = -\frac{3M_1}{bt^2} = -\frac{1}{2}\sigma_1, \quad \sigma'_B = \sigma'_C = 0, \quad \sigma'_D = \frac{3M_1}{bt^2} = \frac{1}{2}\sigma_1$$

PROBLEM 4.203 (Continued)

(a) Final stresses: $\sigma_A = -\sigma_1 - \left(-\frac{1}{2}\sigma_1\right)$ $\sigma_A = -\frac{1}{2}\sigma_1 \blacktriangleleft$

$\sigma_B = \sigma_1 \blacktriangleleft$

$\sigma_C = -\sigma_1 \blacktriangleleft$

$\sigma_D = \sigma_1 - \frac{1}{2}\sigma_1$

$\sigma_D = \frac{1}{2}\sigma_1 \blacktriangleleft$

$$\frac{1}{\rho'} = \frac{M'}{EI'} = \frac{2M_1}{E\frac{2}{3}bt^3} = \frac{3M_1}{Et^3} = \frac{1}{4} \frac{1}{\rho'}$$

(b) Final radius: $\frac{1}{\rho} = \frac{1}{\rho_1} - \frac{1}{\rho'} = \frac{1}{\rho_1} - \frac{1}{4} \frac{1}{\rho_1} = \frac{3}{4} \frac{1}{\rho_1}$

$\rho = \frac{4}{3}\rho_1 \blacktriangleleft$